

Supplementary Information

EcoNiche-Opt: a locked immune-ecology transcriptomic score for multicohort prediction of immune checkpoint blockade response

Pengyuan Xu^{1,3}, Guang Yang^{2,3}, Moyan Li^{3*}

¹ Department of Materials Science and Engineering, Monash University, Clayton, VIC 3800, Australia.

² School of Economics and Management, China University of Mining and Technology, Xuzhou 221116, Jiangsu, China.

³ Hong Kong University of Science and Technology (Guangzhou), Guangzhou 510000, China.

*Correspondence: moyanli@hkust-gz.edu.cn

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Supplementary Tables

Supplementary Table 1 | Data registry roles.

Lists all registered data sources, cohort roles, cancer types, treatment types, assay platforms, data-use categories, and pipeline status. This table supports Figure 1a-b.

Full tabulation (19 rows x 8 columns).

accession	name	layer	cancer_type	therapy	role	access_status	priority
GSE91061	Riaz_antiPD1_melanoma	A	melanoma	anti-PD1	discovery_or_lodo	public	high
GSE78220	Hugo_antiPD1_melanoma	A	melanoma	anti-PD1	classic_external_benchmark	public	high
GSE145996	JerbyArnon_or_related_melanoma_antiPD1	A	melanoma	anti-PD1	external_small	public	medium
GSE168204	Du_pathway_signature_melanoma_antiPD1	A	melanoma	anti-PD1_based	pathway_signature_comparator	public	high
GSE115821	Auslander_IMPRES_melanoma	A	melanoma	ICB	IMPRES_comparator	public	high
GSE93157	Prat_mixed_antiPD1_panel	A_B	melanoma_NSCLC_HNSC	pembrolizumab_or_nivolumab	mixed_tumor_external	public	medium
PRJEB23709	Gide_melanoma_ICB	A	melanoma	anti-PD1_or_combo	locked_external_if_accessible	unknown	high
Liu_DFCI_melanoma	Liu_NatMed_2019_melanoma	A	melanoma	anti-PD1	locked_external_if_accessible	controlled	high
GSE244982	acquired_resistance_melanoma_progression	A_C	melanoma	ICB_progression	mechanism_only	public	medium
GSE136961	NSCLC_antiPD1_DCB_NDB	B	NSCLC	anti-PD1	pan_cancer_external	public	high
GSE176307	urothelial_ICB_real_world	B	urothelial_cancer	ICB	pan_cancer_external	public	high
IMvigor210	IMvigor210_urothelial_atezolizumab	B	urothelial_cancer	anti-PDL1_atezolizumab	pan_cancer_external_large	unknown	high
GSE67501	RCC_nivolumab_small	B	renal_cell_carcinoma	nivolumab	pan_cancer_external_small	public	medium
GSE121810	glioblastoma_pembrolizumab	B	glioblastoma	pembrolizumab	pan_cancer_external_context_specific	public	medium
GSE140901	HCC_ICI_nCounter	B	hepatocellular_carcinoma	anti-PD1_or_anti-PDL1_based	pan_cancer_external	public	medium
GSE165252	PERFECT_esophageal_atezolizumab	B	esophageal_cancer	atezolizumab_plus_chemoradiotherapy	pan_cancer_dynamic	public	medium
GSE183924	esophageal_GEj_durvalumab	B	esophageal_or_GEj_adenocarcinoma	durvalumab	survival_external	public	medium
GSE115978	melanoma_ecosystem_scRNA	C	melanoma	mixed_or_ICB_context	single_cell_mechanism	public	high
GSE123139	melanoma_Tcell_states_scRNA	C	melanoma	ICB_context	single_cell_Tcell_validation	public	high

Supplementary Table 2 | Data access status.

Records the access status of each data source, including public, controlled, and unknown access. This table supports Figure 1d and ensures that controlled data are not replaced by simulated outputs.

Full tabulation (19 rows x 12 columns).

accession	name	layer	cancer_type	therapy	platform	role	access	access_status	priority	warning	is_valid
GSE91061	Riaz_antiPD1_melanoma	A	melanoma	anti-PD1	RNA-seq	discovery_or_lodo	public	public	high		True
GSE78220	Hugo_antiPD1_melanoma	A	melanoma	anti-PD1	RNA-seq	classic_external_benchmark	public	public	high		True
GSE145996	JerbyArnon_or_related_melanoma_antiPD1	A	melanoma	anti-PD1	RNA-seq	external_small	public	public	medium		True
GSE168204	Du_pathway_signature_melanoma_antiPD1	A	melanoma	anti-PD1_based	RNA-seq	pathway_signature_comparator	public	public	high		True
GSE115821	Auslander_IMPRES_melanoma	A	melanoma	ICB	expression	IMPRES_comparator	public	public	high		True
GSE93157	Prat_mixed_antiPD1_panel	A_B	melanoma_NSCLC_HNSC	pembrolizumab_or_nivolumab	NanoString_nCounter_immune_panel	mixed_tumor_external	public	public	medium		True
PRJEB23709	Gide_melanoma_ICB	A	melanoma	anti-PD1_or_combo	RNA-seq	locked_external_if_accessible	public_or_EBI_verify	unknown	high	ACCESS_UNKNOWN_VERIFY_BEFORE_USE	True
Liu_DFCI_melanoma	Liu_NatMed_2019_melanoma	A	melanoma	anti-PD1	RNA-seq_WES	locked_external_if_accessible	dbGaP_or_controlled_verify	controlled	high	ACCESS_RESTRICTED	True
GSE244982	acquired_resistance_melanoma_progression	A_C	melanoma	ICB_progression	RNA-seq	mechanism_only	public	public	medium		True

accession	name	layer	cancer_type	therapy	platform	role	access	access_status	priority	warning	is_valid
GSE136961	NSCLC_antiPD1_DCB_NDB	B	NSCLC	anti-PD1	expression	pan_cancer_external	public	public	high		True
GSE176307	urothelial_ICB_real_world	B	urothelial_cancer	ICB	RNA-seq	pan_cancer_external	public	public	high		True
IMvigor210	IMvigor210_urothelial_atezolizumab	B	urothelial_cancer	anti-PDL1_atezolizumab	RNA-seq	pan_cancer_external_large	public_or_package_verify	unknown	high	ACCESS_UNKNOWN_VERIFY_BEFORE_USE	True
GSE67501	RCC_nivolumab_small	B	renal_cell_carcinoma	nivolumab	expression	pan_cancer_external_small	public	public	medium		True
GSE121810	glioblastoma_pembrolizumab	B	glioblastoma	pembrolizumab	RNA-seq	pan_cancer_external_context_specific	public	public	medium		True
GSE140901	HCC_ICI_nCounter	B	hepatocellular_carcinoma	anti-PD1_or_anti-PDL1_based	NanoString_nCounter	pan_cancer_external	public	public	medium		True
GSE165252	PERFECT_esophageal_atezolizumab	B	esophageal_cancer	atezolizumab_plus_chemoradiotherapy	RNA-seq	pan_cancer_dynamic	public	public	medium		True
GSE183924	esophageal_GEJ_durvalumab	B	esophageal_or_GEJ_adenocarcinoma	durvalumab	RNA-seq	survival_external	public	public	medium		True
GSE115978	melanoma_ecosystem_scRNA	C	melanoma	mixed_or_ICB_context	scRNA-seq	single_cell_mechanism	public	public	high		True
GSE123139	melanoma_Tcell_states_scRNA	C	melanoma	ICB_context	scRNA-seq	single_cell_Tcell_validation	public	public	high		True

Supplementary Table 3 | Expression QC.

Reports n_samples, n_genes, source file, and QC status for each expression matrix. This table supports Figure 1b and Supplementary Figure 2.

Full tabulation (17 rows x 5 columns).

cohort	n_samples	n_genes	status	source_file
GSE91061	62	22187	processed	D:\EcoNiche-Opt\data\raw\GSE91061\suppl\GSE91061_BMS038109Sample.hg19KnownGene.fpkms.csv.gz
GSE78220	27	25265	processed	D:\EcoNiche-Opt\data\raw\GSE78220\suppl\GSE78220_PatientFPKM.xlsx
GSE145996	0	19734	metadata_needs_manual_curation	D:\EcoNiche-Opt\data\raw\GSE145996\suppl\GSE145996_Melanoma_Immune_FPKM.txt.gz
GSE168204	21	75251	processed	D:\EcoNiche-Opt\data\raw\GSE168204\suppl\GSE168204_MGH_counts.csv.gz
GSE115821	8	75249	processed	D:\EcoNiche-Opt\data\raw\GSE115821\suppl\GSE115821_MGH_counts.csv.gz
GSE93157	0	0	unusable_expression_shape	D:\EcoNiche-Opt\data\raw\GSE93157\suppl\GSE93157_raw_data_values.txt.gz
GSE244982	0	18759	metadata_needs_manual_curation	D:\EcoNiche-Opt\data\raw\GSE244982\suppl\GSE244982_ProcessedData_bulkRNAseq.txt.gz
GSE136961	21	390	processed	D:\EcoNiche-Opt\data\raw\GSE136961\suppl\GSE136961_TPM.tsv.gz
GSE176307	88	16383	processed	D:\EcoNiche-Opt\data\raw\GSE176307\suppl\GSE176307_BACI_log_trans_normalized_RNAseq.csv.gz
GSE67501	11	1	unusable_expression_shape	D:\EcoNiche-Opt\data\raw\GSE67501\suppl\GSE67501_Non-normalized_data.txt.gz
GSE121810	0	37036	metadata_needs_manual_curation	D:\EcoNiche-Opt\data\raw\GSE121810\suppl\GSE121810_Prins.PD1NeoAdjv.Jul2018.HUGO.PtID.xlsx
GSE140901	0	0	unusable_expression_shape	D:\EcoNiche-Opt\data\raw\GSE140901\suppl\GSE140901_processed_data.txt.gz
GSE165252	0	62300	metadata_needs_manual_curation	D:\EcoNiche-Opt\data\raw\GSE165252\suppl\GSE165252_vst_PERFECT.txt.gz
GSE183924	0	28052	metadata_needs_manual_curation	D:\EcoNiche-Opt\data\raw\GSE183924\suppl\GSE183924_FPKM.txt.gz
GSE123728	0	0	missing_expression_file	
GSE165745	0	0	missing_expression_file	
GSE122220	0	0	missing_expression_file	

Supplementary Table 4 | Manual curation evidence.

Records response labels, sample IDs, patient IDs, baseline/on-treatment status, treatment regimen, and evidence source. This table supports the main-text description of manual response-label curation.

Full tabulation (17 rows x 17 columns).

accession	name	cancer_type	therapy_registry	role	source_file	qc_status	n_metadata_rows	n_processed_samples	n_genes	response_raw_counts	processed_label_counts	evidence_field	decision	reason	next_action	geo_url
GSE91061	Riaz_antiPD1_melanoma	melanoma	anti-PD1	discovery_or_lodo	D:\EcoNiche-Opt\data\raw\GSE91061\suppl\GSE91061_BMS038109Sample.hg19KnownGene.fpkms.csv.gz	processed	109	49	22187	{"PD": 48, "PR": 23, "SD": 34, "nan": 4}	{"0.0": 10, "1.0": 39}	characteristics_ch1 visit_pre_or_on_treatment and response	usable_response_benchmark	Sample-level response labels and expression matrix are available after harmonization.	Keep in primary response benchmark if endpoint /setting matches registry role.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE91061
GSE78220	Hugo_antiPD1_melanoma	melanoma	anti-PD1	classic_external_benchmark	D:\EcoNiche-Opt\data\raw\GSE78220\suppl\GSE78220_PatientFPKM.xlsx	processed	28	27	25265	{"CR": 5, "PD": 13, "PR": 10}	{"0.0": 14, "1.0": 13}	clinical/FPKM workbook metadata response fields	usable_response_benchmark	Sample-level response labels and expression matrix are available after harmonization.	Keep in primary response benchmark if endpoint /setting matches registry role.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE78220
GSE145996	JerbyArnon_or_related_melanoma_antiPD1	melanoma	anti-PD1	external_small	D:\EcoNiche-Opt\data\raw\GSE145996\suppl\GSE145996_Melanoma_Immune_FPKM.txt.gz	processed	14	14	19734	{"nan": 14}	{"0": 8, "1": 6}	MDPI Cancers supplementary workbook: SuppTable1 Patient Number/Best Response plus SuppTable2 Tumor ID number/RNAseq; expression columns matched by MB tumor ID	usable_melanoma_response	Paper supplement maps RNA-seq tumor IDs to patient-level RECIST best response; expression columns match MB tumor IDs.	Include in melanoma endpoint - stratified analysis with explicit supplement citation/evidence.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE145996
GSE168204	Du_pathway_signature_melanoma_antiPD1	melanoma	anti-PD1_based	pathway_signature_comparator	D:\EcoNiche-Opt\data\raw\GSE168204\suppl\GSE168204_MGH_counts.csv.gz	processed	27	21	75251	{"NR": 18, "R": 9}	{"0.0": 9, "1.0": 12}	series/sample metadata response fields	usable_response_benchmark	Sample-level response labels and expression matrix are available after harmonization.	Keep in primary response benchmark if endpoint /setting matches registry role.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE168204
GSE115821	Auslander_IMPRES_melanoma	melanoma	ICB	IMPRES_comparator	D:\EcoNiche-Opt\data\raw\GSE115821\suppl\GSE115821_MGH_counts.csv.gz	processed	37	8	75249	{"NR": 34, "R": 3}	{"0.0": 2, "1.0": 6}	series/sample metadata response fields	usable_response_benchmark	Sample-level response labels and expression matrix are available after harmonization.	Keep in primary response benchmark if endpoint /setting matches registry role.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE115821
GSE93157	Prat_mixed_antiPD1_panel	melanoma_NSC LC_HNSC	pembrolizumab_or_nivolumab	mixed_tumor_external	D:\EcoNiche-Opt\data\raw\GSE93157\suppl\GSE93157_raw_data_values.txt.gz	processed	65	65	770	{"CR": 3, "PD": 29, "PR": 17, "SD": 16}	{"0.0": 20, "1.0": 45}	characteristics_ch1 best.resp/drug/biopsy; expression columns matched by GEO sample order	usable_pan_cancer_response	Sample-level response labels, therapy, and expression sample matching are traceable from GEO metadata/supplementary files.	Include in real processed data and stratified pan-cancer LODO.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE93157
GSE244982	acquired_resistance_melanoma_progression	melanoma	ICB_progression	mechanism_only	D:\EcoNiche-Opt\data\raw\GSE244982\suppl\GSE244982_ProcessedData_bulkRNAseq.txt.gz	metadata_needs_manual_curation	41	0	18759	{"nan": 41}	{}	response_raw harmonized from available sample	exclude_primary_response	Registry and sample	Keep for mechanism	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE244982

accession	name	cancer_type	therapy_registry	role	source_file	qc_status	n_metadata_rows	n_processed_samples	n_genes	response_raw_counts	processed_label_counts	evidence_field	decision	reason	next_action	geo_url
												metadata		metadata describe acquired resistance/ progression mechanism groups rather than primary binary ICB response.	analyses ; exclude from primary response LODO.	
GSE136961	NSCLC_antiPD1_DCB_NDB	NSCLC	anti-PD1	pan_cancer_external	D:\EcoNiche-Opt\data\raw\GSE136961\suppl\GSE136961_TPM.tsv.gz	processed	21	21	390	{"DCB": 9, "NDB": 12}	{"0.0": 9, "1.0": 12}	GEO sample title D/N durable clinical benefit encoding	usable_response_benchmark	Sample-level response labels and expression matrix are available after harmonization.	Keep in primary response benchmark if endpoint /setting matches registry role.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE136961
GSE176307	urothelial_ICB_real_world	urothelial_cancer	ICB	pan_cancer_external	D:\EcoNiche-Opt\data\raw\GSE176307\suppl\GSE176307_BACI_log_trans_normalized_RNAseq.csv.gz	processed	90	88	16383	{"CR": 7, "PD": 69, "PR": 9, "SD": 4, "nan": 1}	{"0.0": 16, "1.0": 72}	series/sample metadata response fields	usable_response_benchmark	Sample-level response labels and expression matrix are available after harmonization.	Keep in primary response benchmark if endpoint /setting matches registry role.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE176307
GSE67501	RCC_nivolumab_small	renal_cell_carcinoma	nivolumab	pan_cancer_external_small	D:\EcoNiche-Opt\data\raw\GSE67501\suppl\GSE67501_Non-normalized_data.txt.gz	processed	11	11	20819	{"NR": 7, "R": 4}	{"0.0": 4, "1.0": 7}	description (RCC sample ID); characteristics_ch1 response to anti-PD-1 nivolumab; GEO series design states pre-treatment tumors	usable_pan_cancer_response	Sample-level response labels, therapy, and expression sample matching are traceable from GEO metadata/supplementary files.	Include in real processed data and stratified pan-cancer LODO.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE67501
GSE121810	glioblastoma_pembrolizumab	glioblastoma	pembrolizumab	pan_cancer_external_context_specific	D:\EcoNiche-Opt\data\raw\GSE121810\suppl\GSE121810_Prins.PD1NeoAdjv.Jul2018.HUGO.PtID.xlsx	metadata_needs_manual_curation	29	0	37036	{"nan": 29}	{}	response_raw harmonized from available sample metadata	exclude_primary_response	GEO metadata describes neoadjuvant/adjuvant pembrolizumab treatment-arm and survival setting in recurrent glioblastoma, not per-sample RECIST response.	Use only as context-specific survival/mechanism cohort unless a traceable binary response table is added.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE121810
GSE140901	HCC_ICI_nCounter	hepatocellular_carcinoma	anti-PD1_or_anti-PDL1_based	pan_cancer_external	D:\EcoNiche-Opt\data\raw\GSE140901\suppl\GSE140901_processed_data.txt.gz	processed	24	24	784	{"PD": 8, "PR": 6, "SD": 10}	{"0.0": 6, "1.0": 18}	characteristics_ch1 best_response/clinical_benefit_response; title Sxx matched to expression column Sxx	usable_pan_cancer_response	Sample-level response labels, therapy, and expression sample matching are traceable from GEO metadata/s	Include in real processed data and stratified pan-cancer LODO.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE140901

accession	name	cancer_type	therapy_registry	role	source_file	qc_status	n_metadata_rows	n_processed_samples	n_genes	response_raw_counts	processed_label_counts	evidence_field	decision	reason	next_action	geo_url
														upplementary files.		
GSE165252	PERFECT_esophageal_atezolizumab	esophageal_cancer	atezolizumab_plus_chemoradiotherapy	pan_cancer_dynamic	D:\EcoNiche-Opt\data\raw\GSE165252\suppl\GSE165252_vst_PERFECT.txt.gz	processed	77	32	61475	{"NR": 48, "R": 23, "nan": 6}	{"0.0": 12, "1.0": 20}	characteristics_ch1 response; title sample_N baseline/on_treatment/resection; expression columns matched by GEO sample order	usable_secondary_dynamic_confounded	Binary pathologic response labels and baseline samples recovered, but therapy is atezolizumab plus chemoradiotherapy; keep as secondary/context-specific validation.	Use in pan-cancer/secondary analyses; report confounding explicitly.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165252
GSE183924	esophageal_GEJ_durvalumab	esophageal_or_GEJ_adenocarcinoma	durvalumab	survival_external	D:\EcoNiche-Opt\data\raw\GSE183924\suppl\GSE183924_FPKM.txt.gz	metadata_needs_manual_curation	37	0	28052	{"nan": 37}	{}	response_raw harmonized from available sample metadata	survival_only_not_response	Clinical workbook contains relapse/RFS/OS fields but no CR/PR/SD/PD or DCB/NDB response endpoint.	Use in survival analyses, not response classification.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE183924
GSE123728	melanoma_ICB_public_candidate	melanoma	ICB	candidate_external_requires_metadata_curation		missing_expression_file	0	0	0	{}	{}	response_raw harmonized from available sample metadata	missing_expression_file	No public expression matrix is present locally for this accession.	Download/verify expression data before curation.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE123728
GSE165745	melanoma_ICB_public_candidate	melanoma	ICB	candidate_external_requires_metadata_curation		missing_expression_file	0	0	0	{}	{}	response_raw harmonized from available sample metadata	missing_expression_file	No public expression matrix is present locally for this accession.	Download/verify expression data before curation.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165745
GSE12220	melanoma_ICB_public_candidate	melanoma	ICB	candidate_external_requires_metadata_curation		missing_expression_file	0	0	0	{}	{}	response_raw harmonized from available sample metadata	missing_expression_file	No public expression matrix is present locally for this accession.	Download/verify expression data before curation.	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE12220

Supplementary Table 5 | External cohort availability assessment.

Records availability, missing data, and access boundaries for candidate external cohorts. This table supports the external-validation design in Figures 1 and 5.

Full tabulation (9 rows x 16 columns).

accession	priority	status	evidence_strength	local_expression	local_response_map	local_external_evidence	source_url	completed_action	remaining_gap	next_action	local_expression_exists	local_response_map_exists	local_external_evidence_exists	local_expression_bytes	local_response_map_bytes
GSE145996	high	COMPLETED_INCLUDED	strong_sample_level	data/processed/bulk/GSE145996.expr.tsv	data/processed/bulk/GSE145996.metadata.tsv	data/external/GSE145996/cancers-12-01943-s001/cancers-836356-supplementary-final/cancers-836356-supplementary-revised.xlsx	https://res.mdpi.com/d_attachment/cancers/cancers-12-01943/article_deploy/cancers-12-01943-s001.zip	Downloaded MDPI supplement; mapped expression Sample_XX_	Small RNA-seq subset only; SD handling must remain endpoint-stratified	Use in melanoma/end point-stratified analysis; cite exact supplement table evidenc	True	True	True	4566031.0	5834.0

accession	priority	status	evidence_strength	local_expression	local_response_map	local_external_evidence	source_url	completed_action	remaining_gap	next_action	local_expression_exists	local_response_map_exists	local_external_evidence_exists	local_expression_bytes	local_response_map_bytes
								MB_Y YYY to SuppT able2 Tumor ID numbe r and SuppT able1 Best Respo nse.	.	e.					
PR JE B2 370 9	hi g h	COMPLETED_INCL UD_SPLIT_PRE	strong_exp ression_plu s_clinical_ map	data/processed/bulk/PRJEB23709_PD1_PRE.expr.tsv;data/proce ssed/bulk/PRJEB23709_COMBO _PRE.expr.tsv	data/processed/bulk/PRJEB23709_PD1_PRE.metadata .tsv;data/processed/bulk/PRJEB23709_COMBO_PRE_ metadata.tsv;data/external/PRJEB23709/gide_prjeb237 09_response_map.tsv	data/external/TIGER_PRJEB23709/Melano ma- PRJEB23709.expression.tsv;data/external/T IGER_PRJEB23709/Melanoma- PRJEB23709.clinical.tsv;data/external/PRJ EB23709/mmc2.xlsx;data/external/PRJEB2 3709/mmc3.xlsx;data/raw/PRJEB23709/ena _run_metadata.tsv	http://tiger.canceromics.org/tiger /Download/immunotherapy/expr ession/tsv/Melanoma- PRJEB23709.Response.tsv;http ://tiger.canceromics.org/tiger/Do wnload/immunotherapy/clinical/t sv/Melanoma- PRJEB23709.Response.tsv;http s://ars.els- cdn.com/content/image/1-s2.0- S1535610819300376- mmc2.xlsx;https://ars.els- cdn.com/content/image/1-s2.0- S1535610819300376- mmc3.xlsx	Downl oaded TIGER proces sed expres sion/cl inical TSV; match ed ENA run access ion to Gide respon se map; split PRE anti- PD1 monot herapy and combo therap y into separa te proces sed cohort s.	TIGER expressi on is third- party process ed rather than locally quantifie d FASTQ; describe provena nce explicitly and keep combo therapy separat e from melano ma monoth erapy primary.	Use PRJEB2 3709_P D1_PR E in high-evidenc e melano ma primary/ core analysis ; use PRJEB2 3709_C OMBO_ PRE only as seconda ry/confo unded combo transfer.	True	True	True	430036 82.0	98161.0
IMv igor 210	hi g h	EXPRESSION_AND_C LINICAL_PENDING_EN VIRONMENT_BLOCKE D	known_pub lic_packag e_not_local			data/raw/IMvigor210/status.tsv	https://bioconductor.org/package s/IMvigor210CoreBiologies/	Recor ded access blocke r; Rscript /Bioco nducto r packa ge execut ion is unavai lable in this enviro nment.	Need expressi on matrix plus respons e/therap y/sampl e annotati ons from Biocond uctor/Ex perimen tHub or a verified mirror.	Install/e nable Rscript or fetch package RData/R DS assets via a direct verified source; only then add as urothelia l anti- PDL1 validatio n.	False	False	True	0.0	0.0
GS E1 237 28	m e di um	RAW_AND_RESPONS E_PENDING	not_local				https://www.ncbi.nlm.nih.gov/ge o/query/acc.cgi?acc=GSE12372 8	Regist ry candid ate identifi ed; no local raw directo ry or expres	Need verify whether tumor bulk expressi on and binary ICB respons e exist	Downlo ad GEO supplem entary files and sample metadat a; include only if expressi	False	False	False	0.0	0.0

acc ession	pr iority	status	evidence_ strength	local_expression	local_response_map	local_external_evidence	source_url	compl eted_ action	remaini ng_gap	next_ac tion	local_e xpression_ex ists	local_re sponse_map_e xists	local_ext ernal_evid ence_e xists	local_ex pression_b ytes	local_re sponse_map_b ytes
								sion matrix found.	at sample level.	on columns can be matched to respons e evidenc e.					
GS E1 657 45	m e d i u m	RAW_AND_RESPONS E_PENDING	not_local				https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165745	Regist ry candid ate identifi ed; no local raw directo ry or expres sion matrix found.	Need verify expressi on modality . baseline status, therapy, and binary respons e.	Downlo ad GEO supplem entary files and clinical/s ample annotati ons; reject if only single- cell/mec hanism without patient respons e.	False	False	False	0.0	0.0
GS E1 222 20	m e d i u m	RAW_AND_RESPONS E_PENDING	not_local				https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE12220	Regist ry candid ate identifi ed; no local raw directo ry or expres sion matrix found.	Need verify sample- level expressi on and respons e labels.	Downlo ad GEO supplem entary files and audit sample- to- patient/r esponse mapping before use.	False	False	False	0.0	0.0
GS E9 315 7	m e d i u m	INCLUDED_ORDER_M ATCH_NEEDS_EXTR A_SUPPLEMENT_CON FIRMATION	usable_but _order_mat ch	data/processed/bulk/GSE93157. expr.tsv	data/processed/bulk/GSE93157.metadata.tsv	data/metadata/manual_response_curation.t sv	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE93157	Includ ed throug h GEO sampl e- order expres sion/m etadat a match with best respon se fields.	A supplem entary sample map would further harden the expressi on- column to metadat a match.	Search article supplem ent for a NanoStr ing sample sheet; retain current cohort but flag order- match evidenc e in Methods .	True	True	True	891197 .0	136853. 0
GS E1 652 52	m e d i u m	INCLUDED_SECONDAR Y_CONFOUNDED_O RDER_MATCH	usable_sec ondary_but _order_mat ch	data/processed/bulk/GSE165252 .expr.tsv	data/processed/bulk/GSE165252.metadata.tsv	data/metadata/manual_response_curation.t sv	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165252	Includ ed as second ary transfe r cohort with baseli ne sampl es and respon se labels.	Treatme nt combine s chemor adiother apy and atezoliz umab; expressi on columns are GEO- order matched .	Keep out of primary melano ma/pan- cancer headline ; use as seconda ry/confo unded transfer validatio n.	True	True	True	364213 38.0	34809.0
phs 000	hi g	COMPLETED_STRESS _TEST_SECONDARY	TIGER_pro cessed_pat	data/processed/bulk/PHS000452 _LIU_LIKE_PRE.expr.tsv	data/processed/bulk/PHS000452_LIU_LIKE_PRE.meta data.tsv	data/external/TIGER_phs000452/Melanoma - data.tsv	http://tiger.canceromics.org/tiger/Download/immunotherapy/expr	Downl oaded	Baseline /timepoi	Keep as seconda	True	True	True	627795 56.0	98175.0

accession	priority	status	evidence_strength	local_expression	local_response_map	local_external_evidence	source_url	completed_action	remaining_gap	next_action	local_expression_exists	local_response_map_exists	local_external_evidence_exists	local_expression_bytes	local_response_map_bytes
452	h		ient_like_subset			phs000452.expression.tsv;data/external/TIGER_phs000452/Melanoma-phs000452.clinical.tsv	ession/tsv/Melanoma-phs000452.Response.tsv;http://tiger.canceromics.org/tiger/Download/immunotherapy/clinical/tsv/Melanoma-phs000452.Response.tsv	TIGER processed expression/clinical; retained 121 Patient-like anti-PD1 rows and excluded 32 IPI-like rows from primary.	nt and source provenance are inferred from TIGER dataset context; performance as a primary melanoma validation layer is weak.	ry/stress-test until source/timepoint evidence is hardened; do not use for primary melanoma headline.					

Supplementary Table 6 | Endpoint label sensitivity.

Lists n_used, n_dropped, responders, and nonresponders for each cohort under strict RECIST, primary RECIST, and clinical-benefit endpoints. This table supports Figure 4a and Supplementary Figure 5.

Full tabulation (42 rows x 8 columns).

endpoint	cohort	n_total_with_response_raw	n_used	n_dropped	n_responders	n_nonresponders	response_raw_counts
strict_recist	GSE115821	8	8	0	2	6	NR:6;R:2
strict_recist	GSE136961	21	21	0	9	12	DCB:9;NDB:12
strict_recist	GSE140901	24	14	10	6	8	PD:8;PR:6;SD:10
strict_recist	GSE145996	14	13	1	8	5	CR:2;PD:5;PR:6;SD:1
strict_recist	GSE165252	32	32	0	12	20	NR:20;R:12
strict_recist	GSE168204	21	21	0	9	12	NR:12;R:9
strict_recist	GSE176307	88	84	4	16	68	CR:7;PD:68;PR:9;SD:4
strict_recist	GSE67501	11	11	0	4	7	NR:7;R:4
strict_recist	GSE78220	27	27	0	14	13	CR:4;PD:13;PR:10
strict_recist	GSE91061	49	33	16	10	23	PD:23;PR:10;SD:16
strict_recist	GSE93157	65	49	16	20	29	CR:3;PD:29;PR:17;SD:16
strict_recist	PHS000452_LIU_LIKE_PRE	121	105	16	50	55	CR:16;MR:2;PD:55;PR:32;SD:16
strict_recist	PRJEB23709_COMBO_PRE	32	27	5	21	6	CR:10;PD:6;PR:11;SD:5
strict_recist	PRJEB23709_PD1_PRE	41	35	6	19	16	CR:4;PD:16;PR:15;SD:6
primary_recist	GSE115821	8	8	0	2	6	NR:6;R:2
primary_recist	GSE136961	21	21	0	9	12	DCB:9;NDB:12
primary_recist	GSE140901	24	24	0	6	18	PD:8;PR:6;SD:10
primary_recist	GSE145996	14	14	0	8	6	CR:2;PD:5;PR:6;SD:1
primary_recist	GSE165252	32	32	0	12	20	NR:20;R:12
primary_recist	GSE168204	21	21	0	9	12	NR:12;R:9
primary_recist	GSE176307	88	88	0	16	72	CR:7;PD:68;PR:9;SD:4
primary_recist	GSE67501	11	11	0	4	7	NR:7;R:4
primary_recist	GSE78220	27	27	0	14	13	CR:4;PD:13;PR:10
primary_recist	GSE91061	49	49	0	10	39	PD:23;PR:10;SD:16
primary_recist	GSE93157	65	65	0	20	45	CR:3;PD:29;PR:17;SD:16

endpoint	cohort	n_total_with_response_raw	n_used	n_dropped	n_responders	n_nonresponders	response_raw_counts
primary_recist	PHS000452_LIU_LIKE_PRE	121	121	0	50	71	CR:16;MR:2;PD:55;PR:32;SD:16
primary_recist	PRJEB23709_COMBO_PRE	32	32	0	21	11	CR:10;PD:6;PR:11;SD:5
primary_recist	PRJEB23709_PD1_PRE	41	41	0	19	22	CR:4;PD:16;PR:15;SD:6
clinical_benefit	GSE115821	8	8	0	2	6	NR:6;R:2
clinical_benefit	GSE136961	21	21	0	9	12	DCB:9;NDB:12
clinical_benefit	GSE140901	24	24	0	16	8	PD:8;PR:6;SD:10
clinical_benefit	GSE145996	14	14	0	9	5	CR:2;PD:5;PR:6;SD:1
clinical_benefit	GSE165252	32	32	0	12	20	NR:20;R:12
clinical_benefit	GSE168204	21	21	0	9	12	NR:12;R:9
clinical_benefit	GSE176307	88	88	0	20	68	CR:7;PD:68;PR:9;SD:4
clinical_benefit	GSE67501	11	11	0	4	7	NR:7;R:4
clinical_benefit	GSE78220	27	27	0	14	13	CR:4;PD:13;PR:10
clinical_benefit	GSE91061	49	49	0	26	23	PD:23;PR:10;SD:16
clinical_benefit	GSE93157	65	65	0	36	29	CR:3;PD:29;PR:17;SD:16
clinical_benefit	PHS000452_LIU_LIKE_PRE	121	121	0	66	55	CR:16;MR:2;PD:55;PR:32;SD:16
clinical_benefit	PRJEB23709_COMBO_PRE	32	32	0	26	6	CR:10;PD:6;PR:11;SD:5
clinical_benefit	PRJEB23709_PD1_PRE	41	41	0	25	16	CR:4;PD:16;PR:15;SD:6

Supplementary Table 7 | Module gene sets.

Lists EcoNiche-Opt prior modules, gene symbols, module weights, and score directions. This table supports Figure 2a, Figure 6d, and Figure 7b.

Full tabulation (63 rows x 4 columns).

module	gene_symbol	module_weight	score_direction
ifn_t_cell_inflamed	IFNG	1.0	response_high
ifn_t_cell_inflamed	CXCL9	1.0	response_high
ifn_t_cell_inflamed	CXCL10	1.0	response_high
ifn_t_cell_inflamed	CXCL11	1.0	response_high
ifn_t_cell_inflamed	STAT1	1.0	response_high
ifn_t_cell_inflamed	IDO1	1.0	response_high
ifn_t_cell_inflamed	GBP1	1.0	response_high
ifn_t_cell_inflamed	CXCR3	1.0	response_high
ifn_t_cell_inflamed	CCL5	1.0	response_high
ifn_t_cell_inflamed	CD274	1.0	response_high
ifn_t_cell_inflamed	PDCD1LG2	1.0	response_high
cytotoxic_cd8	CD8A	0.5	response_high
cytotoxic_cd8	CD8B	0.5	response_high
cytotoxic_cd8	GZMA	0.5	response_high
cytotoxic_cd8	GZMB	0.5	response_high
cytotoxic_cd8	GZMH	0.5	response_high
cytotoxic_cd8	PRF1	0.5	response_high
cytotoxic_cd8	NKG7	0.5	response_high
cytotoxic_cd8	GNLY	0.5	response_high
exhaustion_checkpoint	PDCD1	0.25	response_high
exhaustion_checkpoint	CTLA4	0.25	response_high
exhaustion_checkpoint	LAG3	0.25	response_high

module	gene_symbol	module_weight	score_direction
exhaustion_checkpoint	HAVCR2	0.25	response_high
exhaustion_checkpoint	TIGIT	0.25	response_high
exhaustion_checkpoint	TOX	0.25	response_high
exhaustion_checkpoint	CXCL13	0.25	response_high
antigen_presentation	HLA-A	0.5	response_high
antigen_presentation	HLA-B	0.5	response_high
antigen_presentation	HLA-C	0.5	response_high
antigen_presentation	HLA-DRA	0.5	response_high
antigen_presentation	HLA-DRB1	0.5	response_high
antigen_presentation	B2M	0.5	response_high
antigen_presentation	TAP1	0.5	response_high
antigen_presentation	TAP2	0.5	response_high
antigen_presentation	PSMB8	0.5	response_high
antigen_presentation	PSMB9	0.5	response_high
myeloid_suppression	CD68	-0.5	nonresponse_high
myeloid_suppression	CD163	-0.5	nonresponse_high
myeloid_suppression	MRC1	-0.5	nonresponse_high
myeloid_suppression	CSF1R	-0.5	nonresponse_high
myeloid_suppression	ITGAM	-0.5	nonresponse_high
myeloid_suppression	S100A8	-0.5	nonresponse_high
myeloid_suppression	S100A9	-0.5	nonresponse_high
myeloid_suppression	IL10	-0.5	nonresponse_high
myeloid_suppression	TGFB1	-0.5	nonresponse_high
stromal_exclusion	COL1A1	-0.5	nonresponse_high
stromal_exclusion	COL1A2	-0.5	nonresponse_high
stromal_exclusion	COL3A1	-0.5	nonresponse_high
stromal_exclusion	FN1	-0.5	nonresponse_high
stromal_exclusion	ACTA2	-0.5	nonresponse_high
stromal_exclusion	VIM	-0.5	nonresponse_high
stromal_exclusion	TGFBI	-0.5	nonresponse_high
stromal_exclusion	POSTN	-0.5	nonresponse_high
stromal_exclusion	LOXL2	-0.5	nonresponse_high
trm_tls	ITGAE	0.25	response_high
trm_tls	CD69	0.25	response_high
trm_tls	CXCR6	0.25	response_high
trm_tls	CXCL13	0.25	response_high
trm_tls	ZNF683	0.25	response_high
trm_tls	MS4A1	0.25	response_high
trm_tls	CD79A	0.25	response_high
trm_tls	BANK1	0.25	response_high
trm_tls	LTB	0.25	response_high

Supplementary Table 8 | Ecological edges.

Lists optimized ecological interaction edges, including source_state, target_state, gene_a, gene_b, and edge_class. This table supports Figure 2d and Figure 6b.

Full tabulation (885 rows x 11 columns).

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector HLA-A B2M pathway	antigen_present ation_mhc	tnk_effector	HLA-A	B2M	pathway
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc HLA-A PSMB8 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	PSMB8	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion PSMB8 COL1A1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	PSMB8	COL1A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion HLA-A COL1A1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-A	COL1A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression PSMB8 CD163 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	PSMB8	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression HLA-A CD163 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	HLA-A	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction PSMB8 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB8	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction HLA-A PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	HLA-A	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector PSMB8 IFNG coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB8	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector HLA-A IFNG coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation PSMB8 IGKV4-1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	PSMB8	IGKV4-1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation PSMB8 VIM coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	PSMB8	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 LOXL2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	LOXL2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression COL1A1 CD163 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	COL1A1	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression LOXL2 CD163 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOXL2	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction APOC1,APOC1P1 ATP5E,SLMO2,SLMO2-ATP5E coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	APOC1,APOC1P1	ATP5E,SLMO2,SLMO2-ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction APOC1,APOC1P1 SLC13A3,TP53RK coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	APOC1,APOC1P1	SLC13A3,TP53RK	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector APOC1,APOC1P1 IFNG coexpression_network	caf_ecm_exclus ion	tnk_effector	APOC1,APOC1P1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector COL1A1 IFNG coexpression_network	caf_ecm_exclus ion	tnk_effector	COL1A1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation COL1A1 VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	COL1A1	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation APOC1,APOC1P1 VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	APOC1,APOC1P1	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression IGHV6-1 TRAJ6 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGHV6-1	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression RILP,SCARF1 S100A9 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	RILP,SCARF1	S100A9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction S100A9 ATP5E,SLMO2,SLMO2-ATP5E coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	S100A9	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction RILP,SCARF1 MIR143,MIR143HG coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	RILP,SCARF1	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector RILP,SCARF1 CUZD1,FAM24B,FAM24B- CUZD1 coexpression_network	myeloid_suppre ssion	tnk_effector	RILP,SCARF1	CUZD1,FAM24B,FAM24B- CUZD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector CD163 IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	CD163	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation RILP,SCARF1 RILP,SCARF1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	RILP,SCARF1	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation TRAJ6 TRAJ6 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	TRAJ6	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction ATP5E,SLMO2,SLMO2- ATP5E SLC13A3,TP53RK coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	ATP5E,SLMO2,SLMO2- ATP5E	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction ATP5E,SLMO2,SLMO2- ATP5E MIR143,MIR143HG coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	ATP5E,SLMO2,SLMO2- ATP5E	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 IFNG coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 GNLY coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	GNLY	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation PDCD1 TRAJ1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation PDCD1 IGKV4-1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	IGKV4-1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CXCL11 IFNG coexpression_network	tnk_effector	tnk_effector	CXCL11	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector GNLY IFNG coexpression_network	tnk_effector	tnk_effector	GNLY	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation CUZD1,FAM24B,FAM24B-CUZD1 CUZD1,FAM24B,FAM24B- CUZD1 coexpression_network	tnk_effector	tumor_dediffe rentiation	CUZD1,FAM24B,FAM24B- CUZD1	CUZD1,FAM24B,FAM24B- CUZD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation IFNG IGKV4-1 coexpression_network	tnk_effector	tumor_dediffe rentiation	IFNG	IGKV4-1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation IGHV6-1 IGKV4-1 coexpression_network	tumor_dediffe rentiation	tumor_dediffe rentiation	IGHV6-1	IGKV4-1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation CUZD1,FAM24B,FAM24B- CUZD1 RILP,SCARF1 coexpression_network	tumor_dediffe rentiation	tumor_dediffe rentiation	CUZD1,FAM24B,FAM24B- CUZD1	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector HLA-A B2M pathway	antigen_present ation_mhc	tnk_effector	HLA-A	B2M	pathway
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tccll_dysfunction PDCD1 LAG3 checkpoint	tccll_dysfunctio n	tccll_dysfunctio n	PDCD1	LAG3	checkpoint
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression S100A8 S100A9 network	myeloid_suppre ssion	myeloid_suppre ssion	S100A8	S100A9	network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc HLA-A TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	TAP1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc TAP1 TRAJ6 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	TAP1	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion RPS8,SNORD55 MIR548N,OSBPL6 coexpression_netw ork	antigen_present ation_mhc	caf_ecm_exclus ion	RPS8,SNORD55	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion TAP1 COL3A1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TAP1	COL3A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression TRAJ6 AL049794 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	TRAJ6	AL049794	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression TAP1 S100A9 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	TAP1	S100A9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tccll_dysfunction TAP1 TOX coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	TAP1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tccll_dysfunction TAP1 HAVCR2 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	TAP1	HAVCR2	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector TAP1 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	TAP1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector RPS8,SNORD55 MIR548N,OSBPL6 coexpression_network	antigen_present ation_mhc	tnk_effector	RPS8,SNORD55	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation TRAJ6 TRAJ1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TRAJ6	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation TAP1 TRAJ1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP1	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion COL1A2 COL3A1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A2	COL3A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 COL1A2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	COL1A2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression MIR548N,OSBPL6 S100A8 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	MIR548N,OSBPL6	S100A8	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression MIR548N,OSBPL6 S100A9 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	MIR548N,OSBPL6	S100A9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction COL1A2 TOX coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	COL1A2	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction COL1A1 TOX coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	COL1A1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector MIR548N,OSBPL6 MIR548N,OSBPL6 coexpression_network	caf_ecm_exclus ion	tnk_effector	MIR548N,OSBPL6	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector CXCL12 GZMA coexpression_network	caf_ecm_exclus ion	tnk_effector	CXCL12	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation COL3A1 VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	COL3A1	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation COL1A2 VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	COL1A2	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression S100A8 S100A9 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	S100A8	S100A9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction S100A8 RILP,SCARF1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	S100A8	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction S100A9 RILP,SCARF1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	S100A9	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector S100A8 RILP,SCARF1 coexpression_network	myeloid_suppre ssion	tnk_effector	S100A8	RILP,SCARF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector S100A8 MIR548N,OSBPL6 coexpression_network	myeloid_suppre ssion	tnk_effector	S100A8	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation AL049794 TRAJ1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	AL049794	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation AL049794 IGHD2-15 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	AL049794	IGHD2-15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tccl_dysfunction LAG3 PDCD1 coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tccl_dysfunction HAVCR2 TOX coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	HAVCR2	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tnk_effector RILP,SCARF1 RILP,SCARF1 coexpression_network	tccl_dysfunctio n	tnk_effector	RILP,SCARF1	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tccl_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tumor_dedifferentiation PDCD1 TRAJ1 coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	PDCD1	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tumor_dedifferentiation TOX TRAJ1 coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	TOX	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CD8A GZMA coexpression_network	tnk_effector	tnk_effector	CD8A	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CD8A IFNG coexpression_network	tnk_effector	tnk_effector	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation CD8A TRAJ1 coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8A	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation GZMA TRAJ1 coexpression_network	tnk_effector	tumor_dediffere ntiation	GZMA	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation IGHD2-15 TRAJ1 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IGHD2-15	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation NGFR VIM coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	NGFR	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-A B2M pathway	antigen_present ation_mhc	tnk_effector	HLA-A	B2M	pathway
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FN1 network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FN1	network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression S100A8 S100A9 network	myeloid_suppre ssion	myeloid_suppre ssion	S100A8	S100A9	network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-A HLA-B coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	HLA-B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-B PSMB9 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-B	PSMB9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion AL049794 AL049794 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	AL049794	AL049794	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion TRAJ10 TRAJ6 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TRAJ10	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression MIR548N,OSBPL6 MIR548N,OSBPL6 coexpression_n etwork	antigen_present ation_mhc	myeloid_suppre ssion	MIR548N,OSBPL6	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression TAP2 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	TAP2	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction MIR548N,OSBPL6 MIR548N,OSBPL6 coexpression_netwo rk	antigen_present ation_mhc	tcell_dysfunctio n	MIR548N,OSBPL6	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction MIR548N,OSBPL6 RPS8,SNORD55 coexpression_networ k	antigen_present ation_mhc	tcell_dysfunctio n	MIR548N,OSBPL6	RPS8,SNORD55	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-B PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-B	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-B IFNG coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-B	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation TAP2 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP2	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation MIR548N,OSBPL6 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	MIR548N,OSBPL6	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 POSTN coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	POSTN	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FN1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression COL1A1 CSF1R coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	COL1A1	CSF1R	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression FN1 CSF1R coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	FN1	CSF1R	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction TRAJ6 TIGIT coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	TIGIT	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction TRAJ6 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector TRAJ6 IFNG coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ6	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector TRAJ6 PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ6	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation FN1 VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	FN1	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression S100A8 S100A9 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	S100A8	S100A9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression ATP5E,SLMO2,SLMO2- ATP5E S100A8 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	ATP5E,SLMO2,SLMO2- ATP5E	S100A8	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction MIR548N,OSBPL6 MIR548N,OSBPL6 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	MIR548N,OSBPL6	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction ATP5E,SLMO2,SLMO2-ATP5E ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	ATP5E,SLMO2,SLMO2- ATP5E	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector CSF1R IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	CSF1R	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector CSF1R GZMA coexpression_network	myeloid_suppre ssion	tnk_effector	CSF1R	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation ATP5E,SLMO2,SLMO2-ATP5E ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	ATP5E,SLMO2,SLMO2- ATP5E	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation S100A8 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	S100A8	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction PDCD1 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TIGIT	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction MIR548N,OSBPL6 RPS8,SNORD55 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	MIR548N,OSBPL6	RPS8,SNORD55	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 GZMA coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation ATP5E,SLMO2,SLMO2-ATP5E ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	ATP5E,SLMO2,SLMO2- ATP5E	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccll_dysfunction tumor_dedifferentiation RPS8,SNORD55 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	tccll_dysfunctio n	tumor_dediffere ntiation	RPS8,SNORD55	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector GZMA PRF1 coexpression_network	tnk_effector	tnk_effector	GZMA	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector GZMA IFNG coexpression_network	tnk_effector	tnk_effector	GZMA	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation IFNG ATP5E,SLMO2,SLMO2-ATP5E coexpression_network	tnk_effector	tumor_dediffere ntiation	IFNG	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tccll_dysfunction tccll_dysfunction PDCD1 LAG3 checkpoint	tccll_dysfunctio n	tccll_dysfunctio n	PDCD1	LAG3	checkpoint
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc HLA-A PSMB8 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	PSMB8	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc PSMB8 TAP2 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB8	TAP2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion HLA-A LAG3 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-A	LAG3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion PSMB8 LAG3 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	PSMB8	LAG3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression HLA-A PDCD1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	HLA-A	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression PSMB8 PDCD1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	PSMB8	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tccll_dysfunction HLA-A PDCD1 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	HLA-A	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tccll_dysfunction PSMB8 PDCD1 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	PSMB8	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector HLA-A PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-A	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector PSMB8 PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB8	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion FAP FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FAP	FN1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion FAP TGFB1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FAP	TGFB1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression TGFB1 TGFB1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	TGFB1	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression LAG3 PDCD1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction LOC101928460 GPR111 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	LOC101928460	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction LAG3 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector LOC101928460 BCMO1 coexpression_network	caf_ecm_exclus ion	tnk_effector	LOC101928460	BCMO1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector LAG3 PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	LAG3	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation LAG3 CASC14,CASC15 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	LAG3	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation LOC101928460 CASC14,CASC15 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	LOC101928460	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression PDCD1 TGFB1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	PDCD1	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG TGFB1 co expression_network	myeloid_suppre ssion	myeloid_suppre ssion	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction PDCD1 PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	PDCD1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction PDCD1 CXCL13 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	PDCD1	CXCL13	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector PDCD1 PRF1 coexpression_network	myeloid_suppre ssion	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector PDCD1 CD8B coexpression_network	myeloid_suppre ssion	tnk_effector	PDCD1	CD8B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG LOC101 928738 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	LOC101928738	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG CASC14 ,CASC15 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction CXCL13 IDO1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CXCL13	IDO1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction CXCL13 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CXCL13	PDCD1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector GPR111 BCMO1 coexpression_network	tcell_dysfunctio n	tnk_effector	GPR111	BCMO1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation LOC101927217 CASC14,CASC15 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	LOC101927217	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation GPR111 MITF coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	GPR111	MITF	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CD8B PRF1 coexpression_network	tnk_effector	tnk_effector	CD8B	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector IFNG PRF1 coexpression_network	tnk_effector	tnk_effector	IFNG	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation BCMO1 MITF coexpression_network	tnk_effector	tumor_dediffere ntiation	BCMO1	MITF	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation BCMO1 CASC14,CASC15 coexpression_network	tnk_effector	tumor_dediffere ntiation	BCMO1	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 MITF coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	MITF	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 VIM coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc HLA-A HLA-C coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	HLA-C	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc HLA-A PSMB9 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	PSMB9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion LOC101928303 LOC101928303 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	LOC101928303	LOC101928303	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion GPR111 GPR111 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	GPR111	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression LOC101928303 LOC101928303 coexpression_networ k	antigen_present ation_mhc	myeloid_suppre ssion	LOC101928303	LOC101928303	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression GPR111 BCMO1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	GPR111	BCMO1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction LOC101928372 LOC101928372 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	LOC101928372	LOC101928372	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction GPR111 GPR111 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	GPR111	GPR111	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector LOC101928372 LOC101928372 coexpression_network	antigen_present ation_mhc	tnk_effector	LOC101928372	LOC101928372	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector HLA-DRA PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-DRA	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation LOC101927322 CASC14,CASC15 coexpression_net work	antigen_present ation_mhc	tumor_dediffere ntiation	LOC101927322	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation LOC101928372 LOC101926975 coexpression_netw ork	antigen_present ation_mhc	tumor_dediffere ntiation	LOC101928372	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion FN1 LOXL2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FN1	LOXL2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion PRF1 TGFB1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	PRF1	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression TGFB1 TGFB1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	TGFB1	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression LOC101928303 LOC101928303 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOC101928303	LOC101928303	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction GPR111 GPR111 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	GPR111	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction GPR111 BCMO1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	GPR111	BCMO1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector PRF1 PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	PRF1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector PRF1 GZMA coexpression_network	caf_ecm_exclus ion	tnk_effector	PRF1	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation GPR111 LOC101926975 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	GPR111	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation GPR111 LOC441178 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	GPR111	LOC441178	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression MRC1 PDCD1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	MRC1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression BCMO1 LOC441178 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	BCMO1	LOC441178	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction PDCD1 PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	PDCD1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction BCMO1 BCMO1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	BCMO1	BCMO1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector PDCD1 PRF1 coexpression_network	myeloid_suppre ssion	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector PDCD1 GZMA coexpression_network	myeloid_suppre ssion	tnk_effector	PDCD1	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation LOC441178 LOC441178 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	LOC441178	LOC441178	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation BCMO1 LOC101926975 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	BCMO1	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tccl_dysfunction BCMO1 GPR111 coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	BCMO1	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tccl_dysfunction PDCD1 TOX coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	PDCD1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tnk_effector LOC101928372 LOC101928372 coexpression_network	tccl_dysfunctio n	tnk_effector	LOC101928372	LOC101928372	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tccl_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tumor_dedifferentiation CASC14,CASC15 CASC14,CASC15 coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	CASC14,CASC15	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tumor_dedifferentiation LOC101928372 LOC101926975 coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	LOC101928372	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CCL5 GZMA coexpression_network	tnk_effector	tnk_effector	CCL5	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector GZMA PRF1 coexpression_network	tnk_effector	tnk_effector	GZMA	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation LOC101926975 LOC101926975 coexpression_network	tnk_effector	tumor_dediffere ntiation	LOC101926975	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation LOC101928372 LOC101926975 coexpression_network	tnk_effector	tumor_dediffere ntiation	LOC101928372	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 LOC101926975 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	LOC101926975	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 VIM coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tccl_dysfunction PDCD1 LAG3 checkpoint	tccl_dysfunctio n	tccl_dysfunctio n	PDCD1	LAG3	checkpoint
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-A HLA-B coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	HLA-B	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-A TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	TAP1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion HLA-B TGFB1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-B	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion HLA-A COL1A1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-A	COL1A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG MIR4 763,MIRLET7A3,MIRLET7B,MIRLET7BHG coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression HLA-A HLA-A coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	HLA-A	HLA-A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction TAP1 TOX coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	TAP1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction TAP1 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	TAP1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-B PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-B	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-B CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-B	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation TAP1 PDCD1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation HLA-A PDCD1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	HLA-A	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 COL3A1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	COL3A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 TGFB1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression TGFB1 TGFB1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	TGFB1	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression LOC441178 LOC441178 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOC441178	LOC441178	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction COL1A1 TOX coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	COL1A1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction COL3A1 TOX coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	COL3A1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector LOC101928372 GPR111 coexpression_network	caf_ecm_exclus ion	tnk_effector	LOC101928372	GPR111	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector LOC441178 GPR111 coexpression_network	caf_ecm_exclus ion	tnk_effector	LOC441178	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation LOC101928372 GPR111 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	LOC101928372	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation LOC441178 GPR111 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	LOC441178	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression HLA-A MRC1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	HLA-A	MRC1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression IL10 MRC1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IL10	MRC1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction HLA-A PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	HLA-A	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction HLA-A LAG3 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	HLA-A	LAG3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector GPR111 GPR111 coexpression_network	myeloid_suppre ssion	tnk_effector	GPR111	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector HLA-A PRF1 coexpression_network	myeloid_suppre ssion	tnk_effector	HLA-A	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation GPR111 GPR111 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	GPR111	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation HLA-A PDCD1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	HLA-A	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction LAG3 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction PDCD1 TOX coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector LAG3 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	LAG3	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation PDCD1 PDCD1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation LAG3 PDCD1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CD8A PRF1 coexpression_network	tnk_effector	tnk_effector	CD8A	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CD8A GZMA coexpression_network	tnk_effector	tnk_effector	CD8A	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation GPR111 GPR111 coexpression_network	tnk_effector	tumor_dediffere ntiation	GPR111	GPR111	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation PRF1 PDCD1 coexpression_network	tnk_effector	tumor_dediffere ntiation	PRF1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc PSMB9 TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB9	TAP1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc IGLV1-40 PSMB9 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	IGLV1-40	PSMB9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion IGHD2-15 IGHD2-15 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	IGHD2-15	IGHD2-15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion IGLV1-40 IGHD2-15 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	IGLV1-40	IGHD2-15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression TAP1 ITGAM coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	TAP1	ITGAM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression IGLV1-40 IGHD2-2 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	IGLV1-40	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction TAP1 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	TAP1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector PSMB9 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector TAP1 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	TAP1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation PSMB9 IFNG coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	PSMB9	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation IGLV1-40 IFNG coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	IGLV1-40	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion IGHD2-15 VIM coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	IGHD2-15	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression IGHD2-15 IGHD2-2 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	IGHD2-15	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression IGHD2-15 TRAJ1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	IGHD2-15	TRAJ1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction IGHD2-15 IGHD2-2 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	IGHD2-15	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction IGHD2-15 IGHD3-10 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	IGHD2-15	IGHD3-10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector IGHD2-15 IGHD2-2 coexpression_network	caf_ecm_exclus ion	tnk_effector	IGHD2-15	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector RP11-33B1 TRAJ10 coexpression_network	caf_ecm_exclus ion	tnk_effector	RP11-33B1	TRAJ10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation RP11-33B1 RP11-428O18 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	RP11-33B1	RP11-428O18	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression IGHD2-2 TRAJ1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGHD2-2	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression TRAJ1 TRAJ6 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	TRAJ1	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction IGHD2-2 IGHD2-2 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IGHD2-2	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction TRAJ1 PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	TRAJ1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector IGHD2-2 IGHD2-2 coexpression_network	myeloid_suppre ssion	tnk_effector	IGHD2-2	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector TRAJ1 CD8A coexpression_network	myeloid_suppre ssion	tnk_effector	TRAJ1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation TRAJ1 IFNG coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	TRAJ1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation ITGAM IFNG coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	ITGAM	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction IGHD2-2 IGHD3-10 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	IGHD2-2	IGHD3-10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction IGHD3-10 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	IGHD3-10	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector IGHD2-2 IGHD2-2 coexpression_network	tcell_dysfunctio n	tnk_effector	IGHD2-2	IGHD2-2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation PDCD1 IFNG coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation IDO1 IFNG coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	IDO1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CD8A IFNG coexpression_network	tnk_effector	tnk_effector	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CD8A GZMA coexpression_network	tnk_effector	tnk_effector	CD8A	GZMA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation IFNG IFNG coexpression_network	tnk_effector	tumor_dediffere ntiation	IFNG	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation CD8A IFNG coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector HLA-A B2M pathway	antigen_present ation_mhc	tnk_effector	HLA-A	B2M	pathway
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc HLA-B HLA-C coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-B	HLA-C	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc HLA-A HLA-C coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	HLA-C	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion IGLV2-11 TRAJ6 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	IGLV2-11	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion IGLV2-11 TRAJ10 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	IGLV2-11	TRAJ10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression LOC102724050 LOC102724050 coexpression_networ k	antigen_present ation_mhc	myeloid_suppre ssion	LOC102724050	LOC102724050	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression PSMB8 MRC1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	PSMB8	MRC1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB8 TOX coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB8	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector PSMB9 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector PSMB9 PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation RP11-444B24 RP11-444B24 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	RP11-444B24	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation PSMB8 SOX10 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	PSMB8	SOX10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion TRAJ10 TRAJ6 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	TRAJ10	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion FN1 LOXL2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FN1	LOXL2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression FN1 MRC1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	FN1	MRC1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression TRAJ6 TRAJ1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	TRAJ6	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction TRAJ6 TRAJ1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction TRAJ6 TIGIT coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	TIGIT	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector TRAJ10 CD8A coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ10	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector TRAJ10 IFNG coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ10	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation TRAJ10 TRAJ10 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	TRAJ10	TRAJ10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression CD68 MRC1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD68	MRC1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression IGHD2-2 TRAJ1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGHD2-2	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction TRAJ1 TRAJ1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	TRAJ1	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction IFNG PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IFNG	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector IFNG IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	IFNG	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector IFNG CD8A coexpression_network	myeloid_suppre ssion	tnk_effector	IFNG	CD8A	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation MRC1 SOX10 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	MRC1	SOX10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation CD68 SOX10 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	CD68	SOX10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction PDCD1 TOX coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TOX	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction TIGIT TRAJ1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	TIGIT	TRAJ1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tumor_dedifferentiation TOX SOX10 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	TOX	SOX10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tumor_dedifferentiation TOX RP11-444B24 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	TOX	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CD8A PRF1 coexpression_network	tnk_effector	tnk_effector	CD8A	PRF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CD8A CD8B coexpression_network	tnk_effector	tnk_effector	CD8A	CD8B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation CD8B RP11-444B24 coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8B	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation CD8A TRAJ10 coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8A	TRAJ10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation AXL SOX10 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	AXL	SOX10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation RP11-444B24 TRAJ10 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	RP11-444B24	TRAJ10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc PSMB9 TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB9	TAP1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-DRB1 TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-DRB1	TAP1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion TRAJ6 IGLV2-11 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TRAJ6	IGLV2-11	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion TRAJ1 IGLV2-11 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TRAJ1	IGLV2-11	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression PSMB9 TGFB1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	PSMB9	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression TRAJ1 IGHD2-15 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	TRAJ1	IGHD2-15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction TAP1 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	TAP1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector TRAJ6 TRAJ6 coexpression_network	antigen_present ation_mhc	tnk_effector	TRAJ6	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector PSMB9 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation TAP1 ZEB1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP1	ZEB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation PSMB9 IFNG coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	PSMB9	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A2 POSTN coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A2	POSTN	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion IGLV2-11 RP11-444B24 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	IGLV2-11	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression IGLV2-11 IGHGP coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	IGLV2-11	IGHGP	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression RP11-444B24 IGHGP coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	RP11-444B24	IGHGP	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction RP11-444B24 RP11-444B24 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	RP11-444B24	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction IGLV2-11 CXCL13 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	IGLV2-11	CXCL13	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector IGLV2-11 TRAJ6 coexpression_network	caf_ecm_exclus ion	tnk_effector	IGLV2-11	TRAJ6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector IGLV2-11 IFNG coexpression_network	caf_ecm_exclus ion	tnk_effector	IGLV2-11	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation RP11-444B24 RP11-444B24 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	RP11-444B24	RP11-444B24	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression IGHD2-15 IGHGP coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGHD2-15	IGHGP	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction IGHGP RP11-444B24 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IGHGP	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction IGHGP CXCL13 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IGHGP	CXCL13	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector IGHGP CD8B coexpression_network	myeloid_suppre ssion	tnk_effector	IGHGP	CD8B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector IGHGP IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	IGHGP	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation IGHGP GKV4-1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IGHGP	IGKV4-1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation IGHGP GLV2-11 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IGHGP	IGLV2-11	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction CXCL13 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CXCL13	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction PDCD1 RP11-444B24 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 CD8B coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation RP11-444B24 RP11-444B24 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	RP11-444B24	RP11-444B24	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation PDCD1 IFNG coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CD8A CD8B coexpression_network	tnk_effector	tnk_effector	CD8A	CD8B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CD8A IFNG coexpression_network	tnk_effector	tnk_effector	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation IFNG IFNG coexpression_network	tnk_effector	tumor_dediffere ntiation	IFNG	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation CD8A IFNG coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation IFNG GLV2-11 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IFNG	IGLV2-11	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	GSE145996	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation IGKV4-1 IGLV2-11 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IGKV4-1	IGLV2-11	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc PSMB9 TAP2 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB9	TAP2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc CUZD1,FAM24B,FAM24B- CUZD1 HPN,SCN1B coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	CUZD1,FAM24B,FAM24B- CUZD1	HPN,SCN1B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion MIR548N,OSBPL6 SLC13A3,TP53RK coexpression_net work	antigen_present ation_mhc	caf_ecm_exclus ion	MIR548N,OSBPL6	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion MIR548N,OSBPL6 CASC14,CASC15 coexpression_net work	antigen_present ation_mhc	caf_ecm_exclus ion	MIR548N,OSBPL6	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression MIR548N,OSBPL6 CD163 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	MIR548N,OSBPL6	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression HLA-DRB1 CD163 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	HLA-DRB1	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction MIR548N,OSBPL6 SLC13A3,TP53RK coexpression_netwo rk	antigen_present ation_mhc	tcell_dysfunctio n	MIR548N,OSBPL6	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction MIR548N,OSBPL6 INTS3,SLC27A3 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	MIR548N,OSBPL6	INTS3,SLC27A3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector HPN,SCN1B MIR143,MIR143HG coexpression_network	antigen_present ation_mhc	tnk_effector	HPN,SCN1B	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector HPN,SCN1B RILP,SCARF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HPN,SCN1B	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation MIR548N,OSBPL6 MIR548N,OSBPL6 coexpression _network	antigen_present ation_mhc	tumor_dediffere ntiation	MIR548N,OSBPL6	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation HPN,SCN1B ITGA3 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	HPN,SCN1B	ITGA3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion CASC14,CASC15 VIM coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	CASC14,CASC15	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion CASC14,CASC15 SLC13A3,TP53RK coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	CASC14,CASC15	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression SLC13A3,TP53RK TGFB1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	SLC13A3,TP53RK	TGFB1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression SLC13A3,TP53RK CD163 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	SLC13A3,TP53RK	CD163	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction SLC13A3,TP53RK SLC13A3,TP53RK coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	SLC13A3,TP53RK	SLC13A3,TP53RK	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction CASC14,CASC15 MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BH G coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	CASC14,CASC15	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector CASC14,CASC15 CD8A coexpression_network	caf_ecm_exclus ion	tnk_effector	CASC14,CASC15	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector CASC14,CASC15 MIR143,MIR143HG coexpression_network	caf_ecm_exclus ion	tnk_effector	CASC14,CASC15	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation SLC13A3,TP53RK MIR548N,OSBPL6 coexpression_networ k	caf_ecm_exclus ion	tumor_dediffere ntiation	SLC13A3,TP53RK	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression CD163 S100A8 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD163	S100A8	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction TGFB1 SLC13A3,TP53RK coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	TGFB1	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction S100A8 INTS3,SLC27A3 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	S100A8	INTS3,SLC27A3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector S100A8 RILP,SCARF1 coexpression_network	myeloid_suppre ssion	tnk_effector	S100A8	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector CD163 CD8A coexpression_network	myeloid_suppre ssion	tnk_effector	CD163	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation CD163 MIR548N,OSBPL6 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	CD163	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation S100A8 MIR548N,OSBPL6 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	S100A8	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction INTS3,SLC27A3 SLC13A3,TP53RK coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	INTS3,SLC27A3	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction IDO1 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	IDO1	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 IFNG coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation SLC13A3,TP53RK MIR548N,OSBPL6 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	SLC13A3,TP53RK	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation INTS3,SLC27A3 MIR548N,OSBPL6 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	INTS3,SLC27A3	MIR548N,OSBPL6	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CD8A IFNG coexpression_network	tnk_effector	tnk_effector	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector MIR143,MIR143HG RILP,SCARF1 coexpression_network	tnk_effector	tnk_effector	MIR143,MIR143HG	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation MIR143,MIR143HG ITGA3 coexpression_network	tnk_effector	tumor_dediffe rentiation	MIR143,MIR143HG	ITGA3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation RILP,SCARF1 MIR548N,OSBPL6 coexpression_network	tnk_effector	tumor_dediffe rentiation	RILP,SCARF1	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation ITGA3 MIR548N,OSBPL6 coexpression_network	tumor_dediffe rentiation	tumor_dediffe rentiation	ITGA3	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc B2M HLA-DRA coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	B2M	HLA-DRA	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion HLA-DRA CTD- 2201118.1,SERINC5 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-DRA	CTD-2201118.1,SERINC5	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion B2M FAP coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	B2M	FAP	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression B2M MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	B2M	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression B2M HPN,SCN1B coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	B2M	HPN,SCN1B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB9 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector B2M GZMB coexpression_network	antigen_present ation_mhc	tnk_effector	B2M	GZMB	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector B2M CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	B2M	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation HLA-DRA CASC14,CASC15 coexpression_network	antigen_present ation_mhc	tumor_dediffe rentiation	HLA-DRA	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation B2M AXL coexpression_network	antigen_present ation_mhc	tumor_dediffe rentiation	B2M	AXL	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion CTD-2201118.1,SERINC5 FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	CTD-2201118.1,SERINC5	FN1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion FAP TGFB coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FAP	TGFB	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression HPN,SCN1B HPN,SCN1B coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	HPN,SCN1B	HPN,SCN1B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression LOC100288974,MBL1P INTS3,SLC27A3 coexpression_netwo	caf_ecm_exclus ion	myeloid_suppre ssion	LOC100288974,MBL1P	INTS3,SLC27A3	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction LOC100288974,MBL1P ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	LOC100288974,MBL1P	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction LOC100288974,MBL1P ZNF595,ZNF718 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	LOC100288974,MBL1P	ZNF595,ZNF718	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector CTD-2201118.1,SERINC5 GZMB coexpression_network	caf_ecm_exclus ion	tnk_effector	CTD-2201118.1,SERINC5	GZMB	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector CTD-2201118.1,SERINC5 CCL5 coexpression_network	caf_ecm_exclus ion	tnk_effector	CTD-2201118.1,SERINC5	CCL5	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation CTD- 2201118.1,SERINC5 CASC14,CASC15 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	CTD-2201118.1,SERINC5	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression HPN,SCN1B MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BH G coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	HPN,SCN1B	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression INTS3,SLC27A3 ZNF595,ZNF718 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	INTS3,SLC27A3	ZNF595,ZNF718	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction ZNF595,ZNF718 ZNF595,ZNF718 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	ZNF595,ZNF718	ZNF595,ZNF718	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction APOC1,APOC1P1 APOC1,APOC1P1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	APOC1,APOC1P1	APOC1,APOC1P1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector APOC1,APOC1P1 IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	APOC1,APOC1P1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector S100A9 GZMB coexpression_network	myeloid_suppre ssion	tnk_effector	S100A9	GZMB	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation APOC1,APOC1P1 APOC1,APOC1P1 coexpression_netw ork	myeloid_suppre ssion	tumor_dediffere ntiation	APOC1,APOC1P1	APOC1,APOC1P1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG CASC14 ,CASC15 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction APOC1,APOC1P1 ATP5E,SLMO2,SLMO2- ATP5E coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	APOC1,APOC1P1	ATP5E,SLMO2,SLMO2- ATP5E	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction APOC1,APOC1P1 SLC13A3,TP53RK coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	APOC1,APOC1P1	SLC13A3,TP53RK	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tccll_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tnk_effector PDCD1 CCL5 coexpression_network	tccll_dysfunctio n	tnk_effector	PDCD1	CCL5	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tumor_dedifferentiation APOC1,APOC1P1 APOC1,APOC1P1 coexpression_network	tccll_dysfunctio n	tumor_dediffere ntiation	APOC1,APOC1P1	APOC1,APOC1P1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tumor_dedifferentiation ZNF595,ZNF718 CASC14,CASC15 coexpression_network	tccll_dysfunctio n	tumor_dediffere ntiation	ZNF595,ZNF718	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CCL5 CD8A coexpression_network	tnk_effector	tnk_effector	CCL5	CD8A	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CD8A IFNG coexpression_network	tnk_effector	tnk_effector	CD8A	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation IFNG APOC1,APOC1P1 coexpression_network	tnk_effector	tumor_dediffere ntiation	IFNG	APOC1,APOC1P1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation CD8A CASC14,CASC15 coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8A	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 MITF coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	MITF	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation MITF SOX10 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	MITF	SOX10	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression S100A8 S100A9 network	myeloid_suppre ssion	myeloid_suppre ssion	S100A8	S100A9	network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc PSMB8 PSMB9 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB8	PSMB9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc CASC14,CASC15 MIR548N,OSBPL6 coexpressi on_network	antigen_present ation_mhc	antigen_present ation_mhc	CASC14,CASC15	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion MIR548N,OSBPL6 MIR548N,OSBPL6 coexpression_net work	antigen_present ation_mhc	caf_ecm_exclus ion	MIR548N,OSBPL6	MIR548N,OSBPL6	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion HPN,SCN1B RILP,SCARF1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HPN,SCN1B	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression CASC14,CASC15 CASC14,CASC15 coexpression_ne twork	antigen_present ation_mhc	myeloid_suppre ssion	CASC14,CASC15	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression CASC14,CASC15 MIR4763,MIRLET7A3,MIRLET7B,M IRLET7BHG coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	CASC14,CASC15	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tccll_dysfunction MIR4469,RNF170 MIR4469,RNF170 coexpression_networ k	antigen_present ation_mhc	tccll_dysfunctio n	MIR4469,RNF170	MIR4469,RNF170	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction PSMB8 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB8	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector CASC14,CASC15 CTD- 2201118.1,SERINC5 coexpression_network	antigen_present ation_mhc	tnk_effector	CASC14,CASC15	CTD-2201118.1,SERINC5	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector MIR548N,OSBPL6 SLC13A3,TP53RK coexpression_network	antigen_present ation_mhc	tnk_effector	MIR548N,OSBPL6	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation CASC14,CASC15 CASC14,CASC15 coexpression_ network	antigen_present ation_mhc	tumor_dediffere ntiation	CASC14,CASC15	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation HPN,SCN1B MIR143,MIR143HG coexpression_netw ork	antigen_present ation_mhc	tumor_dediffere ntiation	HPN,SCN1B	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A2 COL3A1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A2	COL3A1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A2 LOXL2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A2	LOXL2	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression RILP,SCARF1 MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BH G coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	RILP,SCARF1	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression CXCL12 MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG coe xpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	CXCL12	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction RILP,SCARF1 RILP,SCARF1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	RILP,SCARF1	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction CXCL12 TNFSF12,TNFSF12- TNFSF13,TNFSF13 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	CXCL12	TNFSF12,TNFSF12- TNFSF13,TNFSF13	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector MIR548N,OSBPL6 SLC13A3,TP53RK coexpression_network	caf_ecm_exclus ion	tnk_effector	MIR548N,OSBPL6	SLC13A3,TP53RK	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector MIR548N,OSBPL6 RPS8,SNORD55 coexpression_network	caf_ecm_exclus ion	tnk_effector	MIR548N,OSBPL6	RPS8,SNORD55	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation RILP,SCARF1 MIR143,MIR143HG coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	RILP,SCARF1	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression S100A8 S100A9 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	S100A8	S100A9	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression CASC14,CASC15 MIR4763,MIRLET7A3,MIRLET7B,MIRLE T7BHG coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CASC14,CASC15	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction CD68 HAVCR2 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	CD68	HAVCR2	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tccl_dysfunction MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG RILP,SCARF1 coexpression_network	myeloid_suppre ssion	tccl_dysfunctio n	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	RILP,SCARF1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector CASC14,CASC15 CTD-2201118.1,SERINC5 coexpression_network	myeloid_suppre ssion	tnk_effector	CASC14,CASC15	CTD-2201118.1,SERINC5	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector MIR4763,MIRLET7A3,MIRLET7B,MIRLET7BHG CTD- 2201118.1,SERINC5 coexpression_network	myeloid_suppre ssion	tnk_effector	MIR4763,MIRLET7A3,MIRLE T7B,MIRLET7BHG	CTD-2201118.1,SERINC5	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation CASC14,CASC15 CASC14,CASC15 coexpression_netwo rk	myeloid_suppre ssion	tumor_dediffere ntiation	CASC14,CASC15	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation APOC1,APOC1P1 APOC1,APOC1P1 coexpression_netw ork	myeloid_suppre ssion	tumor_dediffere ntiation	APOC1,APOC1P1	APOC1,APOC1P1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tccl_dysfunction CXCL13 PDCD1 coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	CXCL13	PDCD1	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tccl_dysfunction MIR4469,RNF170 TNFSF12,TNFSF12- TNFSF13,TNFSF13 coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	MIR4469,RNF170	TNFSF12,TNFSF12- TNFSF13,TNFSF13	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tnk_effector PDCD1 CD8B coexpression_network	tccl_dysfunctio n	tnk_effector	PDCD1	CD8B	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tnk_effector PDCD1 IFNG coexpression_network	tccl_dysfunctio n	tnk_effector	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tumor_dedifferentiation LIMS3,LIMS3- LOC440895,LOC440895 AXL coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	LIMS3,LIMS3- LOC440895,LOC440895	AXL	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tumor_dedifferentiation RILP,SCARF1 MIR143,MIR143HG coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	RILP,SCARF1	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CD8B IFNG coexpression_network	tnk_effector	tnk_effector	CD8B	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector GZMB IFNG coexpression_network	tnk_effector	tnk_effector	GZMB	IFNG	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation CTD-2201118.1,SERINC5 CASC14,CASC15 coexpression_network	tnk_effector	tumor_dediffere ntiation	CTD-2201118.1,SERINC5	CASC14,CASC15	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation RPS8,SNORD55 ZNF595,ZNF718 coexpression_network	tnk_effector	tumor_dediffere ntiation	RPS8,SNORD55	ZNF595,ZNF718	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 ZNF595,ZNF718 coexpression_netw ork	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	ZNF595,ZNF718	coexpression _network
primary _recist	melanoma_recist_sup ported_primary	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation CASC14,CASC15 MIR143,MIR143HG coexpression_ne twork	tumor_dediffere ntiation	tumor_dediffere ntiation	CASC14,CASC15	MIR143,MIR143HG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tccl_dysfunction tccl_dysfunction PDCD1 LAG3 checkpoint	tccl_dysfunctio n	tccl_dysfunctio n	PDCD1	LAG3	checkpoint

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc IGHD2-2 IGLV1-40 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	IGHD2-2	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc B2M IGLV1-40 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	B2M	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion IGHD2-2 IGHD2-2 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	IGHD2-2	IGHD2-2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion AL049794 AL049794 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	AL049794	AL049794	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression IGLV1-40 IGLV2-11 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	IGLV1-40	IGLV2-11	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression IGHD2-2 IGHD3-10 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	IGHD2-2	IGHD3-10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction IGLV1-40 LAG3 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	IGLV1-40	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction IGLV1-40 TRAJ1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	IGLV1-40	TRAJ1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector AL049794 AL049794 coexpression_network	antigen_present ation_mhc	tnk_effector	AL049794	AL049794	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector IGLV1-40 CCL5 coexpression_network	antigen_present ation_mhc	tnk_effector	IGLV1-40	CCL5	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation IGLV1-40 IGLV1-40 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	IGLV1-40	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation IGHD2-2 IGLV1-40 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	IGHD2-2	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion CARMIL2 IGHD2-2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	CARMIL2	IGHD2-2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 VIM coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression COL1A1 ITGAM coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	COL1A1	ITGAM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression IGHD2-2 IGHD3-10 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	IGHD2-2	IGHD3-10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction CARMIL2 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	CARMIL2	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction CARMIL2 TRAJ1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	CARMIL2	TRAJ1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector AL049794 AL049794 coexpression_network	caf_ecm_exclus ion	tnk_effector	AL049794	AL049794	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector CARMIL2 CCL5 coexpression_network	caf_ecm_exclus ion	tnk_effector	CARMIL2	CCL5	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation CARMIL2 CARMIL2 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	CARMIL2	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression IGHD3-10 IGLV2-11 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGHD3-10	IGLV2-11	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression IGLV2-11 ITGAM coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGLV2-11	ITGAM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction ITGAM HAVCR2 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	ITGAM	HAVCR2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction IGLV2-11 TIGIT coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IGLV2-11	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector IGLV2-11 CCL5 coexpression_network	myeloid_suppre ssion	tnk_effector	IGLV2-11	CCL5	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector IGLV2-11 IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	IGLV2-11	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation ITGAM SOX10 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	ITGAM	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation IGLV2-11 IGLV1-40 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IGLV2-11	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction LAG3 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction PDCD1 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 IFNG coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector TRAJ1 CCL5 coexpression_network	tcell_dysfunctio n	tnk_effector	TRAJ1	CCL5	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation PDCD1 CARMIL2 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation TRAJ1 CARMIL2 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	TRAJ1	CARMIL2	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CCL5 IFNG coexpression_network	tnk_effector	tnk_effector	CCL5	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector GNLY IFNG coexpression_network	tnk_effector	tnk_effector	GNLY	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation CCL5 CARMIL2 coexpression_network	tnk_effector	tumor_dediffere ntiation	CCL5	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation CCL5 IGLV1-40 coexpression_network	tnk_effector	tumor_dediffere ntiation	CCL5	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation CARMIL2 IGLV1-40 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CARMIL2	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation CARMIL2 TRAJ52 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	CARMIL2	TRAJ52	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion AL049794 TRAJ6 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	AL049794	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion PSMB9 TRAJ6 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	PSMB9	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression B2M CD68 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	B2M	CD68	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression PSMB9 CD68 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	PSMB9	CD68	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB9 IDO1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	IDO1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector PSMB9 IFNG coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector PSMB9 IGHV6-1 coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	IGHV6-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation AL049794 TRAJ6 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	AL049794	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation B2M GKV1-17 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	B2M	IGKV1-17	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FAP coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FAP	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion ACTA2 COL1A1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	ACTA2	COL1A1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression COL1A1 CD68 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	COL1A1	CD68	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression FAP LOC101927356 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	FAP	LOC101927356	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction TRAJ6 IGLV1-40 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction TRAJ6 CXCL13 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector TRAJ6 IGHV6-1 coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ6	IGHV6-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector TRAJ6 IFNG coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ6	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation TRAJ6 TRAJ6 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	TRAJ6	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation FAP VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	FAP	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression CD68 S100A9 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD68	S100A9	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction RP11-33B1 RP11-33B1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	RP11-33B1	RP11-33B1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction LOC101927356 LOC102724050 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	LOC101927356	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector RP11-33B1 RP11-428O18 coexpression_network	myeloid_suppre ssion	tnk_effector	RP11-33B1	RP11-428O18	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector CD68 IFNG coexpression_network	myeloid_suppre ssion	tnk_effector	CD68	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation RP11-33B1 RP11-33B1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	RP11-33B1	RP11-33B1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation CD68 VIM coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	CD68	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction CXCL13 IGLV1-40 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CXCL13	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction IGLV1-40 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	IGLV1-40	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tnk_effector RP11-428O18 RP11-428O18 coexpression_network	tcell_dysfunctio n	tnk_effector	RP11-428O18	RP11-428O18	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tnk_effector PDCD1 IFNG coexpression_network	tccll_dysfunctio n	tnk_effector	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tumor_dedifferentiation RP11-33B1 RP11-33B1 coexpression_network	tccll_dysfunctio n	tumor_dediffere ntiation	RP11-33B1	RP11-33B1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tumor_dedifferentiation IGLV1-40 IGKV1-17 coexpression_network	tccll_dysfunctio n	tumor_dediffere ntiation	IGLV1-40	IGKV1-17	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector IFNG IGHV6-1 coexpression_network	tnk_effector	tnk_effector	IFNG	IGHV6-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation IFNG IGKV1-17 coexpression_network	tnk_effector	tumor_dediffere ntiation	IFNG	IGKV1-17	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation IGHV6-1 IGKV1-17 coexpression_network	tnk_effector	tumor_dediffere ntiation	IGHV6-1	IGKV1-17	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation IGKV1-17 TRAJ6 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IGKV1-17	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation RP11-33B1 TRAJ6 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	RP11-33B1	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FN1 network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FN1	network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-B HLA-C coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-B	HLA-C	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc CARMIL2 TRAJ6 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	CARMIL2	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion CARMIL2 CARMIL2 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	CARMIL2	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion TRAJ6 CARMIL2 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TRAJ6	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression CARMIL2 TRAJ1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	CARMIL2	TRAJ1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression IGHD2-2 IGLV1-40 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	IGHD2-2	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tccll_dysfunction TRAJ6 TRAJ6 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	TRAJ6	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tccll_dysfunction CARMIL2 CARMIL2 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	CARMIL2	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector TRAJ6 TRAJ6 coexpression_network	antigen_present ation_mhc	tnk_effector	TRAJ6	TRAJ6	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector TRAJ6 IGLV2-11 coexpression_network	antigen_present ation_mhc	tnk_effector	TRAJ6	IGLV2-11	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation IGHD2-2 IGHD2-2 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	IGHD2-2	IGHD2-2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation B2M B2M coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	B2M	B2M	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion COL1A1 LOXL2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	LOXL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression CARMIL2 TRAJ1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	CARMIL2	TRAJ1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression CARMIL2 IGLV1-40 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	CARMIL2	IGLV1-40	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction CARMIL2 CARMIL2 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	CARMIL2	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction CARMIL2 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	CARMIL2	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector IGHV6-1 IGKV4-1 coexpression_network	caf_ecm_exclus ion	tnk_effector	IGHV6-1	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector CARMIL2 IGKV4-1 coexpression_network	caf_ecm_exclus ion	tnk_effector	CARMIL2	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation COL1A1 ZEB1 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	COL1A1	ZEB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation IGHV6-1 IGKV4-1 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	IGHV6-1	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression IGLV1-40 IGLV2-11 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGLV1-40	IGLV2-11	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression IGLV1-40 TRAJ1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IGLV1-40	TRAJ1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction TRAJ1 CARMIL2 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	TRAJ1	CARMIL2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction TRAJ1 TIGIT coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	TRAJ1	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector IGLV2-11 IGLV2-11 coexpression_network	myeloid_suppre ssion	tnk_effector	IGLV2-11	IGLV2-11	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector IGLV1-40 IGKV4-1 coexpression_network	myeloid_suppre ssion	tnk_effector	IGLV1-40	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation IGLV1-40 IGKV4-1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IGLV1-40	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation IGLV1-40 IGKV1-17 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IGLV1-40	IGKV1-17	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction PDCD1 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction CARMIL2 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CARMIL2	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector TRAJ6 TRAJ6 coexpression_network	tcell_dysfunctio n	tnk_effector	TRAJ6	TRAJ6	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 IFNG coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation PDCD1 IGKV4-1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation TIGIT IGKV4-1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	TIGIT	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CXCL10 CXCL11 coexpression_network	tnk_effector	tnk_effector	CXCL10	CXCL11	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector CXCL10 IFNG coexpression_network	tnk_effector	tnk_effector	CXCL10	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation IGKV4-1 IGKV4-1 coexpression_network	tnk_effector	tumor_dediffere ntiation	IGKV4-1	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation IGKV4-1 IGKV1-17 coexpression_network	tnk_effector	tumor_dediffere ntiation	IGKV4-1	IGKV1-17	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation IGKV1-17 IGKV4-1 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IGKV1-17	IGKV4-1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE91061	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation IGHD2-15 IGHD2-2 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IGHD2-15	IGHD2-2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector HLA-A B2M pathway	antigen_present ation_mhc	tnk_effector	HLA-A	B2M	pathway
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc CD8A PRF1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	CD8A	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc HLA-A HLA-B coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	HLA-B	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion PRF1 PRF1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	PRF1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion IFNG IFNG coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	IFNG	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression IFNG IFNG coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	IFNG	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression CD8A IFNG coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	CD8A	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction CD8A CD8A coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	CD8A	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tcell_dysfunction PRF1 CD8A coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PRF1	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector PRF1 PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	PRF1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector CD8A PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	CD8A	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation HLA-DRB1 SOX10 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	HLA-DRB1	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation HLA-DRB1 AXL coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	HLA-DRB1	AXL	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion IFNG PRF1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	IFNG	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion FAP FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FAP	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression IFNG IFNG coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	IFNG	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression PRF1 IFNG coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	PRF1	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction PRF1 CD8A coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	PRF1	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction PRF1 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	PRF1	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector PRF1 PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	PRF1	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector IFNG PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	IFNG	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation FAP AXL coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	FAP	AXL	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation PRF1 FRMPD3 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	PRF1	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression CD163 ITGAM coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD163	ITGAM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression ITGAM TGFB1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	ITGAM	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction IFNG CD8A coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IFNG	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction IFNG PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IFNG	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector TRAJ10 TRAJ10 coexpression_network	myeloid_suppre ssion	tnk_effector	TRAJ10	TRAJ10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector IFNG PRF1 coexpression_network	myeloid_suppre ssion	tnk_effector	IFNG	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation CD68 SOX10 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	CD68	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation ITGAM SOX10 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	ITGAM	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction CD8A PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CD8A	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction CD8A HAVCR2 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CD8A	HAVCR2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector CD8A PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	CD8A	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation PDCD1 FRMPD3 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation CD8A FRMPD3 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	CD8A	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector PRF1 TOX coexpression_network	tnk_effector	tnk_effector	PRF1	TOX	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector TOX TRAJ10 coexpression_network	tnk_effector	tnk_effector	TOX	TRAJ10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation TOX SOX10 coexpression_network	tnk_effector	tumor_dediffere ntiation	TOX	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation TOX FRMPD3 coexpression_network	tnk_effector	tumor_dediffere ntiation	TOX	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation AXL SOX10 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	AXL	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation FRMPD3 SOX10 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	FRMPD3	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector GZMB PRF1 pathway	tnk_effector	tnk_effector	GZMB	PRF1	pathway
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccll_dysfunction tccll_dysfunction PDCD1 LAG3 checkpoint	tccll_dysfunctio n	tccll_dysfunctio n	PDCD1	LAG3	checkpoint
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector antigen_presentation_mhc IFNG HLA-A regulatory	tnk_effector	antigen_present ation_mhc	IFNG	HLA-A	regulatory
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc HLA-A TAP2 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	TAP2	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc CXCL13 FRMPD3 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	CXCL13	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion HLA-A TGFB1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-A	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion HLA-A FN1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-A	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression FRMPD3 FRMPD3 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	FRMPD3	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression HLA-A PRF1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	HLA-A	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tccll_dysfunction RP11-444B24 RP11-444B24 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	RP11-444B24	RP11-444B24	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tccll_dysfunction FRMPD3 FRMPD3 coexpression_network	antigen_present ation_mhc	tccll_dysfunctio n	FRMPD3	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector FRMPD3 FRMPD3 coexpression_network	antigen_present ation_mhc	tnk_effector	FRMPD3	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector HLA-A PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-A	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation RP11-444B24 RP11-444B24 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	RP11-444B24	RP11-444B24	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation AL049794 AL049794 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	AL049794	AL049794	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion TGFB1 TGFB1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	TGFB1	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion FN1 TGFB1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FN1	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression TGFB1 TGFB1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	TGFB1	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression TGFB1 PRF1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	TGFB1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction TGFB1 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TGFB1	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction TGFB1 TIGIT coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TGFB1	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector TGFB1 PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	TGFB1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector TGFB1 CD8A coexpression_network	caf_ecm_exclus ion	tnk_effector	TGFB1	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation TGFB1 AXL coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	TGFB1	AXL	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation FN1 AXL coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	FN1	AXL	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression FRMPD3 PRF1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	FRMPD3	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression PRF1 TGFB1 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	PRF1	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction FRMPD3 FRMPD3 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	FRMPD3	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction PRF1 PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	PRF1	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector PRF1 PRF1 coexpression_network	myeloid_suppre ssion	tnk_effector	PRF1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector IDO1 IDO1 coexpression_network	myeloid_suppre ssion	tnk_effector	IDO1	IDO1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation IDO1 IDO1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IDO1	IDO1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation PRF1 LAG3 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	PRF1	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tccl_dysfunction FRMPD3 PDCD1 coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	FRMPD3	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tccl_dysfunction TIGIT TRAJ1 coexpression_network	tccl_dysfunctio n	tccl_dysfunctio n	TIGIT	TRAJ1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tnk_effector FRMPD3 FRMPD3 coexpression_network	tccl_dysfunctio n	tnk_effector	FRMPD3	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tccl_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tumor_dedifferentiation RP11-444B24 RP11-444B24 coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	RP11-444B24	RP11-444B24	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tccl_dysfunction tumor_dedifferentiation IGHD2-15 IGHD2-15 coexpression_network	tccl_dysfunctio n	tumor_dediffere ntiation	IGHD2-15	IGHD2-15	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CD8A PRF1 coexpression_network	tnk_effector	tnk_effector	CD8A	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector GZMB PRF1 coexpression_network	tnk_effector	tnk_effector	GZMB	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation IDO1 IDO1 coexpression_network	tnk_effector	tumor_dediffere ntiation	IDO1	IDO1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation PRF1 LAG3 coexpression_network	tnk_effector	tumor_dediffere ntiation	PRF1	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation IFNG LAG3 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IFNG	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation IDO1 IFNG coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	IDO1	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-A B2M pathway	antigen_present ation_mhc	tnk_effector	HLA-A	B2M	pathway
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector GZMB PRF1 pathway	tnk_effector	tnk_effector	GZMB	PRF1	pathway
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tccl_dysfunction tccl_dysfunction PDCD1 LAG3 checkpoint	tccl_dysfunctio n	tccl_dysfunctio n	PDCD1	LAG3	checkpoint
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-A HLA-C coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-A	HLA-C	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-C TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-C	TAP1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion TAP1 FN1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TAP1	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion HLA-C FN1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	HLA-C	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression HLA-C LAG3 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	HLA-C	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression TAP1 LAG3 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	TAP1	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction GZMB GZMB coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	GZMB	GZMB	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction TAP1 TOX coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	TAP1	TOX	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector HLA-C PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	HLA-C	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector GZMB PRF1 coexpression_network	antigen_present ation_mhc	tnk_effector	GZMB	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation TAP1 ZEB1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP1	ZEB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation TAP1 SOX10 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP1	SOX10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion CXCL12 FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	CXCL12	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion FN1 VIM coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	FN1	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression CXCL12 CD163 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	CXCL12	CD163	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression FN1 IL10 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	FN1	IL10	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction TRAJ6 CXCL13 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction TRAJ6 TOX coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	TRAJ6	TOX	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector CXCL12 PRF1 coexpression_network	caf_ecm_exclus ion	tnk_effector	CXCL12	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector TRAJ6 AL049794 coexpression_network	caf_ecm_exclus ion	tnk_effector	TRAJ6	AL049794	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation FN1 ZEB1 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	FN1	ZEB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression CD8B LAG3 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD8B	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression CD8B CXCL13 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD8B	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction CXCL13 CXCL13 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	CXCL13	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction LAG3 PDCD1 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector LAG3 LAG3 coexpression_network	myeloid_suppre ssion	tnk_effector	LAG3	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector CXCL13 CXCL13 coexpression_network	myeloid_suppre ssion	tnk_effector	CXCL13	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation LAG3 LAG3 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	LAG3	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation FRMPD3 FRMPD3 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	FRMPD3	FRMPD3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction PDCD1 TOX coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TOX	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction GZMB PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	GZMB	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector CXCL13 CXCL13 coexpression_network	tcell_dysfunctio n	tnk_effector	CXCL13	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 PRF1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation IDO1 IDO1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	IDO1	IDO1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation PDCD1 LAG3 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector LAG3 PRF1 coexpression_network	tnk_effector	tnk_effector	LAG3	PRF1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector GNLY PRF1 coexpression_network	tnk_effector	tnk_effector	GNLY	PRF1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation LAG3 LAG3 coexpression_network	tnk_effector	tumor_dediffere ntiation	LAG3	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation PRF1 LAG3 coexpression_network	tnk_effector	tumor_dediffere ntiation	PRF1	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation SOX10 ZEB1 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	SOX10	ZEB1	coexpression _network
primary _recist	melanoma_core_high _evidence	GSE78220	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation AXL ZEB1 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	AXL	ZEB1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tccl_dysfunction tccl_dysfunction PDCD1 LAG3 checkpoint	tccl_dysfunctio n	tccl_dysfunctio n	PDCD1	LAG3	checkpoint
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FN1 network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FN1	network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc PSMB9 TAP1 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB9	TAP1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc antigen_presentation_mhc TAP1 TAP2 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	TAP1	TAP2	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion TAP1 LOC102724678 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	TAP1	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc caf_ecm_exclusion PSMB9 VIM coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	PSMB9	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression PSMB9 TGFB1 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	PSMB9	TGFB1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc myeloid_suppression B2M GRAMD3 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	B2M	GRAMD3	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tccl_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tccl_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tccl_dysfunction PSMB9 LAG3 coexpression_network	antigen_present ation_mhc	tccl_dysfunctio n	PSMB9	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector TAP1 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	TAP1	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tnk_effector PSMB9 CD8A coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709_ PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation TAP2 WISP1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP2	WISP1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	antigen_presentation_mhc tumor_dedifferentiation TAP1 LOC102724678 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP1	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 FN1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	FN1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 LOC102724678 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression LOC102724678 LOC102724678 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOC102724678	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion myeloid_suppression LOC102724678 LOC101927697 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOC102724678	LOC101927697	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction LOC102724678 RASGRF2-AS1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	LOC102724678	RASGRF2-AS1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tcell_dysfunction VIM PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	VIM	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tnk_effector LOC102724678 RASGRF2-AS1 coexpression_network	caf_ecm_exclus ion	tnk_effector	LOC102724678	RASGRF2-AS1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	caf_ecm_exclusion tumor_dedifferentiation LOC102724678 LOC102724678 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	LOC102724678	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression LOC101927697 LOC102724678 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	LOC101927697	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression myeloid_suppression CD68 IL10 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD68	IL10	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction IL10 CTLA4 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IL10	CTLA4	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tcell_dysfunction TGFB1 LAG3 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	TGFB1	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector IL10 GZMB coexpression_network	myeloid_suppre ssion	tnk_effector	IL10	GZMB	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tnk_effector CD68 GZMA coexpression_network	myeloid_suppre ssion	tnk_effector	CD68	GZMA	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation WISP1 WISP1 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	WISP1	WISP1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	myeloid_suppression tumor_dedifferentiation LOC102724678 LOC102724678 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	LOC102724678	LOC102724678	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction LAG3 PDCD1 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	LAG3	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tcell_dysfunction CTLA4 LAG3 coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CTLA4	LAG3	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector RASGRF2-AS1 RASGRF2-AS1 coexpression_network	tcell_dysfunctio n	tnk_effector	RASGRF2-AS1	RASGRF2-AS1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tnk_effector PDCD1 CD8A coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8A	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation RASGRF2-AS1 LOC102724050 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	RASGRF2-AS1	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tcell_dysfunction tumor_dedifferentiation RASGRF2-AS1 PAPD7 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	RASGRF2-AS1	PAPD7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector CD8A NKG7 coexpression_network	tnk_effector	tnk_effector	CD8A	NKG7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tnk_effector GZMA NKG7 coexpression_network	tnk_effector	tnk_effector	GZMA	NKG7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation RASGRF2-AS1 LOC102724050 coexpression_network	tnk_effector	tumor_dediffere ntiation	RASGRF2-AS1	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tnk_effector tumor_dedifferentiation H1F0 LOC100507600 coexpression_network	tnk_effector	tumor_dediffere ntiation	H1F0	LOC100507600	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation LOC102724050 PAPD7 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	LOC102724050	PAPD7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordFullGra ph	word_full_gra ph	tumor_dedifferentiation tumor_dedifferentiation LOC102724050 WISP1 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	LOC102724050	WISP1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc antigen_presentation_mhc LOC100996635 LOC102724678 coexpression_n etwork	antigen_present ation_mhc	antigen_present ation_mhc	LOC100996635	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion LOC100996635 H1F0 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	LOC100996635	H1F0	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc caf_ecm_exclusion LOC100996635 WISP1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	LOC100996635	WISP1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression LOC100996635 LOC100996635 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	LOC100996635	LOC100996635	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc myeloid_suppression LOC102724678 LOC101927697 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	LOC102724678	LOC101927697	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tcell_dysfunction LOC102724678 LOC101927697 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	LOC102724678	LOC101927697	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector LOC100996635 LOC100996635 coexpression_network	antigen_present ation_mhc	tnk_effector	LOC100996635	LOC100996635	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tnk_effector PSMB9 CD8B coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	CD8B	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	antigen_presentation_mhc tumor_dedifferentiation LOC102724678 LOC102724050 coexpression_netw ork	antigen_present ation_mhc	tumor_dediffere ntiation	LOC102724678	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion ACTA2 COL1A1 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	ACTA2	COL1A1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion caf_ecm_exclusion COL1A1 LOXL2 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	COL1A1	LOXL2	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression WISP1 WISP1 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	WISP1	WISP1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion myeloid_suppression H1F0 LOC100996635 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	H1F0	LOC100996635	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction H1F0 H1F0 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	H1F0	H1F0	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tcell_dysfunction WISP1 H1F0 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	WISP1	H1F0	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector WISP1 LOC102724050 coexpression_network	caf_ecm_exclus ion	tnk_effector	WISP1	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tnk_effector H1F0 LOC100996635 coexpression_network	caf_ecm_exclus ion	tnk_effector	H1F0	LOC100996635	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	caf_ecm_exclusion tumor_dedifferentiation H1F0 LOC388849 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	H1F0	LOC388849	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression CD163 ITGAM coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD163	ITGAM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression myeloid_suppression IL10 ITGAM coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	IL10	ITGAM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction LOC101927697 LOC101927697 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	LOC101927697	LOC101927697	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tcell_dysfunction ITGAM HAVCR2 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	ITGAM	HAVCR2	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector LOC100996635 LOC100996635 coexpression_network	myeloid_suppre ssion	tnk_effector	LOC100996635	LOC100996635	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tnk_effector WISP1 LOC102724050 coexpression_network	myeloid_suppre ssion	tnk_effector	WISP1	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation LOC101927391 LOC101927391 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	LOC101927391	LOC101927391	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	myeloid_suppression tumor_dedifferentiation WISP1 LOC102724050 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	WISP1	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction PDCD1 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tcell_dysfunction CXCL13 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CXCL13	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tnk_effector PDCD1 CD8B coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	CD8B	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tnk_effector TIGIT CCL5 coexpression_network	tcell_dysfunctio n	tnk_effector	TIGIT	CCL5	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tumor_dedifferentiation PDCD1 PDCD1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	PDCD1	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tcell_dysfunction tumor_dedifferentiation TIGIT PDCD1 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	TIGIT	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CCL5 CD8B coexpression_network	tnk_effector	tnk_effector	CCL5	CD8B	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tnk_effector CCL5 IFNG coexpression_network	tnk_effector	tnk_effector	CCL5	IFNG	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation LOC102724050 LOC102724050 coexpression_network	tnk_effector	tumor_dediffere ntiation	LOC102724050	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tnk_effector tumor_dedifferentiation CD8B PDCD1 coexpression_network	tnk_effector	tumor_dediffere ntiation	CD8B	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation AXL C19orf10 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	AXL	C19orf10	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordUnsign edGraph	word_unsigne d_graph	tumor_dedifferentiation tumor_dedifferentiation LOC102724050 RASGRF2-AS1 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	LOC102724050	RASGRF2-AS1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc HLA-B HLA-C coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	HLA-B	HLA-C	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc antigen_presentation_mhc PSMB9 TAP2 coexpression_network	antigen_present ation_mhc	antigen_present ation_mhc	PSMB9	TAP2	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion GRAMD3 GRAMD3 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	GRAMD3	GRAMD3	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc caf_ecm_exclusion PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	caf_ecm_exclus ion	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression LOC101927356 IL10 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	LOC101927356	IL10	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc myeloid_suppression H1F0 LOC101927843 coexpression_network	antigen_present ation_mhc	myeloid_suppre ssion	H1F0	LOC101927843	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tcell_dysfunction H1F0 LOC388849 coexpression_network	antigen_present ation_mhc	tcell_dysfunctio n	H1F0	LOC388849	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector PSMB9 PDCD1 coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tnk_effector PSMB9 NKG7 coexpression_network	antigen_present ation_mhc	tnk_effector	PSMB9	NKG7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation H1F0 LOC388849 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	H1F0	LOC388849	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	antigen_presentation_mhc tumor_dedifferentiation TAP2 LOC388849 coexpression_network	antigen_present ation_mhc	tumor_dediffere ntiation	TAP2	LOC388849	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion LOC102724050 LOC102724678 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	LOC102724050	LOC102724678	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion caf_ecm_exclusion LOC100507600 LOC102724050 coexpression_network	caf_ecm_exclus ion	caf_ecm_exclus ion	LOC100507600	LOC102724050	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression LOC101927843 LOC101927843 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOC101927843	LOC101927843	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion myeloid_suppression LOC100507600 IL10 coexpression_network	caf_ecm_exclus ion	myeloid_suppre ssion	LOC100507600	IL10	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction LINC00945 LINC00945 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	LINC00945	LINC00945	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tcell_dysfunction PDCD1 PDCD1 coexpression_network	caf_ecm_exclus ion	tcell_dysfunctio n	PDCD1	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector PDCD1 PDCD1 coexpression_network	caf_ecm_exclus ion	tnk_effector	PDCD1	PDCD1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tnk_effector PDCD1 NKG7 coexpression_network	caf_ecm_exclus ion	tnk_effector	PDCD1	NKG7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation VIM VIM coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	VIM	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	caf_ecm_exclusion tumor_dedifferentiation LOC100507600 LOC100507600 coexpression_network	caf_ecm_exclus ion	tumor_dediffere ntiation	LOC100507600	LOC100507600	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression myeloid_suppression CD163 IL10 coexpression_network	myeloid_suppre ssion	myeloid_suppre ssion	CD163	IL10	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction IL10 CTLA4 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	IL10	CTLA4	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tcell_dysfunction LOC101927843 CXCL13 coexpression_network	myeloid_suppre ssion	tcell_dysfunctio n	LOC101927843	CXCL13	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector CD163 GZMA coexpression_network	myeloid_suppre ssion	tnk_effector	CD163	GZMA	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tnk_effector IL10 GZMA coexpression_network	myeloid_suppre ssion	tnk_effector	IL10	GZMA	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation IL10 LOC100507600 coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	IL10	LOC100507600	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	myeloid_suppression tumor_dedifferentiation LOC101927843 AXL coexpression_network	myeloid_suppre ssion	tumor_dediffere ntiation	LOC101927843	AXL	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction PDCD1 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	PDCD1	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tcell_dysfunction CXCL13 TIGIT coexpression_network	tcell_dysfunctio n	tcell_dysfunctio n	CXCL13	TIGIT	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector PDCD1 PDCD1 coexpression_network	tcell_dysfunctio n	tnk_effector	PDCD1	PDCD1	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tnk_effector TIGIT NKG7 coexpression_network	tcell_dysfunctio n	tnk_effector	TIGIT	NKG7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation LOC388849 LOC388849 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	LOC388849	LOC388849	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tcell_dysfunction tumor_dedifferentiation LOC388849 LOC100507600 coexpression_network	tcell_dysfunctio n	tumor_dediffere ntiation	LOC388849	LOC100507600	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector GZMA NKG7 coexpression_network	tnk_effector	tnk_effector	GZMA	NKG7	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche- Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tnk_effector NKG7 PDCD1 coexpression_network	tnk_effector	tnk_effector	NKG7	PDCD1	coexpression _network

endpoi nt	stratum	holdout	model_nam e	selected_mo del	edge_id	source_state	target_state	gene_a	gene_b	edge_class
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche-Opt- WordNoBio Objective	word_no_bio _objective	tnk_effector tumor_dedifferentiation PDCD1 VIM coexpression_network	tnk_effector	tumor_dediffere ntiation	PDCD1	VIM	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche-Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation AXL LOC100507600 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	AXL	LOC100507600	coexpression _network
primary _recist	melanoma_core_high _evidence	PRJEB23709 _PD1_PRE	EcoNiche-Opt- WordNoBio Objective	word_no_bio _objective	tumor_dedifferentiation tumor_dedifferentiation LOC100507600 LOC388849 coexpression_network	tumor_dediffere ntiation	tumor_dediffere ntiation	LOC100507600	LOC388849	coexpression _network

Supplementary Table 9 | Optimizer history.

Records optimizer objective, best score, best AUROC, best AUPRC, best ECE, and model variant. This table supports Figure 2e and Supplementary Figure 6.

Full tabulation (28 rows x 11 columns).

endpoint	stratum	holdout	model_name	selected_model	generation	best_score	best_AUROC	best_AUPRC	best_ECE	backend
primary_recist	melanoma_recist_supported_primary	GSE91061	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.8913029221173216	0.7593013711434763	0.7549851090272796	0.20095525045402485	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE91061	EcoNiche-Opt-WordNoInteraction	word_no_interaction	1.0	0.6890771015401055	0.6545554226475279	0.6471587171035145	0.12829701837639235	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE91061	EcoNiche-Opt-WordUnsignedGraph	word_unsigned_graph	1.0	0.6667319350326216	0.5406773416641838	0.6034179696824569	0.13448870032942414	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE91061	EcoNiche-Opt-WordNoBioObjective	word_no_bio_objective	1.0	0.7630534380413939	0.6960873556268292	0.720694529590018	0.1584830632934151	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE78220	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.8269493491622834	0.7117687502555924	0.6025751655599777	0.2713748215479362	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE78220	EcoNiche-Opt-WordNoInteraction	word_no_interaction	1.0	0.7825513752419002	0.7672402159244265	0.6831670741783631	0.23776372522239328	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE78220	EcoNiche-Opt-WordUnsignedGraph	word_unsigned_graph	1.0	0.7070380491815047	0.6702702633623687	0.5790726121017032	0.3268587373621095	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE78220	EcoNiche-Opt-WordNoBioObjective	word_no_bio_objective	1.0	0.7103715837780357	0.6937400318979267	0.6639769319553349	0.2709479829157706	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE145996	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.8713356428788018	0.7266102318733898	0.6884031670995232	0.25809680549017916	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE145996	EcoNiche-Opt-WordNoInteraction	word_no_interaction	1.0	0.5908142044636103	0.570757312862576	0.5204828597190337	0.2526144282315625	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE145996	EcoNiche-Opt-WordUnsignedGraph	word_unsigned_graph	1.0	0.6744047782147747	0.6054614975667607	0.5009443097976933	0.29626271752844174	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	GSE145996	EcoNiche-Opt-WordNoBioObjective	word_no_bio_objective	1.0	0.5791768195393435	0.5996810207336524	0.5366745560577447	0.2664486412955563	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.7012884397184597	0.5600503663003663	0.49392842074992943	0.2164479249571637	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordNoInteraction	word_no_interaction	1.0	0.6771697889333712	0.6297008547008547	0.5961246795436572	0.236570540987355	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordUnsignedGraph	word_unsigned_graph	1.0	0.7512356063039878	0.6640949328449328	0.6154862348585988	0.29792152440166497	xgboost_cuda
primary_recist	melanoma_recist_supported_primary	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordNoBioObjective	word_no_bio_objective	1.0	0.7327610797301994	0.7351648351648351	0.6662352997879314	0.2706495970562683	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE91061	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.7942145444717982	0.6719661917030337	0.6523721361240786	0.18148914488350473	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE91061	EcoNiche-Opt-WordNoInteraction	word_no_interaction	1.0	0.7409662386799988	0.6318747042431253	0.6061701336137426	0.11663873013119062	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE91061	EcoNiche-Opt-WordUnsignedGraph	word_unsigned_graph	1.0	0.7943370092247595	0.6274514958725484	0.6364356522098213	0.17471291217700574	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE91061	EcoNiche-Opt-WordNoBioObjective	word_no_bio_objective	1.0	0.821546824020543	0.7750473210999527	0.7362017106602624	0.15534745516169685	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE78220	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.8110359884832083	0.6847073978652927	0.5612760929162294	0.28224953116024826	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE78220	EcoNiche-Opt-	word_no_interaction	1.0	0.757017083430797	0.6758250521408418	0.5426518030841339	0.17812637409919407	xgboost_cuda

endpoint	stratum	holdout	model_name	selected_model	generation	best_score	best_AUROC	best_AUPRC	best_ECE	backend
primary_recist	melanoma_core_high_evidence	GSE78220	WordNoInteraction	word_unsigned_graph	1.0	0.7794517674341581	0.6976352594773647	0.5276442725255768	0.21756778591423262	xgboost_cuda
primary_recist	melanoma_core_high_evidence	GSE78220	EcoNiche-Opt-WordUnsignedGraph	word_no_bio_objective	1.0	0.6666077133694059	0.6745245982088086	0.48117154675825025	0.22819159701355302	xgboost_cuda
primary_recist	melanoma_core_high_evidence	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordFullGraph	word_full_graph	1.0	0.6652564917907996	0.5713369963369963	0.43491049163082507	0.30079406889568017	xgboost_cuda
primary_recist	melanoma_core_high_evidence	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordNoInteraction	word_no_interaction	1.0	0.7350959948211603	0.6517399267399268	0.540046018407363	0.2803713215327768	xgboost_cuda
primary_recist	melanoma_core_high_evidence	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordUnsignedGraph	word_unsigned_graph	1.0	0.6777782749498128	0.5755494505494505	0.4160892678749821	0.2133064760304903	xgboost_cuda
primary_recist	melanoma_core_high_evidence	PRJEB23709_PD1_PRE	EcoNiche-Opt-WordNoBioObjective	word_no_bio_objective	1.0	0.6576621180119602	0.6445054945054944	0.4499561297949131	0.2800254975857378	xgboost_cuda

Supplementary Table 10 | Melanoma benchmark summary.

Summarizes pooled AUROC, AUPRC, balanced accuracy, ECE, and fold statistics for EcoNiche-Opt and baseline signatures in the primary melanoma benchmark. This table supports Figure 3.

Full tabulation (38 rows x 17 columns).

endpoint	stratum	model_name	n_samples	n_cohorts	pooled_AUROC	pooled_AUPRC	pooled_balanced_accuracy	pooled_Brier	pooled_ECE	mean_fold_AUROC	mean_fold_AUPRC	mean_fold_balanced_accuracy	mean_fold_ECE	min_fold_AUROC	max_fold_AUROC	evaluation_modes
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	117	3	0.7049025769956003	0.5584727461504684	0.6315210559396606	0.250038321836014	0.2348619118541941	0.7122567490988544	0.6357707545878359	0.618025249604197	0.29592142820492645	0.6128205128205129	0.8755980861244019	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	117	3	0.7049025769956003	0.5584727461504684	0.6315210559396606	0.250038321836014	0.2348619118541941	0.7122567490988544	0.6357707545878359	0.618025249604197	0.29592142820492645	0.6128205128205129	0.8755980861244019	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleFNConsensus	117	3	0.6778755499685732	0.5245434755898563	0.6363922061596481	0.23694073620276537	0.1514650082968751	0.6720554299501668	0.5782013374198035	0.5818024081181976	0.25476768021917473	0.5494505494505495	0.8564593301435406	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	117	3	0.6731615336266499	0.5196335270591229	0.6363922061596481	0.23586473738326025	0.1368921744590639	0.6666678350888877	0.5822261126363547	0.6361413440360809	0.25353205028114967	0.5274725274725276	0.84688995215311	LODO
primary_recist	melanoma_core_high_evidence	CXCL9	117	3	0.6656191074795725	0.5169521608871197	0.6528912633563797	0.23801152234931466	0.13808986252382452	0.655404244877929	0.5681166544859496	0.6399319978267347	0.23446733021447463	0.49999999999999994	0.8277511961722488	LODO
primary_recist	melanoma_core_high_evidence	TIG	117	3	0.6656191074795725	0.51688752653089108	0.6228786926461345	0.2390586042378697	0.1366348275497397	0.6549730970783602	0.5826841639541903	0.5404104667262563	0.2183739798787705	0.5164835164835164	0.8253588516746412	LODO
primary_recist	melanoma_core_high_evidence	IFNG	117	3	0.6643620364550596	0.5120861576692182	0.6345065996228787	0.23877482236375455	0.13531512901356266	0.6568425726320463	0.5689353224494074	0.5624302597986808	0.2224581534423448	0.510989010989011	0.8492822966507176	LODO
primary_recist	melanoma_core_high_evidence	APM	117	3	0.6637335009428034	0.48783872828045893	0.6258642363293526	0.2405737891862596	0.13641855938444358	0.6870702397018187	0.568588568801435	0.6279153594943069	0.2335062686851002	0.5258410256410255	0.8157894736842105	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	117	3	0.6552482715273412	0.4877455053893849	0.6382778126964174	0.23934611047808996	0.14865713902293257	0.7122567490988544	0.6357707545878359	0.618025249604197	0.25178482833564675	0.6128205128205129	0.8755980861244019	LODO
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	117	3	0.6546197360150849	0.5271690269213777	0.5917661847894407	0.24133386366172327	0.12717488628172327	0.6543322174901122	0.5951931325025758	0.6440915809336862	0.24680210393288463	0.5054945054945055	0.80622009569379	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	117	3	0.6316781898177247	0.4780554067404326	0.6180075424261471	0.2472227411855556	0.15067365469527264	0.643316332790017	0.5708102010189485	0.5487567987567988	0.26414175516605776	0.4945054945054945	0.8277511961722488	LODO
primary_recist	melanoma_core_high_evidence	CYT	117	3	0.6297925832809553	0.48703958290421456	0.5861093651791326	0.24993623239416308	0.13835256442885546	0.6171641224272802	0.5402938287365687	0.583668962616331	0.21844717216070422	0.4175824175824175	0.8133971291866029	LODO
primary_recist	melanoma_core_high_evidence	PDCD1LG2	117	3	0.5914519170333123	0.44399720830576167	0.5493400377121307	0.26275564443123955	0.1716014375495769	0.5908559276980329	0.5079743869860308	0.5612054612054611	0.21588754735946858	0.46153846153846156	0.7392344497607655	LODO
primary_recist	melanoma_core_high_evidence	IPRES	117	3	0.5669390320553112	0.4291849220847015	0.5658390949088623	0.27263539990208385	0.18024643962755493	0.5825396825396826	0.5116573952664597	0.5636293531030373	0.23814010409125644	0.4948717948717949	0.7527472527472527	LODO
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	117	3	0.5436832181018227	0.38164509071064057	0.5193274670018856	0.28105469040300723	0.16930397228588195	0.5608514292724819	0.4689728210967974	0.5229927966770073	0.2210421144820699	0.4615384615384615	0.6923076923076923	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	117	3	0.5194846008799497	0.5069277791385192	0.46715901948460087	0.30640600225171666	0.30043088340384266	0.49860607229028275	0.5154834500740808	0.5344871502766239	0.31969231932514497	0.3406593406593406	0.6602870813397129	LODO
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	117	3	0.5028284098051541	0.38342873400056515	0.5020427404148334	0.3073571853697838	0.25073198318640844	0.4696958012747487	0.4455062226516266	0.4941161178003283	0.2867625683972192	0.3571428571428571	0.562200956937799	LODO

endpoint	stratum	model_name	n_sam ples	n_coh orts	pooled_AURO C	pooled_AUPR C	pooled_balanced_a ccuracy	pooled_Brier	pooled_ECE	mean_fold_AU ROC	mean_fold_AU PRC	mean_fold_balanced_ accuracy	mean_fold_EC E	min_fold_AUR OC	max_fold_AU ROC	evaluation_ modes
primary_r ecist	melanoma_core_high_evi dence	EcoNiche-Opt- WordUnsigned Graph	117	3	0.48774355751 09994	0.38999811868 077994	0.503928346951602 7	0.30603221803 63592	0.28142311820 920274	0.47596964439 069706	0.45410313186 83049	0.4633892423366108	0.29863947570 9693	0.40769230769 23077	0.5824175824 175823	LODO
primary_r ecist	melanoma_core_high_evi dence	EcoNiche-Opt- WordNoBioObj ective	117	3	0.46668761785 04086	0.35574433404 44494	0.434475172847265 9	0.35760342791 728184	0.35378347305 20795	0.50141904878 74698	0.47557849350 622616	0.4924739003686372	0.33132292146 17219	0.37948717948 71795	0.5769230769 230769	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- HeuristicEcolog y	131	4	0.68455882352 94118	0.58047302020 48082	0.612867647058823 5	0.25993137706 48727	0.23858416431 842683	0.66440089515 74741	0.65224518354 54945	0.5586395512382354	0.32241365107 924796	0.52083333333 33334	0.8755980861 244019	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- ModulePriorFix ed	131	4	0.68455882352 94118	0.58047302020 48082	0.612867647058823 5	0.25993137706 48727	0.23858416431 842683	0.66440089515 74741	0.65224518354 54945	0.5586395512382354	0.32241365107 924796	0.52083333333 33334	0.8755980861 244019	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- ModuleFNCCon sensus	131	4	0.65686274509 80392	0.54435479661 42261	0.606617647058823 6	0.24306732875 351095	0.13728703635 307532	0.61341657246 2625	0.59928276713 41167	0.5050987937501095	0.27481550347 77452	0.4375	0.8564593301 435406	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- AdaptiveConse nsus	131	4	0.65441176470 58824	0.54595379057 68007	0.606617647058823 6	0.24108732491 55878	0.12310390205 1317	0.60937587631 66658	0.60263521606 78976	0.5416655712708345	0.26054518791 67006	0.4375	0.8468899521 5311	LODO
primary_r ecist	melanoma_recist_support ed_primary	CXCL9	131	4	0.64926470588 2353	0.53712255565 19088	0.639583333333333 4	0.24297858153 27824	0.13456030368 42726	0.60613651699 17801	0.58979617801 06493	0.5879618188828715	0.26073510446 525255	0.45833333333 33333	0.8277511961 722488	LODO
primary_r ecist	melanoma_recist_support ed_primary	TIDE_dysfuncti on	131	4	0.64558823529 41177	0.54781253992 64986	0.570098039215686 2	0.24312804014 090106	0.11415110795 75842	0.63137416311 78408	0.62492907556 7408	0.5658992323466008	0.25153876140 480363	0.50549450549 45055	0.8062200956 9378	LODO
primary_r ecist	melanoma_recist_support ed_primary	TIG	131	4	0.64436274509 80393	0.544027501666 94281	0.594117647058823 5	0.24488926780 877937	0.11949029910 582962	0.60581315614 21035	0.60468330001 08198	0.5325598743361901	0.24012129188 036682	0.45833333333 33333	0.8253588516 746412	LODO
primary_r ecist	melanoma_recist_support ed_primary	APM	131	4	0.64191176470 58823	0.50256904760 13558	0.602083333333333 3	0.24812291955 157373	0.13856848077 764605	0.64551101310 17079	0.60646955819 17079	0.6358645192526772	0.26505708109 59388	0.52083333333 33334	0.8157894736 842105	LODO
primary_r ecist	melanoma_recist_support ed_primary	IFNG	131	4	0.63970588235 29411	0.53132262845 32585	0.603921568627450 9	0.24584069941 702053	0.13524042281 735324	0.59679859614 07014	0.58901452602 50896	0.5047499649473333	0.22755324121 387033	0.41666666666 666663	0.8492822966 507176	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- ModulePriorFix ed-Platt	131	4	0.63039215686 27452	0.51744604825 82935	0.625367647058823 5	0.24013020200 933854	0.12076221874 679825	0.66440089515 74741	0.65224518354 54945	0.5586395512382354	0.27256849441 953396	0.52083333333 33334	0.8755980861 244019	LODO
primary_r ecist	melanoma_recist_support ed_primary	CYT	131	4	0.62426470588 23529	0.51167370686 83719	0.587867647058823 6	0.25118840957 504796	0.12143659252 064323	0.61391475848 7127	0.59218262804 5933	0.5176433215906899	0.23610162189 809106	0.41758241758 241754	0.8133971291 866029	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- ImmuneCompo site	131	4	0.61985294117 64706	0.50326246177 11886	0.596813725490196 1	0.24942756497 044724	0.13328941742 116335	0.61790391625 91794	0.60267943314 84938	0.5955605797711061	0.23620310899 526903	0.49450549450 54945	0.8277511961 722488	LODO
primary_r ecist	melanoma_recist_support ed_primary	PDCD1LG2	131	4	0.58897058823 52941	0.47487468153 96976	0.555759803921568 7	0.26119595229 318887	0.14133191364 890493	0.58897527910 68581	0.55795251642 99992	0.5749375624375624	0.24088321158 506387	0.46153846153 846156	0.7392344497 607655	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- WordNoInteract ion	131	4	0.58455882352 94118	0.50239365033 02233	0.510906862745098	0.30566493248 50579	0.26306314793 308894	0.53562166999 667	0.53161681591 75181	0.4743818681318681	0.27986030291 355446	0.43406593406 5934	0.6818181818 181819	LODO
primary_r ecist	melanoma_recist_support ed_primary	IPRES	131	4	0.57328431372 54903	0.47080769428 251634	0.578921568627450 9	0.26857585924 38269	0.17402200379 843538	0.59315476190 47619	0.56842711238 39107	0.6301578355854671	0.24475781723 139717	0.49487179487 17949	0.7527472527 472527	LODO
primary_r ecist	melanoma_recist_support ed_primary	TIDE_exclusion	131	4	0.53921568627 45099	0.39508881504 616233	0.533455882352941 2	0.28222769799 674907	0.15619176434 98541	0.54563857195 43614	0.50322447512 99574	0.49672334244702665	0.23722146589 167517	0.46153846153 84615	0.6923076923 076923	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- WordUnsigned Graph	131	4	0.50049019607 84314	0.41642908500 261183	0.485171568627450 9	0.31592288288 624604	0.26322504198 714874	0.48529524203 866303	0.51033296864 27148	0.4223381333578702	0.33429536535 932464	0.32820512820 51282	0.7153110047 84689	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- WordFullGraph	131	4	0.47818627450 980394	0.37529207118 28562	0.438725490196078 4	0.37638898701 592133	0.37021337573 504476	0.46590454501 63871	0.51244078178 23635	0.3887019230769231	0.41081772190 87791	0.30219780219 78022	0.6578947368 421052	LODO
primary_r ecist	melanoma_recist_support ed_primary	EcoNiche-Opt- WordNoBioObj ective	131	4	0.42892156862 7451	0.35962659485 742166	0.418259803921568 64	0.40483875466 35722	0.38195556172 685213	0.41961996336 99634	0.44318821301 11228	0.4224587912087912	0.40191199891 078566	0.26410256410 25641	0.5	LODO

Supplementary Table 11 | Signature family FDR.

Reports the strong-signature family omnibus test, including target AUROC, family mean AUROC, mean delta, confidence interval, P value, and FDR q. This table supports Figure 3b and Figure 3f.

Full tabulation (2 rows x 16 columns).

endpoint	stratum	target_mod el	baseline_family	n_sampl es	target_AUROC	mean_signature_A UROC	mean_delta_vs_signature _family	ci_low	ci_high	one_side d_p	two_side d_p	n_signatu res	two_sided_f dr_q	one_sided_f dr_q	claim_level
primary_re	melanoma_core_high_eviden	EcoNiche-	eight_strong_signa	117	0.704902576995	0.631796040226272	0.07341550358799613	0.0315031315812	0.1121055808865	0.0006	0.0012	8	0.002	0.001	family_two_sided_FDR_su

endpoint	stratum	target_model	baseline_family	n_samples	target_AUROC	mean_signature_AUROC	mean_delta_vs_signature_family	ci_low	ci_high	one_side_d_p	two_side_d_p	n_signature	two_sided_fdr_q	one_sided_fdr_q	claim_level
cist	ce	Opt-HeuristicEcology	tures		6003	8		5656	7187						pported
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	eight_strong_signals	131	0.6845588235294118	0.6196997549019608	0.06449025519724563	0.02452079325838018	0.10265347439297784	0.001	0.002	8	0.002	0.001	family_two_sided_FDR_supported

Supplementary Table 12 | LODO metrics.

Lists leave-one-dataset-out metrics by holdout cohort, endpoint, model, and stratum. This table supports Figure 3d, Figure 4e, and Supplementary Figure 3.

Full tabulation (133 rows x 29 columns).

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
0.612820129	0.443052059052059	0.5064102564102564	0.5102040816326531	0.010336227882434037	0.29411764705882354	0.5	0.51282012820205128	0.20833333333333334	0.8	0.31423500343198435	0.33997949256796905	0.24712001488480964	-1.4350441545460417	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNiche-Opt-HeuristicEcology	49	10	39	module_prior_composite	0.420954778669646	0.6815943559364612	0.7219762250433064	0.5813189003978478	0.7467591122085189	GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.3410254104	0.16424694194333417	0.2935897435897436	0.40816326530612246	-0.3362055740676945	0.06451612903225806	0.1	0.48717948717948717	0.047619047619047616	0.6785714285714286	0.40001152525438144	0.438032152342243	-0.3520080006662343	-1.4618817482562398	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNiche-Opt-WordFullGraph	49	10	39	word_full_graph	0.5463259021422	0.7593013711434763	0.7549851090272796	0.7118141361562413	0.8913029221173216	GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.4641025641025641	0.23251447876447878	0.466666666666667	0.3877551020408163	-0.05644709478621483	0.2857142857142857	0.6	0.33333333333333333	0.1875	0.7647058823529411	0.30746773918542253	0.364166289929839	-0.03991109641999178	-1.3611535320472516	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNiche-Opt-WordNoInteraction	49	10	39	word_no_interaction	0.46133744964339696	0.654554226475279	0.6471587121035145	0.5327432070853124	0.6890771015401055	GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.3282051282051282	0.1686083136497767	0.30512820512820515	0.3673469387755102	-0.31422132762599475	0.11428571428571428	0.2	0.41025641025641024	0.08	0.6666666666666666	0.35832884236731893	0.42032283958428594	-0.310929875424823	-1.376806140474016	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNiche-Opt-WordUnsignedGraph	49	10	39	word_unsigned_graph	0.5295958648727994	0.5406773416641838	0.603417696824569	0.4768389505231611	0.6667319350326216	GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.2641025641025641	0.149982535902411	0.3166666666666665	0.32653061224489793	-0.3006697504747937	0.15384615384615385	0.3	0.33333333333333333	0.10344827586206896	0.65	0.418234715056669	0.4910321399912952	-1.1216200790676436	-1.2817825030496266	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNiche-Opt-WordNoBioObjective	49	10	39	word_no_bio_objective	0.5382813322110971	0.6960873556268292	0.720694529590018	0.6256313131313131	0.7630534380413939	GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.6483516483516484	0.6310725135349196	0.6263736263736264	0.6296296296296297	0.2570193398539127	0.6666666666666666	0.7142857142857143	0.5384615384615384	0.625	0.6363636363636364	0.2909107984360428	0.32640283009776705	0.13949154944492717	0.0771320292345374	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNiche-Opt-HeuristicEcology	27	14	13	module_prior_composite	0.40980104728815997	0.6697506440927494	0.6593027402156862	0.4846813274444853	0.726949847475758	GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.3021978021978022	0.429333204307773	0.3653846153846154	0.37037037037037035	-0.27857101420419017	0.45161290322580644	0.5	0.23076923076923078	0.4117647058823529	0.3	0.4810876064168923	0.4886819953218409	-0.30544396015048525	0.4083616113444496	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNiche-Opt-WordFullGraph	27	14	13	word_full_graph	0.40031942802194476	0.7117687502555924	0.60257561655599777	0.5303418803418803	0.8269493491622834	GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.434065934065934	0.5224891169282359	0.4725274725274725	0.48148148148148145	-0.06264700137907352	0.5882352941176471	0.7142857142857143	0.23076923076923078	0.5	0.42857142857142855	0.3237770417167709	0.24689543204577887	-0.249929432045786053622368	0.2789881370779128	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNiche-Opt-WordNoInteraction	27	14	13	word_no_interaction	0.5381728695733254	0.7672402159244265	0.6831670741783631	0.5785256410256411	0.7825513752419002	GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.33516483516483514	0.4564640726040549	0.3653846153846154	0.37037037037037035	-0.27857101420419017	0.45161290322580644	0.5	0.23076923076923078	0.4117647058823529	0.3	0.3927781580384706	0.3574764906203303	-0.40910263121806395	0.3187277473963717	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNiche-Opt-WordUnsignedGraph	27	14	13	word_unsigned_graph	0.4189121177666395	0.6702702633623687	0.5790726121017032	0.48034188034188036	0.7070380491815047	GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.45604395604395603	0.484218060437918	0.41483516483516486	0.4074074074074074	-0.18638210221950421	0.2727272727272727	0.21428571428571427	0.6153846153846154	0.375	0.42105263157894735	0.40908984336789667	0.38949361588515813	-0.12677893405785787	0.06739811807633919	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNiche-Opt-WordNoBioObjective	27	14	13	word_no_bio_objective	0.5129004561971433	0.6937400318979267	0.6639769319553349	0.5392094017094017	0.7103715837780357	GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.5208333333333334	0.7016684704184704	0.35416666666666666	0.35714285714285715	-0.28867513459481287	0.28867513459481287	0.4	0.33333333333333333	0.42857142857142855	0.2857142857142857	0.3426092590604876	0.4018903197022125	0.16905098599105375	0.30535128873521733	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNiche-Opt-HeuristicEcology	14	8	6	module_prior_composite	0.36866837649370504	0.71122567490988544	0.6357707545878359	0.618025249604197	0.7749465493084561	GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.5625	0.7793612637362637	0.39583333333333333	0.42857142857142855	-0.22821773229381923	0.5555555555555555	0.625	0.16666666666666666	0.5	0.25	0.3056979276738546013	0.44871687882136013	-0.001487954142890588	0.2881542131103825	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNiche-Opt-WordFullGraph	14	8	6	word_full_graph	0.48482319555748143	0.7266102318733898	0.6884031670995232	0.54456009945609945	0.8713356428780818	GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.5625	0.694047619047619	0.45833333333333333	0.5	-0.10050378152592121	0.613578947368421	0.75	0.16666666666666666	0.5454545454545454	0.33333333333333333	0.258608097778387825	0.22354568778387825	0.3305497295401968	0.13037983146683507	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNiche-Opt-WordNoInteraction	14	8	6	word_no_interaction	0.49841365521978553	0.507057312862576	0.5204828597190337	0.5333333333333333	0.5908142044636103	GSE91061,GSE78220,PRJEB23709_PD1_PRE

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
																		n										
0.5625	0.6884386446886447	0.4375	0.5	-0.2401923070763068	0.6666666666666666	0.875	0.0	0.5384615384615384	0.0	0.28915410171060346	0.42090631238020054	0.2412259834375539	0.021366017628071686	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich e-Opt-WordUnsignedGraph	14	8	6	word_undefined_graph	0.46068808719231596	0.6054614975667607	0.5009443097976933	0.5346153846153846	0.6744047782147747	GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.4583333333333333	0.6671092796092796	0.4583333333333333	0.42857142855	-0.09128709291752768	0.3333333333333333	0.25	0.6666666666666666	0.5	0.4	0.31264325374321605	0.3325088708762167	0.013283312892195656	0.2929095051204936	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich e-Opt-WordNoBioObjective	14	8	6	word_no_bio_objective	0.4451054347682488	0.5996810207336524	0.5366745560577447	0.6157447815342552	0.5791768195393435	GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8755980861244019	0.8331876911765292	0.7476076555023923	0.75609756098	0.5126841617284027	0.7058823529411765	0.631578947368421	0.8636363636363636	0.8	0.7307692307692307	0.1463994372895915	0.2213819619490431	0.7739163054922703	-0.12331167315888543	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-HeuristicEcology	41	19	22	module_prior_composite	0.6731694744274304	0.5940018315018315	0.5919310143351496	0.5492979242979243	0.651063278434852	GSE91061,GSE78220,GSE145996
0.6578947368421052	0.6768217171420832	0.5	0.4634146341463415	0.0	0.6333333333333333	1.0	0.0	0.4634146341463415		0.30334771304770053	0.26783986114967234	1.5069705066585073	-1.731658977463252	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordFullGraph	41	19	22	word_full_graph	0.4342136859179444	0.560050366303663	0.49392842074992943	0.5010683760683761	0.7012884397184597	GSE91061,GSE78220,GSE145996
0.6818181818181819	0.6774160489297387	0.5	0.4634146341463415	0.0	0.6333333333333333	1.0	0.0	0.4634146341463415		0.307651059528131	0.2830338018947187	1.806617551115118	-2.1388963160805305	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordNoInteraction	41	19	22	word_no_interaction	0.47380487574224567	0.6297008547008547	0.5961246795436572	0.5119658119658119	0.6771697889333712	GSE91061,GSE78220,GSE145996
0.715311004784689	0.7278208436283831	0.5813397129186603	0.58536585366	0.16346342761154994	0.5405405406	0.5263157894736842	0.6363636363636364	0.5555555555555555	0.6086956521739131	0.223771388076209	0.13847631885248188	1.0579505739150388	-0.41536100823655214	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordUnsignedGraph	41	19	22	word_undefined_graph	0.5747045470535836	0.66409493284493284	0.6154862348585988	0.5556626931623932	0.7512356063039878	GSE91061,GSE78220,GSE145996
0.5	0.4714429760948826	0.5	0.4634146341463415	0.0	0.6333333333333333	1.0	0.0	0.4634146341463415		0.41751084145885226	0.3946133688904725	-0.019376790195264948	-0.10661325183051795	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordNoBioObjective	41	19	22	word_no_bio_objective	0.4822565595341214	0.7351648351648351	0.6662352997879314	0.6512820512820513	0.7327610797301994	GSE91061,GSE78220,GSE145996
0.5256410256410255	0.2772983818726853	0.48205128205128206	0.5306122448979592	-0.029235267310234306	0.25806451612903225	0.4	0.5641025641025641	0.19047619047619047	0.7857142857142857	0.294804440873912	0.3005540674652811	0.09864082696350163	-1.3636368228818594	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	APM	49	10	39	fixed_score	0.5837231490420055					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.7197802197802198	0.6773333180439501	0.7060439560439561	0.7037037037037037	0.4143709650823724	0.6923076923076923	0.6428571428571429	0.7692307692307693	0.75	0.6666666666666666	0.21969492614842054	0.255150338358452	0.7219225188256037	0.08162544900412584	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	APM	27	14	13	fixed_score	0.5664684498898013					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.5208333333333334	0.7201125263625263	0.6041666666666667	0.5714285714285714	0.28281773229381923	0.5	0.375	0.8333333333333333	0.75	0.5	0.311212080461699	0.35970951832845444	-0.1517827936885337	0.2892486307557867	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	APM	14	8	6	fixed_score	0.5936414374526191					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8157894736842105	0.7511340064876697	0.7511961722488039	0.75609756098	0.5085118434825329	0.7222222222222222	0.6842105263157895	0.8181818181818182	0.75	0.18951104209666667	0.1448140023156747	1.239730751536342	-0.2298790751536342	0.2298790751536342	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	APM	41	19	22	fixed_score	0.5878277087491882					GSE91061,GSE78220,GSE145996
0.6384615384615385	0.4140607840077617	0.5692307692307692	0.5510204081632653	0.11163126113028761	0.3529417647058826	0.6	0.5384615384615384	0.25	0.84	0.2656572413596917	0.2972697038926877	0.520556470865696	1.4395025731818942	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	CXCL9	49	10	39	fixed_score	0.49858596841501407					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.4999999999999999	0.5161586204921446	0.5549450549450555	0.5555555555555555	0.10989010989010989	0.5714285714285714	0.5714285714285714	0.5384615384615384	0.5714285714285714	0.5384615384615384	0.2902349611105584	0.23338037385529548	0.02340744589380973	0.07411533015811612	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	CXCL9	27	14	13	fixed_score	0.457118884941497					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.4583333333333333	0.6548347485847485	0.4166666666666666	0.42857142855	-0.1666666666666666	0.5	0.5	0.3333333333333333	0.5	0.3333333333333333	0.284489004708905	0.339538897847398637	0.1254904272175256	0.2888893732806983	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	CXCL9	14	8	6	fixed_score	0.41129569938869637					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8277511961722488	0.7741305589579424	0.811004784688979	0.8048780487804879	0.6250072232206321	0.8095238095238095	0.8947368421052632	0.7272727272727273	0.7391304347826086	0.8888888888888888	0.17057991888158	0.17275237542844077	1.5385405517564854	-0.14446774662917447	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	CXCL9	41	19	22	fixed_score	0.4167356344682064					GSE91061,GSE78220,GSE145996
0.6205128205128205	0.3713724968988127	0.47820512820512817	0.3469387755102041	-0.0397918097332813	0.30434782608695654	0.7	0.2564102564102564	0.19444444444444444	0.7692307692307693	0.27063408633827907	0.31289676702016206	0.4204856869806787	-1.412417681727277	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	CYT	49	10	39	fixed_score	0.31442219163651575					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.41758241758241754	0.48000905199932176	0.48351648351648346	0.481481814814815	-0.033149677206589796	0.46153846153846156	0.42857142857142855	0.5384615384615384	0.5	0.4666666666666667	0.319920833565531	0.25029470222321404	-0.3558083008911333	0.07517346736062233	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	CYT	27	14	13	fixed_score	0.5229439070417874					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.6041666666666667	0.747849025974026	0.5	0.5	0.0	0.5333333333333333	0.5	0.5	0.5714285714285714	0.42857142855	0.2616530285828912	0.2890649711102516	0.3711608253869752	0.3012506927107412	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	CYT	14	8	6	fixed_score	0.39590262327880005					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8133971291866	0.7694999373115	0.6088516746411	0.6341463414634	0.30714510906463	0.4	0.2631578947368	0.9545454545454545	0.8333333333333333	0.6	0.1791123540811	0.0921500472387	1.34754463997445	-0.144250	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1	CYT	41	19	22	fixed_score	0.7560730923619					GSE91061,GSE78220,GSE145996

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
029	716	483	146	827		421	46	34		1686	3652	7	42434910398	cist	primary		_PRE						394					
0.6256410256410256	0.4010549592526336	0.5064102564102564	0.5102040816326531	0.010336227882434037	0.29411764705882354	0.5	0.5128205128205128	0.2083333333333333	0.8	0.2680383933452674	0.3256095934904411	0.4729566365946398	-1.4295052465385998	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNich e-Opt-Adaptive Consensus	49	10	39	fixed_score	0.46540592612585713					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.5274725274725276	0.5570796697280986	0.5521978021978022	0.555555555555556	0.10616016211357263	0.6	0.6428571428571429	0.46153846153846156	0.5625	0.5454545454545454	0.28923419525003286	0.2392676650301868	0.04010728612747618	0.07455069675094378	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNich e-Opt-Adaptive Consensus	27	14	13	fixed_score	0.4619608275477277					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.4375	0.6638625263625264	0.4166666666666663	0.42857142857142855	-0.16666666666666666	0.5	0.5	0.3333333333333333	0.5	0.3333333333333333	0.2847332350071821	0.2815846008233535	0.20789320154769975	0.298760389825214	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich e-Opt-Adaptive Consensus	14	8	6	fixed_score	0.3727111653856482					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.84688995215311	0.7885437089283318	0.691387559806124	0.7073170731707317	0.43079918382479443	0.6	0.47368421052631576	0.9090909090909091	0.8181818181818182	0.6666666666666666	0.16226755434566978	0.1957188923228211	1.5243098891425035	-0.12753493976528335	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-Adaptive Consensus	41	19	22	fixed_score	0.6880091194324769					GSE91061,GSE78220,GSE145996
0.6076923076923078	0.4537276822276822	0.5435897435897435	0.5102040816326531	0.07040378719222165	0.6	0.3333333333333333	0.48717948717948717	0.23076923076923078	0.8260869565217391	0.27139998945096044	0.2968354137979493	0.4137881733634225	-1.405667250750155	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNich e-Opt-Immune Composite	49	10	39	fixed_score	0.4568308350097927					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.4945054945054945	0.4986391195950196	0.5164835164835164	0.5185185185185185	0.033149677206589796	0.5517241379310345	0.5714285714285714	0.46153846153846156	0.5333333333333333	0.5	0.3029772435272014	0.3187456875116468	-0.1478380312832747	0.07440832242048911	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNich e-Opt-Immune Composite	27	14	13	fixed_score	0.4568308350097927					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.5416666666666666	0.6982871295371296	0.6666666666666667	0.6428571428571429	0.34426518632954817	0.6153846153846154	0.5	0.8333333333333333	0.8	0.5555555555555556	0.2678535923156133	0.15238717048290285	0.3024075064320741	0.296619602787951	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich e-Opt-Immune Composite	14	8	6	fixed_score	0.45224223528105273					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8277511961722488	0.7600638012341614	0.6555023923444976	0.6341463414634146	0.3746786679274571	0.7058823529411765	0.9473684210526315	0.36363636363636365	0.5625	0.8888888888888888	0.1816116014726465	0.1768441641885772	1.5630230277240795	-0.1618437560230179	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-Immune Composite	41	19	22	fixed_score	0.293654617086527					GSE91061,GSE78220,GSE145996
0.6102564102564103	0.36111490812710323	0.4935897435897436	0.4897959183673469	-0.010336227882434037	0.2857142857142857	0.5	0.48717948717948717	0.2	0.7916666666666666	0.27850490730466393	0.3541229797259613	0.3658875326675255	-1.4086148967090257	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNich e-Opt-Module FNConsensus	49	10	39	fixed_score	0.4553944273366236					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.5494505494505495	0.5792263061604493	0.4505494505494506	0.4444444444444444	-0.10482848367219182	0.34782608695652173	0.2857142857142857	0.6153846153846154	0.4444444444444444	0.4444444444444444	0.28698823236514304	0.20846093275910518	0.12275418756143773	0.0764190988109209	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNich e-Opt-Module FNConsensus	27	14	13	fixed_score	0.6610119912722886					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.4375	0.6625270562770562	0.4375	0.42857142857142855	-0.125	0.42857142857142855	0.375	0.5	0.5	0.375	0.29426813792759926	0.3349589732534565	0.17798659362189825	0.301542606887871	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich e-Opt-Module FNConsensus	14	8	6	fixed_score	0.4205089245306464					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8564593301435406	0.7942627979718577	0.638755980861244	0.6585365853658537	0.3343283526194	0.5	0.3684210526315789	0.9090909090909091	0.7777777777777778	0.625	0.15430837570575975	0.20171912817245777	1.4291264222181228	-1.1208733684699504	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-Module FNConsensus	41	19	22	fixed_score	0.7492590687199302					GSE91061,GSE78220,GSE145996
0.6128205128205129	0.443052059052059	0.5064102564102564	0.5102040816326531	0.010336227882434037	0.29411764705882354	0.5	0.5128205128205128	0.2083333333333333	0.8	0.31423500343198435	0.33997949256796905	0.24712001488480964	-1.4350441545460417	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNich e-Opt-ModuleP riorFixed	49	10	39	fixed_score	0.4209547786696646					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.6483516483516484	0.6310725135349196	0.62637366263736264	0.6296296296296297	0.2570193398539127	0.6666666666666666	0.7142857142857143	0.5384615384615384	0.625	0.6363636363636364	0.2909107984360428	0.32640283009776705	0.13949154944492717	0.0771320292345374	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNich e-Opt-ModuleP riorFixed	27	14	13	fixed_score	0.40980104728815997					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.5208333333333334	0.7016684704184704	0.3541666666666663	0.35714285715715	-0.28867513459481287	0.4	0.375	0.3333333333333333	0.42857142857142855	0.2857142857142857	0.34260905290604876	0.4018903197022125	0.16905095899105375	0.30535128873521733	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich e-Opt-ModuleP riorFixed	14	8	6	fixed_score	0.36866837649370504					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8755980861244019	0.8331876911765292	0.7476076555023923	0.756097565609756098	0.5126841617284027	0.7058823529411765	0.631578947368421	0.8636363636363636	0.8	0.7307692307692307	0.1463994372895915	0.2213819619490431	0.7739163054922703	-0.12331167315888543	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-ModuleP riorFixed	41	19	22	fixed_score	0.6731694744274304					GSE91061,GSE78220,GSE145996

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
0.6102564102564103	0.36229801620676305	0.491025641025641	0.3673469387755102	-0.01601281538058715	0.311111111111111	0.7	0.28205128205128205	0.2	0.7857142857142857	0.2696007530618479	0.3154597757751346	0.44555241721203787	-1.418406747751206	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE91061	IFNG	49	10	39	fixed_score	0.3547864411957176					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.510989010989011	0.5374488389882454	0.48351648351648346	0.48148148148145	-0.033149677206589796	0.46153846153846156	0.42857142857142855	0.5384615384615384	0.5	0.4666666666666667	0.2966811645497343	0.19221788633288597	-0.04263475853613909	0.07413659865158102	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE78220	IFNG	27	14	13	fixed_score	0.5778335807179699					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.4166666666666663	0.6492521367521367	0.4583333333333333	0.42857142857142855	-0.09128709291752768	0.3333333333333333	0.25	0.6666666666666666	0.5	0.4	0.30489124336217144	0.2428385045284469	-0.015320410563705651	0.287659804730614	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE14596	IFNG	14	8	6	fixed_score	0.5495439873464955					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.8492822966507176	0.8070591121532196	0.5861244019138756	0.6097560975609756	0.228273699383534	0.38461538461538464	0.2631578947368421	0.9090909090909091	0.7142857142857143	0.5882352941176471	0.16380063106551384	0.15969679821901384	1.649924936838622	-0.1696127007739159	primary_reclist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	IFNG	41	19	22	fixed_score	0.7274400208878603					GSE91061,GSE78220,GSE145996
0.4948717948717949	0.2537692702289013	0.5179487179487179	0.46938775510204084	0.029235267310234306	0.3157894736842105	0.6	0.4358974358974359	0.21428571428571427	0.8095238095238095	0.2992518783112473	0.3394012579111751	-0.02086805073750401	-1.360929402374976	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE91061	IPRES	49	10	39	fixed_score	0.42896580046397587					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.7527472527472527	0.803599198420627	0.7005494505494505	0.7037037037037037	0.4078785175942527	0.7333333333333333	0.7857142857142857	0.6153846153846154	0.6875	0.7272727272727273	0.20157487691528142	0.11668357990418274	1.011888926876875	0.08235290545034116	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE78220	IPRES	27	14	13	fixed_score	0.4587714153480886					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.625	0.7387362637362636	0.729166666666667	0.7142857142857143	0.4583333333333333	0.7142857142857143	0.625	0.8333333333333333	0.8333333333333333	0.625	0.2346496980283938	0.26461095665181933	0.6281989454885814	0.3241552889735655	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE14596	IPRES	14	8	6	fixed_score	0.4414052849261511					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.5	0.47760371714985095	0.5729665071770335	0.5609756097560976	0.15346021213308497	0.6086956521739131	0.7368421052631579	0.4090909090909091	0.5185185185185185	0.6428571428571429	0.287621464531731	0.25833560102452115	0.09736660102452684	-0.14694328986182256	primary_reclist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	IPRES	41	19	22	fixed_score	0.4398409119361582					GSE91061,GSE78220,GSE145996
0.5717948717948718	0.34565597571540496	0.5051282051282051	0.4489795918367347	0.008410342670623601	0.3076923076923077	0.6	0.41025641025641024	0.20689655172413793	0.8	0.28163218631733916	0.3025107663043654	0.2550272731776755	-1.3795972261184755	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE91061	PDCD1LG2	49	10	39	fixed_score	0.38584093302343564					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.46153846153846156	0.4846665848075622	0.5521978021978022	0.555555555555556	0.10616016211357263	0.6	0.6428571428571429	0.46153846153846156	0.5625	0.5454545454545454	0.31167931218932654	0.22831736939231637	-0.2406708597502211	0.07424011117369435	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE78220	PDCD1LG2	27	14	13	fixed_score	0.4037209671641261					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.583333333333333334	0.7078869047619047	0.5833333333333333	0.5714285714285714	0.16666666666666666	0.5714285714285714	0.5	0.6666666666666666	0.6666666666666666	0.5	0.24816138228233633	0.31587020426184975	0.5726622055879511	0.33018127973049277	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE14596	PDCD1LG2	14	8	6	fixed_score	0.40913263180790194					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.7392344497607655	0.6936006004351252	0.6590909090909091	0.6341463414634146	0.4216722502501392	0.7169811320754716	1.0	0.31818181818181818	0.5588235294117647	1.0	0.2079778497510632	0.11683450638172393	1.072234992745312	-0.16249549950554965	primary_reclist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	PDCD1LG2	41	19	22	fixed_score	0.27065119821452344					GSE91061,GSE78220,GSE145996
0.6512820512820513	0.4783725718725719	0.5923076923076923	0.46938775510204084	0.15631503171567185	0.38095238095238093	0.8	0.38461538461538464	0.25	0.8823529411764706	0.2926091600355944	0.6318011201077683	-1.4754118887365568	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE91061	TIDE_dysfunction	49	10	39	fixed_score	0.37157683058513313					GSE78220,GSE145996,PRJEB23709_PD1_PRE	
0.5054945054945055	0.5301352252836542	0.5851648351648352	0.5925925925925926	0.18638210221950421	0.6666666666666666	0.7857142857142857	0.38461538461538464	0.5789473684210527	0.625	0.2999548078226442	0.2959394422682456	-0.10115963127418932	0.07422961162508371	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE78220	TIDE_dysfunction	27	14	13	fixed_score	0.3651584443389399					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.5625	0.7141369047619047	0.5	0.5	0.0	0.5333333333333333	0.5	0.5	0.5714285714285714	0.42857142857142855	0.25812222928831574	0.26574873382056063	0.4743556421091657	0.31417577112834094	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE14596	TIDE_dysfunction	14	8	6	fixed_score	0.4035875243049953					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.80622009596378	0.777071600351501	0.5861244019138756	0.6097560975609756	0.228273699383534	0.38461538461538461	0.2631578947368421	0.9090909090909091	0.7142857142857143	0.5882352941176471	0.18638481404011397	0.15185770949485	1.299091469300085	-0.10626228239340658	primary_reclist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	TIDE_dysfunction	41	19	22	fixed_score	0.762866231047587					GSE91061,GSE78220,GSE145996
0.4615384615384615	0.19521690119892943	0.4897435897435898	0.30612244897959184	-0.021350420507344953	0.32	0.8	0.1794871794871795	0.2	0.7777777777777778	0.306773381056998	0.2942063459686146	-0.157585736800474362928168	-1.36800426265775	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE91061	TIDE_exclusion	49	10	39	fixed_score	0.2611868174117531					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.6923076923076923	0.7444006071568182	0.5412087912087913	0.555555555555556	0.13103560459023977	0.6842105263157895	0.9285714285714286	0.15384615384615385	0.5416666666666666	0.22797171802345456	0.16597827613670726	0.0720948628689383	0.08481890854012547	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE78220	TIDE_exclusion	27	14	13	fixed_score	0.2277840702515037					GSE91061,GSE145996,PRJEB23709_PD1_PRE	
0.5	0.6059794372294373	0.4583333333333333	0.5	-0.10050378152592121	0.631578947368421	0.75	0.16666666666666666	0.5454545454545454	0.3333333333333333	0.2920306900301637	0.285759520120491	-0.09667236987242783	0.28825542657705167	primary_reclist	melanoma_recist_supported_primary	LO DO	GSE14596	TIDE_exclusion	14	8	6	fixed_score	0.35223974519059004					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.5287081339712918	0.4673009549346446	0.49760765550239233	0.4878048780487805	-0.004953470161627079	0.5333333333333333	0.631578947368421	0.36363636363636365	0.46153846153846156	0.5333333333333333	0.28527479801769706	0.202941135770588784	-0.08046613577058243	-0.1468108463986748	primary_reclist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	TIDE_exclusion	41	19	22	fixed_score	0.4017809302622677					GSE91061,GSE78220,GSE145996

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_accuracy	selection_score	train_cohorts
0.623076 9230769 231	0.419292 8887721 892	0.516666 6666666 666	0.408163 2653061 2246	0.028647 98545634 7027	0.325581 3953488 3723	0.7	0.333333 3333333 33	0.212121 2121212 213	0.8125	0.265345 5786355 9327	0.295491 7737676 4635	0.509902 90438408 14	-1.436406 40689378 16	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	TIG	49	10	39	fixed_score	0.383432 5870510 621					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.516483 5164835 164	0.549666 8841010 274	0.516483 5164835 164	0.518518 5185185 185	0.033149 67720658 9796	0.551724 1379310 345	0.571428 5714285 714	0.533333 3333333 33	0.5	0.300071 8415601 082	0.215967 2304983 6508	-0.095178 92769549 889	0.074258 33815209 912	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	TIG	27	14	13	fixed_score	0.471014 8269172 358					GSE91061,GSE145996,PRJEB23709_PD1_PRE	
0.458333 3333333 333	0.670680 7081807 082	0.458333 3333333 333	0.428571 4285714 2855	-0.091287 09291752 768	0.333333 3333333 333	0.25	0.666666 6666666 66	0.5	0.4	0.293616 9562228 105	0.305363 2278851 558	0.098536 29274356 097	0.288399 16609032 074	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	TIG	14	8	6	fixed_score	0.536333 7481604 222					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.825358 8516746 412	0.779092 1789893 544	0.638755 9808612 44	0.658536 5853658 537	0.334328 65753526 537	0.5	0.368421 0526315 789	0.909090 9090909 01	0.777777 7777777 78	0.625	0.167463 0151357 0147	0.143662 9353703 001	1.521914 67427083 63	-0.173476 93750341 58	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	TIG	41	19	22	fixed_score	0.720050 6505402 119					GSE91061,GSE78220,GSE145996
0.612820 5128205 129	0.443052 0590520 59	0.506410 2564102 564	0.510204 0816326 531	0.010336 22788243 4037	0.294117 6470588 2354	0.5	0.512820 51282051 28	0.208333 3333333 334	0.8	0.258458 1841125 3474	0.312923 6394513 3955	0.630822 31071035 93	-1.452827 53777939 49	primary_recist	melanoma_recist_supported_primary	LO DO	GSE91061	EcoNich-e-Opt-ModulePriorFixed-Platt	49	10	39	fixed_score_platt	0.475696 2805207 5186					GSE78220,GSE145996,PRJEB23709_PD1_PRE
0.648351 6483516 484	0.631072 5135349 196	0.626373 6263736 264	0.629629 6296296 297	0.257019 33985391 27	0.666666 6666666 666	0.714285 7142857 143	0.538461 53846153 84	0.625	0.636363 6363636 364	0.253310 1524233 757	0.161038 5941785 887	0.370772 40082735 84	0.102929 34655716 177	primary_recist	melanoma_recist_supported_primary	LO DO	GSE78220	EcoNich-e-Opt-ModulePriorFixed-Platt	27	14	13	fixed_score_platt	0.448430 8486844 715					GSE91061,GSE145996,PRJEB23709_PD1_PRE
0.520833 3333333 334	0.701668 4704184 704	0.354166 6666666 6663	0.357142 8571428 5715	-0.288675 13459481 287	0.4	0.375	0.333333 3333333 33	0.428571 42857142 855	0.285714 2857142 857	0.258050 4201131 9107	0.387343 2573222 005	0.485526 46870541 344	0.335844 08983784 403	primary_recist	melanoma_recist_supported_primary	LO DO	GSE145996	EcoNich-e-Opt-ModulePriorFixed-Platt	14	8	6	fixed_score_platt	0.437792 2748557 1015					GSE91061,GSE78220,PRJEB23709_PD1_PRE
0.875598 0861244 019	0.833187 6911765 292	0.747607 6555023 923	0.756097 5609756 098	0.512684 16172840 27	0.705882 3529411 765	0.631578 9473684 21	0.863636 3636363 36	0.8	0.730769 2307692 307	0.203427 4498705 69	0.228968 7183160 07	4.905624 18541843 6	-0.045801 30460231 5684	primary_recist	melanoma_recist_supported_primary	LO DO	PRJEB23709_PD1_PRE	EcoNich-e-Opt-ModulePriorFixed-Platt	41	19	22	fixed_score_platt	0.524520 9639120 226					GSE91061,GSE78220,GSE145996
0.612820 5128205 129	0.443052 0590520 59	0.506410 2564102 564	0.510204 0816326 531	0.010336 22788243 4037	0.294117 6470588 2354	0.5	0.512820 51282051 28	0.208333 3333333 334	0.8	0.314235 0034319 8435	0.339979 4925679 6905	0.247120 01488480 964	1.435044 15454604 17	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich-e-Opt-HeuristicEcology	49	10	39	module_prior_comp_osite	0.436049 1969306 4953	0.76197 4867238 0251	0.732130 1023557 244	0.638118460 4868816	0.83048 7295380 1553	GSE78220,PRJEB23709_PD1_PRE
0.494871 7948717 949	0.382337 2018052 8685	0.516666 6666666 666	0.408163 2653061 2246	0.028647 98545634 7027	0.325581 3953488 3723	0.7	0.333333 3333333 33	0.212121 2121212 213	0.8125	0.298720 2081853 2175	0.405632 5374345 7894	0.259477 60061130 476	-1.436065 94479348 47	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich-e-Opt-WordFullGraph	49	10	39	word_full_graph	0.469175 6035060 841	0.67196 6191703 0337	0.652372 1361240 786	0.654615121 7203848	0.79421 4544471 7982	GSE78220,PRJEB23709_PD1_PRE
0.489743 5897435 898	0.297673 7821565 408	0.466666 6666666 667	0.387755 1020408 163	-0.056447 09478621 483	0.285714 2857142 857	0.6	0.333333 3333333 33	0.1875	0.764705 8823529 411	0.328966 7423935 543	0.363777 6948999 192	0.082188 32092434 693	-1.367583 97998439	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich-e-Opt-WordNoInteraction	49	10	39	word_no_interaction	0.345665 1580766 7486	0.63187 4704243 1253	0.606170 1336137 426	0.605250013 14475	0.74096 6238679 9988	GSE78220,PRJEB23709_PD1_PRE
0.407692 3076923 077	0.194363 9368403 1117	0.380769 2307692 3075	0.428571 4285714 2855	-0.192253 83861327 31	0.176470 5882352 9413	0.3	0.461538 46153846 156	0.125	0.72	0.311840 0730086 4425	0.380754 8998786 86	-0.175355 64203522 433	-1.455264 03135554 6	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich-e-Opt-WordUnsignedGraph	49	10	39	word_unsigned_graph	0.447137 2752985 2365	0.62745 1495872 5484	0.636435 6522098 213	0.567912350 8070876	0.79433 7009224 7595	GSE78220,PRJEB23709_PD1_PRE
0.379487 1794871 795	0.194253 4032298 81	0.416666 6666666 6663	0.367346 9387755 102	-0.139336 19725752 808	0.243902 4390243 9024	0.5	0.333333 3333333 33	0.161290 32258064 516	0.722222 2222222 222	0.449951 3342112 017	0.505129 2442924 482	-0.253510 68260185 894	-1.279948 73757743 37	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich-e-Opt-WordNoBioObjective	49	10	39	word_no_bio_objective	0.483702 9057743 3067	0.77504 7321099 9527	0.736201 7106602 624	0.742461485 8825385	0.82154 6824020 543	GSE78220,PRJEB23709_PD1_PRE
0.648351 6483516 484	0.631072 5135349 196	0.626373 6263736 264	0.629629 6296296 297	0.257019 33985391 27	0.666666 6666666 666	0.714285 7142857 143	0.538461 53846153 84	0.625	0.636363 6363636 364	0.290910 7984360 428	0.326402 8300977 6705	0.139491 54944492 717	0.077132 02923453 74	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE78220	EcoNich-e-Opt-HeuristicEcology	27	14	13	module_prior_comp_osite	0.409801 0472881 5997	0.74420 9299472 4574	0.638119 8751142 941	0.578983375 0460066	0.80506 4961980 4724	GSE91061,PRJEB23709_PD1_PRE
0.340659 3406593 406	0.466885 6514431 0484	0.532967 0329670 33	0.518518 5185185 185	0.104828 48367219 183	0.235294 1176470 5882	0.142857 1428571 4285	0.923076 92307692 31	0.666666 6666666 66	0.5	0.339323 7390285 1887	0.272512 4784426 242	-0.214145 62526103 817	-0.028904 27683314 697	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE78220	EcoNich-e-Opt-WordFullGraph	27	14	13	word_full_graph	0.557559 8927612 759	0.68470 7397865 2927	0.561276 0829162 294	0.526923076 9230769	0.81103 5988483 2083	GSE91061,PRJEB23709_PD1_PRE
0.357142 8571428 571	0.529406 7282498 96	0.491758 2417582 418	0.481481 4814814 8145	-0.019810 72129375 818	0.3	0.214285 7142857 1427	0.769230 76923076 93	0.5	0.476190 4761904 7616	0.319452 8194125 1413	0.307405 0812014 897	-0.365484 65957738 463	-0.083253 52467576 445	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE78220	EcoNich-e-Opt-WordNoInteraction	27	14	13	word_no_interaction	0.521578 7248774 352	0.67582 5052140 8418	0.542651 8030841 339	0.5	0.75701 7083430 797	GSE91061,PRJEB23709_PD1_PRE
0.582417 5824175 823	0.652004 7431714 787	0.535714 2857142 857	0.518518 5185185 185	0.188982 23650461 36	0.133333 3333333 3333	0.071428 5714285 7142	1.0	1.0	0.5	0.299112 6193895 906	0.293581 7816164 003	0.438166 34040889 313	0.480316 30357931 99	primary_recist	melanoma_cor_e_high_evidence	LO DO	GSE78220	EcoNich-e-Opt-WordUnsignedGraph	27	14	13	word_unsigned_graph	0.559476 5446331 229	0.69763 5259477 3647	0.527644 2725255 768	0.5	0.77945 1767434 1581	GSE91061,PRJEB23709_PD1_PRE
0.576923	0.633505	0.590659	0.592592	0.182323	0.620689	0.642857	0.538461	0.6	0.583333	0.264420	0.178205	0.395162	0.223672	prima	melanoma_cor	LO	GSE78220	EcoNich	27	14	13	word_no_	0.409872	0.67452	0.481171	0.523076923	0.66660	GSE91061,PRJEB2

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
0769230769	106005106	3406593407	5925925926	22463624387	6551724138	1428571429	5384615384		3333333334	2829391909	60924818007	195595779	67485068132	ry_ricist	e_high_evidence	DO	0	e-Opt-WordNoBioObjective				bio_objective	75350966446	45982088086	54675825025	0769231	77133694059	3709_PD1_PRE
0.8755980861244019	0.8331876911765292	0.7212918660287082	0.7317073170731707	0.46541211876427413	0.6666666666666666	0.5789473684210527	0.8636363636363636	0.7857142857142857	0.7037037037037037	0.1463994372895915	0.2213819619490431	0.7739163054922703	0.12331167315888543	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-Heuristic Ecology	41	19	22	module_prior_comp osite	0.7106589134139443	0.6305860805862862934893	0.5370628827839	0.5338827839	0.684133340948942	GSE91061,GSE78220
0.6602870813397129	0.6972274969738508	0.553827511961722	0.53658536586	0.12116227045072343	0.6122448979591837	0.7894736842105263	0.3181818181818182	0.5	0.6363636363636364	0.29391392947805056	0.2809319420982318	0.3215314785376362	-0.6049067625941468	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordFullGraph	41	19	22	word_full_graph	0.4628979958822678	0.5713369963369963	0.43491049163082507	0.4576923076923077	0.6652564917907996	GSE91061,GSE78220
0.562200956937799	0.5094381575484431	0.5239234449760766	0.5121951219512195	0.0503148236501918	0.5652173913043478	0.6842105263157895	0.3636363636363636	0.4814814814814815	0.5714285714285714	0.2735657118742112	0.1891049290902488	0.406477054565419	-0.3682424135810566	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordNoInteraction	41	19	22	word_no_interaction	0.5725013467493818	0.6517399267399268	0.540046018407363	0.5480769230769231	0.7350959948211603	GSE91061,GSE78220
0.43779904306220097	0.515940715593125	0.47368421052631576	0.43902439024390244	-0.17013926184468015	0.6101694915254238	0.9473684210526315	0.0	0.45	0.0	0.30364793193442724	0.22158174563399277	0.01383796335505951	-0.15609464339086643	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordUnsignedGraph	41	19	22	word_unsigned_graph	0.41733185767908365	0.5755494505494505	0.4160892678749821	0.5	0.6777782749498128	GSE91061,GSE78220
0.547846889521531	0.598976129836937799914	0.4700956937799043	0.43902439024390244	-0.11452477365506437	0.5964912280701754	0.8947368421052632	0.04545454545454545	0.4473684210526316	0.3333333333333333	0.308600928063788	0.31063391084453734	0.19260665195400262	-0.29940883030421833	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-WordNoBioObjective	41	19	22	word_no_bio_objective	0.45872917489924625	0.6445054945054944	0.4499561297949131	0.5346153846153846	0.6576621180119602	GSE91061,GSE78220
0.5256410256410255	0.2772983818726853	0.4948717948717948717	0.551020481632342670623601	-0.008410342670623601	0.2666666666666666	0.4	0.5897435897435898	0.2	0.7931034482758621	0.294804440873911	0.300554067465281	0.09864082696350163	-1.3636363628818594	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE91061	APM	49	10	39	fixed_score	0.5913291473453558					GSE78220,PRJEB23709_PD1_PRE
0.7197802197802198	0.6773333180439501	0.6675824175824177	0.6666666666666666	0.33516483516483514	0.6666666666666666	0.6428571428571429	0.6923076923076923	0.6923076923076923	0.6428571428571429	0.21969492614842054	0.255150338358452	0.7219225188256037	0.08162544900412584	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE78220	APM	27	14	13	fixed_score	0.5374246967100874					GSE91061,PRJEB23709_PD1_PRE
0.8157894736842105	0.751134006487697	0.7212918660287082	0.7317073170731707	0.46541211876427413	0.6666666666666666	0.5789473684210527	0.8636363636363636	0.7857142857142857	0.7037037037037037	0.1895114002315667	0.1448144002315667	1.2397330751536342	-0.229879833366525	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	APM	41	19	22	fixed_score	0.6018063574612967					GSE91061,GSE78220
0.6384615384615385	0.41406077617	0.5538461538461539	0.40816326530612246	0.09607689228305227	0.3555555555555555	0.8	0.3076923076923077	0.22857142857142856	0.8571428571428571	0.2656577038926917	0.2972692413596877	0.520556470865696	-1.4395025731818942	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE91061	CXCL9	49	10	39	fixed_score	0.34719785582968593					GSE78220,PRJEB23709_PD1_PRE
0.4999999999999999	0.5161586204921446	0.5549455555555555	0.5555555555555555	0.10989010989010989	0.5714285714285714	0.5714285714285714	0.5384615384615384	0.5714285714285714	0.5384615384615384	0.29023496111055624	0.23338037385529548	0.02340744589380973	0.07411533015811612	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE78220	CXCL9	27	14	13	fixed_score	0.5011820589293959					GSE91061,PRJEB23709_PD1_PRE
0.8277511961722488	0.7741307589579424	0.8110047846889952	0.8048784878048789	0.6250072322206321	0.80952384028095	0.8947368421052632	0.7272727272727273	0.7391304347826086	0.8888888888888888	0.1705799188815344	0.17275237542844077	1.5385405517564854	-0.14446774662917447	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	CXCL9	41	19	22	fixed_score	0.39575758236301534					GSE91061,GSE78220
0.6205128205128205	0.3713724968988127	0.5423076923076923	0.4489795918367347	0.07073991553074502	0.341463141463414637	0.7	0.38461538461538464	0.225806611290322	0.8333333333333333	0.27063408633827907	0.31289676702016206	0.4204856869806787	-1.412417681727277	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE91061	CYT	49	10	39	fixed_score	0.43014437982993					GSE78220,PRJEB23709_PD1_PRE
0.41758241758241754	0.48000905199932176	0.43956043956043955	0.4444444444444444	-0.12507270025494252	0.5161290322580645	0.5714285714285714	0.3076923076923077	0.47058823529411764	0.4	0.31992083356551404	0.25029470222321404	-0.355808300891133	0.07517346736062233	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE78220	CYT	27	14	13	fixed_score	0.4530396433661013					GSE91061,PRJEB23709_PD1_PRE
0.8133971291866029	0.7694999373115716	0.7691387559808612	0.7560975609756098	0.5660417660646577	0.782608695652174	0.9473684210526315	0.5909090909090909	0.6666666666666666	0.9285714285714286	0.1791123540811686	0.09215004723873652	1.374574639974457	-0.14425042434910398	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	CYT	41	19	22	fixed_score	0.3627664642326902					GSE91061,GSE78220
0.6256410256410256	0.40105456794819	0.5679489592526336	0.4897959183673469	0.11143704038576271	0.358974358974359	0.7	0.4358974358974359	0.2413793103448276	0.85	0.2680383933452674	0.3256095934904411	0.4729566365946398	-1.4295052465385998	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich e-Opt-AdaptiveConsensus	49	10	39	fixed_score	0.37629636442292536					GSE78220,PRJEB23709_PD1_PRE
0.5274725274725276	0.5570796697280986	0.5521978021978022	0.5555555555555555	0.10616016211357263	0.6	0.6428571428571429	0.46153846153846156	0.5625	0.5454545454545454	0.28923419525003286	0.2392676650301868	0.04010728612747618	0.07455069675094378	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE78220	EcoNich e-Opt-AdaptiveConsensus	27	14	13	fixed_score	0.4646785522112534					GSE91061,PRJEB23709_PD1_PRE
0.8468895215311	0.7885437089283318	0.7882775119617225	0.7804878048780488	0.5835778775204306	0.7906976744186046	0.8947368421052632	0.6818181818181818	0.7083333333333333	0.8823529411764706	0.1622675843456978	0.1957188923228211	1.5243098891425035	-0.12753493976528335	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNich e-Opt-AdaptiveConsensus	41	19	22	fixed_score	0.361450277663899					GSE91061,GSE78220
0.6076923076923	0.4537276822276	0.5064102564102	0.5102040816326	0.10133622788243	0.2941176470588	0.5	0.51282051282051	0.2083333333333333	0.8	0.2713999894509	0.2968354137979	0.41378817336342	-1.405667	prima_ry_ricist	melanoma_cor_e_high_evidence	LO DO	GSE91061	EcoNich e-Opt-	49	10	39	fixed_score	0.4848019292585					GSE78220,PRJEB23709_PD1_PRE

AUROC	AUPRC	balance d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
078	822	564	531	4037	2354		28	334		6044	493	25	250750155	cist	ce			Immune Composite					35					
0.494505 4945054 945	0.498639 1195950 0196	0.480769 2307692 308	0.481481 4814814 8145	- 0.038461 53846153 8464	0.5	0.5	0.461538 46153846 156	0.5	0.461538 4615384 6156	0.302977 2435272 014	0.318745 6875116 468	- 0.147838 03128327 47	0.074408 32242048 911	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE7822 0	EcoNich e-Opt-Immune Composite	27	14	13	fixed_sco re	0.479220 0600302 3104					GSE91061,PRJEB2 3709_PD1_PRE
0.827751 1961722 488	0.760063 8012341 614	0.659090 9090909 091	0.634146 3414634 146	0.421672 25025013 92	0.716981 1320754 716	1.0	0.318181 81818181 82	0.558823 52941176 47	1.0	0.181611 6014726 465	0.176844 1641885 772	1.563023 02772407 95	- 0.161843 75602301 79	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	PRJEB23 709_PD1 _PRE	EcoNich e-Opt-Immune Composite	41	19	22	fixed_sco re	0.283203 0736627 923					GSE91061,GSE782 20
0.610256 4102564 103	0.361114 9081271 0323	0.580769 2307692 307	0.510204 0816326 531	0.131558 70289605 437	0.368421 0526315 789	0.7	0.461538 46153846 156	0.25	0.857142 8571428 571	0.278504 9073046 6393	0.354122 9797259 613	0.365887 53266752 55	- 1.408614 89670902 57	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE9106 1	EcoNich e-Opt-Module1 FNConsensus	49	10	39	fixed_sco re	0.402068 6474856 396					GSE78220,PRJEB2 3709_PD1_PRE
0.549450 5494505 495	0.579226 3061604 493	0.552197 8021978 022	0.555555 5555555 556	0.106160 16211357 263	0.6	0.642857 1428571 429	0.461538 46153846 156	0.5625	0.545454 5454545 454	0.286988 2323651 4304	0.208460 9327591 0518	0.122754 18756143 773	0.076419 09881092 09	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE7822 0	EcoNich e-Opt-Module1 FNConsensus	27	14	13	fixed_sco re	0.401570 3167492 957					GSE91061,PRJEB2 3709_PD1_PRE
0.856459 3301435 406	0.794262 7979718 577	0.612440 1913875 598	0.634146 3414634 146	0.282968 27628207 755	0.444444 4444444 444	0.315789 4736842 105	0.909090 90909090 91	0.75	0.606060 6060606 061	0.154308 3757057 5975	0.201719 1281724 5777	1.429126 42221812 28	- 0.120873 36846995 04	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	PRJEB23 709_PD1 _PRE	EcoNich e-Opt-Module1 FNConsensus	41	19	22	fixed_sco re	0.763036 1741533 921					GSE91061,GSE782 20
0.612820 5128205 129	0.443052 0590520 59	0.506410 2564102 564	0.510204 0816326 531	0.101036 272788243 4037	0.294117 6470588 2354	0.5	0.512820 51282051 28	0.208333 33333333 334	0.8	0.314235 0034319 8435	0.339979 4925679 6905	0.247120 01488480 964	- 1.435044 15454604 17	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE9106 1	EcoNich e-Opt-ModuleP riorFixed	49	10	39	fixed_sco re	0.436049 1969306 4953					GSE78220,PRJEB2 3709_PD1_PRE
0.648351 6483516 484	0.631072 5135349 196	0.626373 6263736 264	0.629629 6296296 297	0.257019 33985391 666	0.666666 6666666 666	0.714285 7142857 143	0.538461 64846153 84	0.625	0.636363 6363636 364	0.290910 7984360 6705	0.326402 8300977 6705	0.139491 54944492 717	0.077132 02923453 74	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE7822 0	EcoNich e-Opt-ModuleP riorFixed	27	14	13	fixed_sco re	0.409801 0472881 5997					GSE91061,PRJEB2 3709_PD1_PRE
0.875598 0691244 019	0.833187 6911765 292	0.721291 8660287 082	0.731707 3170731 707	0.465412 11876427 413	0.666666 6666666 666	0.578947 3684210 527	0.863636 36363636 36	0.785714 28571428 57	0.703703 7037037 037	0.146399 4372895 915	0.221381 9619490 431	0.773916 30549227 03	- 0.123311 67315888 543	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	PRJEB23 709_PD1 _PRE	EcoNich e-Opt-ModuleP riorFixed	41	19	22	fixed_sco re	0.710658 9134139 443					GSE91061,GSE782 20
0.610256 4102564 103	0.362298 0162067 6305	0.532051 2820512 82	0.551020 4081632 653	0.051941 49944491 429	0.3125	0.5	0.564102 56410256 41	0.227272 72727272 727	0.814814 8148148 148	0.269600 7530618 479	0.315459 7757751 346	0.445552 41721203 787	- 1.418406 74775120 6	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE9106 1	IFNG	49	10	39	fixed_sco re	0.534211 8039098 275					GSE78220,PRJEB2 3709_PD1_PRE
0.510989 0109890 11	0.537448 8389882 454	0.516483 5164835 164	0.518518 5185185 185	0.033149 67720658 9796	0.551724 1379310 345	0.571428 5714285 714	0.461538 46153846 156	0.533333 33333333 33	0.5	0.296681 1645497 343	0.192217 8863328 8597	- 0.042634 75853613 909	0.074136 59865158 102	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE7822 0	IFNG	27	14	13	fixed_sco re	0.538702 9512148 751					GSE91061,PRJEB2 3709_PD1_PRE
0.849282 2966507 176	0.807059 1121532 136	0.638755 9808612 44	0.658536 5853658 537	0.334328 65753526 94	0.5	0.368421 0526315 789	0.909090 90909090 91	0.777777 77777777 777	0.625	0.163800 6310655 1004	0.159696 7982190 1384	1.649924 93683862 2	- 0.169612 70077391 59	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	PRJEB23 709_PD1 _PRE	IFNG	41	19	22	fixed_sco re	0.677816 1938984 679					GSE91061,GSE782 20
0.494871 7948717 949	0.253769 2702289 013	0.480769 2307692 308	0.469387 7551020 4084	- 0.031060 49434950 955	0.277777 7777777 778	0.5	0.461538 46153846 156	0.192307 69230769 232	0.782608 6956521 74	0.299251 8783112 473	0.339401 2579111 751	- 0.020868 05073750 401	- 1.360929 40237497 6	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE9106 1	IPRES	49	10	39	fixed_sco re	0.464945 9590308 5245					GSE78220,PRJEB2 3709_PD1_PRE
0.752747 2527472 527	0.803599 1984206 27	0.700549 4505494 505	0.703703 7037037 037	0.407878 5179425 27	0.733333 3333333 333	0.785714 2857142 857	0.615384 61538461 54	0.6875	0.727272 7272727 273	0.201574 8769152 8142	0.116683 5799041 8274	1.011888 92687687 5	0.082352 90545034 116	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE7822 0	IPRES	27	14	13	fixed_sco re	0.442383 7034272 867					GSE91061,PRJEB2 3709_PD1_PRE
0.5	0.477603 7171498 5095	0.509569 3779904 306	0.512195 1219512 195	0.019138 75598086 1243	0.473684 2105263 1576	0.473684 2105263 1576	0.545454 54545454 54	0.473684 21052631 576	0.545454 5454545 454	0.287621 4164531 731	0.258335 60102452 115	0.097366 60102452 684	- 0.146943 28986182 256	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	PRJEB23 709_PD1 _PRE	IPRES	41	19	22	fixed_sco re	0.547425 5671615 024					GSE91061,GSE782 20
0.571794 8717948 718	0.345655 9757154 0496	0.530769 2307692 308	0.489795 9183673 469	0.049863 83946711 771	0.324324 3243243 2434	0.6	0.461538 46153846 156	0.222222 2222222 22	0.818181 8181818 182	0.281632 1863173 3916	0.302510 7663043 654	0.255027 27317767 55	- 1.379597 22611847 55	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE9106 1	PDCD1L G2	49	10	39	fixed_sco re	0.404158 9672519 375					GSE78220,PRJEB2 3709_PD1_PRE
0.461538 4615384 6156	0.484666 5848075 622	0.516483 5164835 164	0.518518 5185185 185	0.033149 67720658 9796	0.551724 1379310 345	0.571428 5714285 714	0.461538 46153846 156	0.533333 33333333 33	0.5	0.311679 3121893 2654	0.228317 3693923 1637	- 0.240670 85975022 11	0.074240 11117369 435	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE7822 0	PDCD1L G2	27	14	13	fixed_sco re	0.421225 5340744 388					GSE91061,PRJEB2 3709_PD1_PRE
0.739234 4497607 655	0.693600 6004351 252	0.636363 6363636 364	0.609756 0975609 756	0.384775 19157548 0917	0.703703 7037037 037	1.0	0.272727 2727272 27	0.542857 14285714 28	1.0	0.207977 8497510 632	0.116834 5063817 2393	1.072234 99274531 2	- 0.162495 49950554 965	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	PRJEB23 709_PD1 _PRE	PDCD1L G2	41	19	22	fixed_sco re	0.262103 9443243 9077					GSE91061,GSE782 20
0.651282 0512820 513	0.478372 5718725 719	0.605128 2051282 052	0.489795 9183673 469	0.175777 97192488 157	0.390243 9024390 244	0.8	0.410256 41025641 024	0.258064 51612903 225	0.888888 8888888 888	0.255010 3032972 02	0.292609 1600355 944	0.631801 12010776 83	- 1.475411 88873655	primary_re cist	melanoma_cor e_high_eviden ce	LO DO	GSE9106 1	TIDE_dy sfunction	49	10	39	fixed_sco re	0.399553 5104313 3397					GSE78220,PRJEB2 3709_PD1_PRE

AUROC	AUPRC	balance_d_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	endpoint	stratum	evaluation	cohort	model_name	n_samples	n_responders	n_nonresponders	selected_model	threshold	inner_mean_AUROC	inner_mean_AUPRC	inner_mean_balanced_accuracy	selection_score	train_cohorts
0.505494 5054945 055	0.530135 2252836 542	0.587912 0879120 879	0.592592 5925925 926	0.181923 92764355 278	0.645161 2903225 806	0.714285 7142857 143	0.461538 46153846 156	0.588235 29411764 71	0.6	0.299954 8078226 442	0.295939 4422682 456	- 0.101159 63127418 932	0.074229 61162508 371	primary_recist	melanoma_core_high_evidence	LO DO	GSE78220	TIDE dysfunction	27	14	13	fixed_score	0.400220 4421855 477					GSE91061,PRJEB23709_PD1_PRE
0.806220 0956937 8	0.777071 6003515 01	0.739234 4497607 655	0.731707 3170731 707	0.484296 99379288 84	0.744186 0465116 279	0.842105 2631578 947	0.636363 63636363 64	0.666666 66666666 66	0.823529 4117647 058	0.186384 8140401 7884	0.151857 7094948 1397	1.299091 - 0.106262 46930008 5	primary_recist	melanoma_core_high_evidence	LO DO	PRJEB23709_PD1_PRE	TIDE dysfunction	41	19	22	fixed_score	0.398398 5028699 2345					GSE91061,GSE78220	
0.461538 4615384 615	0.195216 9011989 2943	0.503846 1538461 538	0.387755 1020408 163	0.006726 72793996 3125	0.318181 8181818 182	0.7	0.307692 30769230 77	0.205882 35294117 646	0.8	0.306773 3810569 998	0.294206 3459686 146	- 0.157585 74362928 168	primary_recist	melanoma_core_high_evidence	LO DO	GSE91061	TIDE_exclusion	49	10	39	fixed_score	0.357985 0265039 886					GSE78220,PRJEB23709_PD1_PRE	
0.692307 6923076 923	0.744400 6071568 182	0.541208 7912087 913	0.555555 5555555 556	0.131035 60459023 977	0.684210 5263157 895	0.928571 4285714 286	0.153846 15384615 385	0.541666 66666666 66	0.666666 66666666 666	0.227971 7180234 5456	0.165978 2761367 0726	0.720944 86286893 83	primary_recist	melanoma_core_high_evidence	LO DO	GSE78220	TIDE_exclusion	27	14	13	fixed_score	0.229020 0307152 0008					GSE91061,PRJEB23709_PD1_PRE	
0.528708 1339712 918	0.467300 9549346 446	0.523923 4449760 766	0.512195 1219512 195	0.050314 32635019 18	0.565217 3913043 478	0.684210 5263157 895	0.363636 36363636 365	0.481481 48148148 145	0.571428 5714285 714	0.285274 7980176 9706	0.202941 7213408 8784	0.080466 13577058 243	primary_recist	melanoma_core_high_evidence	LO DO	PRJEB23709_PD1_PRE	TIDE_exclusion	41	19	22	fixed_score	0.386951 8443353 871					GSE91061,GSE78220	
0.623076 9230769 231	0.419292 8887721 892	0.492307 6923076 9234	0.428571 4285714 2855	- 0.012725 69525951 5555	0.3	0.6	0.384615 38461538 464	0.2	0.789473 6842105 263	0.265345 5786355 9327	0.295491 7737676 4635	0.509902 90438408 14	primary_recist	melanoma_core_high_evidence	LO DO	GSE91061	TIG	49	10	39	fixed_score	0.386335 4192286 923					GSE78220,PRJEB23709_PD1_PRE	
0.516483 5164835 164	0.549666 8841010 274	0.516483 5164835 164	0.518518 5185185 185	0.033149 67720658 9796	0.551724 7379310 345	0.571428 5714285 714	0.461538 46153846 156	0.533333 33333333 33	0.5	0.300071 8415601 082	0.215967 2304983 6508	- 0.095178 92769549 889	primary_recist	melanoma_core_high_evidence	LO DO	GSE78220	TIG	27	14	13	fixed_score	0.485867 9229861 1914					GSE91061,PRJEB23709_PD1_PRE	
0.825358 8516746 412	0.779092 7189893 544	0.612440 1913875 598	0.634146 3414634 146	0.282968 27628207 755	0.444444 4444444 444	0.315789 4736842 105	0.909090 90909090 91	0.75	0.606060 6060606 061	0.167463 0151357 0147	0.143662 9353703 001	1.521914 67427083 63	primary_recist	melanoma_core_high_evidence	LO DO	PRJEB23709_PD1_PRE	TIG	41	19	22	fixed_score	0.727429 5730156 389					GSE91061,GSE78220	
0.612820 5128205 129	0.443052 0590520 59	0.506410 2564102 564	0.510204 0816326 531	0.010336 22788243 4037	0.294117 6470588 2354	0.5	0.512820 51282051 28	0.208333 33333333 334	0.8	0.260795 6708959 116	0.299473 7822020 228	0.557600 16438911 71	primary_recist	melanoma_core_high_evidence	LO DO	GSE91061	EcoNiche-Opt-ModulePriorFixed-Platt	49	10	39	fixed_score_platt	0.474100 2379335 7814					GSE78220,PRJEB23709_PD1_PRE	
0.648351 6483516 484	0.631072 5135349 196	0.626373 6263736 264	0.629629 6296296 297	0.257019 33985391 27	0.666666 6666666 666	0.714285 7142857 143	0.538461 53846153 84	0.625	0.636363 6363636 364	0.259660 3442220 278	0.233123 9129251 1558	0.309583 84249576 79	primary_recist	melanoma_core_high_evidence	LO DO	GSE78220	EcoNiche-Opt-ModulePriorFixed-Platt	27	14	13	fixed_score_platt	0.425741 2706299 257					GSE91061,PRJEB23709_PD1_PRE	
0.875598 0861244 019	0.833187 6911765 292	0.721291 8660287 082	0.731707 3170731 707	0.465412 11876427 413	0.666666 6666666 666	0.578947 3684210 527	0.863636 36363636 36	0.785714 28571428 57	0.703703 7037037 037	0.200333 6038546 8568	0.222756 7898798 0188	4.541936 87621107 9	primary_recist	melanoma_core_high_evidence	LO DO	PRJEB23709_PD1_PRE	EcoNiche-Opt-ModulePriorFixed-Platt	41	19	22	fixed_score_platt	0.532155 6593934 782					GSE91061,GSE78220	

Supplementary Table 13 | Aligned locked-panel ablation.

Reports ablations matched to the locked module-panel scoring family used for the main melanoma result, including no-response-module, no-resistance-module, unsigned-direction, IFNG-only, equal-weight response-module, and calibrated variants. The table reports full AUROC, ablation AUROC, delta AUROC, ECE, confidence interval, FDR q value, and claim level. This table supports Figure 4b.

Full tabulation (12 rows x 17 columns).

endpoint	stratum	target_model	ablation_model	n_samples	target_AUROC	ablation_AUROC	delta_AUROC	target_ECE	ablation_ECE	delta_ECE	mean_delta	ci_low	ci_high	p_value	fdr_q	claim_level
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AlignedPanelNoCalibration	EcoNiche-Opt-NoResponseModules	117	0.704902576 9956003	0.553739786 2979259	0.15116279069 767447	0.2348619118 541941	0.1493290312 817757	0.0855328805 7241842	0.15016379056 580875	0.00879613361 9065453	0.2956324442 4855933	0.028	0.028	FDR_supported_component_gain
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AlignedPanelNoCalibration	EcoNiche-Opt-UnsignedStateDirection	117	0.704902576 9956003	0.563482086 7379007	0.14142049025 769965	0.2348619118 541941	0.1661301695 395132	0.0687317423 146809	0.14166927109 49993	0.06853413638 33008	0.2106591251 3861076	0.0	0.0	FDR_supported_component_gain
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AlignedPanelNoCalibration	IFNG_ModuleOnly	117	0.704902576 9956003	0.590509113 7649278	0.11439346323 067257	0.2348619118 541941	0.1512809760 2475243	0.0835809358 2944168	0.11452876378 351629	0.05869556943 684865	0.1749051044 1598597	0.0	0.0	FDR_supported_component_gain
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AlignedPanelNoCalibration	EcoNiche-Opt-NoResistanceModules	117	0.704902576 9956003	0.611565053 4255186	0.09333752357 008174	0.2348619118 541941	0.1327495485 0322207	0.1021123633 5097204	0.09297652079 953406	0.04060498726 2394726	0.1458996137 205843	0.0	0.0	FDR_supported_component_gain

endpoin t	stratum	target_model	ablation_model	n_sam ples	target_AURO C	ablation_AU ROC	delta_AUROC	target_ECE	ablation_ECE	delta_ECE	mean_delta	ci_low	ci_high	p_va lue	fdr_q	claim_level
primary_ recist	melanoma_core_high_e vidence	EcoNiche-Opt- AlignedPanelNo Calibration	ResponseModulesE qualWeight	117	0.704902576 9956003	0.615021998 742929	0.08988057825 267137	0.2348619118 541941	0.1505283841 5576308	0.0843335276 9843103	0.08994494972 049527	0.03683113767 500639	0.1434532510 0515166	0.0	0.0	FDR_supported_component_gain
primary_ recist	melanoma_core_high_e vidence	EcoNiche-Opt- AlignedPanelNo Calibration	EcoNiche-Opt- AlignedPanelCalibra ted	117	0.704902576 9956003	0.655248271 5273412	0.04965430546 825911	0.2348619118 541941	0.1486571390 2293257	0.0862047728 3126154	0.04900376508 7609626	0.01967898193 7602688	0.0799108679 1966949	0.00 2	0.00240000000 00000002	calibration_improves_ECE_with_dis crimination_tradeoff
primary_ recist	melanoma_recist_supp orted_primary	EcoNiche-Opt- AlignedPanelNo Calibration	EcoNiche-Opt- NoResponseModule s	131	0.684558823 5294118	0.538725490 1960784	0.14583333333 333337	0.2385841643 1842683	0.1220187277 2394376	0.1165654365 9448307	0.14470755926 913245	0.00401256187 5408552	0.2775249494 067695	0.04 2	0.042	FDR_supported_component_gain
primary_ recist	melanoma_recist_supp orted_primary	EcoNiche-Opt- AlignedPanelNo Calibration	EcoNiche-Opt- UnsignedStateDirec tion	131	0.684558823 5294118	0.557107843 1372548	0.12745098039 215697	0.2385841643 1842683	0.1074228578 8507754	0.1311613064 3334928	0.12602989989 699886	0.06200384786 781856	0.1908576329 3310443	0.0	0.0	FDR_supported_component_gain
primary_ recist	melanoma_recist_supp orted_primary	EcoNiche-Opt- AlignedPanelNo Calibration	IFNG_ModuleOnly	131	0.684558823 5294118	0.5625	0.12205882352 941178	0.2385841643 1842683	0.1120876073 1973418	0.1264965569 9869267	0.12000499225 227589	0.06715427559 912854	0.1715791214 8894386	0.0	0.0	FDR_supported_component_gain
primary_ recist	melanoma_recist_supp orted_primary	EcoNiche-Opt- AlignedPanelNo Calibration	EcoNiche-Opt- NoResistanceModul es	131	0.684558823 5294118	0.599019607 8431372	0.08553921568 627454	0.2385841643 1842683	0.1322571466 0030596	0.1063270177 077927	0.08452347885 8528235	0.04321242998 158002	0.1272436466 158002	0.0	0.0	FDR_supported_component_gain
primary_ recist	melanoma_recist_supp orted_primary	EcoNiche-Opt- AlignedPanelNo Calibration	ResponseModulesE qualWeight	131	0.684558823 5294118	0.604901960 7843137	0.07965686274 509809	0.2385841643 1842683	0.1136239412 842009	0.1249602230 3422592	0.07894995759 278456	0.03455772607 550482	0.1237632014 528299	0.00 2	0.00240000000 00000002	FDR_supported_component_gain
primary_ recist	melanoma_recist_supp orted_primary	EcoNiche-Opt- AlignedPanelNo Calibration	EcoNiche-Opt- AlignedPanelCalibra ted	131	0.684558823 5294118	0.630392156 8627452	0.05416666666 6666585	0.2385841643 1842683	0.1207622187 4679825	0.1178219455 7162858	0.05427828446 502271	0.02868370249 949197	0.0843189775 9103645	0.0	0.0	calibration_improves_ECE_with_dis crimination_tradeoff

Supplementary Table 14 | Decision curve.

Lists threshold probability, model net benefit, treat-all net benefit, and treat-none net benefit. This table supports Figure 3e and Figure 4c.

Full tabulation (722 rows x 11 columns).

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	APM	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.1	0.2991452991452992	0.2972459639126306	0.0	0.0018993352326686086	0.2991452991452992	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.15	0.2564102564102564	0.2559074912016089	0.0	0.0005027652086475043	0.2564102564102564	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.2	0.2222222222222222	0.20940170940170943	0.0	0.012820512820512775	0.2222222222222222	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.25	0.1737891737891738	0.15669515669515674	0.0	0.01709401709401706	0.1737891737891738	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.3	0.1355311355311355	0.09645909645909645	0.0	0.039072039072039044	0.1355311355311355	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.35	0.0775805391190007	0.026955950032873155	0.0	0.05062458908612755	0.0775805391190007	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.4	0.05982905982905978	-0.054131054131054124	0.0	0.1139601139601139	0.05982905982905978	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.45	0.006216006216006265	-0.14996114996114984	0.0	0.1561771561771561	0.006216006216006265	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.5	-0.02564102564102566	-0.2649572649572649	0.0	0.23931623931623924	-0.02564102564102566	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.55	-0.04273504273504283	-0.40550807217473894	0.0	0.36277302943969614	-0.04273504273504283	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.6	-0.05555555555555555	-0.581196581196581	0.0	0.5256410256410254	-0.05555555555555555	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.65	-0.11599511599511603	-0.8070818070818071	0.0	0.6910866910866911	-0.11599511599511603	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.7	-0.1367521367521367	-1.108262108262108	0.0	0.9715099715099713	-0.1367521367521367	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.75	-0.09401709401709402	-1.5299145299145298	0.0	1.4358974358974357	-0.09401709401709402	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.8	-0.07692307692307696	-2.1623931623931627	0.0	2.0854700854700856	-0.07692307692307696	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.85	-0.09686609686609686	-3.216524216524216	0.0	3.1196581196581192	-0.09686609686609686	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.9	-0.07692307692307694	-5.324786324786325	0.0	5.247863247863249	-0.07692307692307694	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	APM	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.15	0.25741578682755156	0.2559074912016089	0.0	0.0015082956259426794	0.25741578682755156	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.2	0.2115384615384615	0.20940170940170943	0.0	0.0021367521367520736	0.2115384615384615	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.25	0.17663817663817666	0.15669515669515674	0.0	0.01994301994301992	0.17663817663817666	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.3	0.1343101343101343	0.09645909645909645	0.0	0.037851037851037855	0.1343101343101343	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.35	0.11439842209072981	0.026955950032873155	0.0	0.08744242705785666	0.11439842209072981	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	CXCL9	0.4	0.09401709401709399	-0.054131054131054124	0.0	0.1481481481481481	0.09401709401709399	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.45	0.04273504273504278	-0.14996114996114984	0.0	0.19269619269619262	0.04273504273504278	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.5	0.008547008547008517	-0.2649572649572649	0.0	0.2735042735042734	0.008547008547008517	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.55	-0.0683760683760685	-0.40550807217473894	0.0	0.3371320037986705	-0.0683760683760685	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.6	-0.1025641025641025	-0.581196581196581	0.0	0.4786324786324785	-0.1025641025641025	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.65	-0.12454212454212457	-0.8070818070818071	0.0	0.6825396825396826	-0.12454212454212457	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.7	-0.15669515669515663	-1.108262108262108	0.0	0.9515669515669514	-0.15669515669515663	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.75	-0.18803418803418803	-1.5299145299145298	0.0	1.3418803418803418	-0.18803418803418803	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.8	-0.16239316239316243	-2.1623931623931627	0.0	2.0000000000000004	-0.16239316239316243	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.85	-0.03988603988603988	-3.216524216524216	0.0	3.176638176638176	-0.03988603988603988	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.9	0.008547008547008548	-5.324786324786325	0.0	5.333333333333334	0.008547008547008548	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CXCL9	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.15	0.248868778280543	0.2559074912016089	0.0	-0.007038712921065893	0.248868778280543	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.2	0.2115384615384615	0.20940170940170943	0.0	0.0021367521367520736	0.2115384615384615	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.25	0.14245014245014243	0.15669515669515674	0.0	-0.014245014245014315	0.14245014245014243	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.3	0.11721611721611722	0.09645909645909645	0.0	0.020757020757020767	0.11721611721611722	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.35	0.10059171597633138	0.026955950032873155	0.0	0.07363576594345822	0.10059171597633138	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.4	0.04843304843304841	-0.054131054131054124	0.0	0.10256410256410253	0.04843304843304841	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.45	0.006216006216006265	-0.14996114996114984	0.0	0.1561771561771561	0.006216006216006265	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.5	-0.05128205128205132	-0.2649572649572649	0.0	0.21367521367521358	-0.05128205128205132	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.55	-0.06267806267806275	-0.40550807217473894	0.0	0.3428300094966762	-0.06267806267806275	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.6	-0.11538461538461536	-0.581196581196581	0.0	0.46581196581196566	-0.11538461538461536	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.65	-0.12576312576312582	-0.8070818070818071	0.0	0.6813186813186813	-0.12576312576312582	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.7	-0.14814814814814808	-1.108262108262108	0.0	0.9601139601139599	-0.14814814814814808	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.75	-0.11965811965811968	-1.5299145299145298	0.0	1.4102564102564101	-0.11965811965811968	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.8	-0.10256410256410259	-2.1623931623931627	0.0	2.05982905982906	-0.10256410256410259	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.85	-0.07977207977207976	-3.216524216524216	0.0	3.1367521367521363	-0.07977207977207976	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.9	-0.07692307692307694	-5.3247863247863249	0.0	5.247863247863249	-0.07692307692307694	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	CYT	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.15	0.2604323780794369	0.2559074912016089	0.0	0.004524886877828038	0.2604323780794369	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.2	0.2158119658119658	0.20940170940170943	0.0	0.00641025641025636	0.2158119658119658	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.25	0.1623931623931624	0.15669515669515674	0.0	0.005698005698005659	0.1623931623931624	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.3	0.1294261294261294	0.09645909645909645	0.0	0.03296703296703296	0.1294261294261294	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.35	0.11045364891518739	0.026955950032873155	0.0	0.08349769888231423	0.11045364891518739	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.4	0.09686609686609685	-0.054131054131054124	0.0	0.15099715099715097	0.09686609686609685	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.45	0.049728049728049756	-0.14996114996114984	0.0	0.1996891996891996	0.049728049728049756	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.5	0.0	-0.2649572649572649	0.0	0.2649572649572649	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.55	-0.05792972459639131	-0.40550807217473894	0.0	0.3475783475783476	-0.05792972459639131	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.6	-0.08974358974358973	-0.581196581196581	0.0	0.49145299145299126	-0.08974358974358973	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-	0.65	-0.12332112332112335	-0.8070818070818071	0.0	0.6837606837606838	-0.12332112332112335	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	AdaptiveConsensus								
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.7	-0.13390313390313388	-1.108262108262108	0.0	0.9743589743589741	-0.13390313390313388	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.75	-0.13675213675213677	-1.5299145299145298	0.0	1.393162393162393	-0.13675213675213677	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.8	-0.11111111111111113	-2.1623931623931627	0.0	2.0512820512820515	-0.11111111111111113	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.85	-0.07122507122507121	-3.216524216524216	0.0	3.145299145299145	-0.07122507122507121	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.9	-0.0683760683760684	-5.324786324786325	0.0	5.256410256410257	-0.0683760683760684	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-AdaptiveConsensus	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.05	0.3288349077822762	0.3342330184435448	0.0	-0.005398110661268607	0.3288349077822762	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.1	0.2905982905982906	0.2972459639126306	0.0	-0.0066476733143399636	0.2905982905982906	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.15	0.2523881347410759	0.2559074912016089	0.0	-0.003519356460532974	0.2523881347410759	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.2	0.22649572649572652	0.20940170940170943	0.0	0.01709401709401709	0.22649572649572652	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.25	0.19088319088319092	0.15669515669515674	0.0	0.03418803418803418	0.19088319088319092	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.3	0.1391941391941392	0.09645909645909645	0.0	0.04273504273504275	0.1391941391941392	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.35	0.12097304404996714	0.026955950032873155	0.0	0.09401709401709399	0.12097304404996714	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.4	0.11680911680911679	-0.054131054131054124	0.0	0.17094017094017092	0.11680911680911679	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.45	0.05205905205905206	-0.14996114996114984	0.0	0.2020202020202019	0.05205905205905206	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.5	0.0	-0.2649572649572649	0.0	0.2649572649572649	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.55	-0.01994301994302	-0.40550807217473894	0.0	0.38556505223171894	-0.01994301994302	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.6	-0.05982905982905978	-0.581196581196581	0.0	0.5213675213675213	-0.05982905982905978	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.65	-0.11111111111111116	-0.8070818070818071	0.0	0.6959706959706959	-0.11111111111111116	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.7	-0.1339031339031339	-1.108262108262108	0.0	0.9743589743589741	-0.1339031339031339	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.75	-0.19658119658119655	-1.5299145299145298	0.0	1.3333333333333333	-0.19658119658119655	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.8	-0.35897435897435903	-2.1623931623931627	0.0	1.8034188034188037	-0.35897435897435903	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.85	-0.566951566951567	-3.216524216524216	0.0	2.649572649572649	-0.566951566951567	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.9	-0.5213675213675215	-5.324786324786325	0.0	4.803418803418804	-0.5213675213675215	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-HeuristicEcology	0.95	-0.5726495726495722	-11.649572649572637	0.0	11.076923076923064	-0.5726495726495722	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.15	0.25892408245349424	0.2559074912016089	0.0	0.003016591251885359	0.25892408245349424	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.2	0.21153846153846154	0.20940170940170943	0.0	0.0021367521367521014	0.21153846153846154	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.25	0.1794871794871795	0.15669515669515674	0.0	0.022792022792022748	0.1794871794871795	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.3	0.14041514041514042	0.09645909645909645	0.0	0.043956043956043966	0.14041514041514042	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.35	0.07889546351084817	0.026955950032873155	0.0	0.05193951347797501	0.07889546351084817	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.4	0.03703703703703701	-0.054131054131054124	0.0	0.09116809116809113	0.03703703703703701	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.45	0.011655011655011649	-0.14996114996114984	0.0	0.1616161616161615	0.011655011655011649	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.5	-0.01709401709401709	-0.2649572649572649	0.0	0.24786324786324782	-0.01709401709401709	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.55	-0.09591642924976265	-0.40550807217473894	0.0	0.3095916429249763	-0.09591642924976265	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.6	-0.1367521367521367	-0.581196581196581	0.0	0.4444444444444443	-0.1367521367521367	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.65	-0.1318681318681319	-0.8070818070818071	0.0	0.6752136752136753	-0.1318681318681319	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.7	-0.15669515669515663	-1.108262108262108	0.0	0.9515669515669514	-0.15669515669515663	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.75	-0.1452991452991453	-1.5299145299145298	0.0	1.3846153846153846	-0.1452991452991453	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.8	-0.1025641025641026	-2.1623931623931627	0.0	2.05982905982906	-0.1025641025641026	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.85	-0.03988603988603988	-3.216524216524216	0.0	3.176638176638176	-0.03988603988603988	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.9	0.008547008547008548	-5.324786324786325	0.0	5.333333333333334	0.008547008547008548	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ImmuneComposite	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.15	0.2634489693313223	0.2559074912016089	0.0	0.007541478129713397	0.2634489693313223	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.2	0.22435897435897434	0.20940170940170943	0.0	0.014957264957264904	0.22435897435897434	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.25	0.17094017094017094	0.15669515669515674	0.0	0.014245014245014204	0.17094017094017094	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.3	0.13186813186813187	0.09645909645909645	0.0	0.03540903540903542	0.13186813186813187	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.35	0.11111111111111116	0.026955950032873155	0.0	0.084155161078238	0.11111111111111116	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.4	0.09401709401709399	-0.054131054131054124	0.0	0.1481481481481481	0.09401709401709399	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.45	0.04817404817404819	-0.14996114996114984	0.0	0.19813519813519803	0.04817404817404819	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.5	0.0	-0.2649572649572649	0.0	0.2649572649572649	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.55	-0.04938271604938277	-0.40550807217473894	0.0	0.3561253561253562	-0.04938271604938277	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.6	-0.0641025641025641	-0.581196581196581	0.0	0.5170940170940169	-0.0641025641025641	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.65	-0.12210012210012214	-0.8070818070818071	0.0	0.684981684981685	-0.12210012210012214	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.7	-0.13675213675213668	-1.108262108262108	0.0	0.9715099715099713	-0.13675213675213668	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.75	-0.1282051282051282	-1.5299145299145298	0.0	1.4017094017094016	-0.1282051282051282	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.8	-0.2051282051282052	-2.1623931623931627	0.0	1.9572649572649574	-0.2051282051282052	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.85	-0.10256410256410255	-3.216524216524216	0.0	3.1139601139601134	-0.10256410256410255	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.9	-0.0683760683760684	-5.324786324786325	0.0	5.256410256410257	-0.0683760683760684	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModuleIFNConsensus	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.05	0.3288349077822762	0.3342330184435448	0.0	-0.005398110661268607	0.3288349077822762	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.1	0.2905982905982906	0.2972459639126306	0.0	-0.0066476733143399636	0.2905982905982906	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.15	0.2523881347410759	0.2559074912016089	0.0	-0.003519356460532974	0.2523881347410759	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.2	0.22649572649572652	0.20940170940170943	0.0	0.01709401709401709	0.22649572649572652	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.25	0.19088319088319092	0.15669515669515674	0.0	0.03418803418803418	0.19088319088319092	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.3	0.1391941391941392	0.09645909645909645	0.0	0.04273504273504275	0.1391941391941392	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.35	0.12097304404996714	0.026955950032873155	0.0	0.09401709401709399	0.12097304404996714	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.4	0.11680911680911679	-0.054131054131054124	0.0	0.17094017094017092	0.11680911680911679	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.45	0.05205905205905206	-0.14996114996114984	0.0	0.2020202020202019	0.05205905205905206	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.5	0.0	-0.2649572649572649	0.0	0.2649572649572649	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.55	-0.01994301994302	-0.40550807217473894	0.0	0.38556505223171894	-0.01994301994302	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.6	-0.05982905982905978	-0.581196581196581	0.0	0.5213675213675213	-0.05982905982905978	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.65	-0.11111111111111116	-0.8070818070818071	0.0	0.6959706959706959	-0.11111111111111116	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.7	-0.1339031339031339	-1.108262108262108	0.0	0.9743589743589741	-0.1339031339031339	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.75	-0.19658119658119655	-1.5299145299145298	0.0	1.3333333333333333	-0.19658119658119655	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.8	-0.35897435897435903	-2.1623931623931627	0.0	1.8034188034188037	-0.35897435897435903	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.85	-0.566951566951567	-3.216524216524216	0.0	2.649572649572649	-0.566951566951567	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.9	-0.5213675213675215	-5.324786324786325	0.0	4.803418803418804	-0.5213675213675215	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed	0.95	-0.5726495726495722	-11.649572649572637	0.0	11.076923076923064	-0.5726495726495722	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.15	0.25037707390648567	0.2559074912016089	0.0	-0.005530417295123213	0.25037707390648567	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.2	0.21153846153846154	0.20940170940170943	0.0	0.0021367521367521014	0.21153846153846154	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.25	0.1566951566951567	0.15669515669515674	0.0	-2.7755575615628914e-17	0.1566951566951567	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.3	0.10134310134310132	0.09645909645909645	0.0	0.004884004884004867	0.10134310134310132	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.35	0.04865220249835639	0.026955950032873155	0.0	0.021696252465483234	0.04865220249835639	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.4	0.04558404558404555	-0.054131054131054124	0.0	0.09971509971509968	0.04558404558404555	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.45	0.03574203574203577	-0.14996114996114984	0.0	0.1857031857031856	0.03574203574203577	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.5	0.008547008547008572	-0.2649572649572649	0.0	0.2735042735042735	0.008547008547008572	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.55	0.006647673314339908	-0.40550807217473894	0.0	0.41215574548907885	0.006647673314339908	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.6	-0.04700854700854695	-0.581196581196581	0.0	0.5341880341880341	-0.04700854700854695	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.65	-0.12210012210012211	-0.8070818070818071	0.0	0.684981684981685	-0.12210012210012211	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.7	-0.13675213675213677	-1.108262108262108	0.0	0.9715099715099713	-0.13675213675213677	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.75	-0.042735042735042736	-1.5299145299145298	0.0	1.487179487179487	-0.042735042735042736	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.8	-0.07692307692307696	-2.1623931623931627	0.0	2.0854700854700856	-0.07692307692307696	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.85	-0.031339031339031334	-3.216524216524216	0.0	3.1851851851851847	-0.031339031339031334	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.9	-0.0683760683760684	-5.324786324786325	0.0	5.256410256410257	-0.0683760683760684	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-ModulePriorFixed-Platt	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.05	0.3346828609986505	0.3342330184435448	0.0	0.00044984255510571725	0.3346828609986505	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.1	0.30009496676163344	0.2972459639126306	0.0	0.0028490028490028574	0.30009496676163344	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.15	0.25339366515837103	0.2559074912016089	0.0	-0.0025138260432378545	0.25339366515837103	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.2	0.188034188034188	0.20940170940170943	0.0	-0.02136752136752143	0.188034188034188	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.25	0.09401709401709402	0.15669515669515674	0.0	-0.06267806267806272	0.09401709401709402	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.3	0.032967032967032905	0.09645909645909645	0.0	-0.06349206349206354	0.032967032967032905	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.35	-0.00986193293885601	0.026955950032873155	0.0	-0.036817882971729166	-0.00986193293885601	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.4	-0.04558404558404558	-0.054131054131054124	0.0	0.008547008547008544	-0.04558404558404558	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.45	-0.108003108003108	-0.14996114996114984	0.0	0.04195804195804184	-0.108003108003108	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.5	-0.17094017094017092	-0.2649572649572649	0.0	0.09401709401709399	-0.17094017094017092	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.55	-0.1756885090218425	-0.40550807217473894	0.0	0.22981956315289645	-0.1756885090218425	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.6	-0.20085470085470084	-0.581196581196581	0.0	0.3803418803418802	-0.20085470085470084	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.65	-0.15750915750915756	-0.8070818070818071	0.0	0.6495726495726495	-0.15750915750915756	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.7	-0.1396011396011395	-1.108262108262108	0.0	0.9868609686609685	-0.1396011396011395	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.75	-0.17094017094017094	-1.5299145299145298	0.0	1.3589743589743588	-0.17094017094017094	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.8	-0.19658119658119666	-2.1623931623931627	0.0	1.965811965811966	-0.19658119658119666	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.85	-0.23646723646723647	-3.216524216524216	0.0	2.9800569800569794	-0.23646723646723647	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.9	-0.35897435897435903	-5.324786324786325	0.0	4.965811965811966	-0.35897435897435903	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordFullGraph	0.95	-0.7264957264957257	-11.649572649572637	0.0	10.923076923076911	-0.7264957264957257	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.05	0.3351327035537562	0.3342330184435448	0.0	0.000899685110211379	0.3351327035537562	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.1	0.30104463437796775	0.2972459639126306	0.0	0.0037986704653371617	0.30104463437796775	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.15	0.22071392659627953	0.2559074912016089	0.0	-0.03519356460532935	0.22071392659627953	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.2	0.1752136752136752	0.20940170940170943	0.0	-0.03418803418803423	0.1752136752136752	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.25	0.12535612535612536	0.15669515669515674	0.0	-0.031339031339031376	0.12535612535612536	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.3	0.06349206349206349	0.09645909645909645	0.0	-0.03296703296703296	0.06349206349206349	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.35	-0.002629848783694877	0.026955950032873155	0.0	-0.029585798816568032	-0.002629848783694877	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.4	-0.03988603988603989	-0.054131054131054124	0.0	0.014245014245014231	-0.03988603988603989	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.45	-0.11577311577311572	-0.14996114996114984	0.0	0.03418803418803412	-0.11577311577311572	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.5	-0.23076923076923078	-0.2649572649572649	0.0	0.03418803418803412	-0.23076923076923078	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.55	-0.2820512820512822	-0.40550807217473894	0.0	0.12345679012345673	-0.2820512820512822	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.6	-0.3376068376068375	-0.581196581196581	0.0	0.2435897435897435	-0.3376068376068375	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.65	-0.3858363858363859	-0.8070818070818071	0.0	0.4212454212454212	-0.3858363858363859	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.7	-0.4900284900284899	-1.108262108262108	0.0	0.6182336182336181	-0.4900284900284899	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.75	-0.5042735042735043	-1.5299145299145298	0.0	1.0256410256410255	-0.5042735042735043	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.8	-0.52991452991453	-2.1623931623931627	0.0	1.6324786324786327	-0.52991452991453	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.85	-0.5612535612535612	-3.216524216524216	0.0	2.655270655270655	-0.5612535612535612	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.9	-0.3504273504273505	-5.324786324786325	0.0	4.9743589743589745	-0.3504273504273505	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoBioObjective	0.95	-0.6410256410256405	-11.649572649572637	0.0	11.008547008546996	-0.6410256410256405	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.05	0.3351327035537562	0.3342330184435448	0.0	0.000899685110211379	0.3351327035537562	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.1	0.30009496676163344	0.2972459639126306	0.0	0.0028490028490028574	0.30009496676163344	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.15	0.24484665661136246	0.2559074912016089	0.0	-0.011060834590246427	0.24484665661136246	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.2	0.19017094017094016	0.20940170940170943	0.0	-0.019230769230769273	0.19017094017094016	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.25	0.13105413105413105	0.15669515669515674	0.0	-0.02564102564102569	0.13105413105413105	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.3	0.058608058608058566	0.09645909645909645	0.0	-0.03785103785103788	0.058608058608058566	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.35	0.0032873109796186517	0.026955950032873155	0.0	-0.023668639053254503	0.0032873109796186517	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.4	-0.034188034188034205	-0.054131054131054124	0.0	0.01994301994301992	-0.034188034188034205	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.45	-0.09401709401709399	-0.14996114996114984	0.0	0.055944055944055854	-0.09401709401709399	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.5	-0.14529914529914528	-0.2649572649572649	0.0	0.11965811965811962	-0.14529914529914528	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.55	-0.18803418803418814	-0.40550807217473894	0.0	0.2174738841405508	-0.18803418803418814	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.6	-0.23076923076923067	-0.581196581196581	0.0	0.35042735042735035	-0.23076923076923067	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.65	-0.21245421245421248	-0.8070818070818071	0.0	0.5946275946275946	-0.21245421245421248	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.7	-0.22792022792022792	-1.108262108262108	0.0	0.8803418803418801	-0.22792022792022792	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.75	-0.3162393162393162	-1.5299145299145298	0.0	1.2136752136752136	-0.3162393162393162	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.8	-0.41025641025641035	-2.1623931623931627	0.0	1.7521367521367524	-0.41025641025641035	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.85	-0.32193732193732194	-3.216524216524216	0.0	2.894586894586894	-0.32193732193732194	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.9	-0.14529914529914534	-5.324786324786325	0.0	5.17948717948718	-0.14529914529914534	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordNoInteraction	0.95	0.008547008547008548	-11.649572649572637	0.0	11.658119658119645	0.008547008547008548	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.05	0.32703553756185333	0.3342330184435448	0.0	-0.007197480881691476	0.32703553756185333	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.1	0.28584995251661915	0.2972459639126306	0.0	-0.01139601139601143	0.28584995251661915	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.15	0.24534942182001004	0.2559074912016089	0.0	-0.010558069381598839	0.24534942182001004	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.2	0.1965811965811966	0.20940170940170943	0.0	-0.01282051282051283	0.1965811965811966	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.25	0.1082621082621083	0.15669515669515674	0.0	-0.04843304843304844	0.1082621082621083	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.3	0.05128205128205124	0.09645909645909645	0.0	-0.04517704517704521	0.05128205128205124	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.35	-0.013149243918474662	0.026955950032873155	0.0	-0.04010519395134782	-0.013149243918474662	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.4	-0.07692307692307696	-0.054131054131054124	0.0	-0.02279202279202283	-0.07692307692307696	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.45	-0.11111111111111108	-0.14996114996114984	0.0	0.038850038850038765	-0.11111111111111108	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.5	-0.13675213675213674	-0.2649572649572649	0.0	0.12820512820512817	-0.13675213675213674	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.55	-0.17378917378917386	-0.40550807217473894	0.0	0.2317188983855651	-0.17378917378917386	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.6	-0.23931623931623922	-0.581196581196581	0.0	0.3418803418803418	-0.23931623931623922	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.65	-0.3015873015873016	-0.8070818070818071	0.0	0.5054945054945055	-0.3015873015873016	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.7	-0.2706552706552706	-1.108262108262108	0.0	0.8376068376068374	-0.2706552706552706	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.75	-0.1965811965811966	-1.5299145299145298	0.0	1.3333333333333333	-0.1965811965811966	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.8	-0.13675213675213677	-2.1623931623931627	0.0	2.025641025641026	-0.13675213675213677	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.85	-0.06267806267806267	-3.216524216524216	0.0	3.1538461538461533	-0.06267806267806267	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.9	-0.0683760683760684	-5.324786324786325	0.0	5.256410256410257	-0.0683760683760684	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	EcoNiche-Opt-WordUnsignedGraph	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.15	0.25037707390648567	0.2559074912016089	0.0	-0.005530417295123213	0.25037707390648567	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.2	0.22008547008547008	0.20940170940170943	0.0	0.010683760683760646	0.22008547008547008	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.25	0.16524216524216523	0.15669515669515674	0.0	0.008547008547008489	0.16524216524216523	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.3	0.1294261294261294	0.09645909645909645	0.0	0.03296703296703296	0.1294261294261294	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.35	0.08809993425378043	0.026955950032873155	0.0	0.06114398422090728	0.08809993425378043	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.4	0.05698005698005698	-0.054131054131054124	0.0	0.11111111111111111	0.05698005698005698	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.45	0.028749028749028793	-0.14996114996114984	0.0	0.17871017871017864	0.028749028749028793	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.5	-0.008547008547008517	-0.2649572649572649	0.0	0.2564102564102564	-0.008547008547008517	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.55	-0.021842355175688583	-0.40550807217473894	0.0	0.3836657169990504	-0.021842355175688583	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.6	-0.06837606837606836	-0.581196581196581	0.0	0.5128205128205127	-0.06837606837606836	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.65	-0.11599511599511603	-0.8070818070818071	0.0	0.6910866910866911	-0.11599511599511603	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.7	-0.09971509971509969	-1.108262108262108	0.0	1.0085470085470083	-0.09971509971509969	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.75	-0.17094017094017092	-1.5299145299145298	0.0	1.3589743589743588	-0.17094017094017092	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.8	-0.12820512820512825	-2.1623931623931627	0.0	2.0341880341880345	-0.12820512820512825	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.85	-0.07122507122507121	-3.216524216524216	0.0	3.145299145299145	-0.07122507122507121	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.9	0.0	-5.324786324786325	0.0	5.324786324786325	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IFNG	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.1	0.2991452991452992	0.2972459639126306	0.0	0.0018993352326686086	0.2991452991452992	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.15	0.25037707390648567	0.2559074912016089	0.0	-0.005530417295123213	0.25037707390648567	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	IPRES	0.2	0.20085470085470086	0.20940170940170943	0.0	-0.008547008547008572	0.20085470085470086	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.25	0.15384615384615385	0.15669515669515674	0.0	-0.002849002849002885	0.15384615384615385	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.3	0.09768009768009767	0.09645909645909645	0.0	0.0012210012210012167	0.09768009768009767	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.35	0.04799474030243264	0.026955950032873155	0.0	0.021038790269559488	0.04799474030243264	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.4	0.014245014245014231	-0.054131054131054124	0.0	0.06837606837606836	0.014245014245014231	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.45	-0.009324009324009341	-0.14996114996114984	0.0	0.1406371406371405	-0.009324009324009341	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.5	-0.07692307692307693	-0.2649572649572649	0.0	0.18803418803418798	-0.07692307692307693	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.55	-0.11680911680911685	-0.40550807217473894	0.0	0.28869895536562207	-0.11680911680911685	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.6	-0.16666666666666666	-0.581196581196581	0.0	0.4145299145299144	-0.16666666666666666	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.65	-0.2136752136752137	-0.8070818070818071	0.0	0.5934065934065934	-0.2136752136752137	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.7	-0.27350427350427353	-1.108262108262108	0.0	0.8347578347578345	-0.27350427350427353	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.75	-0.22222222222222222	-1.5299145299145298	0.0	1.3076923076923075	-0.22222222222222222	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.8	-0.1025641025641026	-2.1623931623931627	0.0	2.05982905982906	-0.1025641025641026	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.85	-0.031339031339031334	-3.216524216524216	0.0	3.1851851851851847	-0.031339031339031334	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.9	0.0	-5.324786324786325	0.0	5.324786324786325	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	IPRES	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.15	0.25892408245349424	0.2559074912016089	0.0	0.003016591251885359	0.25892408245349424	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.2	0.22008547008547008	0.20940170940170943	0.0	0.010683760683760646	0.22008547008547008	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.25	0.1794871794871795	0.15669515669515674	0.0	0.022792022792022748	0.1794871794871795	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.3	0.11721611721611722	0.09645909645909645	0.0	0.020757020757020762	0.11721611721611722	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.35	0.06508875739644973	0.026955950032873155	0.0	0.038132807363576576	0.06508875739644973	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.4	0.03988603988603989	-0.054131054131054124	0.0	0.09401709401709402	0.03988603988603989	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.45	-0.020979020979020935	-0.14996114996114984	0.0	0.1289821289821289	-0.020979020979020935	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.5	-0.0854700854700855	-0.2649572649572649	0.0	0.1794871794871794	-0.0854700854700855	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.55	-0.1462488129154797	-0.40550807217473894	0.0	0.25925925925925924	-0.1462488129154797	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.6	-0.17521367521367512	-0.581196581196581	0.0	0.4059829059829059	-0.17521367521367512	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.65	-0.2136752136752137	-0.8070818070818071	0.0	0.5934065934065934	-0.2136752136752137	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.7	-0.14529914529914525	-1.108262108262108	0.0	0.9629629629629628	-0.14529914529914525	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.75	-0.15384615384615385	-1.5299145299145298	0.0	1.376068376068376	-0.15384615384615385	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.8	-0.15384615384615388	-2.1623931623931627	0.0	2.008547008547009	-0.15384615384615388	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.85	-0.08831908831908832	-3.216524216524216	0.0	3.1282051282051277	-0.08831908831908832	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.9	-0.0683760683760684	-5.324786324786325	0.0	5.256410256410257	-0.0683760683760684	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	PDCD1LG2	0.95	0.008547008547008548	-11.649572649572637	0.0	11.658119658119645	0.008547008547008548	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.15	0.2473604826546003	0.2559074912016089	0.0	-0.008547008547008572	0.2473604826546003	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.2	0.21153846153846154	0.20940170940170943	0.0	0.0021367521367521014	0.21153846153846154	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.25	0.1794871794871795	0.15669515669515674	0.0	0.022792022792022748	0.1794871794871795	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.3	0.1343101343101343	0.09645909645909645	0.0	0.037851037851037855	0.1343101343101343	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.35	0.10059171597633138	0.026955950032873155	0.0	0.07363576594345822	0.10059171597633138	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.4	0.0655270655270655	-0.054131054131054124	0.0	0.11965811965811962	0.0655270655270655	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.45	0.01554001554001555	-0.14996114996114984	0.0	0.1655011655011654	0.01554001554001555	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.5	-0.025641025641025633	-0.2649572649572649	0.0	0.23931623931623927	-0.025641025641025633	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.55	-0.06932573599240272	-0.40550807217473894	0.0	0.3361823361823362	-0.06932573599240272	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.6	-0.09829059829059825	-0.581196581196581	0.0	0.48290598290598274	-0.09829059829059825	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.65	-0.10989010989010992	-0.8070818070818071	0.0	0.6971916971916972	-0.10989010989010992	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.7	-0.12820512820512817	-1.108262108262108	0.0	0.9800569800569798	-0.12820512820512817	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.75	-0.10256410256410259	-1.5299145299145298	0.0	1.4273504273504272	-0.10256410256410259	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.8	-0.10256410256410259	-2.1623931623931627	0.0	2.05982905982906	-0.10256410256410259	117	0.36752136752136755

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.85	0.011396011396011402	-3.216524216524216	0.0	3.2279202279202273	0.011396011396011402	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.9	-0.0683760683760684	-5.324786324786325	0.0	5.256410256410257	-0.0683760683760684	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_dysfunction	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.15	0.2619406737053796	0.2559074912016089	0.0	0.006033182503770718	0.2619406737053796	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.2	0.21581196581196582	0.20940170940170943	0.0	0.0064102564102563875	0.21581196581196582	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.25	0.1737891737891738	0.15669515669515674	0.0	0.01709401709401706	0.1737891737891738	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.3	0.10134310134310132	0.09645909645909645	0.0	0.004884004884004867	0.10134310134310132	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.35	0.06969099276791588	0.02695590032873155	0.0	0.04273504273504272	0.06969099276791588	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.4	0.002849002849002802	-0.054131054131054124	0.0	0.056980056980056926	0.002849002849002802	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.45	-0.07459207459207456	-0.14996114996114984	0.0	0.07536907536907528	-0.07459207459207456	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.5	-0.1111111111111111	-0.2649572649572649	0.0	0.1538461538461538	-0.1111111111111111	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.55	-0.1614434947768282	-0.40550807217473894	0.0	0.24406457739791074	-0.1614434947768282	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.6	-0.22222222222222215	-0.581196581196581	0.0	0.35897435897435886	-0.22222222222222215	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.65	-0.2063492063492064	-0.8070818070818071	0.0	0.6007326007326007	-0.2063492063492064	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.7	-0.28205128205128205	-1.108262108262108	0.0	0.826210826210826	-0.28205128205128205	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.75	-0.28205128205128205	-1.5299145299145298	0.0	1.2478632478632479	-0.28205128205128205	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.8	-0.2564102564102565	-2.1623931623931627	0.0	1.9059829059829063	-0.2564102564102565	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.85	-0.17663817663817663	-3.216524216524216	0.0	3.0398860398860394	-0.17663817663817663	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.9	-0.07692307692307694	-5.324786324786325	0.0	5.247863247863249	-0.07692307692307694	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIDE_exclusion	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.05	0.3342330184435448	0.3342330184435448	0.0	0.0	0.3342330184435448	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.1	0.2972459639126306	0.2972459639126306	0.0	0.0	0.2972459639126306	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.15	0.248868778280543	0.2559074912016089	0.0	-0.007038712921065893	0.248868778280543	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.2	0.20940170940170938	0.20940170940170943	0.0	-5.551115123125783e-17	0.20940170940170938	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.25	0.15954415954415957	0.15669515669515674	0.0	0.0028490028490028296	0.15954415954415957	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.3	0.1343101343101343	0.09645909645909645	0.0	0.037851037851037855	0.1343101343101343	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.35	0.09138724523339911	0.02695590032873155	0.0	0.06443129520052596	0.09138724523339911	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.4	0.0626780626780627	-0.054131054131054124	0.0	0.11680911680911682	0.0626780626780627	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.45	0.034188034188034205	-0.14996114996114984	0.0	0.18414918414918405	0.034188034188034205	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.5	-0.01709401709401706	-0.2649572649572649	0.0	0.24786324786324784	-0.01709401709401706	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.55	-0.028490028490028574	-0.40550807217473894	0.0	0.37701804368471037	-0.028490028490028574	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.6	-0.08974358974358973	-0.581196581196581	0.0	0.49145299145299126	-0.08974358974358973	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.65	-0.12332112332112335	-0.8070818070818071	0.0	0.6837606837606838	-0.12332112332112335	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.7	-0.12250712250712242	-1.108262108262108	0.0	0.9857549857549857	-0.12250712250712242	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.75	-0.17094017094017092	-1.5299145299145298	0.0	1.3589743589743588	-0.17094017094017092	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.8	-0.11111111111111113	-2.1623931623931627	0.0	2.0512820512820515	-0.11111111111111113	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.85	-0.02279202279202279	-3.216524216524216	0.0	3.193732193732193	-0.02279202279202279	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.9	-0.07692307692307694	-5.324786324786325	0.0	5.247863247863249	-0.07692307692307694	117	0.36752136752136755
primary_recist	melanoma_core_high_evidence	TIG	0.95	0.0	-11.649572649572637	0.0	11.649572649572637	0.0	117	0.36752136752136755
primary_recist	melanoma_recist_supported_primary	APM	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.1	0.3231552162849872	0.32145886344359625	0.0	0.001696352841390969	0.3231552162849872	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.15	0.2819937135159407	0.28154467894027835	0.0	0.00044903457566236593	0.2819937135159407	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.2	0.23282442748091603	0.23664122137404578	0.0	-0.0038167938931297496	0.23282442748091603	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.25	0.18575063613231552	0.18575063613231552	0.0	0.0	0.18575063613231552	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.3	0.14285714285714285	0.12758996728462374	0.0	0.01526717557251911	0.14285714285714285	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.35	0.08690546095126253	0.060481503229594835	0.0	0.026423957721667696	0.08690546095126253	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.4	0.06615776081424934	-0.017811704834605646	0.0	0.08396946564885499	0.06615776081424934	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.45	0.011103400416377529	-0.11034004163775157	0.0	0.1214434420541291	0.011103400416377529	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.5	-0.022900763358778664	-0.22137404580152675	0.0	0.1984732824427481	-0.022900763358778664	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	APM	0.55	-0.035623409669211265	-0.3570822731128077	0.0	0.3214588634435964	-0.035623409669211265	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.6	-0.03816793893129769	-0.5267175572519083	0.0	0.4885496183206106	-0.03816793893129769	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.65	-0.10250817884405675	-0.7448200654307527	0.0	0.642311886586696	-0.10250817884405675	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.7	-0.12468193384223915	-1.0356234096692112	0.0	0.910941475826972	-0.12468193384223915	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.75	-0.0916030534351145	-1.4427480916030535	0.0	1.351145038167939	-0.0916030534351145	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.8	-0.05343511450381681	-2.0534351145038174	0.0	2.0000000000000004	-0.05343511450381681	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.85	-0.08651399491094146	-3.0712468193384224	0.0	2.984732824427481	-0.08651399491094146	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.9	-0.0687022900763359	-5.106870229007635	0.0	5.0381679389312986	-0.0687022900763359	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	APM	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.15	0.28289178266726533	0.28154467894027835	0.0	0.0013471037269869868	0.28289178266726533	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.2	0.23091603053435114	0.23664122137404578	0.0	-0.005725190839694638	0.23091603053435114	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.25	0.19592875318066158	0.18575063613231552	0.0	0.010178117048346064	0.19592875318066158	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.3	0.15376226826608505	0.12758996728462374	0.0	0.02617230098146131	0.15376226826608505	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.35	0.1274221961244862	0.060481503229594835	0.0	0.06694069289489135	0.1274221961244862	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.4	0.08905852417302795	-0.017811704834605646	0.0	0.1068702290076336	0.08905852417302795	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.45	0.04996530187369885	-0.11034004163775157	0.0	0.16030534351145043	0.04996530187369885	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.5	0.02290076335877861	-0.22137404580152675	0.0	0.24427480916030536	0.02290076335877861	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.55	-0.047497879558948325	-0.3570822731128077	0.0	0.30958439355385936	-0.047497879558948325	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.6	-0.08778625954198468	-0.5267175572519083	0.0	0.43893129770992356	-0.08778625954198468	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.65	-0.1101417666303163	-0.7448200654307527	0.0	0.6346782988004365	-0.1101417666303163	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.7	-0.14249363867684472	-1.0356234096692112	0.0	0.8931297709923665	-0.14249363867684472	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.75	-0.18320610687022904	-1.4427480916030535	0.0	1.2595419847328244	-0.18320610687022904	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.8	-0.1374045801526718	-2.0534351145038174	0.0	1.9160305343511457	-0.1374045801526718	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.85	-0.027989821882951647	-3.0712468193384224	0.0	3.0432569974554706	-0.027989821882951647	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.9	0.015267175572519083	-5.106870229007635	0.0	5.122137404580154	0.015267175572519083	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CXCL9	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.15	0.27525819488100584	0.28154467894027835	0.0	-0.006286484059272512	0.27525819488100584	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.2	0.2385496183206107	0.23664122137404578	0.0	0.0019083969465649164	0.2385496183206107	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.25	0.16539440203562344	0.18575063613231552	0.0	-0.020356234096692072	0.16539440203562344	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.3	0.14176663031624862	0.12758996728462374	0.0	0.014176663031624875	0.14176663031624862	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.35	0.11156782149148564	0.060481503229594835	0.0	0.051086318261890806	0.11156782149148564	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.4	0.05852417302798982	-0.017811704834605646	0.0	0.07633587786259546	0.05852417302798982	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.45	0.023594725884802242	-0.11034004163775157	0.0	0.13393476752255382	0.023594725884802242	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.5	-0.02290076335877861	-0.22137404580152675	0.0	0.19847328244274814	-0.02290076335877861	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.55	-0.04240882103477531	-0.3570822731128077	0.0	0.31467345207803243	-0.04240882103477531	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.6	-0.09923664122137402	-0.5267175572519083	0.0	0.4274809160305343	-0.09923664122137402	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.65	-0.11123227917121051	-0.7448200654307527	0.0	0.6335877862595423	-0.11123227917121051	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.7	-0.1348600508905852	-1.0356234096692112	0.0	0.900763358778626	-0.1348600508905852	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.75	-0.11450381679389314	-1.4427480916030535	0.0	1.3282442748091603	-0.11450381679389314	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.8	-0.10687022900763361	-2.0534351145038174	0.0	1.946564885496184	-0.10687022900763361	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.85	-0.055979643765903295	-3.0712468193384224	0.0	3.015267175572519	-0.055979643765903295	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.9	-0.0687022900763359	-5.106870229007635	0.0	5.0381679389312986	-0.0687022900763359	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	CYT	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.15	0.2855859901212393	0.28154467894027835	0.0	0.00404131118096096	0.2855859901212393	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.2	0.24236641221374047	0.23664122137404578	0.0	0.005725190839694694	0.24236641221374047	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.25	0.183206106870229	0.18575063613231552	0.0	-0.002544529262086509	0.183206106870229	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.3	0.14940021810250817	0.12758996728462374	0.0	0.02181025081788443	0.14940021810250817	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.35	0.12037580739870818	0.060481503229594835	0.0	0.059894304169113344	0.12037580739870818	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.4	0.094147582697201	-0.017811704834605646	0.0	0.11195928753180664	0.094147582697201	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.45	0.05482303955586401	-0.11034004163775157	0.0	0.16516308119361559	0.05482303955586401	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.5	0.0	-0.22137404580152675	0.0	0.22137404580152675	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.55	-0.05513146734520785	-0.3570822731128077	0.0	0.30195080576759986	-0.05513146734520785	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.6	-0.08778625954198468	-0.5267175572519083	0.0	0.43893129770992356	-0.08778625954198468	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.65	-0.09487459105779722	-0.7448200654307527	0.0	0.6499454743729556	-0.09487459105779722	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.7	-0.10432569974554703	-1.0356234096692112	0.0	0.9312977099236641	-0.10432569974554703	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.75	-0.1068702290076336	-1.4427480916030535	0.0	1.33587786259542	-0.1068702290076336	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.8	-0.09160305343511455	-2.0534351145038174	0.0	1.9618320610687028	-0.09160305343511455	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.85	-0.055979643765903295	-3.0712468193384224	0.0	3.015267175572519	-0.055979643765903295	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.9	-0.053435114503816813	-5.106870229007635	0.0	5.053435114503818	-0.053435114503816813	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-AdaptiveConsensus	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.05	0.3523503415026115	0.3571715548413017	0.0	-0.004821213338690178	0.3523503415026115	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.1	0.3155216284987277	0.32145886344359625	0.0	-0.00593723494486853	0.3155216284987277	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.15	0.265828468792097	0.28154467894027835	0.0	-0.015716210148181364	0.265828468792097	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.2	0.24045801526717558	0.23664122137404578	0.0	0.003816793893129805	0.24045801526717558	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.25	0.183206106870229	0.18575063613231552	0.0	-0.002544529262086509	0.183206106870229	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.3	0.1341330425299891	0.12758996728462374	0.0	0.0065430752453653485	0.1341330425299891	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.35	0.11450381679389315	0.060481503229594835	0.0	0.05402231356429832	0.11450381679389315	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.4	0.1068702290076336	-0.017811704834605646	0.0	0.12468193384223925	0.1068702290076336	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.45	0.050659264399722403	-0.11034004163775157	0.0	0.16099930603747398	0.050659264399722403	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.5	0.007633587786259527	-0.22137404580152675	0.0	0.22900763358778628	0.007633587786259527	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.55	-0.004240882103477617	-0.3570822731128077	0.0	0.3528413910093301	-0.004240882103477617	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.6	-0.04961832061068697	-0.5267175572519083	0.0	0.47709923664122134	-0.04961832061068697	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.65	-0.09814612868047987	-0.7448200654307527	0.0	0.6466739367502728	-0.09814612868047987	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.7	-0.12213740458015265	-1.0356234096692112	0.0	0.9134860050890585	-0.12213740458015265	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.75	-0.183206106870229	-1.4427480916030535	0.0	1.2595419847328244	-0.183206106870229	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.8	-0.33587786259541996	-2.0534351145038174	0.0	1.7175572519083975	-0.33587786259541996	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.85	-0.4910941475826972	-3.0712468193384224	0.0	2.5801526717557253	-0.4910941475826972	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.9	-0.4503816793893131	-5.106870229007635	0.0	4.656488549618322	-0.4503816793893131	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-HeuristicEcology	0.95	-0.4961832061068696	-11.213740458015256	0.0	10.717557251908385	-0.4961832061068696	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.15	0.284238863942523	0.28154467894027835	0.0	0.0026942074539739735	0.284238863942523	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.2	0.23854961832061072	0.23664122137404578	0.0	0.0019083969465649442	0.23854961832061072	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.25	0.19847328244274812	0.18575063613231552	0.0	0.0127226463104326	0.19847328244274812	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.3	0.15921483097055614	0.12758996728462374	0.0	0.0316248636859324	0.15921483097055614	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.35	0.09160305343511452	0.060481503229594835	0.0	0.031121550205519682	0.09160305343511452	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.4	0.05089058524173026	-0.017811704834605646	0.0	0.06870229007633591	0.05089058524173026	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.45	0.028452463566967373	-0.11034004163775157	0.0	0.13879250520471895	0.028452463566967373	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.5	-0.007633587786259527	-0.22137404580152675	0.0	0.21374045801526723	-0.007633587786259527	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.55	-0.07972858354537751	-0.3570822731128077	0.0	0.27735368956743023	-0.07972858354537751	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.6	-0.11832061068702288	-0.5267175572519083	0.0	0.4083969465648854	-0.11832061068702288	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.65	-0.1166848418756816	-0.7448200654307527	0.0	0.6281352235550711	-0.1166848418756816	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.7	-0.14249363867684472	-1.0356234096692112	0.0	0.8931297709923665	-0.14249363867684472	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.75	-0.12213740458015268	-1.4427480916030535	0.0	1.3206106870229009	-0.12213740458015268	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.8	-0.08396946564885499	-2.0534351145038174	0.0	1.9694656488549624	-0.08396946564885499	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.85	-0.027989821882951647	-3.0712468193384224	0.0	3.0432569974554706	-0.027989821882951647	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.9	0.015267175572519083	-5.106870229007635	0.0	5.122137404580154	0.015267175572519083	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ImmuneComposite	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.15	0.2882801975752133	0.28154467894027835	0.0	0.006735518634934934	0.2882801975752133	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.2	0.25	0.23664122137404578	0.0	0.01335877862595422	0.25	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.25	0.1908396946564886	0.18575063613231552	0.0	0.0050890585241730735	0.1908396946564886	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.3	0.1472191930207197	0.12758996728462374	0.0	0.019629225736095962	0.1472191930207197	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.35	0.11332941867293012	0.060481503229594835	0.0	0.05284791544333528	0.11332941867293012	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.4	0.08651399491094144	-0.017811704834605646	0.0	0.10432569974554709	0.08651399491094144	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.45	0.05343511450381683	-0.11034004163775157	0.0	0.1637751561415684	0.05343511450381683	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.5	0.0	-0.22137404580152675	0.0	0.22137404580152675	0.0	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.55	-0.047497879558948325	-0.3570822731128077	0.0	0.30958439355385936	-0.047497879558948325	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.6	-0.06488549618320605	-0.5267175572519083	0.0	0.4618320610687022	-0.06488549618320605	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.65	-0.09378407851690299	-0.7448200654307527	0.0	0.6510359869138498	-0.09378407851690299	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.7	-0.10687022900763354	-1.0356234096692112	0.0	0.9287531806615776	-0.10687022900763354	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.75	-0.09923664122137404	-1.4427480916030535	0.0	1.3435114503816794	-0.09923664122137404	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.8	-0.1755725190839695	-2.0534351145038174	0.0	1.877862595419848	-0.1755725190839695	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.85	-0.08396946564885496	-3.0712468193384224	0.0	2.9872773536895676	-0.08396946564885496	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.9	-0.053435114503816813	-5.106870229007635	0.0	5.053435114503818	-0.053435114503816813	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModuleIFNConsensus	0.95	0.007633587786259542	-11.213740458015256	0.0	11.221374045801515	0.007633587786259542	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.05	0.3523503415026115	0.3571715548413017	0.0	-0.004821213338690178	0.3523503415026115	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.1	0.3155216284987277	0.32145886344359625	0.0	-0.00593723494486853	0.3155216284987277	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.15	0.265828468792097	0.28154467894027835	0.0	-0.015716210148181364	0.265828468792097	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.2	0.24045801526717558	0.23664122137404578	0.0	0.003816793893129805	0.24045801526717558	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.25	0.183206106870229	0.18575063613231552	0.0	-0.002544529262086509	0.183206106870229	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.3	0.1341330425299891	0.12758996728462374	0.0	0.0065430752453653485	0.1341330425299891	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.35	0.11450381679389315	0.060481503229594835	0.0	0.05402231356429832	0.11450381679389315	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.4	0.1068702290076336	-0.017811704834605646	0.0	0.12468193384223925	0.1068702290076336	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.45	0.050659264399722403	-0.11034004163775157	0.0	0.16099930603747398	0.050659264399722403	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.5	0.007633587786259527	-0.22137404580152675	0.0	0.22900763358778628	0.007633587786259527	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.55	-0.004240882103477617	-0.3570822731128077	0.0	0.3528413910093301	-0.004240882103477617	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.6	-0.04961832061068697	-0.5267175572519083	0.0	0.47709923664122134	-0.04961832061068697	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.65	-0.09814612868047987	-0.7448200654307527	0.0	0.6466739367502728	-0.09814612868047987	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.7	-0.12213740458015265	-1.0356234096692112	0.0	0.9134860050890585	-0.12213740458015265	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.75	-0.183206106870229	-1.4427480916030535	0.0	1.2595419847328244	-0.183206106870229	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.8	-0.33587786259541996	-2.0534351145038174	0.0	1.7175572519083975	-0.33587786259541996	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.85	-0.4910941475826972	-3.0712468193384224	0.0	2.5801526717557253	-0.4910941475826972	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.9	-0.4503816793893131	-5.106870229007635	0.0	4.656488549618322	-0.4503816793893131	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed	0.95	-0.4961832061068696	-11.213740458015256	0.0	10.717557251908385	-0.4961832061068696	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.15	0.28154467894027835	0.28154467894027835	0.0	0.0	0.28154467894027835	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.2	0.23473282442748095	0.23664122137404578	0.0	-0.0019083969465648332	0.23473282442748095	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.25	0.1959287531806616	0.18575063613231552	0.0	0.010178117048346091	0.1959287531806616	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.3	0.1406761177753544	0.12758996728462374	0.0	0.01308615049073067	0.1406761177753544	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.35	0.06870229007633588	0.060481503229594835	0.0	0.008220786846741046	0.06870229007633588	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.4	0.01781170483460559	-0.017811704834605646	0.0	0.03562340966921124	0.01781170483460559	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.45	0.04024982650936848	-0.11034004163775157	0.0	0.15058986814712005	0.04024982650936848	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.5	0.02290076335877861	-0.22137404580152675	0.0	0.24427480916030536	0.02290076335877861	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.55	-0.005089058524173046	-0.3570822731128077	0.0	0.35199321458863464	-0.005089058524173046	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.6	-0.04580152671755722	-0.5267175572519083	0.0	0.48091603053435106	-0.04580152671755722	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.65	-0.09378407851690297	-0.7448200654307527	0.0	0.6510359869138498	-0.09378407851690297	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.7	-0.07888040712468192	-1.0356234096692112	0.0	0.9567430025445293	-0.07888040712468192	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.75	-0.03053435114503817	-1.4427480916030535	0.0	1.4122137404580153	-0.03053435114503817	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.8	-6.938893903907228e-18	-2.0534351145038174	0.0	2.0534351145038174	-6.938893903907228e-18	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.85	-0.020356234096692103	-3.0712468193384224	0.0	3.05089058524173	-0.020356234096692103	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.9	0.007633587786259542	-5.106870229007635	0.0	5.114503816793895	0.007633587786259542	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-ModulePriorFixed-Platt	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.05	0.3499397348332664	0.3571715548413017	0.0	-0.007231820008035295	0.3499397348332664	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.1	0.3078880407124682	0.32145886344359625	0.0	-0.01357082273112803	0.3078880407124682	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.15	0.24247867085765606	0.28154467894027835	0.0	-0.03906600808262228	0.24247867085765606	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.2	0.20610687022900764	0.23664122137404578	0.0	-0.030534351145038136	0.20610687022900764	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.25	0.15267175572519084	0.18575063613231552	0.0	-0.03307888040712467	0.15267175572519084	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.3	0.08069792802617229	0.12758996728462374	0.0	-0.04689203925845145	0.08069792802617229	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.35	0.008220786846741046	0.060481503229594835	0.0	-0.05226071638285379	0.008220786846741046	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.4	-0.06106870229007638	-0.017811704834605646	0.0	-0.043256997455470736	-0.06106870229007638	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.45	-0.12144344205412905	-0.11034004163775157	0.0	-0.011103400416377474	-0.12144344205412905	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.5	-0.21374045801526717	-0.22137404580152675	0.0	0.0076335877862595825	-0.21374045801526717	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.55	-0.26717557251908414	-0.3570822731128077	0.0	0.08990670059372358	-0.26717557251908414	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.6	-0.3664122137404579	-0.5267175572519083	0.0	0.16030534351145037	-0.3664122137404579	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.65	-0.43184296619411133	-0.7448200654307527	0.0	0.3129770992366414	-0.43184296619411133	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.7	-0.4427480916030534	-1.0356234096692112	0.0	0.5928753180661578	-0.4427480916030534	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.75	-0.5267175572519084	-1.4427480916030535	0.0	0.9160305343511451	-0.5267175572519084	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.8	-0.5801526717557253	-2.0534351145038174	0.0	1.4732824427480922	-0.5801526717557253	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.85	-0.580152671755725	-3.0712468193384224	0.0	2.4910941475826975	-0.580152671755725	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.9	-0.786259541984733	-5.106870229007635	0.0	4.320610687022902	-0.786259541984733	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordFullGraph	0.95	-0.8549618320610679	-11.213740458015256	0.0	10.358778625954187	-0.8549618320610679	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.05	0.3575733226195259	0.3571715548413017	0.0	0.0004017677782242046	0.3575733226195259	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.1	0.2926208651399491	0.32145886344359625	0.0	-0.02883799830364714	0.2926208651399491	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.15	0.24247867085765606	0.28154467894027835	0.0	-0.03906600808262228	0.24247867085765606	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.2	0.1717557251908397	0.23664122137404578	0.0	-0.06488549618320608	0.1717557251908397	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.25	0.12213740458015268	0.18575063613231552	0.0	-0.06361323155216284	0.12213740458015268	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.3	0.0534351145038168	0.12758996728462374	0.0	-0.07415485278080694	0.0534351145038168	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.35	-0.012331180270111541	0.060481503229594835	0.0	-0.07281268349970638	-0.012331180270111541	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.4	-0.07379134860050895	-0.017811704834605646	0.0	-0.05597964376590331	-0.07379134860050895	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.45	-0.16585704371963914	-0.11034004163775157	0.0	-0.05551700208188756	-0.16585704371963914	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.5	-0.22137404580152673	-0.22137404580152675	0.0	2.7755575615628914e-17	-0.22137404580152673	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.55	-0.2883799830364717	-0.3570822731128077	0.0	0.06870229007633599	-0.2883799830364717	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.6	-0.34732824427480913	-0.5267175572519083	0.0	0.17938931297709915	-0.34732824427480913	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.65	-0.37731733914940035	-0.7448200654307527	0.0	0.3675027262813524	-0.37731733914940035	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.7	-0.49109414758269715	-1.0356234096692112	0.0	0.5445292620865141	-0.49109414758269715	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.75	-0.5801526717557252	-1.4427480916030535	0.0	0.8625954198473283	-0.5801526717557252	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.8	-0.7022900763358781	-2.0534351145038174	0.0	1.3511450381679393	-0.7022900763358781	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.85	-1.010178117048346	-3.0712468193384224	0.0	2.0610687022900764	-1.010178117048346	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.9	-1.221374045801527	-5.106870229007635	0.0	3.885496183206108	-1.221374045801527	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoBioObjective	0.95	-1.267175572519083	-11.213740458015256	0.0	9.946564885496173	-1.267175572519083	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.05	0.3575733226195259	0.3571715548413017	0.0	0.0004017677782242046	0.3575733226195259	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.1	0.3231552162849872	0.32145886344359625	0.0	0.001696352841390969	0.3231552162849872	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.15	0.2855859901212393	0.28154467894027835	0.0	0.00404131118096096	0.2855859901212393	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.2	0.23664122137404583	0.23664122137404578	0.0	5.551115123125783e-17	0.23664122137404583	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.25	0.18575063613231552	0.18575063613231552	0.0	0.0	0.18575063613231552	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.3	0.13413304252998906	0.12758996728462374	0.0	0.006543075245365321	0.13413304252998906	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.35	0.07868467410452146	0.060481503229594835	0.0	0.018203170874926622	0.07868467410452146	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.4	-0.005089058524173018	-0.017811704834605646	0.0	0.012722646310432628	-0.005089058524173018	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.45	-0.0652324774462179	-0.11034004163775157	0.0	0.04510756419153367	-0.0652324774462179	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.5	-0.16030534351145037	-0.22137404580152675	0.0	0.06106870229007638	-0.16030534351145037	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.55	-0.2171331636980493	-0.3570822731128077	0.0	0.1399491094147584	-0.2171331636980493	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.6	-0.30534351145038163	-0.5267175572519083	0.0	0.22137404580152664	-0.30534351145038163	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.65	-0.3446019629225738	-0.7448200654307527	0.0	0.40021810250817896	-0.3446019629225738	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.7	-0.3333333333333326	-1.0356234096692112	0.0	0.7022900763358779	-0.3333333333333326	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.75	-0.35877862595419846	-1.4427480916030535	0.0	1.083969465648855	-0.35877862595419846	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.8	-0.2061068702290077	-2.0534351145038174	0.0	1.8473282442748098	-0.2061068702290077	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.85	-0.09923664122137404	-3.0712468193384224	0.0	2.9720101781170483	-0.09923664122137404	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.9	-0.04580152671755727	-5.106870229007635	0.0	5.061068702290077	-0.04580152671755727	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordNoInteraction	0.95	0.007633587786259542	-11.213740458015256	0.0	11.221374045801515	0.007633587786259542	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.05	0.3575733226195259	0.3571715548413017	0.0	0.0004017677782242046	0.3575733226195259	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.1	0.3231552162849872	0.32145886344359625	0.0	0.001696352841390969	0.3231552162849872	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.15	0.2779524023349798	0.28154467894027835	0.0	-0.003592276605298539	0.2779524023349798	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.2	0.22137404580152673	0.23664122137404578	0.0	-0.015267175572519054	0.22137404580152673	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.25	0.1704834605597964	0.18575063613231552	0.0	-0.01526717557251911	0.1704834605597964	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.3	0.09923664122137402	0.12758996728462374	0.0	-0.028353326063249723	0.09923664122137402	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.35	0.03816793893129772	0.060481503229594835	0.0	-0.022313564298297117	0.03816793893129772	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.4	0.005089058524173018	-0.017811704834605646	0.0	0.022900763358778664	0.005089058524173018	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.45	-0.09021512838306728	-0.11034004163775157	0.0	0.020124913254684296	-0.09021512838306728	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.5	-0.15267175572519082	-0.22137404580152675	0.0	0.06870229007633594	-0.15267175572519082	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.55	-0.22731128074639537	-0.3570822731128077	0.0	0.12977099236641235	-0.22731128074639537	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.6	-0.23282442748091597	-0.5267175572519083	0.0	0.2938931297709923	-0.23282442748091597	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.65	-0.2431842966194112	-0.7448200654307527	0.0	0.5016357688113415	-0.2431842966194112	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.7	-0.24427480916030525	-1.0356234096692112	0.0	0.7913486050089059	-0.24427480916030525	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.75	-0.2900763358778626	-1.4427480916030535	0.0	1.1526717557251909	-0.2900763358778626	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.8	-0.3435114503816795	-2.0534351145038174	0.0	1.709923664122138	-0.3435114503816795	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.85	-0.34351145038167935	-3.0712468193384224	0.0	2.727735368956743	-0.34351145038167935	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.9	-0.38931297709923673	-5.106870229007635	0.0	4.717557251908398	-0.38931297709923673	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	EcoNiche-Opt-WordUnsignedGraph	0.95	-0.2748091603053432	-11.213740458015256	0.0	10.938931297709912	-0.2748091603053432	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.15	0.2766052986079928	0.28154467894027835	0.0	-0.004939380332285526	0.2766052986079928	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.2	0.24618320610687025	0.23664122137404578	0.0	0.009541984732824471	0.24618320610687025	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.25	0.18575063613231552	0.18575063613231552	0.0	0.0	0.18575063613231552	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.3	0.14176663031624862	0.12758996728462374	0.0	0.014176663031624875	0.14176663031624862	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.35	0.10804462712859661	0.060481503229594835	0.0	0.047563123899001775	0.10804462712859661	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.4	0.06106870229007633	-0.017811704834605646	0.0	0.0788040712468197	0.06106870229007633	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	IFNG	0.45	0.015961138098542688	-0.11034004163775157	0.0	0.12630117973629426	0.015961138098542688	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.5	-0.007633587786259527	-0.221374045801526723	0.0	0.21374045801526723	-0.007633587786259527	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.55	-0.02290076335877872	-0.3570822731128077	0.0	0.334181509754029	-0.02290076335877872	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.6	-0.06870229007633585	-0.5267175572519083	0.0	0.4580152671755724	-0.06870229007633585	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.65	-0.10250817884405675	-0.7448200654307527	0.0	0.642311886586696	-0.10250817884405675	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.7	-0.09160305343511448	-1.0356234096692112	0.0	0.9440203562340967	-0.09160305343511448	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.75	-0.1450381679389313	-1.4427480916030535	0.0	1.2977099236641223	-0.1450381679389313	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.8	-0.10687022900763363	-2.0534351145038174	0.0	1.946564885496184	-0.10687022900763363	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.85	-0.055979643765903295	-3.0712468193384224	0.0	3.015267175572519	-0.055979643765903295	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.9	0.007633587786259542	-5.106870229007635	0.0	5.114503816793895	0.007633587786259542	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IFNG	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.1	0.3231552162849872	0.32145886344359625	0.0	0.001696352841390969	0.3231552162849872	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.15	0.2766052986079928	0.28154467894027835	0.0	-0.004939380332285526	0.2766052986079928	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.2	0.22900763358778625	0.23664122137404578	0.0	-0.007633587786259527	0.22900763358778625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.25	0.16793893129770993	0.18575063613231552	0.0	-0.01781170483460559	0.16793893129770993	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.3	0.1166848418756816	0.12758996728462374	0.0	-0.010905125408942146	0.1166848418756816	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.35	0.07222548443922489	0.060481503229594835	0.0	0.01174398120963005	0.07222548443922489	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.4	0.04834605597964378	-0.017811704834605646	0.0	0.06615776081424943	0.04834605597964378	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.45	0.023594725884802242	-0.11034004163775157	0.0	0.13393476752255382	0.023594725884802242	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.5	-0.03816793893129772	-0.22137404580152675	0.0	0.18320610687022904	-0.03816793893129772	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.55	-0.08312128922815953	-0.3570822731128077	0.0	0.2739609838846482	-0.08312128922815953	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.6	-0.12977099236641212	-0.5267175572519083	0.0	0.3969465648854962	-0.12977099236641212	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.65	-0.18211559432933483	-0.7448200654307527	0.0	0.5627044711101418	-0.18211559432933483	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.7	-0.23918575063613223	-1.0356234096692112	0.0	0.796437659033079	-0.23918575063613223	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.75	-0.19083969465648853	-1.4427480916030535	0.0	1.251908396946565	-0.19083969465648853	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.8	-0.08396946564885499	-2.0534351145038174	0.0	1.9694656488549624	-0.08396946564885499	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.85	-0.020356234096692103	-3.0712468193384224	0.0	3.05089058524173	-0.020356234096692103	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.9	0.007633587786259542	-5.106870229007635	0.0	5.114503816793895	0.007633587786259542	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	IPRES	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.15	0.2842388863942523	0.28154467894027835	0.0	0.0026942074539739735	0.2842388863942523	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.2	0.24618320610687025	0.23664122137404578	0.0	0.009541984732824471	0.24618320610687025	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.25	0.20610687022900767	0.18575063613231552	0.0	0.020356234096692155	0.20610687022900767	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.3	0.13849509269356594	0.12758996728462374	0.0	0.010905125408942201	0.13849509269356594	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.35	0.08338226658837347	0.060481503229594835	0.0	0.022900763358778636	0.08338226658837347	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.4	0.05089058524173026	-0.017811704834605646	0.0	0.06870229007633591	0.05089058524173026	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.45	0.0055517002081887645	-0.11034004163775157	0.0	0.11589174184594034	0.0055517002081887645	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.5	-0.0534351145038168	-0.22137404580152675	0.0	0.16793893129770995	-0.0534351145038168	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.55	-0.11704834605597966	-0.3570822731128077	0.0	0.24003392705682805	-0.11704834605597966	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.6	-0.1526717557251908	-0.5267175572519083	0.0	0.3740458015267175	-0.1526717557251908	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.65	-0.18974918211559438	-0.7448200654307527	0.0	0.5550708833151583	-0.18974918211559438	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.7	-0.13994910941475824	-1.0356234096692112	0.0	0.8956743002544529	-0.13994910941475824	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.75	-0.1297709923664122	-1.4427480916030535	0.0	1.3129770992366412	-0.1297709923664122	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.8	-0.12977099236641224	-2.0534351145038174	0.0	1.9236641221374051	-0.12977099236641224	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.85	-0.07124681933842238	-3.0712468193384224	0.0	3.0	-0.07124681933842238	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.9	-0.053435114503816813	-5.106870229007635	0.0	5.053435114503818	-0.053435114503816813	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	PDCD1LG2	0.95	0.015267175572519083	-11.213740458015256	0.0	11.229007633587775	0.015267175572519083	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.15	0.2739110911540189	0.28154467894027835	0.0	-0.007633587786259442	0.2739110911540189	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.2	0.23854961832061072	0.23664122137404578	0.0	0.0019083969465649442	0.23854961832061072	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.25	0.20610687022900767	0.18575063613231552	0.0	0.020356234096692155	0.20610687022900767	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.3	0.15376226826608505	0.12758996728462374	0.0	0.02617230098146131	0.15376226826608505	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.35	0.11920140927774514	0.060481503229594835	0.0	0.058719906048150305	0.11920140927774514	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.4	0.0737913486005089	-0.017811704834605646	0.0	0.09160305343511455	0.0737913486005089	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.45	0.02428868841082585	-0.11034004163775157	0.0	0.13462873004857742	0.02428868841082585	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.5	-0.02290076335877861	-0.22137404580152675	0.0	0.19847328244274814	-0.02290076335877861	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.55	-0.055979643765903364	-0.3570822731128077	0.0	0.30110262934690435	-0.055979643765903364	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.6	-0.08396946564885494	-0.5267175572519083	0.0	0.44274809160305334	-0.08396946564885494	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.65	-0.09705561613958563	-0.7448200654307527	0.0	0.6477644492911672	-0.09705561613958563	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.7	-0.11704834605597961	-1.0356234096692112	0.0	0.9185750636132316	-0.11704834605597961	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.75	-0.07633587786259542	-1.4427480916030535	0.0	1.3664122137404582	-0.07633587786259542	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.8	-0.07633587786259545	-2.0534351145038174	0.0	1.977099236641222	-0.07633587786259545	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.85	0.025445292620865152	-3.0712468193384224	0.0	3.0966921119592876	0.025445292620865152	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.9	-0.053435114503816813	-5.106870229007635	0.0	5.053435114503818	-0.053435114503816813	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_dysfunction	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.15	0.2869330938482263	0.28154467894027835	0.0	0.005388414907947947	0.2869330938482263	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.2	0.2366412213740458	0.23664122137404578	0.0	2.7755575615628914e-17	0.2366412213740458	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.25	0.19592875318066158	0.18575063613231552	0.0	0.010178117048346064	0.19592875318066158	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.3	0.11995637949836424	0.12758996728462374	0.0	-0.007633587786259499	0.11995637949836424	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.35	0.087492660011744	0.060481503229594835	0.0	0.027071115678214916	0.087492660011744	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.4	0.025445292620865118	-0.017811704834605646	0.0	0.043256997455470764	0.025445292620865118	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.45	-0.04094378903539206	-0.11034004163775157	0.0	0.06939625260235951	-0.04094378903539206	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.5	-0.07633587786259544	-0.22137404580152675	0.0	0.14503816793893132	-0.07633587786259544	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.55	-0.13231552162849877	-0.3570822731128077	0.0	0.22476675148430894	-0.13231552162849877	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.6	-0.19847328244274806	-0.5267175572519083	0.0	0.3282442748091602	-0.19847328244274806	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.65	-0.1973827699018539	-0.7448200654307527	0.0	0.5474372955288989	-0.1973827699018539	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.7	-0.26208651399491084	-1.0356234096692112	0.0	0.7735368956743003	-0.26208651399491084	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.75	-0.27480916030534347	-1.4427480916030535	0.0	1.16793893129771	-0.27480916030534347	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.8	-0.25954198473282447	-2.0534351145038174	0.0	1.793893129770993	-0.25954198473282447	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.85	-0.2010178117048346	-3.0712468193384224	0.0	2.8702290076335877	-0.2010178117048346	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.9	-0.1374045801526718	-5.106870229007635	0.0	4.969465648854963	-0.1374045801526718	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIDE_exclusion	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.05	0.3571715548413017	0.3571715548413017	0.0	0.0	0.3571715548413017	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.1	0.32145886344359625	0.32145886344359625	0.0	0.0	0.32145886344359625	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.15	0.27525819488100584	0.28154467894027835	0.0	-0.006286484059272512	0.27525819488100584	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.2	0.2366412213740458	0.23664122137404578	0.0	2.7755575615628914e-17	0.2366412213740458	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.25	0.18066157760814253	0.18575063613231552	0.0	-0.00508905852417299	0.18066157760814253	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.3	0.15376226826608505	0.12758996728462374	0.0	0.02617230098146131	0.15376226826608505	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.35	0.11098062243100412	0.060481503229594835	0.0	0.05049911920140929	0.11098062243100412	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.4	0.05852417302798979	-0.017811704834605646	0.0	0.07633587786259544	0.05852417302798979	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.45	0.040943789035392086	-0.11034004163775157	0.0	0.15128383067314366	0.040943789035392086	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.5	-0.015267175572519082	-0.22137404580152675	0.0	0.20610687022900767	-0.015267175572519082	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.55	-0.028837998303647222	-0.3570822731128077	0.0	0.3282442748091605	-0.028837998303647222	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.6	-0.08778625954198468	-0.5267175572519083	0.0	0.43893129770992356	-0.08778625954198468	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.65	-0.12322791712104697	-0.7448200654307527	0.0	0.6215921483097058	-0.12322791712104697	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.7	-0.11195928753180656	-1.0356234096692112	0.0	0.9236641221374047	-0.11195928753180656	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.75	-0.13740458015267176	-1.4427480916030535	0.0	1.3053435114503817	-0.13740458015267176	131	0.3893129770992366

endpoint	stratum	model_name	threshold	net_benefit	treat_all_net_benefit	treat_none_net_benefit	net_benefit_vs_treat_all	net_benefit_vs_treat_none	n_samples	prevalence
primary_recist	melanoma_recist_supported_primary	TIG	0.8	-0.09160305343511455	-2.0534351145038174	0.0	1.9618320610687028	-0.09160305343511455	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.85	-0.012722646310432562	-3.0712468193384224	0.0	3.05852417302799	-0.012722646310432562	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.9	-0.061068702290076354	-5.106870229007635	0.0	5.045801526717558	-0.061068702290076354	131	0.3893129770992366
primary_recist	melanoma_recist_supported_primary	TIG	0.95	0.0	-11.213740458015256	0.0	11.213740458015256	0.0	131	0.3893129770992366

Supplementary Table 15 | Locked external metrics.

Reports AUROC, AUPRC, balanced accuracy, ECE, threshold, and calibration metrics for each endpoint and model in locked external and panel-transfer cohorts. This table supports Figure 5b-c and Supplementary Figure 7.

Full tabulation (135 rows x 29 columns).

endp oint	cohort	analysis_type	model _nam e	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specific ity	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.43779 2274855 71015	discove ry_only _platt	0.70490 257699 56003	0.67426147 07730987		Nivo:PE M;Pem	0.52083 3333333 3334	0.70166 8470418 4704	0.35416 6666666 66663	0.35714 2857142 85715	- 0.288675 13459481 287	0.4	0.375	0.33333 3333333 3333	0.42857 1428571 42855	0.28571 4285714 2857	0.25805 0420113 19107	0.387343 0257322 005	0.485526 4687054 1344	0.33584 4089837 84403
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	IFNG	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.51791 7807859 5991	discove ry_only _platt	0.66436 203645 50596	0.65666247 64299183		Nivo:PE M;Pem	0.41666 6666666 66663	0.64925 2136752 1367	0.45833 3333333 3333	0.42857 1428571 42855	- 0.091287 09291752 768	0.33333 3333333 3333	0.25	0.66666 6666666 6666	0.5	0.4	0.27351 6737998 1562	0.203526 4541185 6737	- 0.026526 8762718 37037	0.28661 4076331 9738
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	CXCL 9	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.43501 7063942 71966	discove ry_only _platt	0.66561 910747 95725	0.67049025 76995599		Nivo:PE M;Pem	0.45833 3333333 3333	0.65483 4748584 7485	0.41666 6666666 66663	0.42857 1428571 42855	- 0.166666 66666666 666	0.5	0.5	0.33333 3333333 3333	0.5	0.33333 3333333 3333	0.26398 2251225 1046	0.147876 6530585 0695	0.208021 7460652 075	0.29814 9620385 27987
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	TIG	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.51016 6643517 3705	discove ry_only _platt	0.66561 910747 95725	0.63827781 26964174		Nivo:PE M;Pem	0.45833 3333333 3333	0.67068 0708180 7082	0.45833 3333333 3333	0.42857 1428571 42855	- 0.091287 09291752 768	0.33333 3333333 3333	0.25	0.66666 6666666 6666	0.5	0.4	0.26512 9986203 66013	0.212968 2209877 9425	0.173745 3127043 6952	0.29543 7877066 54945
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	TIDE_ dysfun ction	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.44017 6933060 8997	discove ry_only _platt	0.65461 973601 50849	0.63372093 02325582		Nivo:PE M;Pem	0.5625	0.71413 6904761 9047	0.5	0.5	0.0	0.53333 3333333 3333	0.5	0.5	0.57142 8571428 5714	0.42857 1428571 42855	0.24273 0501678 10492	0.241493 8080427 0903	0.889191 4317729 48	0.34249 3174861 01594
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	APM	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.54358 7746904 6431	discove ry_only _platt	0.66373 903205 28034	0.66043368 95034569		Nivo:PE M;Pem	0.52083 3333333 3334	0.72011 2526362 5263	0.60416 6666666 6667	0.57142 8571428 5714	0.228217 33229381 923	0.5	0.375	0.83333 3333333 3334	0.75	0.5	0.28064 4298206 76456	0.209684 3463063 3448	- 0.257940 1546221 5414	0.27662 4435468 6012
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	CYT	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.44839 3258678 51143	discove ry_only _platt	0.62979 258328 09553	0.62994971 71590195		Nivo:PE M;Pem	0.60416 6666666 6667	0.74784 8925974 026	0.5	0.5	0.0	0.53333 3333333 3333	0.5	0.5	0.57142 8571428 5714	0.42857 1428571 42855	0.24420 4820632 86617	0.190555 0815126 903	0.857387 4970249 903	0.32190 1129623 39497
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	IPRES	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.48321 0906587 7373	discove ry_only _platt	0.56693 903205 53112	0.59695160 27655563		Nivo:PE M;Pem	0.625	0.73873 6263736 2638	0.72916 6666666 6667	0.71428 5714285 7143	0.458333 3333333 33	0.71428 5714285 7143	0.625	0.83333 3333333 3334	0.83333 3333333 3334	0.625	0.3685 4378534 45546	0.161273 6894931 696	2.514639 8678964 625	0.34517 0639005 41045
prima ry_re cist	GSE14599 6	locked_external_m elanoma_pd1_recis t	TIDE_ exclusi on	14	8	6	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.47584 6535812 14104	discove ry_only _platt	0.54368 321810 18227	0.57102451 2884978		Nivo:PE M;Pem	0.5	0.60597 9437229 4373	0.45833 3333333 3333	0.5	- 0.100503 78152592 121	0.63157 8947368 421	0.75	0.16666 6666666 66666	0.54545 4545454 5454	0.33333 3333333 3333	0.25331 9374342 21136	0.111539 1238607 7125	- 0.630514 3562531 956	0.28642 9321716 75334
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	121	50	71	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.43779 2274855 71015	discove ry_only _platt	0.70490 257699 56003	0.67426147 07730987		anti-PD- 1	0.53492 9577464 7889	0.49439 3215651 3664	0.53352 5619834 5634	0.52892 5619834 7108	0.066081 28948033 426	0.49557 5221238 93805	0.56	0.50704 2253521 1268	0.44444 4444444 4444	0.62068 9655172 4138	0.26196 38765	0.144751 6559291 5976	0.244682 5162711 654	- 0.33766 6907858 9453
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	IFNG	121	50	71	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.51791 7807859 5991	discove ry_only _platt	0.66436 203645 50596	0.65666247 64299183		anti-PD- 1	0.53859 1549295 7747	0.51443 5278487 4647	0.51577 4647887 324	0.52892 5619834 7108	0.031460 79827697 43564	0.43564 3564356 43564	0.44	0.59154 9295774 6479	0.43137 2549019 60786	0.6	0.25564 7940992 0815	0.148503 8747882 9374	0.302170 6995076 5746	- 0.34017 1006217 3274
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	CXCL 9	121	50	71	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.43501 7063942 71966	discove ry_only _platt	0.66561 910747 95725	0.67049025 76995599		anti-PD- 1	0.56591 5492957 7465	0.56064 0992010 7641	0.51014 0845070 4226	0.48760 3305785 12395	0.020663 56577005 964	0.50793 6507936 5079	0.64	0.38028 1690140 84506	0.42105 2631578 94735	0.6	0.25175 9012951 85926	0.145048 6789054 7748	0.408405 3157609 579	- 0.33689 9434538 3088
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	TIG	121	50	71	discov ery_onl y	GSE91061.GSE78 220,PRJEB23709_ PD1_PRE	0.51016 6643517 3705	discove ry_only _platt	0.66561 910747 95725	0.63827781 26964174		anti-PD- 1	0.53859 1549295 7747	0.50901 8297037 2071	0.51985 9154929 5775	0.53719 0082644 6281	0.039978 91787952 7265	0.42857 1428571 42855	0.42	0.61971 8309859 1549	0.4375	0.60273 9726027 3972	0.25586 4002448 3236	0.150197 2994864 7706	0.290603 0875263 4345	- 0.34074 1554890 3941
prima	PHS00045	locked_external_m	TIDE_	121	50	71	discov	GSE91061.GSE78	0.44017	discove	0.65461	0.63372093		anti-PD-	0.50957	0.50753	0.50535	0.49586	0.010603	0.47863	0.56	0.45070	0.41791	0.59259	0.25335	0.121270	0.359219	-

endp oint	cohort	analysis_type	model_ name	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
ry_re cist	2_LIU_LIK E_PRE	elanoma_pd1_like	dysfun ction				ery_onl y	220,PRJEB23709_ PD1_PRE	6933060 8997	ry_only _platt	973601 50849	02325582			7464788 7324	4783365 8351	2112676 0564	7768595 04134	15570145 3471	2478632 47865		4225352 1127	0447761 194	2592592 5926	5681015 22443	8895239 0476	3704852 103	0.34149 7568663 6525
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	APM	121	50	71	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54358 7746904 6431	discove ry_only _platt	0.66373 350094 28034	0.66043368 95034569		anti-PD- 1	0.52901 0484507 0422	0.48199 3888897 0336	0.51802 4169014 0844	0.54545 4545454 5454	0.037511 01979388 627	0.39560 4395604 3956	0.36	0.67605 6338028 169	0.43902 4390243 90244	0.6	0.25680 4667062 1317	0.132617 7427477 987	0.294065 7758190 281	- 0.33907 1273967 3017
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	CYT	121	50	71	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44839 3258678 51143	discove ry_only _platt	0.62979 258328 09553	0.62994971 71590195		anti-PD- 1	0.53267 6056338 0282	0.51191 9894334 7887	0.49901 4084507 04227	0.47107 4380165 28924	- 0.002051 2307692 3077	0.50769 2307692 3077	0.66	0.33802 8169014 0845	0.4125	0.58536 5853658 5366	0.25021 2361575 87223	0.086209 8123085 1635	0.455866 5631625 155	- 0.34254 1828205 4174
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	IPRES	121	50	71	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48321 0906587 7373	discove ry_only _platt	0.56693 903205 53112	0.59695160 27655563		anti-PD- 1	0.47014 0845070 4226	0.41158 1171082 17784	0.46309 8591549 2958	0.44628 0991735 5372	- 0.074285 67364848 329	0.45528 4552845 52843	0.56	0.36619 7183098 59156	0.38356 1643835 6164	0.54166 6666666 6666	0.25479 3634463 1119	0.099730 9440242 898	- 0.178263 6745835 856	- 0.35245 6808913 6754
prima ry_re cist	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	TIDE_ exclusi on	121	50	71	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47584 6535812 14104	discove ry_only _platt	0.54368 321810 18227	0.57102451 2884978		anti-PD- 1	0.46338 0281690 14084	0.40698 2238666 3579	0.47676 0563380 28167	0.43801 6528925 6198	- 0.051389 15323706 4875	0.50724 6376811 5942	0.7	0.25352 1126760 5634	0.39772 7272727 2727	0.54545 4545454 5454	0.25314 8840493 5265	0.099234 6146868 5188	- 0.638107 7639441 07	- 0.35377 3596008 46936
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43779 2274855 71015	discove ry_only _platt	0.70490 257699 56003	0.67426147 07730987		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.74025 9740259 7402	0.86774 8148542 9872	0.65151 5151515 1515	0.65625	0.290129 42659282 975	0.71794 8717948 718	0.66666 6666666 6666	0.63636 3636363 6364	0.77777 7777777 7778	0.5	0.22075 7172795 13635	0.169562 3997528 0474	1.146379 3432121 687	0.83643 9534853 1751
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	IFNG	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.51791 7807859 5991	discove ry_only _platt	0.66436 203645 50596	0.65666247 64299183		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.82683 9826839 8269	0.91960 6045039 1932	0.80952 3809523 8095	0.75	0.598661 83273609 26	0.76470 5882352 9411	0.61904 7619047 6191	1.0	1.0	0.57894 7368421 0527	0.20150 9727168 58123	0.223826 9750264 9546	2.965122 6338260 43	1.23077 6431289 363
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	CXCL 9	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43501 7063942 71966	discove ry_only _platt	0.66561 910747 95725	0.67049025 76995599		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.83116 8831168 8312	0.91594 6063145 324	0.74458 8744588 7446	0.75	0.473068 44125299 624	0.8	0.76190 4761904 7619	0.72727 2727272 7273	0.84210 5263157 8947	0.61538 4615384 6154	0.20137 1743452 43392	0.185624 6048910 5087	2.747104 1600178 35	1.18896 2833102 3634
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIG	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.51016 6643517 3705	discove ry_only _platt	0.66561 910747 95725	0.63827781 26964174		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.82683 9826839 8268	0.91437 9799057 6703	0.71861 4718614 7187	0.6875	0.416145 58708189 84	0.72222 2222222 2222	0.61904 7619047 6191	0.81818 1818181 8182	0.86666 6666666 6667	0.52941 1764705 8824	0.20490 7615663 7072	0.166818 0579649 3038	2.666848 3202491 364	1.11279 9532080 6319
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIDE_ dysfun ction	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44017 6933060 8997	discove ry_only _platt	0.65461 973601 50849	0.63372093 02325582		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.68398 2683982 684	0.82694 3166025 7991	0.65151 5151515 1515	0.65625	0.290129 42659282 975	0.71794 8717948 718	0.66666 6666666 6666	0.63636 3636363 6364	0.77777 7777777 7778	0.5	0.23010 1131190 58948	0.198376 9411972 5938	1.431589 7355979 745	0.78756 8406159 105
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	APM	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54358 7746904 6431	discove ry_only _platt	0.66373 350094 28034	0.66043368 95034569		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.84848 4848484 8484	0.92203 7368908 2225	0.74025 9740259 7402	0.6875	0.464695 54848745 65	0.70588 2352941 1765	0.57142 8571428 5714	0.90909 0909090 9091	0.92307 6923076 9231	0.52631 5789473 6842	0.19825 7589464 6914	0.231397 6425925 6052	2.677378 3886154 607	1.05460 7585882 6094
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	CYT	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44839 3258678 51143	discove ry_only _platt	0.62979 258328 09553	0.62994971 71590195		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.76623 3766233 7663	0.86248 6373457 279	0.65151 5151515 1515	0.65625	0.290129 42659282 975	0.71794 8717948 718	0.66666 6666666 6666	0.63636 3636363 6364	0.77777 7777777 7778	0.5	0.22023 7161685 76551	0.198650 4590121 9858	2.620335 3216202 703	0.91984 9579287 6544
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	IPRES	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48321 0906587 7373	discove ry_only _platt	0.56693 903205 53112	0.59695160 27655563		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.36363 6363636 36365	0.58917 8106146 963	0.42207 7922077 92205	0.46875	- 0.152894 15743128 767	0.58536 5853658 5366	0.57142 8571428 5714	0.27272 7272727 2727	0.6	0.25	0.26881 6911975 2848	0.199055 2796900 358	- 2.233700 2471392 102	0.67276 2949915 9165
prima ry_re cist	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIDE_ exclusi on	32	21	11	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47584 6535812 14104	discove ry_only _platt	0.54368 321810 18227	0.57102451 2884978		Ipilimum ab + Nivolum ab;Ipilim umab +	0.32034 6320346 3203	0.58070 3844633 8544	0.38095 2380952 38093	0.5	- 0.311451 06143410 45	0.66666 6666666 6666	0.76190 4761904 7619	0.0	0.59259 2592592 5926	0.0	0.26300 1158028 35256	0.157001 9377590 8716	- 5.088830 5479742 89	0.71243 6939607 6162

endp oint	cohort	analysis_type	model_ nam e	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept	
														Pembrol izumab															
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43779 2274855 71015	discove ry_only _platt	0.70490 257699 56003	0.67426147 07730987	NanoString_n Counter_immu ne_panel		0.57222 2222222 2222	0.51796 3560945 8979	0.5	0.46153 8461538 46156	0.0	0.40677 9661016 9492	0.6	0.4	0.30769 2307692 3077	0.69230 7692307 6923	0.25282 8308567 08655	0.203149 1548647 0905	0.364486 7204776 5157	- 0.79993 1922684 6928	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	IFNG	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.51791 7807859 5991	discove ry_only _platt	0.66436 203645 50596	0.65666247 64299183	NanoString_n Counter_immu ne_panel		0.56444 4444444 4444	0.47421 4491277 02736	0.47777 7777777 7778	0.50769 2307692 3077	- 0.041424 65106674 667	0.33333 3333333 3333	0.4	0.55555 5555555 5556	0.28571 4285714 2857	0.67567 5675675 6757	0.24941 7377084 10142	0.207033 7991588 3013	0.442602 9705291 7846	- 0.80401 4818520 3234	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	CXCL 9	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43501 7063942 71966	discove ry_only _platt	0.66561 910747 95725	0.67049025 76995599	NanoString_n Counter_immu ne_panel		0.60333 3333333 3333	0.45927 7253730 98893	0.63333 3333333 3333	0.56923 0769230 7692	0.252982 21281347 033	0.53333 3333333 3333	0.8	0.46666 6666666 6667	0.4	0.84	0.24323 3228200 2929	0.211864 1595335 684	0.611138 5955401 87	- 0.80762 2217599 6994	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIG	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.51016 6643517 3705	discove ry_only _platt	0.66561 910747 95725	0.63827781 26964174	NanoString_n Counter_immu ne_panel		0.55555 5555555 5555	0.45558 5435212 2755	0.51666 6666666 6666	0.52307 6923076 9231	0.030802 05518168 487	0.39215 6862745 09803	0.5	0.53333 3333333 3333	0.32258 0645161 2903	0.70588 2352941 1765	0.25150 6413294 7006	0.207927 9766781 3825	0.391069 9766781 3002430 678	- 0.80393 3002430 678	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIDE_ dysfun ction	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44017 6933060 8997	discove ry_only _platt	0.65461 973601 50849	0.63372093 02325582	NanoString_n Counter_immu ne_panel		0.56888 8888888 889	0.50600 6895196 117	0.51666 6666666 6666	0.44615 58474798 845	0.032897 58474798 845	0.4375	0.7	0.33333 3333333 3333	0.31818 1818181 8182	0.71428 5714285 7143	0.25057 2749378 8071	0.226302 1403134 2227	0.442899 8297301 2815	- 0.80727 7904676 0225	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	APM	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54358 7746904 6431	discove ry_only _platt	0.66373 350094 28034	0.66043368 95034569	NanoString_n Counter_immu ne_panel		0.54222 2222222 2223	0.38762 4884304 027	0.58333 3333333 3333	0.61538 4615384 6154	0.158113 88300841 897	0.44444 4444444 4444	0.5	0.66666 6666666 6666	0.4	0.75	0.26097 4373892 7075	0.232731 4444770 4394	0.031194 5330029 27325	- 0.80949 1936928 0467	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	CYT	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44839 3258678 51143	discove ry_only _platt	0.62979 258328 09553	0.62994971 71590195	NanoString_n Counter_immu ne_panel		0.51333 3333333 3333	0.42560 0940386 8896	0.50555 5555555 5555	0.43078 9230769 2308	0.011111 1111111 1112	0.43076 9230769 2308	0.7	0.31111 1111111 1111	0.31111 1111111 1111	0.7	0.25450 0613946 0599	0.239556 0861216 573	0.230674 6109880 4453	- 0.80702 6014274 075	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	IPRES	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48321 0906587 7373	discove ry_only _platt	0.56693 903205 53112	0.59695160 27655563	NanoString_n Counter_immu ne_panel		0.47	0.30448 9052996 0482	0.51944 4444444 4445	0.50769 2307692 3077	0.035935 73104529 901	0.40740 7407407 4074	0.55	0.48888 8888888 8889	0.32352 9411764 7059	0.70967 7419354 8387	0.25634 9687243 14683	0.189919 7782156 5237	- 0.593887 3041410 067	- 0.82021 8340760 7206	
prima ry_re cist	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIDE_ exclusi on	65	20	45	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47584 6535812 14104	discove ry_only _platt	0.54368 321810 18227	0.57102451 2884978	NanoString_n Counter_immu ne_panel		0.41555 5555555 55557	0.28187 3626750 767	0.48333 3333333 3333	0.4	- 0.034380 70820862 6445	0.41791 0447761 194	0.7	0.26666 6666666 66666	0.29787 2340425 5319	0.66666 6666666 66666	0.25408 2741570 462	0.191466 3456899 6834	0.151466 1241609 6218771 593	- 1.241609 0038250 4983	- 0.82181 0038250 4983
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43779 2274855 71015	discove ry_only _platt	0.70490 257699 56003	0.67426147 07730987	NanoString_n Counter		0.75925 9259259 2593	0.53184 7319347 3194	0.72222 2222222 2222	0.58333 3333333 3334	0.408248 29046386 3	0.54545 4545454 5454	1.0	0.44444 4444444 4444	0.375	1.0	0.22358 3002947 70103	0.271318 0020818 3693	1.311813 6422890 65	- 1.17050 0172048 1447	
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	IFNG	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.51791 7807859 5991	discove ry_only _platt	0.66436 203645 50596	0.65666247 64299183	NanoString_n Counter		0.80555 5555555 5556	0.56974 5069745 0697	0.77777 7777777 7778	0.75	0.487950 03647426 66	0.625	0.83333 3333333 3334	0.72222 2222222 2222	0.5	0.92857 1428571 4286	0.20776 5798781 38657	0.296147 6163354 59	2.447576 0692495 414	- 1.43125 3545036 0202	
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	CXCL 9	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43501 7063942 71966	discove ry_only _platt	0.66561 910747 95725	0.67049025 76995599	NanoString_n Counter		0.79166 6666666 6666	0.50188 7001887 0019	0.80555 5555555 5556	0.70833 3333333 3334	0.531085 00454379 44	0.63157 8947368 421	1.0	0.61111 1111111 1112	0.46153 8461538 46156	1.0	0.21310 8423447 55976	0.237133 1592408 7341	1.621923 2146044 813	- 1.23388 6019672 577	
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIG	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.51016 6643517 3705	discove ry_only _platt	0.66561 910747 95725	0.63827781 26964174	NanoString_n Counter		0.75	0.45695 9706959 707	0.69444 4444444 4444	0.70833 3333333 3334	0.347832 79649996 73	0.53333 3333333 3333	0.66666 6666666 66666	0.72222 2222222 2222	0.44444 4444444 4444	0.86666 9361136 38297	0.22036 9361136 103	0.239491 4116581 103	1.588838 2434835 47	- 1.21035 8670335 8094	
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIDE_ dysfun ction	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44017 6933060 8997	discove ry_only _platt	0.65461 973601 50849	0.63372093 02325582	NanoString_n Counter		0.80555 5555555 5555	0.71459 2760180 9954	0.66666 6666666 6667	0.58333 3333333 3334	0.292770 02188455 997	0.5	0.83333 3333333 3334	0.5	0.35714 2857142 85715	0.9	0.20144 7302269 8695	0.286134 6323114 62	2.741419 8885292 764	- 1.34414 4825575 3428	
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	APM	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54358 7746904 6431	discove ry_only _platt	0.66373 350094 28034	0.66043368 95034569	NanoString_n Counter		0.87037 0370370 3705	0.58829 3650793 6507	0.83333 3333333 3334	0.83333 3333333 3334	0.612372 43569579 45	0.71428 5714285 7143	0.83333 3333333 3334	0.83333 3333333 3334	0.625	0.9375	0.21066 2966822 28583	0.249035 4922678 8138	3.669671 7919275 34	- 1.61398 3496878 7885	
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	CYT	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44839 3258678 51143	discove ry_only _platt	0.62979 258328 09553	0.62994971 71590195	NanoString_n Counter		0.54629 6296296 2963	0.30442 8964568 28344	0.52777 7777777 7778	0.45833 3333333 3333	0.049690 39949999 5326	0.38095 2380952 38093	0.66666 6666666 66666	0.26666 6666666 66666	0.77777 7777777 7778	0.25145 3690393 88185	0.254259 7775897 243	0.338705 1389167 775	- 1.09595 6845842 8086		
prima ry_re cist	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	IPRES	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48321 0906587 7373	discove ry_only _platt	0.56693 903205 53112	0.59695160 27655563	NanoString_n Counter		0.26851 8518518 5185	0.22184 6963423 05038	0.36111 1111111 1111	0.45833 3333333 3333	- 0.248451 99749997 662	0.13333 3333333 33333	0.16666 6666666 66666	0.55555 5555555 5556	0.11111 1111111 1111	0.66666 6666666 66666	0.26328 7405021 9944	0.247486 5530345 1078	- 2.548981 9065122 24	- 1.21776 9615225 2842	

endp oint	cohort	analysis_type	model_ nam e	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
prima ry_re cist	GSE14090 1	nanostri ng_hcc_c linical_ assay_tr ansfer	TIDE_ exclusi on	24	6	18	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47584 6535812 14104	discove ry_only _platt	0.54368 321810 18227	0.57102451 2884978	NanoString_n Counter		0.17592 5925925 92593	0.17990 2042402 04238	0.25	0.20833 3333333 33334	- 0.476330 15162246 68	0.17391 3043478 26086	0.33333 3333333 3333	0.16666 6666666 66666	0.11764 7058823 52941	0.42857 1428571 42855	0.26593 5606149 7205	0.249213 3339214 2175	- 9.520025 252708 77	- 1.60021 5710798 9858
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44644 8519721 7447	discove ry_only _platt	0.71019 677996 42218	0.70102862 25402504		Nivo;PE M;Pem	0.575	0.77230 9981684 9817	0.3875	0.38461 5384615 38464	- 0.219577 151641341 997	0.42857 1428571 42855	0.375	0.4	0.5	0.28571 4285714 2857	0.24872 7260824 10315	0.369638 6537515 642	0.828418 3759374 933	0.63039 2954046 0782
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	IFNG	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.53434 3452324 6161	discove ry_only _platt	0.66279 069767 44187	0.63886404 29338104		Nivo;PE M;Pem	0.45000 0000000 00007	0.69522 6648351 6484	0.525	0.46153 8461538 46156	0.057735 02691896 257	0.36363 6363636 36365	0.25	0.8	0.66666 6666666 6666	0.4	0.26890 6331701 25827	0.174552 5131988 165	0.171669 9824387 049	0.48321 6862413 52635
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	CXCL 9	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.39407 3788309 8779	discove ry_only _platt	0.66010 733452 59392	0.65138640 4293381		Nivo;PE M;Pem	0.47500 0000000 00003	0.69636 7521367 5214	0.475	0.53846 1538461 5384	- 0.057735 02691896 257	0.66666 6666666 6666	0.75	0.2	0.6	0.33333 3333333 3333	0.26468 4925181 86265	0.175094 3029509 4733	0.240032 8793710 6282	0.48398 7145941 35717
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	TIG	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42718 0894909 34624	discove ry_only _platt	0.65966 010733 4526	0.62611806 79785331		Nivo;PE M;Pem	0.50000 0000000 0001	0.72647 6648351 6484	0.45	0.46153 8461538 46156	- 0.097590 00729485 333	0.53333 3333333 3333	0.5	0.4	0.57142 8571428 5714	0.33333 3333333 3333	0.25548 3667372 37965	0.230956 2036982 996	0.582196 5304251 671	0.53202 5282967 6648
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	TIDE_ dysfun ction	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43978 2201621 358	discove ry_only _platt	0.64937 388193 20214	0.64736135 95706619		Nivo;PE M;Pem	0.625	0.78325 4245754 2457	0.55	0.53846 1538461 5384	0.097590 00729485 333	0.57142 8571428 5714	0.5	0.6	0.66666 6666666 6666	0.42857 1428571 42855	0.23305 2185511 68408	0.212601 6365822 4912	1.961561 7297545 0294	0.77172 0178970 0294
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	APM	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42927 9223503 1947	discove ry_only _platt	0.68515 205724 50804	0.64736135 95706619		Nivo;PE M;Pem	0.525	0.74671 4743589 7437	0.4125	0.46153 8461538 46156	- 0.184466 19684315 546	0.58823 5294117 6471	0.625	0.2	0.55555 5555555 5556	0.25	0.28622 7306669 2202	0.257622 3862686 019	- 0.221816 7720917 6678	0.46570 8849716 6119
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	CYT	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44940 7548612 8113	discove ry_only _platt	0.61135 957066 18963	0.60889982 11091234		Nivo;PE M;Pem	0.675	0.83704 2124542 1246	0.6125	0.61538 4615384 6154	0.219577 51641341 997	0.66666 6666666 6666	0.625	0.6	0.71428 5714285 7143	0.5	0.23241 8597523 14545	0.212860 5065134 733	2.989349 6148423 454913	0.86362 4554913 4637
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	IPRES	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48070 3042659 90343	discove ry_only _platt	0.55143 112701 25223	0.56238819 32021466		Nivo;PE M;Pem	0.6	0.74983 9743589 7437	0.7125	0.69230 7692307 6923	0.414757 53100312 66	0.71428 5714285 7143	0.625	0.8	0.83333 3333333 3334	0.57142 6921403 5714	0.24100 8921403 16288	0.148779 9768932 6466	2.494186 9269000 53	0.50193 9189402 0551
strict _reci st	GSE14599 6	locked_external_m elanoma_p d1_recis t	TIDE_ exclusi on	13	8	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.45498 2089494 3985	discove ry_only _platt	0.54427 549194 99105	0.57491055 45617174		Nivo;PE M;Pem	0.55	0.67521 8531468 5315	0.5375	0.61538 4615384 6154	0.101129 7936948 631	0.73684 2105263 1579	0.875	0.2	0.63636 3636363 6364	0.5	0.25149 4399806 6105	0.161324 5740855 5493	0.184520 5740855 0399	0.47213 6222291 006
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44644 8519721 7447	discove ry_only _platt	0.71019 677996 42218	0.70102862 25402504		anti-PD- 1	0.57310 6060606 0606	0.57874 6569424 2914	0.56174 2424242 4243	0.56310 6796116 5048	0.123344 76352939 013	0.53608 2474226 8041	0.54166 6666666 6666	0.58181 8181818 1818	0.53061 2244897 9592	0.59259 2592592 5926	0.25317 9282211 34817	0.122089 5749511 2494	0.421578 4414811 772	- 0.10105 3863581 0717
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	IFNG	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.53434 3452324 6161	discove ry_only _platt	0.66279 069767 44187	0.63886404 29338104		anti-PD- 1	0.57992 4242424 2423	0.60483 0163111 5528	0.56420 4545454 5455	0.57281 5533980 5825	0.132754 50207479 308	0.48837 2093023 2558	0.4375	0.69090 9090909 0909	0.55263 1578947 3685	0.58461 5384615 3846	0.24592 4653930 9407	0.107516 9343899 6862	0.573281 9101822 799	- 0.10088 0386482 70943
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	CXCL 9	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.39407 3788309 8779	discove ry_only _platt	0.66010 733452 59392	0.65138640 4293381		anti-PD- 1	0.60492 4242424 2425	0.64948 7267465 1259	0.51780 3030303 0303	0.50485 4368932 48083189	0.038381 8571428 5714	0.70833 3333333 3334	0.32727 2727272 72727	0.47887 3239436 6197	0.5625	0.24220 1280452 05183	0.128884 0173607 8856	0.654356 3190627 666	- 0.09752 6033727 24887	
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	TIG	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42718 0894909 34624	discove ry_only _platt	0.65966 010733 4526	0.62611806 79785331		anti-PD- 1	0.57916 6666666 6667	0.60249 8914707 6832	0.52556 8181818 1819	0.51456 3106796 1165	0.053857 14488351 4015	0.56896 5517241 3793	0.6875	0.36363 6363636 36365	0.48529 4117647 0588	0.57142 8571428 5714	0.24601 7236765 4647	0.101707 7538807 0427	0.591823 7380497 118	- 0.09905 4299204 38215
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	TIDE_ dysfun ction	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43978 2201621 358	discove ry_only _platt	0.64937 388193 20214	0.64736135 95706619		anti-PD- 1	0.56060 6060606 0606	0.60037 3117421 9152	0.54753 7878787 8787	0.54368 9320388 3495	0.095401 64639193 3369	0.55238 0952380 9524	0.60416 6666666 6666	0.49090 9090909 0909	0.50877 1929824 5614	0.58695 6521739 1305	0.24345 3317404 4572	0.053155 0815563 767	0.712334 0309285 767	- 0.09728 1582101 70271
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	APM	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42927 9223503 1947	discove ry_only _platt	0.68515 205724 50804	0.64736135 95706619		anti-PD- 1	0.56969 6969696 9697	0.56616 1482405 769	0.51382 5757575 7577	0.50485 4368932 0388	0.028587 25265917 3733	0.54867 2566371 6814	0.64583 3333333 3334	0.38181 8181818 18183	0.47692 3076923 07695	0.55263 1578947 3685	0.25076 6717697 8311	0.106852 8111326 2772	0.947409 8284554 9784	- 0.11743 3169920 9023
strict _reci st	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_p d1_like	CYT	103	48	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44940 7548612 8113	discove ry_only _platt	0.61135 957066 18963	0.60889982 11091234		anti-PD- 1	0.57159 0909090 9091	0.60862 9009594 5454	0.53465 9090909 0909	0.52427 44737313 1942	0.072520 44737313 997	0.57391 3043478 2609	0.6875	0.38181 8						

endp oint	cohort	analysis_type	model_ name	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
																											83	6689
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44644 8519721 7447	discove ry_only _platt	0.71019 677996 42218	0.70102862 25402504		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.76190 4761904 7619	0.93011 1648388 9595	0.66666 6666666 6666	0.66666 6666666 6666	0.282038 03740888 31	0.75675 6756756 7568	0.66666 6666666 6666	0.66666 6666666 6666	0.875	0.36363 6363636 36365	0.22487 1126025 7452	0.276851 1998932 794	1.294628 8806132 813	1.51925 2683278 961
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	IFNG	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.53434 3452324 6161	discove ry_only _platt	0.66279 069767 44187	0.63886404 29338104		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.86507 9365079 3651	0.96118 2946862 956	0.80952 3809523 8095	0.70370 3703703 7037	0.515078 75363771 27	0.76470 5882352 9411	0.61904 7619047 6191	1.0	1.0	0.42857 1428571 42855	0.20800 9494029 6893	0.295670 0487886 8036	3.373622 1405037 683	2.10815 0442165 6683
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	CXCL 9	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.39407 3786309 8779	discove ry_only _platt	0.66010 733452 59392	0.65138640 4293381		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.85714 2857142 8572	0.95668 1228301 135	0.73809 5238095 2381	0.77777 7777777 7778	0.433554 98476206	0.85	0.80952 3809523 8095	0.66666 6666666 6666	0.89473 6842105 2632	0.5	0.20823 8385286 4658	0.294252 1125301 1496	3.068567 3026942 45	1.97698 5029225 6756
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIG	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42718 0894909 34624	discove ry_only _platt	0.65966 010733 4526	0.62611806 79785331		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.85714 2857142 8572	0.95672 5336853 6468	0.71428 5714285 7142	0.74074 0740740 7407	0.377964 47300922 725	0.82051 2820512 8205	0.76190 4761904 7619	0.66666 6666666 6666	0.88888 8888888 8888	0.44444 4444444 4444	0.21331 6441421 90917	0.274397 0565086 0763	3.012108 0735999 692	1.86307 3706661 0958
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIDE_ dysfun ction	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43978 2201621 358	discove ry_only _platt	0.64937 388193 20214	0.64736135 95706619		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.63492 0634920 6349	0.87385 4268272 3789	0.58333 3333333 3333	0.62962 9629629 6297	0.143486 01079588 788	0.73684 2105263 1579	0.66666 6666666 6666	0.5	0.82352 9411764 7058	0.3	0.23860 8332430 64103	0.273449 5816377 756	1.218142 9537521 1	1.33347 0706591 735
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	APM	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42927 9223503 1947	discove ry_only _platt	0.68515 205724 50804	0.64736135 95706619		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.84126 9841269 8413	0.95132 1508655 701	0.73809 5238095 2381	0.77777 7777777 7778	0.433554 98476206	0.85	0.80952 3809523 8095	0.66666 6666666 6666	0.89473 6842105 2632	0.5	0.20020 2484453 76976	0.293643 9075415 565	2.376603 6727264 988	1.59555 3569626 0372
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	CYT	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44940 7548612 8113	discove ry_only _platt	0.61135 957066 18963	0.60889982 11091234		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.76984 1269841 2699	0.91293 2078536 9477	0.66666 6666666 6666	0.66666 6666666 6666	0.282038 03740888 31	0.75675 6756756 7568	0.66666 6666666 6666	0.66666 6666666 6666	0.875	0.36363 6363636 36365	0.22767 8863674 68015	0.356676 4030949 352	3.072403 3891510 802	1.59481 0091090 1721
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	IPRES	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48070 3042659 90343	discove ry_only _platt	0.55143 112701 25223	0.56238819 32021466		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.39682 5396825 39686	0.75799 9541697 0207	0.45238 0952380 95233	0.51851 8518518 5185	- 0.080582 29640253 803	0.64864 8648648 6487	0.57142 8571428 5714	0.33333 3333333 3333	0.75	0.18181 8181818 18182	0.26248 3052977 4582	0.285810 5798401 122	- 2.025865 7394526 97	1.22662 8984557 653
strict _reci st	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIDE_ exclusi on	27	21	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.45498 2089494 3985	discove ry_only _platt	0.54427 549194 99105	0.57491055 45617174		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.30158 7301587 30157	0.70475 0848257 1364	0.42857 1428571 42855	0.66666 6666666 6666	- 0.188982 23650461 36	0.8	0.85714 2857142 8571	0.0	0.75	0.0	0.26302 7137278 40393	0.280483 3995067 335	- 5.409661 7967888 24	1.36383 0112389 7811
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44644 8519721 7447	discove ry_only _platt	0.71019 677996 42218	0.70102862 25402504	NanoString_n Counter_immu ne_panel		0.58620 6896551 7241	0.60963 8932415 6446	0.52413 7931034 4827	0.51020 4081632 6531	0.047946 35428161 841	0.5	0.6	0.44827 5862068 9655	0.42857 1428571 42855	0.61904 7619047 6191	0.24976 5019248 11123	0.146080 7990553 508	0.434769 8753515 8435	- 0.35731 1914641 5255
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	IFNG	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.53434 3452324 6161	discove ry_only _platt	0.66279 069767 44187	0.63886404 29338104	NanoString_n Counter_immu ne_panel		0.57758 6206896 5517	0.57012 9938368 8246	0.51034 4827586 2069	0.53061 2244897 9592	0.020870 35428161 895	0.41025 6410256 41024	0.4	0.62068 9655172 4138	0.42105 2631578 94735	0.6	0.24849 3448458 4783	0.147279 4300636 7788	0.471599 3411312 0366	- 0.35731 5242108 25544
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	CXCL 9	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.39407 3788309 8779	discove ry_only _platt	0.66010 733452 59392	0.65138640 4293381	NanoString_n Counter_immu ne_panel		0.62413 7931034 4828	0.57364 7061010 7911	0.58965 5172413 7931	0.55102 0408163 2653	0.191220 41415175 975	0.59259 2592592 5926	0.8	0.37931 0344827 5862	0.47058 8235294 11764	0.73333 3333333 3333	0.23591 5342163 7258	0.125634 9957810 336	0.790538 8198070 237	- 0.34479 7381004 74455

endp oint	cohort	analysis_type	model_ name	n_s amp les	n_re spon ders	n_non respon ders	thresh old_source	threshold_trainin g_cohorts	threshold	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIG	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42718 0894909 34624	discove ry_only _platt	0.65966 010733 4526	0.62611806 79785331	NanoString_n Counter_immu ne_panel		0.58275 8620689 6551	0.57034 9075710 6922	0.56465 5172413 7931	0.53061 2244897 9592	0.135528 39816239 234	0.56603 7735849 0566	0.75	0.37931 0344827 5862	0.45454 5454545 45453	0.6875	0.24547 2111051 26826	0.139885 4585227 054	0.548192 2414145 458	- 0.34461 3389557 72286
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIDE_ dysfun ction	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43978 2201621 358	discove ry_only _platt	0.64937 388193 20214	0.64736135 95706619	NanoString_n Counter_immu ne_panel		0.58103 4482758 6207	0.60749 1679646 573	0.52241 3723103 4483	0.48979 5918367 3469	0.046983 17802962 934	0.52830 1886792 4528	0.7	0.34482 7586206 8966	0.42424 2424242 42425	0.625	0.24717 3189574 18294	0.132625 3985262 7014	0.530987 3683674 816	- 0.35999 4249939 8961
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	APM	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42927 9223503 1947	discove ry_only _platt	0.68515 205726 50804	0.64736135 95706619	NanoString_n Counter_immu ne_panel		0.51034 4827586 2069	0.44800 1149760 67873	0.48793 1034482 7586	0.44897 9591836 7347	- 0.026261 0909090 909	0.50909 2068965 51724	0.7	0.27586 2068965 51724	0.4	0.57142 8571428 5714	0.571428 9638579 23004	0.219638 6638579 9577	- 0.086224 8879406 6287	- 0.37122 5015043 1741
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	CYT	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44940 7548612 8113	discove ry_only _platt	0.61135 957066 18963	0.60889982 11091234	NanoString_n Counter_immu ne_panel		0.55172 4137931 0345	0.55740 2233510 0072	0.53965 5172413 793	0.51020 4081632 6531	0.081892 54821003 5487	0.53846 1538461 5384	0.7	0.37931 0344827 5862	0.4375	0.64705 8823529 4118	0.24592 3124880 38745	0.197327 8805294 4708	0.610720 647595 023	- 0.34214 8701121 88117
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	IPRES	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48070 3042659 90343	discove ry_only _platt	0.55143 112701 25223	0.56238819 32021466	NanoString_n Counter_immu ne_panel		0.48103 4482758 62065	0.41345 2634919 6965	0.50689 6551724 138	0.48979 5918367 3469	0.013793 10344827 5862	0.48979 5918367 3469	0.6	0.41379 3103448 27586	0.41379 3103448 27586	0.6	0.25260 2176207 325	0.086035 1911351 5319	- 0.403938 8640771 4485	- 0.38149 6593931 4122
strict _reci st	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIDE_ exclusi on	49	20	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.45498 2089494 3985	discove ry_only _platt	0.54427 549194 99105	0.57491055 45617174	NanoString_n Counter_immu ne_panel		0.43275 8620689 6552	0.38586 7783461 25555	0.51896 5517241 3793	0.44897 9591836 7347	0.056872 04319328 977	0.57142 8571428 5714	0.9	0.13793 1034482 75862	0.451162 7907	0.66666 6666666 6666	0.25270 3077036 1867	0.119187 0617799 3019	- 0.764098 5503385 872	- 0.38231 1281111 1344
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44644 8519721 7447	discove ry_only _platt	0.71019 677996 42218	0.70102862 25402504	NanoString_n Counter		0.83333 3333333 3334	0.79960 3174603 1745	0.75	0.71428 5714285 7143	0.547722 55750516 61	0.75	1.0	0.5	0.6	1.0	0.19128 6328166 84904	0.165574 3377966 0394	2.846788 8213772 56	- 0.16785 8014039 0072
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	IFNG	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.53434 3453234 6161	discove ry_only _platt	0.66279 069767 44187	0.63886404 29338104	NanoString_n Counter		0.91666 6666666 6666	0.87341 2698412 6983	0.77083 3333333 3333	0.78571 4285714 2857	0.559430 92778551 58	0.72727 2727272 7273	0.66666 6666666 6666	0.875	0.8	0.77777 7777777 7778	0.17338 8932075 2688	0.284037 1289354 138	5.514073 0888134 168	- 0.70168 7687809 1448
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	CXCL 9	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.39407 2883309 8779	discove ry_only _platt	0.66010 733452 59392	0.65138640 4293381	NanoString_n Counter		0.91666 6666666 6667	0.87341 2698412 6983	0.8125	0.78571 4285714 2857	0.645497 22436790 22436790	0.8	1.0	0.625	0.66666 6666666 6666	1.0	0.17541 4934745 372	0.287324 0909237 236	3.908450 8926993 236	0.05751 5991909 43964
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIG	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42718 0894909 34624	discove ry_only _platt	0.65966 010733 4526	0.62611806 79785331	NanoString_n Counter		0.91666 6666666 6666	0.91071 4285714 2856	0.8125	0.78571 4285714 2857	0.645497 22436790 28	0.8	1.0	0.625	0.66666 6666666 6666	1.0	0.18463 1926272 79845	0.177009 1119420 9945	5.874273 7284133 31527	0.18868 7769925 31527
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIDE_ dysfun ction	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.43978 2201621 358	discove ry_only _platt	0.64937 388193 20214	0.64736135 95706619	NanoString_n Counter		0.85416 3333333 6667	0.87083 3333333 6667	0.66666 18632954 6667	0.64285 7142857 1429	0.344265 3333333 817	0.66666 6666666 6666	0.83333 3333333 3334	0.5	0.55555 5555555 5556	0.8	0.18253 5539521 81356	0.253179 1389556 2444	3.633578 7150147 305	- 0.31585 6931765 6562
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	APM	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.42927 9223503 1947	discove ry_only _platt	0.68515 205724 50804	0.64736135 95706619	NanoString_n Counter		0.91666 6666666 6666	0.87341 2698412 6983	0.75	0.71428 5714285 7143	0.547722 55750516 61	0.75	1.0	0.5	0.6	1.0	0.18849 3886549 63686	0.267542 9408938 479	7.793804 3312277 42	- 1.63029 1810952 0157
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	CYT	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44940 7548612 8113	discove ry_only _platt	0.61135 957066 18963	0.60889982 11091234	NanoString_n Counter		0.60416 6666666 6666	0.54276 0942760 9428	0.52083 3333333 3333	0.5	0.043033 14829119 352	0.53333 3333333 3333	0.66666 6666666 6666	0.375	0.44444 4444444 4444	0.6	0.23607 6552470 61762	0.102521 3091401 2306	1.403810 0094219 434	- 0.19460 3880663 3053
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	IPRES	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48070 3042659 90343	discove ry_only _platt	0.55143 112701 25223	0.56238819 32021466	NanoString_n Counter		0.29166 6666666 66663	0.38958 9577089 5771	0.27083 3333333 3333	0.28571 4285714 2857	- 0.458333 33333333 33	0.16666 6666666 66666	0.16666 6666666 66666	0.375	0.16666 6666666 66666	0.375	0.26253 9858311 47375	0.063626 8220109 3461	2.386342 3290619 616	- 0.38706 9897624 94776
strict _reci st	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIDE_ exclusi on	14	6	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.45498 2089494 3985	discove ry_only _platt	0.54427 549194 99105	0.57491055 45617174	NanoString_n Counter		0.27083 3333333 3333	0.35075 7575757 57577	0.54166 6666666 6667	0.5	0.105053 78152592 121	0.58823 5294117 6471	0.83333 3333333 3334	0.25	0.45454 5454545 45453	0.66666 6666666 6666	0.26395 8858679 4911	0.174888 5131410 8733	- 7.211223 8912277 65	- 0.77758 3144752 7933
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48663 8814790 76476	discove ry_only _platt	0.64289 940828 40236	0.64230769 23076922		Nivo;PE M;Pem	0.62222 2222222 22222	0.81797 7392977 393	0.52222 2222222 2221	0.5	0.043033 14829119 352	0.53333 3333333 3333	0.44444 4444444 4444	0.6	0.66666 6666666 6666	0.375	0.23253 4775280 29615	0.168558 8670170 8286	1.575679 0046632 296	0.68311 2765326 3712
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	IFNG	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54357 8790905 6865	discove ry_only _platt	0.60384 615384 615384	0.59038461 53846154		Nivo;PE M;Pem	0.48888 8888888 8889	0.73014 8555148 5551	0.56666 6666666 6667	0.5	0.141421 35623730 95	0.46153 8461538 46156	0.33333 3333333 3333	0.8	0.75	0.4	0.25134 2372466 2435	0.142303 7267097 96073	0.476826 6640130 5671	0.58886 6480055 5671
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	CXCL 9	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44615 4129187 2711	discove ry_only _platt	0.60917 159763 31361	0.60000000 00000001		Nivo;PE M;Pem	0.51111 1111111 1111	0.72717 8377178 3773	0.48888 8888888 88893	0.57142 8571428 5714	- 0.025949 96480538 4102	0.7	0.77777 7777777 7778	0.2	0.63636 3636363 6364	0.33333 3333333 3333	0.25227 1226961 42323	0.197593 7018882 8396	0.399587 1533490 3633	0.58785 9309962 6814

endp oint	cohort	analysis_type	model_ nam e	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	TIG	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.58064 4847552 3057	discove ry_onl _platt	0.59733 727810 65088	0.58076923 07692307		Nivo;PE M;Pem	0.55555 5555555 5556	0.78332 3158323 1584	0.61111 111111 1112	0.5	0.304290 30972509 23	0.36363 6363636 2222	0.22222 2222222 2222	1.0	1.0	0.41666 6666666 6667	0.24228 2343036 67198	0.162870 6421265 494	1.327645 6156401 875	0.61088 9612366 3559
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	TIDE_ dysfun ction	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46881 1248482 587	discove ry_onl _platt	0.59852 071005 91717	0.58846153 84615385		Nivo;PE M;Pem	0.66666 6666666 6666	0.83007 0546737 2136	0.57777 7777777 7777	0.57142 8571428 5714	0.149071 19849998 599	0.625	0.55555 5555555 5556	0.6	0.71428 5714285 7143	0.42857 1428571 42855	0.22910 1705995 35568	0.142091 4649878 4486	3.601663 3249862 915	0.81805 9477097 3695
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	APM	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47387 0240819 63003	discove ry_onl _platt	0.65295 857988 16568	0.64038461 53846153		Nivo;PE M;Pem	0.51111 1111111 1111	0.74847 4673474 6734	0.47777 7777777 7778	0.5	- 0.043033 14829119 352	0.58823 5294117 6471	0.55555 5555555 5556	0.4	0.625	0.33333 3333333 3333	0.27204 1162641 9588	0.146701 0258631 7524	- 0.267659 9589077 1021	0.59364 9589077 1021
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	CYT	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46726 2058219 7981	discove ry_onl _platt	0.56035 502958 57989	0.57692307 6923077		Nivo;PE M;Pem	0.71111 1111111 1111	0.66890 3318903 3191	0.53333 3333333 3333	0.57142 8571428 5714	0.66666 6666666 6666	0.66666 6666666 6666	0.66666 6666666 6666	0.4	0.66666 6666666 6666	0.4	0.23300 9753352 76278	0.172630 0374341 8415	6.113170 2029284 67	0.84340 1969448 9955
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	IPRES	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47891 0886258 67356	discove ry_onl _platt	0.52426 503502 95859	0.55769230 76923077		Nivo;PE M;Pem	0.55555 5555555 5555	0.74384 5043845 0439	0.38888 8888888 8889	0.5	- 0.304290 30972509 23	0.66666 6666666 6666	0.77777 7777777 7778	0.0	0.58333 3333333 3334	0.0	0.24691 3012709 62482	0.142775 8430164 9696	4.991144 1754003 383981	0.61075 383981 2005
clinic al_be nefit	GSE14599 6	locked_external_m elanoma_pd1_recis t	TIDE_ exclusi on	14	9	5	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48097 5283094 2262	discove ry_onl _platt	0.53254 437869 82249	0.55384615 38461539		Nivo;PE M;Pem	0.57777 7777777 7777	0.70589 2255892 256	0.48888 8888888 88893	0.57142 8571428 5714	- 0.025949 96480538 4102	0.7	0.77777 7777777 7778	0.2	0.63636 3636363 6364	0.33333 3333333 3333	0.24932 3056343 76672	0.195230 3972199 889	1.183547 5023096 981	0.58891 4395246 2643
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48663 8814790 76476	discove ry_onl _platt	0.64289 940828 40236	0.64230769 23076922		anti-PD- 1	0.58292 0110192 8375	0.64950 0679444 5877	0.57272 7272727 2728	0.57024 7933884 2975	0.144976 07525854 994	0.58064 5161290 3226	0.54545 4545454 5454	0.6	0.62068 9655172 4138	0.52380 9523809 5238	0.24464 0134864 20055	0.074863 6104438 3906	0.715316 1228029 349	0.17742 2291360 86476
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	IFNG	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54357 8790905 6865	discove ry_onl _platt	0.60384 615384 61539	0.59038461 53846154		anti-PD- 1	0.59311 2947658 4023	0.67224 9820001 0829	0.57575 7575757 5758	0.56198 3471074 3802	0.157625 99747031 963	0.51376 1467889 9083	0.42424 2424242 42425	0.72727 2727272 7273	0.65116 2790697 6745	0.51282 0512820 5128	0.24226 6145270 85763	0.045675 2421561 052	0.996531 4022854 45743	0.18120 4772700 45743
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	CXCL 9	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44615 4129187 2711	discove ry_onl _platt	0.60917 159763 13161	0.60000000 00000001		anti-PD- 1	0.61349 8622589 5317	0.69936 1608271 6241	0.52878 7878787 8788	0.54545 4545454 5454	0.061767 79941180 28	0.63087 2483221 4765	0.71212 1212121 2122	0.34545 4545454 54546	0.56626 5060240 9639	0.5	0.24013 3064245 40498	0.074237 5395774 7974	1.072460 2010441 229	0.18080 7105830 06185
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	TIG	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.58064 4847552 3057	discove ry_onl _platt	0.59733 727810 65088	0.58076923 07692307		anti-PD- 1	0.58953 1680440 7713	0.66955 8487464 1041	0.56818 1818181 8181	0.53719 0082644 6281	0.182800 04333033 72	0.34883 7209302 3256	0.22727 2727272 72727	0.90909 0909090 9091	0.75	0.49504 9504950 49505	0.24295 1632464 2034	0.060586 9001921 1212	1.074631 1604078 90473	0.18224 8319289 15708
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	TIDE_ dysfun ction	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46881 1248482 587	discove ry_onl _platt	0.59852 071005 91717	0.58846153 84615385		anti-PD- 1	0.58264 4628099 1736	0.67304 7902380 3051	0.55757 5757575 7576	0.56198 3471074 3802	0.115342 32143838 787	0.60150 3759398 4962	0.60606 0606060 6061	0.50909 0909090 909	0.59701 4925373 1343	0.51851 8518518 5185	0.24159 8896989 6826	0.047975 9256855 0218	1.208173 8796316 945	0.18653 7059538 14678
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	APM	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47387 0240819 63003	discove ry_onl _platt	0.65295 857988 16568	0.64038461 53846153		anti-PD- 1	0.57382 9201101 9285	0.63203 7600500 0303	0.53030 3030303 0303	0.53719 0082644 6281	0.061113 23313521 409	0.58823 5294117 6471	0.60606 0606060 6061	0.45454 5454545 45453	0.57142 8571428 5714	0.49019 6078431 93253	0.24543 4932423 07007	0.096966 1654926 8473	0.17949 6672517 81562	
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	CYT	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46726 2058219 7981	discove ry_onl _platt	0.56035 502958 57989	0.57692307 6923077		anti-PD- 1	0.58925 6198347 1074	0.67263 1437396 336	0.57727 2727272 7273	0.59504 1322314 0496	0.168325 08230603 463	0.67549 6688741 7219	0.77727 7272727 72727	0.38181 1818181 18183	0.6	0.58333 3333333 3334	0.24356 5247211 71163	0.044952 0248065 3769	0.18628 4788585 5059479	0.18628 5059479 2365
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	IPRES	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47891 0886258 67356	discove ry_onl _platt	0.52426 035502 95859	0.55769230 76923077		anti-PD- 1	0.43443 5261707 98896	0.51396 7987835 9975	0.51969 6969696 9697	0.55371 9008264 4629	0.059522 91104443 1914	0.68604 6511627 907	0.89393 9393939 9399	0.14545 4545454 54545	0.55680 3773584 9056	0.53333 3333333 3333	0.25183 4244290 2237	0.093521 8839356 3640224	- 1.809453 6402224 428	0.18378 6250159 61675
clinic al_be nefit	PHS00045 2_LIU_LIK E_PRE	locked_external_m elanoma_pd1_like	TIDE_ exclusi on	121	66	55	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48097 5283094 2262	discove ry_onl _platt	0.53254 437869 82249	0.55384615 38461539		anti-PD- 1	0.42066 1157024 7934	0.50322 8668629 5545	0.49090 9090909 09096	0.52066 1157024 7934	- 0.023904 57218668 7872	0.65060 2409638 5542	0.81818 1818181 8182	0.16363 6363636 36364	0.54	0.42857 1428571 42855	0.25327 9268908 31404	0.097170 8086754 6354	- 2.587862 8083633 9677	0.18613 5480363 9677
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48663 8814790 76476	discove ry_onl _platt	0.64289 940828 40236	0.64230769 23076922		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.72435 8974358 9743	0.93068 7538250 8506	0.62179 4871794 8718	0.59375	0.190524 66523879 25	0.69767 4418604 6512	0.57692 3076923 0769	0.66666 6666666 6666	0.88235 2941176 4706	0.26666 6666666 66666	0.23357 6579146 84565	0.308476 6371105 0777	1.685596 4032171 37	1.65724 6062255 2487
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	IFNG	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54357 8790905 6865	discove ry_onl _platt	0.60384 615384 61539	0.59038461 53846154		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.82692 3076923 0769	0.95589 4773750 9909	0.71153 8461538 4616	0.53125	0.347676 74768255 77	0.59459 4594594 5946	0.42307 6923076 9231	1.0	1.0	0.28571 4285714 2857	0.22754 6222382 76973	0.310997 0263084 3096	4.966293 3077498 24	2.25429 0973383 7098
clinic al_be nefit																												

endp oint	cohort	analysis_type	model_ nam e	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AUROC	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIG	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.58064 4847552 3057	discove ry_only _platt	0.59733 727810 65088	0.58076923 07692307		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.82051 2820512 8205	0.95210 9042675 0181	0.61538 4615384 6154	0.375	0.230769 23076923 078	0.375	0.23076 9230769 23078	1.0	1.0	0.23076 9230769 23078	0.23159 1735518 44944	0.360558 1663760 734	4.269461 6876544 77	1.97201 6943593 6532
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIDE_ dysfun ction	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46881 1248482 587	discove ry_only _platt	0.59852 071005 91717	0.58846153 84615385		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.58974 3589743 5896	0.87669 8708348 5616	0.53846 1538461 5384	0.5625	0.060522 75326688 0246	0.68181 8181818 1818	0.57692 3076923 0769	0.5	0.83333 3333333 3334	0.21428 5714285 71427	0.24532 1079502 69244	0.310860 4304599 988	1.329635 7377829 127	1.51774 1769219 5332
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	APM	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47387 0240819 63003	discove ry_only _platt	0.65295 857988 16568	0.64038461 53846153		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.782051 1282051 282	0.94226 8313600 63	0.67948 7179487 1795	0.6875	0.289414 38070815 306	0.78260 8695652 174	0.69230 7692307 6923	0.66666 6666666 6666	0.9	0.33333 3333333 3333	0.22585 3691022 91153	0.342952 5076625 186	2.174893 5326454 57	1.79958 1289145 9633
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	CYT	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46726 2058219 7981	discove ry_only _platt	0.56035 502958 57989	0.57692307 6923077		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.73717 9487179 4872	0.91389 0550632 1165	0.59615 3846153 8461	0.65625	0.158034 88531025 35	0.76595 7446808 5106	0.69230 7692307 6923	0.5	0.85714 2857142 8571	0.27272 7272727 2727	0.23979 9253291 48954	0.311947 7689397 872	4.999977 9814688 07	1.75645 4629621 3903
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	IPRES	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47891 0886258 67356	discove ry_only _platt	0.52426 035502 95859	0.55769230 76923077		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.43589 7435897 4359	0.81983 1735867 994	0.42307 6923076 9231	0.6875	- 0.181568 25980064 073	0.81481 4814814 8148	0.84615 3846153 8461	0.0	0.78571 4285714 2857	0.0	0.25149 6201666 42417	0.312412 9742644 9186	- 2.349464 2430243 32	1.47810 7980636 3682
clinic al_be nefit	PRJEB237 09_COMB O_PRE	locked_external_m elanoma_combinati on_transfer	TIDE_ exclusi on	32	26	6	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48097 5283094 2262	discove ry_only _platt	0.53254 437869 82249	0.55384615 38461539		Ipilimum ab + Nivolum ab;Ipilim umab + Pembrol izumab	0.32051 2820512 8205	0.74922 0763474 1694	0.44230 7692307 6923	0.71875	- 0.154507 86078738 143	0.83636 3636363 6363	0.88461 5384615 3846	0.0	0.79310 3448275 8621	0.0	0.25527 5706091 0293	0.312374 9742644 947	- 8.147266 6338221 58	1.62883 0941814 953
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	EcoNi che- Opt- Heuris ticEcol ogy- Locke dPanel	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48663 8814790 76476	discove ry_only _platt	0.64289 940828 40236	0.64230769 23076922	NanoString_n Counter_immu ne_panel		0.57662 8352490 4214	0.68051 8424702 9877	0.52969 3486590 0383	0.53846 1538461 5384	0.059905 64654909 1425	0.59459 4594594 5946	0.61111 1111111 1112	0.44827 5862068 9655	0.57894 7368421 0527	0.48148 1481481 48145	0.24681 5519080 2023	0.062290 4731243 93626	0.617725 6549491 684	0.21013 0939498 18873
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	IFNG	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54357 8790905 6865	discove ry_only _platt	0.60384 615384 61539	0.59038461 53846154	NanoString_n Counter_immu ne_panel		0.55651 3409961 6859	0.64245 6755560 8633	0.55986 5900383 5384	0.53846 1538461 5384	0.128955 09595628 601	0.46428 5714285 7143	0.36111 1111111 1111	0.75862 0689655 1724	0.65	0.48888 8888888 8889	0.24778 5778033 72456	0.092302 4861949 431	0.627291 2718965 765	0.21431 6992115 64198
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	CXCL 9	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44615 4129187 2711	discove ry_only _platt	0.60917 159763 31361	0.60000000 00000001	NanoString_n Counter_immu ne_panel		0.59386 9731800 7663	0.67490 9892637 952	0.55076 6283524 9042	0.56923 0769230 7692	0.107924 39687475 158	0.65	0.72222 2222222 2222	0.37931 0344827 5862	0.59090 9090909 0909	0.52380 9523809 5238	0.24097 5302439 8109	0.051926 9880691 4997	1.025090 2914285 438	0.21585 3474647 37192
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIG	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.58064 4847552 3057	discove ry_only _platt	0.59733 727810 65088	0.58076923 07692307	NanoString_n Counter_immu ne_panel		0.57854 4061302 682	0.66547 6321303 5716	0.59051 7241379 3103	0.55384 6153846 5384	0.240003 91846330 954	0.38297 8723404 2553	0.25	0.93103 4482758 6207	0.81818 1818181 8182	0.5	0.24449 9481841 08006	0.102756 4527227 3175	0.924097 0324427 137	0.21505 5503874 77462
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIDE_ dysfun ction	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46881 1248482 587	discove ry_only _platt	0.59852 071005 91717	0.58846153 84615385	NanoString_n Counter_immu ne_panel		0.56034 4827586 207	0.67601 3169978 2478	0.51963 6015325 6705	0.53846 1538461 5384	0.041744 34218740 3916	0.625	0.69444 4444444 4444	0.34482 7586206 8966	0.56818 1818181 8182	0.47619 0476190 47616	0.247891 5760271 36736	0.104804 7977588 1638	0.623311 0495112 397	0.21351 3766227 067
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	APM	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47387 0240819 63003	discove ry_only _platt	0.65295 857988 16568	0.64038461 53846153	NanoString_n Counter_immu ne_panel		0.46455 9386973 18006	0.53456 0440160 7837	0.48132 1839080 4598	0.49230 7692307 69234	- 0.037904 90217894 517	0.56	0.58333 3333333 3334	0.37931 0344827 5862	0.53846 1538461 5384	0.42307 6923076 9231	0.27152 2713242 6181	0.155808 - 0.273857 5984424 2297	- 0.273857 5984424 2297	0.22067 6297553 3344
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	CYT	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46726 2058219 7981	discove ry_only _platt	0.56035 502958 57989	0.57692307 6923077	NanoString_n Counter_immu ne_panel		0.58141 7624521 0728	0.68036 9734065 6227	0.54406 1302681 9924	0.56923 0769230 7692	0.099676 3543603 256	0.66666 6666666 6666	0.77777 7777777 7778	0.31034 4827586 2069	0.58333 3333333 3334	0.52941 1764705 8824	0.24471 2300286 3921	0.064782 6489060 3921	1.523375 1659060 5252	0.21632 5880894 5252
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	IPRES	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47891 0886258 67356	discove ry_only _platt	0.52426 035502 95859	0.55769230 76923077	NanoString_n Counter_immu ne_panel		0.49233 7164750 9579	0.57649 8432029 4171	0.52729 8850574 7126	0.56923 0769230 7692	0.087551 09648855 28	0.70212 7659574 4681	0.91666 6666666 6666	0.13793 1034482 3793	0.56896 8571428 5714	0.24981 1385974 09938	0.053770 4456017 2782	0.747162 2483531 667	0.21628 2483531 6654	
clinic al_be nefit	GSE93157	nanosttring_melano ma_clinical_assay_ transfer	TIDE_ exclusi on	65	36	29	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48097 5283094 2262	discove ry_only _platt	0.53254 437869 82249	0.55384615 38461539	NanoString_n Counter_immu ne_panel		0.48659 0038314 17624	0.56152 8552111 5289	0.53400 3631417 6245	0.56923 0769230 7692	0.090160 20217404 751	0.68888 8888888 8889	0.86111 1111111 1112	0.20689 6551724 13793	0.57407 4074074 0741	0.54545 4545454 5454	0.25015 7528616 80794	0.070770 9052457 5098	0.361101 5584526 547	0.21639 0819252 04323
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	EcoNi che- Opt- Heuris ticEcol	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48663 8814790 76476	discove ry_only _platt	0.64289 940828 40236	0.64230769 23076922	NanoString_n Counter		0.71093 75	0.85090 1253566 6948	0.59375	0.625	0.182574 18583505 539	0.70967 7419354 8387	0.6875	0.73333 3333333 3333	0.44444 4444444 4444	0.22594 5328616 80794	0.198895 7248417 1936	2.081287 3732094 985	0.76998 9732920 5445	

endp oint	cohort	analysis_type	model _nam e	n_s amp les	n_re spon ders	n_non respon ders	thresh old_so urce	threshold_trainin g_cohorts	threshol d	calibrat ion	training_AURO C	training_b alanced_a ccuracy	platform	treatme nt	AUROC	AUPRC	balance d_accu racy	accurac y	MCC	F1	sensitivi ty	specifici ty	PPV	NPV	Brier	ECE	calibrati on_slop e	calibrati on_inter cept
			ogy- Locke dPanel																									
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	IFNG	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.54357 8790905 6865	discove ry_only _platt	0.60384 615384 61539	0.59038461 53846154	NanoString_n Counter		0.76562 5	0.87771 5403693 2019	0.6875	0.625	0.365148 37167011 077	0.64	0.5	0.875	0.88888 8888888 8888	0.46666 6666666 6667	0.22216 2409169 77067	0.169913 8666435 6167	2.810127 2734791 03	0.83199 8956678 4911
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	CXCL 9	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.44615 4129187 2711	discove ry_only _platt	0.60917 159763 31361	0.60000000 00000001	NanoString_n Counter		0.77343 75	0.88560 8055139 3052	0.6875	0.70833 3333333 3334	0.365148 37167011 077	0.77419 3548387 0968	0.75	0.625	0.8	0.55555 5555555 5556	0.22046 0663633 7988	0.178041 9238468 9366	3.274822 7391879 99	0.94197 5770954 2966
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIG	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.58064 4847552 3057	discove ry_only _platt	0.59733 727810 65088	0.58076923 07692307	NanoString_n Counter		0.75781 25	0.89078 9925534 9914	0.625	0.5	0.316227 76601683 794	0.4	0.25	1.0	1.0	0.4	0.22559 5991673 74841	0.165350 9664149 5975	3.256086 3033246 794	0.84790 3008987 1182
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIDE_ dysfun ction	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46881 1248482 587	discove ry_only _platt	0.59852 071005 91717	0.58846153 84615385	NanoString_n Counter		0.71093 75	0.85203 5362513 3037	0.5625	0.58333 3333333 3334	0.119522 86093343 936	0.66666 6666666 6666	0.625	0.5	0.71428 5714285 7143	0.4	0.22740 3662282 8666	0.195417 4831462 3558	3.420384 7182623 917	0.89316 6054056 3507
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	APM	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47387 0240819 63003	discove ry_only _platt	0.65295 857988 16568	0.64038461 53846153	NanoString_n Counter		0.71875	0.85895 7086622 5278	0.59375	0.625	0.182574 18583505 539	0.70967 7419354 8387	0.6875	0.5	0.73333 3333333 3333	0.44444 4444444 4444	0.23841 9175507 48581	0.174947 7937713 042	0.693368 8043158 0606	
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	CYT	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.46726 2058219 7981	discove ry_only _platt	0.56035 502958 57989	0.57692307 6923077	NanoString_n Counter		0.60937 5	0.76892 0955044 6973	0.59375	0.66666 6666666 6666	0.204124 14523193 15	0.76470 5882352 9411	0.8125	0.375	0.72222 2222222 2222	0.5	0.24213 1330152 47798	0.166159 0958950 9997	2.449168 5172530 087	0.73131 6238671 9445
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	IPRES	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.47891 0886258 67356	discove ry_only _platt	0.52426 035502 95859	0.55769230 76923077	NanoString_n Counter		0.4375	0.67159 5919931 5178	0.46875	0.625	- 0.147441 95615489 714	0.76923 0769230 7693	0.9375	0.0	0.65217 3913043 4783	0.0	0.25165 9680682 3547	0.166594 7046964 9265	- 1.747001 1675119 819	0.69660 3019607 413
clinic al_be nefit	GSE14090 1	nanosttring_hcc_clin ical_assay_transfer	TIDE_ exclusi on	24	16	8	discov ery_onl y	GSE91061,GSE78 220,PRJEB23709_ PD1_PRE	0.48097 5283094 2262	discove ry_only _platt	0.53254 437869 82249	0.55384615 38461539	NanoString_n Counter		0.49218 75	0.66793 8870662 3614	0.46875	0.54166 6666666 6666	- 0.064820 37235521 644	0.66666 6666666 6666	0.6875	0.25	0.64705 8823529 4118	0.28571 4285714 2857	0.25063 2421182 54164	0.166549 4902862 7003	- 0.134621 1333474 6706	0.69341 6268770 5477

Supplementary Table 16 | NanoString panel transfer.

Reports n_module_genes, n_available_genes, coverage_fraction, and available genes for each module in NanoString panel-transfer cohorts. This table supports Figure 5d.

Full tabulation (36 rows x 8 columns).

cohort	analysis_type	module	n_module_genes	n_available_genes	coverage_fraction	available_genes	is_nanostring_panel_transfer
GSE145996	locked_external_melanoma_pd1_recist	ifn_t_cell_inflamed	11.0	11.0	1.0	IFNG,CXCL9,CXCL10,CXCL11,STAT1,IDO1,GBP1,CXCR3,CCL5,CD274, PDCD1LG2	False
GSE145996	locked_external_melanoma_pd1_recist	cytotoxic_cd8	8.0	8.0	1.0	CD8A,CD8B,GZMA,GZMB,GZMH,PRF1,NKG7,GNLY	False
GSE145996	locked_external_melanoma_pd1_recist	exhaustion_checkpoint	7.0	7.0	1.0	PDCD1,CTLA4,LAG3,HAVCR2,TIGIT,TOX,CXCL13	False
GSE145996	locked_external_melanoma_pd1_recist	antigen_presentation	10.0	6.0	0.6	HLA-C,HLA-DRA,HLA-DRB1,B2M,TAP2,PSMB9	False
GSE145996	locked_external_melanoma_pd1_recist	myeloid_suppression	9.0	8.0	0.8888888888888888	CD163,MRC1,CSF1R,ITGAM,S100A8,S100A9,IL10,TGFB1	False
GSE145996	locked_external_melanoma_pd1_recist	stromal_exclusion	9.0	7.0	0.7777777777777778	COL1A1,COL1A2,FN1,VIM,TGFBI,POSTN,LOXL2	False
GSE145996	locked_external_melanoma_pd1_recist	trm_tis	9.0	9.0	1.0	ITGAE,CD69,CXCR6,CXCL13,ZNF683,MS4A1,CD79A,BANK1,LTB	False
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	ifn_t_cell_inflamed	11.0	11.0	1.0	IFNG,CXCL9,CXCL10,CXCL11,STAT1,IDO1,GBP1,CXCR3,CCL5,CD274, PDCD1LG2	False
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	cytotoxic_cd8	8.0	8.0	1.0	CD8A,CD8B,GZMA,GZMB,GZMH,PRF1,NKG7,GNLY	False
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	exhaustion_checkpoint	7.0	7.0	1.0	PDCD1,CTLA4,LAG3,HAVCR2,TIGIT,TOX,CXCL13	False
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	antigen_presentation	10.0	10.0	1.0	HLA-A,HLA-B,HLA-C,HLA-DRA,HLA-DRB1,B2M,TAP1,TAP2,PSMB8,PSMB9	False
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	myeloid_suppression	9.0	9.0	1.0	CD68,CD163,MRC1,CSF1R,ITGAM,S100A8,S100A9,IL10,TGFB1	False
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	stromal_exclusion	9.0	9.0	1.0	COL1A1,COL1A2,COL3A1,FN1,ACTA2,VIM,TGFBI,POSTN,LOXL2	False

cohort	analysis_type	module	n_module_genes	n_available_genes	coverage_fraction	available_genes	is_nanostring_panel_transfer
PHS000452_LIU_LIKE_PRE	locked_external_melanoma_pd1_like	trm_tls	9.0	9.0	1.0	ITGAE,CD69,CXCR6,CXCL13,ZNF683,MS4A1,CD79A,BANK1,LTB	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	ifn_t_cell_inflamed	11.0	11.0	1.0	IFNG,CXCL9,CXCL10,CXCL11,STAT1,IDO1,GBP1,CXCR3,CCL5,CD274,PDCD1LG2	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	cytotoxic_cd8	8.0	8.0	1.0	CD8A,CD8B,GZMA,GZMB,GZMH,PRF1,NKG7,GNLY	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	exhaustion_checkpoint	7.0	7.0	1.0	PDCD1,CTLA4,LAG3,HAVCR2,TIGIT,TOX,CXCL13	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	antigen_presentation	10.0	10.0	1.0	HLA-A,HLA-B,HLA-C,HLA-DRA,HLA-DRB1,B2M,TAP1,TAP2,PSMB8,PSMB9	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	myeloid_suppression	9.0	9.0	1.0	CD68,CD163,MRC1,CSF1R,ITGAM,S100A8,S100A9,IL10,TGFB1	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	stromal_exclusion	9.0	9.0	1.0	COL1A1,COL1A2,COL3A1,FN1,ACTA2,VIM,TGFB1,POSTN,LOXL2	False
PRJEB23709_COMBO_PRE	locked_external_melanoma_combination_transfer	trm_tls	9.0	9.0	1.0	ITGAE,CD69,CXCR6,CXCL13,ZNF683,MS4A1,CD79A,BANK1,LTB	False
GSE93157	nanostring_melanoma_clinical_assay_transfer	ifn_t_cell_inflamed	11.0	10.0	0.90909090909091	IFNG,CXCL9,CXCL10,CXCL11,STAT1,IDO1,CXCR3,CCL5,CD274,PDCD1LG2	True
GSE93157	nanostring_melanoma_clinical_assay_transfer	cytotoxic_cd8	8.0	7.0	0.875	CD8A,CD8B,GZMA,GZMB,GZMH,PRF1,GNLY	True
GSE93157	nanostring_melanoma_clinical_assay_transfer	exhaustion_checkpoint	7.0	6.0	0.8571428571428571	PDCD1,CTLA4,LAG3,HAVCR2,TIGIT,CXCL13	True
GSE93157	nanostring_melanoma_clinical_assay_transfer	antigen_presentation	10.0	8.0	0.8	HLA-A,HLA-B,HLA-C,HLA-DRA,TAP1,TAP2,PSMB8,PSMB9	True
GSE93157	nanostring_melanoma_clinical_assay_transfer	myeloid_suppression	9.0	8.0	0.8888888888888888	CD68,CD163,MRC1,CSF1R,ITGAM,S100A8,IL10,TGFB1	True
GSE93157	nanostring_melanoma_clinical_assay_transfer	stromal_exclusion	9.0	2.0	0.2222222222222222	COL3A1,FN1	True
GSE93157	nanostring_melanoma_clinical_assay_transfer	trm_tls	9.0	6.0	0.6666666666666666	ITGAE,CXCR6,CXCL13,MS4A1,CD79A,LTB	True
GSE140901	nanostring_hcc_clinical_assay_transfer	ifn_t_cell_inflamed	11.0	10.0	0.90909090909091	IFNG,CXCL9,CXCL10,CXCL11,STAT1,IDO1,CXCR3,CCL5,CD274,PDCD1LG2	True
GSE140901	nanostring_hcc_clinical_assay_transfer	cytotoxic_cd8	8.0	7.0	0.875	CD8A,CD8B,GZMA,GZMB,GZMH,PRF1,GNLY	True
GSE140901	nanostring_hcc_clinical_assay_transfer	exhaustion_checkpoint	7.0	6.0	0.8571428571428571	PDCD1,CTLA4,LAG3,HAVCR2,TIGIT,CXCL13	True
GSE140901	nanostring_hcc_clinical_assay_transfer	antigen_presentation	10.0	8.0	0.8	HLA-A,HLA-B,HLA-C,HLA-DRA,TAP1,TAP2,PSMB8,PSMB9	True
GSE140901	nanostring_hcc_clinical_assay_transfer	myeloid_suppression	9.0	8.0	0.8888888888888888	CD68,CD163,MRC1,CSF1R,ITGAM,S100A8,IL10,TGFB1	True
GSE140901	nanostring_hcc_clinical_assay_transfer	stromal_exclusion	9.0	2.0	0.2222222222222222	COL3A1,FN1	True
GSE140901	nanostring_hcc_clinical_assay_transfer	trm_tls	9.0	6.0	0.6666666666666666	ITGAE,CXCR6,CXCL13,MS4A1,CD79A,LTB	True
__all__	module_feature_builder_summary	__builder_rows__					False

Supplementary Table 17 | PD1-like rescue.

Reports AUROC, AUPRC, balanced accuracy, ECE, and threshold sensitivity for locked panel scoring and PD1LikeTransferHead in GSE145996, PHS000452_LIU_LIKE_PRE, and the pooled stress cohort. This table supports Figure 5e-f and Supplementary Figure 8.

Full tabulation (6 rows x 23 columns).

endpoint	cohort	model_name	analysis_type	claim_status	n_samples	n_responders	n_nonresponders	threshold	AUROC	AUPRC	balanced_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept
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endp oint	cohort	model_na me	analysis_type	claim_status	n_sa mple s	n_res ponders	n_nonres ponders	threshold	AUROC	AUPRC	balanced_ accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_ slope	calibration_i ntercept
strict_ recist	GSE145996	EcoNiche- Opt- HeuristicEc ology- LockedPan el	pd1_like_external _rescue	locked_primary	13	8	5	0.39077794 189971365	0.575	0.77230998 16849817	0.3875	0.38461538 461538464	- 0.219577516 41341997	0.42857142 857142855	0.375	0.4	0.5	0.28571428 57142857	0.31721181 89403036	0.36971053 92890441	0.28875798 09668471	0.580429371 904649
strict_ recist	PHS000452_LIU_LIKE _PRE	EcoNiche- Opt- HeuristicEc ology- LockedPan el	pd1_like_external _rescue	locked_primary	103	48	55	0.39077794 189971365	0.57310606 06060606	0.57874656 94242914	0.56174242 42424243	0.56310679 61165048	0.123344763 52939013	0.53608247 42268041	0.54166666 66666666	0.58181818 18181818	0.53061224 48979592	0.59259259 25925926	0.31953534 325436284	0.26558137 383762903	0.14702087 454878315	- 0.126370543 58979427
strict_ recist	GSE145996+PHS000 452_LIU_LIKE_PRE	EcoNiche- Opt- HeuristicEc ology- LockedPan el	pd1_like_external _rescue_pooled	locked_primary	116	56	60	0.39077794 189971365	0.57113095 23809525	0.60008898 01278199	0.54226190 47619048	0.54310344 82758621	0.084586769 53963362	0.52252252 25225225	0.51785714 28571429	0.56666666 66666667	0.52727272 72727272	0.55737704 91803278	0.31927494 82881321	0.26129599 89566861	0.15610781 61647438	- 0.055609318 992563765
strict_ recist	GSE145996	EcoNiche- Opt- PD1LikeTra nsferHead	pd1_like_external _rescue	secondary_model_de velopment_rescue	13	8	5	0.43179023 673026123	0.575	0.75975378 78787879	0.45	0.46153846 153846156	- 0.097590007 29485333	0.53333333 33333333	0.5	0.4	0.57142857 14285714	0.33333333 33333333	0.24147861 527575393	0.12654339 642745133	1.17652745 83705074	0.559835536 4743072
strict_ recist	PHS000452_LIU_LIKE _PRE	EcoNiche- Opt- PD1LikeTra nsferHead	pd1_like_external _rescue	secondary_model_de velopment_rescue	103	48	55	0.43179023 673026123	0.59734848 48484849	0.61511329 67076572	0.52291666 66666667	0.51456310 67961165	0.047136910 74625972	0.55357142 85714286	0.64583333 33333334	0.4	0.484375	0.56410256 41025641	0.24154313 48925524	0.16212016 378097932	0.76831143 66615312	- 0.098502793 52574239
strict_ recist	GSE145996+PHS000 452_LIU_LIKE_PRE	EcoNiche- Opt- PD1LikeTra nsferHead	pd1_like_external _rescue_pooled	secondary_model_de velopment_rescue	116	56	60	0.43179023 673026123	0.59523809 52380951	0.62690796 27979104	0.5125	0.50862068 96551724	0.025637413 974602907	0.55118110 23622047	0.625	0.4	0.49295774 647887325	0.53333333 33333333	0.24153590 42458422	0.14499609 824034138	0.79122614 21679782	- 0.028736891 886128543

Supplementary Table 18 | Single-cell enrichment.

Reports cell_type, state, mean, median, and count for single-cell module localization. This table supports Figure 6a and Supplementary Figure 9.

Full tabulation (60 rows x 5 columns).

cell_type	state	mean	median	count
?	antigen_presentation_mhc	3.11578359033079	3.3140540681570227	307
?	caf_ecm_exclusion	0.6850252769480438	0.21076128873915798	307
?	myeloid_suppression	0.6394886937210953	0.268397149445724	307
?	tcell_dysfunction	0.5476949750916642	0.1547992650222346	307
?	tnk_effector	0.6111916574587489	0.0	307
?	tumor_dedifferentiation	0.9333661848579975	0.8057011004964199	307
B.cell	antigen_presentation_mhc	4.0763810748116756	4.1323831697478255	818
B.cell	caf_ecm_exclusion	0.8712172362351864	0.7644534443232289	818
B.cell	myeloid_suppression	0.35086392555040236	0.318527142978392	818
B.cell	tcell_dysfunction	0.35232443662783247	0.1885216672223328	818
B.cell	tnk_effector	0.1482891403115956	0.0	818
B.cell	tumor_dedifferentiation	1.0264875860638591	1.080542774069151	818
CAF	antigen_presentation_mhc	2.5939674808419313	2.69372100457377	106
CAF	caf_ecm_exclusion	0.301244741854516	0.0	106
CAF	myeloid_suppression	0.213627829097511	0.1613515419207101	106
CAF	tcell_dysfunction	0.1285418798864109	0.0514882741969952	106
CAF	tnk_effector	0.02222936273282603	0.0	106
CAF	tumor_dedifferentiation	1.1271717792767484	1.169465071964285	106
Endo.	antigen_presentation_mhc	4.373573725455506	4.476712073168872	104
Endo.	caf_ecm_exclusion	0.6601042645224937	0.6344841466000773	104
Endo.	myeloid_suppression	0.2961540605416332	0.20812257618660499	104

cell_type	state	mean	median	count
Endo.	tcell_dysfunction	0.20981959357830646	0.12877122318174689	104
Endo.	tnk_effector	0.04436183007558328	0.0	104
Endo.	tumor_dedifferentiation	0.9974652501667235	0.9216655078825203	104
Macrophage	antigen_presentation_mhc	5.001398103646217	5.227600983517206	420
Macrophage	caf_ecm_exclusion	1.0754214534497604	1.0265795308815981	420
Macrophage	myeloid_suppression	2.9507259285077323	2.8437094905270204	420
Macrophage	tcell_dysfunction	0.875074670207021	0.830944402073886	420
Macrophage	tnk_effector	0.5073914060795234	0.0	420
Macrophage	tumor_dedifferentiation	1.464765806501472	1.4534060004807436	420
Mal	antigen_presentation_mhc	2.9877837603862214	3.174676043571909	2018
Mal	caf_ecm_exclusion	0.38990517876611935	0.0	2018
Mal	myeloid_suppression	0.1755434461702751	0.0996501554280945	2018
Mal	tcell_dysfunction	0.12125910356170203	0.038059758325319	2018
Mal	tnk_effector	0.05144280179115324	0.0	2018
Mal	tumor_dedifferentiation	0.5673548421069978	0.4203038282185294	2018
NK	antigen_presentation_mhc	5.379257977837196	5.205364560880325	92
NK	caf_ecm_exclusion	1.708090182273928	1.4608456072024358	92
NK	myeloid_suppression	0.4807298074128988	0.34017669755540303	92
NK	tcell_dysfunction	1.446870229946974	1.336417910673284	92
NK	tnk_effector	5.681872558461493	5.83153069501032	92
NK	tumor_dedifferentiation	1.582679825788054	1.543360371656958	92
T.CD4	antigen_presentation_mhc	5.439158853269247	5.516819951468012	856
T.CD4	caf_ecm_exclusion	2.7502183953058585	2.813004799820302	856
T.CD4	myeloid_suppression	0.35893218256715753	0.289726947283353	856
T.CD4	tcell_dysfunction	1.8764501100578643	1.6312860954245894	856
T.CD4	tnk_effector	0.4703812880890118	0.0	856
T.CD4	tumor_dedifferentiation	1.8186094325829192	1.804858946362365	856
T.CD8	antigen_presentation_mhc	6.081113145777197	6.210703954787964	1759
T.CD8	caf_ecm_exclusion	3.2044347950970558	3.3106069904890196	1759
T.CD8	myeloid_suppression	0.34526720664563126	0.300364253084182	1759
T.CD8	tcell_dysfunction	3.1306978355766333	3.134697047371065	1759
T.CD8	tnk_effector	3.6576563613788196	3.5695613461822844	1759
T.CD8	tumor_dedifferentiation	2.277560347400637	2.336243645592794	1759
T.cell	antigen_presentation_mhc	5.113372649511914	5.150515086011968	706
T.cell	caf_ecm_exclusion	2.4919675544263153	2.586235108964273	706
T.cell	myeloid_suppression	0.3648426890888865	0.28665668552427315	706
T.cell	tcell_dysfunction	1.6765472253482412	1.341017013942191	706
T.cell	tnk_effector	1.086672158950994	0.0	706
T.cell	tumor_dedifferentiation	1.7345363226725927	1.6429393683322733	706

Supplementary Table 19 | Perturbation hypotheses.

Lists perturbation_name, target_gene, mechanism, reversal_score, depmap_score, DGIdb evidence, priority_score, and status. This table supports Figure 6e.

Full tabulation (100 rows x 11 columns).

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
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perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
LJP005;LJP007;LJP008;LJP009	dasatinib	CD163;GZMB;HLA-B;PRF1;PSMB9;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	8399.260227178529	-0.022772188362304956	AXL:interaction:DTC	EcoNiche module-derived resistance signature	8409.260227178529	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	KIN001-043	GNLY;GZMB;NKG7;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	7561.105067928693	-0.009918188361694395		EcoNiche module-derived resistance signature	7561.105067928693	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005;LJP006;LJP008	mitoxantrone	AXL;GNLY;GZMB;HLA-A;HLA-B;NKG7;PSMB9;S100A8;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	5343.710784556607	0.005511323700645923		EcoNiche module-derived resistance signature	5343.7658977936135	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	AZ-628	CD163;GNLY;GZMB;PRF1;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	5304.722910900314	-0.0010443942517725016		EcoNiche module-derived resistance signature	5304.722910900314	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	AZD7762	GNLY;GZMB;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	3987.7797937326095	0.05642166864966527		EcoNiche module-derived resistance signature	3988.3440104191063	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	NVP-AUY922	GZMB;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	3917.0422623651525	-0.013680949509360468		EcoNiche module-derived resistance signature	3917.0422623651525	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005;LJP006;LJP007;LJP008;LJP009	CGP-60474	AXL;CD163;CD3D;CD3E;GNLY;GZMB;HLA-B;NKG7;PSMB9;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	3797.349435107101	-0.009718999110302148		EcoNiche module-derived resistance signature	3797.349435107101	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP008;LJP009	olaparib	HLA-B;PSMB9;S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	3442.2284199038268	-0.0082302093933272		EcoNiche module-derived resistance signature	3442.2284199038268	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	BMS-387032	AXL;CD3D;GNLY;HLA-B;NKG7;PSMB9;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	3413.1161874251093	-0.0037974730530125414		EcoNiche module-derived resistance signature	3413.1161874251093	hypothesis_only	candidate perturbation hypothesis only; requires

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
										experimental validation
LJP006	AZD-5438	AXL;GNLY;GZMB;ITGB7;NKG7;PSMB9;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	3295.7666321619367	-0.00823791804541741		EcoNiche module-derived resistance signature	3295.7666321619367	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	AT-7519	AXL;CD3D;HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	2481.7829244491027	0.03548159869336113		EcoNiche module-derived resistance signature	2482.1377404360364	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	CYT387	GNLY;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	2320.241457120753	0.020214526625620545		EcoNiche module-derived resistance signature	2320.443602387009	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	alvocidib	AXL;CD3D;GZMB;HLA-B;NKG7;PSMB9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	2234.932148170935	-0.0021689243027112116		EcoNiche module-derived resistance signature	2234.932148170935	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	SAR-245408	GNLY;NKG7;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	2147.639817881552	-0.031223524084279728		EcoNiche module-derived resistance signature	2147.639817881552	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005;LJP006;LJP008;LJP009	PD-0325901	CD3D;GZMB;HLA-A;HLA-B;PSMB9;S100A8;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	2141.665561090455	0.039524828566494294		EcoNiche module-derived resistance signature	2142.06080937612	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	TG-101348	GNLY;GZMB	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	2108.670534915746	0.0851698687060263		EcoNiche module-derived resistance signature	2109.5222336028064	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	linsitinib	GZMB;HLA-B;PRF1;PSMB9;S100A9;TAP1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	2108.49735634511	-0.010393459979561766		EcoNiche module-derived resistance signature	2108.49735634511	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	PF-562271	AXL;HLA-B;NKG7;S100A9;TAP1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	2080.3432217084614	-0.053893715872956	AXL:interaction:DTC	EcoNiche module-	2090.3432217084614	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_state	priority_score	status	interpretation
			down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)		09		derived resistance signature			n hypothesis only; requires experimental validation
LJP006	OSI-027	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	2015.5024106854	-0.04752033366707151		EcoNiche module-derived resistance signature	2015.5024106854	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	GSK-429286A	S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	1994.7293337992112	0.0121064062395271		EcoNiche module-derived resistance signature	1994.8503978616066	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	selumetinib	GNLY;HLA-A;HLA-B;NKG7;PRF1;PSMB9;S100A8;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	1989.8552872658195	0.0036488418438134105		EcoNiche module-derived resistance signature	1989.8917756842577	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	A443654	AXL;CD163;CD3D;HLA-B;PSMB9;S100A8;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	1813.6847261710009	-0.0003386812096094443		EcoNiche module-derived resistance signature	1813.6847261710009	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	dabrafenib	GNLY;GZMB;PRF1;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1788.1338797886938	0.023338035148561472		EcoNiche module-derived resistance signature	1788.3672601401795	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	YM-201636	GNLY;GZMB;PRF1;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1781.6692644151392	0.023338035148561472		EcoNiche module-derived resistance signature	1781.902644766625	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	WZ-7043	GNLY;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1775.908514460057	0.020214526625620545		EcoNiche module-derived resistance signature	1776.110659726313	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	AS-605240	GNLY;PSMB9;S100A8;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	1731.134539862112	0.0001556643843102404		EcoNiche module-derived resistance signature	1731.136096505955	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgldb_evidence	target_status	priority_score	status	interpretation
LJP007	KIN001-266	GNLY;GZMB;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1715.913639205 3172	0.056421668649665 27		EcoNiche module- derived resistance signature	1716.477855891 8138	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP007	tivantinib	GNLY;GZMB	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1707.144059920 3274	0.085169868706026 3		EcoNiche module- derived resistance signature	1707.995758607 3876	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP008	I-BET151	CD163;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1637.544059049 697	- 0.058510457153210 266		EcoNiche module- derived resistance signature	1637.544059049 697	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP009	PF 573228	GNLY;GZMB;HLA-A;HLA-B	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	1473.094427521 7976	0.083212943936272 24		EcoNiche module- derived resistance signature	1473.926556961 1603	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP007	SB-525334	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1334.187512688 9731	- 0.047520333667071 51		EcoNiche module- derived resistance signature	1334.187512688 9731	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP005	XMD16-144	GNLY;GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1282.315870553 066	0.031475624019100 9		EcoNiche module- derived resistance signature	1282.630626793 2569	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP008	AZ20	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1268.042169137 566	- 0.078201784740118 85		EcoNiche module- derived resistance signature	1268.042169137 566	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP006	ZSTK-474	GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1253.436507233 37	- 0.010957523274344 152		EcoNiche module- derived resistance signature	1253.436507233 37	hypothesis_ only	validation candidate perturbation hypothesis only; requires experimental validation
LJP005	TWS-119	GNLY;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1181.011696443 937	0.057633593571467 1		EcoNiche module- derived resistance	1181.588032379 6516	hypothesis_ only	validation candidate perturbation hypothesis

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgldb_evidence	target_status	priority_score	status	interpretation
							signature			only; requires experimental validation
LJP006	afatinib	NKG7;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1147.5328828164716	-0.14242531237526165		EcoNiche module-derived resistance signature	1147.5328828164716	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005;LJP006	geldanamycin	HLA-B;IFNG;PSMB9;S100A8;S100A9;TGFB1;ZEB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	1130.8671297344092	-0.033417942750203725		EcoNiche module-derived resistance signature	1130.8671297344092	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	BMS-345541	AXL;CD3D;HLA-B;PSMB9;S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	1092.3087904493457	0.030806560202594323		EcoNiche module-derived resistance signature	1092.6168560513715	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	regorafenib	GZMB;HLA-B;PSMB9;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	1060.2324757072302	-0.01467802024204925		EcoNiche module-derived resistance signature	1060.2324757072302	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC013	BRD-K18163752	CD163;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1056.5728485287273	-0.04980925305244045		EcoNiche module-derived resistance signature	1056.5728485287273	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	XMD11-85H	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1056.5728485287273	-0.07820178474011885		EcoNiche module-derived resistance signature	1056.5728485287273	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP008	MK-1775	GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1056.5728485287273	-0.010957523274344152		EcoNiche module-derived resistance signature	1056.5728485287273	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	ALW-II-38-3	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1029.0547394171365	-0.07820178474011885		EcoNiche module-derived resistance signature	1029.0547394171365	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
LJP008	CX-5461	GNLY;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1002.788738497 8784	0.020214526625620 545		EcoNiche module- derived resistance signature	1002.990883764 1347	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP007	NVP-TAE226	GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	1002.788738497 8784	- 0.010957523274344 152		EcoNiche module- derived resistance signature	1002.788738497 8784	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP005	SB590885	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	977.6829662439 23	- 0.078201784740118 85		EcoNiche module- derived resistance signature	977.6829662439 23	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP005	PF-431396	AXL;HLA-B;NKG7;PSMB9;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	944.5409441435 55	- 0.059904793501697 515		EcoNiche module- derived resistance signature	944.5409441435 55	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP005	trametinib	PSMB9;S100A8;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	942.7574683183 478	- 0.028890899171109 95		EcoNiche module- derived resistance signature	942.7574683183 478	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP007	pazopanib	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	930.6854451805 837	- 0.078201784740118 85	HLA-B:interaction:FDA;PharmGKB	EcoNiche module- derived resistance signature	940.6854451805 837	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP009	SYK-inhibitor	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	930.6854451805 837	- 0.078201784740118 85		EcoNiche module- derived resistance signature	930.6854451805 837	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP005	LDN-193189	CD3D;HLA-B;PSMB9;ZEB1	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	914.7538278781 002	0.024602275653682 826		EcoNiche module- derived resistance signature	914.9998506346 37	hypothesis_ only	candidate perturbation hypothesis only; requires experimental validation
LJP005	enzastaurin	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	887.5482208279 581	- 0.047520333667071 51		EcoNiche module- derived resistance signature	887.5482208279 581	hypothesis_ only	candidate perturbation hypothesis only; requires

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
										experimental validation
LJP009	GSK-2334470	NKG7;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	847.8218659162171	-0.14242531237526165		EcoNiche module-derived resistance signature	847.8218659162171	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	BMS-777607	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	829.1295407023977	-0.04752033366707151	AXL.inhibitor:INHIBITORY:GuideToPharmacology	EcoNiche module-derived resistance signature	839.1295407023977	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP008	(-)-JQ1	GZMB;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	829.1295407023977	0.026461543671502403		EcoNiche module-derived resistance signature	829.3941561391127	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006;LJP009	BI-2536	HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	777.6838487944024	-0.01839851720975435		EcoNiche module-derived resistance signature	777.6838487944024	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	JW-7-24-1	HLA-B;PSMB9;S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	766.3598195229283	-0.032959425438412925		EcoNiche module-derived resistance signature	766.3598195229283	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	OTSSP167	CD3E;HLA-A;HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	740.3431510192247	0.02144197708267625		EcoNiche module-derived resistance signature	740.5575707900515	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC006	vorinostat	GZMB;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	703.6051178996023	0.01743500841333425		EcoNiche module-derived resistance signature	703.7794679837357	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC006	mln2238	GZMB;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	675.8946680832967	0.01743500841333425		EcoNiche module-derived resistance signature	676.06901816743	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC014	BRD-K83972459	CD163;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;	675.8946680832967	-0.049809253052440		EcoNiche module-	675.8946680832967	hypothesis_only	candidate perturbation

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_state	priority_score	status	interpretation
			LINCS_L1000_Chem_Pert_down)		45		derived resistance signature			n hypothesis only; requires experimental validation
LJP007	motesanib	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	663.3215252321327	-0.07820178474011885		EcoNiche module-derived resistance signature	663.3215252321327	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	DCC-2036	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	663.3215252321327	-0.04752033366707151		EcoNiche module-derived resistance signature	663.3215252321327	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	AS-601245	CD3D;GZMB;PRF1;S100A8	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	657.3268168597881	0.055407307778485175		EcoNiche module-derived resistance signature	657.880889937573	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	fostamatinib	NKG7;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	639.36534926789	-0.1050062454294151		EcoNiche module-derived resistance signature	639.36534926789	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	GSK-1904529A	PRF1;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	627.9545623331182	-0.03849379840890335		EcoNiche module-derived resistance signature	627.9545623331182	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	A66	CD163;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	627.9545623331182	-0.07820178474011885		EcoNiche module-derived resistance signature	627.9545623331182	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	AT7867	GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	606.1729409700409	-0.010957523274344152		EcoNiche module-derived resistance signature	606.1729409700409	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	dovitinib	CD3D;S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	593.7115498483283	0.08435307188546795	AXL:interaction:DTC	EcoNiche module-derived resistance signature	604.555080567183	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
LJP006	QL-XII-47	GZMB;S100A8;S100A9;ZEB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes;upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_down;LINCS_L1000_Chem_Pert_up)	597.6066270799162	-0.007800560201582501		EcoNiche module-derived resistance signature	597.6066270799162	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
LJP005	WH-4-025	HLA-B;S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	575.708138298268	0.003969353704921149		EcoNiche module-derived resistance signature	575.7478318353171	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
CPC002	beclomethasone dipropionate	CD163;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	566.3522452274439	-0.04980925305244045		EcoNiche module-derived resistance signature	566.3522452274439	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
LJP006	nintedanib	AXL;HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	563.9559820929466	-0.004891114252654767		EcoNiche module-derived resistance signature	563.9559820929466	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
LJP005	ruxolitinib	HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	558.6362708435369	-0.01839851720975435		EcoNiche module-derived resistance signature	558.6362708435369	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
LJP005	radicicol	GNLY;GZMB	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	548.1302390811057	0.0851698687060263		EcoNiche module-derived resistance signature	548.981937768166	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
LJP009	JNK-IN-5A	GNLY;NKG7	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	539.40376141366	-0.046297920394891195		EcoNiche module-derived resistance signature	539.40376141366	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
CPC010	itavastatin ca	CD163;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	539.40376141366	-0.04980925305244045		EcoNiche module-derived resistance signature	539.40376141366	hypothesis_only	validation candidate perturbation hypothesis only; requires experimental validation
LJP009	KIN001-265	GZMB;PRF1;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	532.5279286720727	-0.007663259337248367		EcoNiche module-derived resistance	532.5279286720727	hypothesis_only	validation candidate perturbation hypothesis

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
							signature			only; requires experimental validation
LJP005	nilotinib	HLA-A;HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	493.565058915453	0.043294234247737505		EcoNiche module-derived resistance signature	493.99800125793035	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	NU-7441	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	484.5243374089621	-0.04752033366707151		EcoNiche module-derived resistance signature	484.5243374089621	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	canertinib	NKG7;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	477.4765813516793	-0.11403278068758324		EcoNiche module-derived resistance signature	477.4765813516793	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	WZ-3105	HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	475.0663787936278	-0.01839851720975435		EcoNiche module-derived resistance signature	475.0663787936278	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	AZD-7762	HLA-B;S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	448.69594101976713	0.003969353704921149		EcoNiche module-derived resistance signature	448.73563455681636	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	PI-103	GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	426.9383319477178	-0.010957523274344152		EcoNiche module-derived resistance signature	426.9383319477178	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP008	(+)-JQ1	GZMB;PRF1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	415.7481041425479	0.026461543671502403		EcoNiche module-derived resistance signature	416.0127195792629	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	pelitinib	S100A9;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	389.8675963773491	-0.04752033366707151		EcoNiche module-derived resistance signature	389.8675963773491	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgldb_evidence	target_status	priority_score	status	interpretation
CPC014	BRD-A52193669	CD163;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	375.64502004307246	-0.04980925305244045		EcoNiche module-derived resistance signature	375.64502004307246	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP008	tanespimycin	GNLY;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	371.1055001106239	0.020214526625620545		EcoNiche module-derived resistance signature	371.3076453768801	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC006	hdac6 inhibitor isox	GZMB;TGFB1	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	341.8061637333013	0.01743500841333425		EcoNiche module-derived resistance signature	341.9805138174346	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC006	BRD-K31342827	CD163;GZMB	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	334.1585333066493	-0.013246442659713097		EcoNiche module-derived resistance signature	334.1585333066493	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP007	INK-128	GZMB;S100A9	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	330.4467876087468	-0.010957523274344152		EcoNiche module-derived resistance signature	330.4467876087468	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
CPC012	BRD-K72238567	CD163;GZMB	LINCS Enrichr signature reversal (downregulate_resistance_up_genes; LINCS_L1000_Chem_Pert_down)	323.23297793596987	-0.013246442659713097		EcoNiche module-derived resistance signature	323.23297793596987	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	sorafenib	AXL;HLA-A;HLA-B	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	304.4552133478669	0.06154524333152694	AXL:interaction:DTC	EcoNiche module-derived resistance signature	315.0706657811822	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	serdemetan	HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	292.0827310778606	-0.01839851720975435		EcoNiche module-derived resistance signature	292.0827310778606	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	PHA-793887	AXL;HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	251.29044920067383	-0.004891114252654767		EcoNiche module-derived resistance signature	251.29044920067383	hypothesis_only	candidate perturbation hypothesis only; requires

perturbation_id	perturbation_name	target_gene	mechanism	reversal_score	depmap_score	dgidb_evidence	target_status	priority_score	status	interpretation
										experimental validation
LJP006	JNK-9L	AXL;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	230.55577976215676	-0.00525282196413975		EcoNiche module-derived resistance signature	230.55577976215676	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP006	MK-2206	HLA-A;HLA-B	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	201.77998097385256	0.0812560191665182		EcoNiche module-derived resistance signature	202.59254116551773	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	RAF 265	HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	193.517528458574	-0.01839851720975435		EcoNiche module-derived resistance signature	193.517528458574	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP009	BX-795	HLA-B;PSMB9	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	183.36401607846983	-0.01839851720975435		EcoNiche module-derived resistance signature	183.36401607846983	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation
LJP005	AZD-8330	HLA-B;S100A8	LINCS Enrichr signature reversal (upregulate_resistance_down_genes; LINCS_L1000_Chem_Pert_up)	148.82592893800745	0.003969353704921149		EcoNiche module-derived resistance signature	148.86562247505665	hypothesis_only	candidate perturbation hypothesis only; requires experimental validation

Supplementary Table 20 | Algorithmic reproducibility and independent-scoring boundary.

Summarizes the reproducible analysis boundary from public ICB cohort curation, endpoint harmonization, signed-rank ecological modules, interaction features, heuristic optimization, threshold locking, external scoring, coverage QC, signature-family benchmarking, and the FDR-aware claim gate to open scoring code. This table clarifies which information is learned in the discovery layer, which information is read only during external evaluation, and how EcoNiche-Opt becomes an independently scorable method without using external labels for model selection.

Full tabulation (19 rows x 5 columns).

domain	reproducible_element	status	analysis_boundary	scientific_role
cohort evidence	public ICB expression cohorts	reported	retrospective public cohorts	Defines the multicohort response-prediction setting used for discovery and external evaluation.
cohort evidence	patient-sample-response linkage	reported	sample-level curation	Links each expression profile to patient, treatment timepoint and response evidence before model fitting.
endpoint definition	primary RECIST endpoint	reported	predefined response mapping	Treats CR/PR as response and SD/PD as non-response for the main binary endpoint.
endpoint definition	strict RECIST endpoint	reported	sensitivity analysis	Compares CR/PR against PD after excluding

domain	reproducible_element	status	analysis_boundary	scientific_role
				intermediate SD/MR samples.
endpoint definition	clinical benefit endpoint	reported	sensitivity analysis	Evaluates CR/PR/SD or durable benefit against progression/non-benefit where supported by cohort metadata.
feature construction	signed-rank ecological module scores	reported	training-defined directions	Transforms expression into response-high and resistance-high module activities without using held-out labels.
feature construction	ecological interaction features	reported	training-only graph construction	Aggregates biologically signed state-state gene interactions as model features.
model fitting	class-balanced logistic model	reported	discovery cohorts only	Estimates response probability from module and interaction features under class imbalance.
optimizer	heuristic ecological module search	reported	inner training folds only	Searches module composition and interaction edges with biological compactness and leakage-safety constraints.
thresholding	endpoint-specific locked thresholds	reported	discovery cohorts only	Selects fixed decision thresholds before external cohort scoring.
calibration	calibration object	reported	discovery cohorts only	Keeps probability calibration separated from locked external outcome labels.
external scoring	one-way independent cohort scoring	reported	no model refit	Scores external cohorts once and evaluates discrimination and calibration only after predictions are fixed.
external scoring	gene coverage assessment	reported	assay-aware QC	Reports module coverage so platform transfer does not silently change the biological score.
benchmarking	signature-family comparison	reported	paired evaluation	Compares EcoNiche-Opt against IFNG/CXCL9/TIG/APM/TIDE-like immune signatures under shared samples.
claim boundary	FDR-aware superiority gate	reported	paired bootstrap or family test	Separates FDR-supported statements from point-estimate-only improvements.
mechanism	single-cell ecological enrichment	reported	post hoc interpretation	Maps bulk ecological modules to immune, stromal and myeloid cell-state patterns.
mechanism	hypothesis-only perturbation ranking	reported	not treatment recommendation	Uses external perturbation resources only to generate mechanistic hypotheses.
reproducibility	open scoring code	available	same frozen algorithm	Allows independent users to apply the published scoring rule to normalized expression matrices.
reproducibility	controlled-data boundary	reported	no substitute data	Marks access-restricted cohorts as non-shareable while preserving interfaces and analysis rules.

Supplementary Table 21 | GPU biological-prior rescue combo.

Reports the frozen GPU lipid/PI3K rescue-combo candidate, biological prior, transform policy, GPU device, primary melanoma LODO metrics, strict melanoma external metrics, family-level delta, FDR q value, claim level, and no-leakage selection boundary. This table supports Figure 5e.

Full tabulation (1 rows x 18 columns).

selection_id	candidate	prior	transform_policy	device	gpu_name	selection_boundary	primary_AUROC	primary_AUPRC	primary_balanced_accuracy	strict_external_AUROC	strict_external_AUPRC	strict_external_balanced_accuracy	strict_external_AECE	family_mean_AUROC	delta_vs_family_mean	two_sided_fdr_q	claim_level
gpu_lipid_pi3k_primary_selected_rescue_combo	0.80*base+0.20*rz_PLA2G2D	lipid_pi3k	robust_only	cuda	NVIDIA GeForce RTX 4070 Laptop GPU	candidate_and_weight_selected_by_primary_lodo_only_with_biological_prior	0.7774984286612194	0.6645221072897123	0.6585480829666877	0.7125	0.7260085416533733	0.6285714285714286	0.12765595267109317	0.5666666666666667	0.14649167160228566	0.0	strict_external_family_FDR_supported_numeric_target_met

Supplementary Table 22 | GPU rescue component ablation.

Compares the selected GPU lipid/PI3K rescue combo with the ablated MAP4K1/TBX3/AXL base rescue axis in primary melanoma LODO and strict melanoma external settings. It reports AUROC, AUPRC, balanced accuracy, paired bootstrap intervals, P values, FDR q values, and claim levels. This table supports Figure 5e and the component-ablation claim boundary.

Full tabulation (6 rows x 16 columns).

context	comparison_id	target_candidate	ablated_candidate	metric	target_value	ablated_value	delta	bootstrap_mean_delta	ci_low	ci_high	two_sided_p	n_samples	claim_boundary	two_sided_fdr_q	claim_level
primary_melanoma_lodo	selected_gpu_combo_vs_base_rescue	0.80*base+0.20*rz_PLA2G2D	base_rescue_robust	AUROC	0.7774984286612194	0.7724701445631679	0.005028284098051561	0.005059892898517561	-0.026856231231231285	0.036841069269225826	0.742	117	component_ablation_evaluation_not_used_for_candidate_selection	0.785	point_estimate_component_gain
primary_melanoma_lodo	selected_gpu_combo_vs_base_rescue	0.80*base+0.20*rz_PLA2G2D	base_rescue_robust	AUPRC	0.6645221072897123	0.657990756879542	0.006531350410170278	0.00660487143041365	-0.035999926035953825	0.053917530124004374	0.785	117	component_ablation_evaluation_not_used_for_candidate_selection	0.785	point_estimate_component_gain
primary_melanoma_lodo	selected_gpu_combo_vs_base_rescue	0.80*base+0.20*rz_PLA2G2D	base_rescue_robust	balanced_accuracy	0.7302011313639221	0.7253299811439347	0.004871150219987408					117	threshold_metric_reported_without_bootstrap_claim		point_estimate_component_gain
strict_melanoma_external	selected_gpu_combo_vs_base_rescue	0.80*base+0.20*rz_PLA2G2D	base_rescue_robust	AUROC	0.7125	0.6860119047619048	0.026488095238095255	0.026661824362333605	-0.016727530939338726	0.07390311004784686	0.24	116	component_ablation_evaluation_not_used_for_candidate_selection	0.48	point_estimate_component_gain
strict_melanoma_external	selected_gpu_combo_vs_base_rescue	0.80*base+0.20*rz_PLA2G2D	base_rescue_robust	AUPRC	0.7260085416533733	0.6892418361322834	0.03676670552108985	0.03547334231284006	-0.01793143100225034	0.09242319289478022	0.198	116	component_ablation_evaluation_not_used_for_candidate_selection	0.48	point_estimate_component_gain
strict_melanoma_external	selected_gpu_combo_vs_base_rescue	0.80*base+0.20*rz_PLA2G2D	base_rescue_robust	balanced_accuracy	0.6285714285714286	0.6738095238095239	-0.0452380952380953					116	threshold_metric_reported_without_bootstrap_claim		component_tradeoff_or_no_gain

Supplementary Table 23 | cBioPortal GPU rescue external validation.

Reports cBioPortal Liu/DFCI and cBioPortal Liu plus GSE145996 strict RECIST metrics for the frozen GPU lipid/PI3K rescue combo, including AUROC, AUPRC, balanced accuracy, ECE, Brier score, sensitivity, specificity, PPV, NPV, calibration slope/intercept, and no-leakage scoring boundary. This table supports Figure 5f.

Full tabulation (2 rows x 21 columns).

group_id	model_name	endoint	n_samples	n_responders	n_nonresponders	AUROC	AUPRC	balanced_accuracy	accuracy	MCC	F1	sensitivity	specificity	PPV	NPV	Brier	ECE	calibration_slope	calibration_intercept	selection_boundary
cbio_liu_dfci_only	EcoNiche-Opt-GPU-LipidPI3K-RescueCombo	strict_recist	103	47	56	0.6736322188449847	0.6793733927288442	0.611322188449848	0.5922330097087378	0.2440802368853342	0.65	0.8297872340425532	0.39285714285714285	0.5342465753424658	0.7333333333333333	0.22197684043721158	0.08948088437046826	0.9303055184278347	0.027665329486126322	locked_gpu_bioprior_package_no_cbio_label_fit
strict_cbio_liu_plus_gse145996	EcoNiche-Opt-GPU-LipidPI3K-RescueCombo	strict_recist	116	55	61	0.6858420268256333	0.7067143698235946	0.6131147540983607	0.603448275862069	0.24237125183447317	0.6567164179104478	0.8	0.4262295081967213	0.5569620253164557	0.7027027027027027	0.21811932480405544	0.0889918384488029	0.947113757369138	0.11901806313456241	locked_gpu_bioprior_package_no_cbio_label_fit

Supplementary Table 24 | GPU lipid/PI3K pair rescue.

Reports the component-dominant PLA2G2D/PIK3CD lipid/PI3K pair rescue selected by primary melanoma LODO with a balanced-accuracy guardrail. It includes the locked candidate formula, GPU device, primary AUROC/AUPRC/balanced accuracy, strict melanoma external metrics, cBioPortal Liu/DFCI metrics, family-level delta, FDR q value, claim level, and no-leakage selection boundary. This table supports Figure 5e-f.

Full tabulation (1 rows x 25 columns).

selection_id	candidate	prior	base_weight_policy	selection_policy	min_primary_balanced_accuracy_guardrail	device	gpu_name	selection_boundary	primary_AUROC	primary_AUPRC	primary_balanced_accuracy	primary_min_fold_AUROC	strict_external_AUROC	strict_external_AUPRC	strict_external_balanced_accuracy	strict_external_ECE	cbio_liu_AUROC	cbio_liu_AUPRC	cbio_liu_balanced_accuracy	cbio_liu_ECE	cbio_family_mean_AUROC	cbio_delta_vs_family_mean	cbio_two_sided_fdr_q	cbio_claim_level
gpu_lipid_pair_component_dominant_primary_selected_rescue	0.35*base+0.65*(0.35*pct_PLA2G2D+0.65*rz_PIK3CD)	lipid_pi3k_pair	component_dominant	balanced_accuracy_guardrail	0.65	cuda	NVIDIA GeForce	pair_candidate_and_weights_selected_by_primary_lodo_only_lipid_pi3k_prior_no_external_labels	0.7715273412947832	0.6992237965455167	0.6855751099937146	0.5604395604395604	0.7196428571428571	0.7458297351553689	0.6452380952380953	0.0803501927371498	0.7006079027355624	0.7047481346214961	0.5655395845242638728	0.07672845242638728	0.5645896656534954	0.13623041031476163	0.0	strict_external_family_FDR_supported_numeric_target_met

selection_id	candidate	prior	base_weight_policy	select_ion_policy	min_primary_balanced_accuracy_guardrail	d_evice	gpu_name	selection_boundary	primary_AUROC	primary_AUPRC	primary_balanced_accuracy	primary_min_fold_AUROC	strict_external_AUROC	strict_external_AUPRC	strict_external_balanced_accuracy	strict_external_ECE	cbio_liu_AUROC	cbio_liu_AUPRC	cbio_liu_balanced_accuracy	cbio_liu_ECE	cbio_family_mean_AUROC	cbio_delta_vs_family_mean	cbio_two_sided_fdr_q	cbio_claim_level
							ce RT X 407 0 Lap top GPU																	

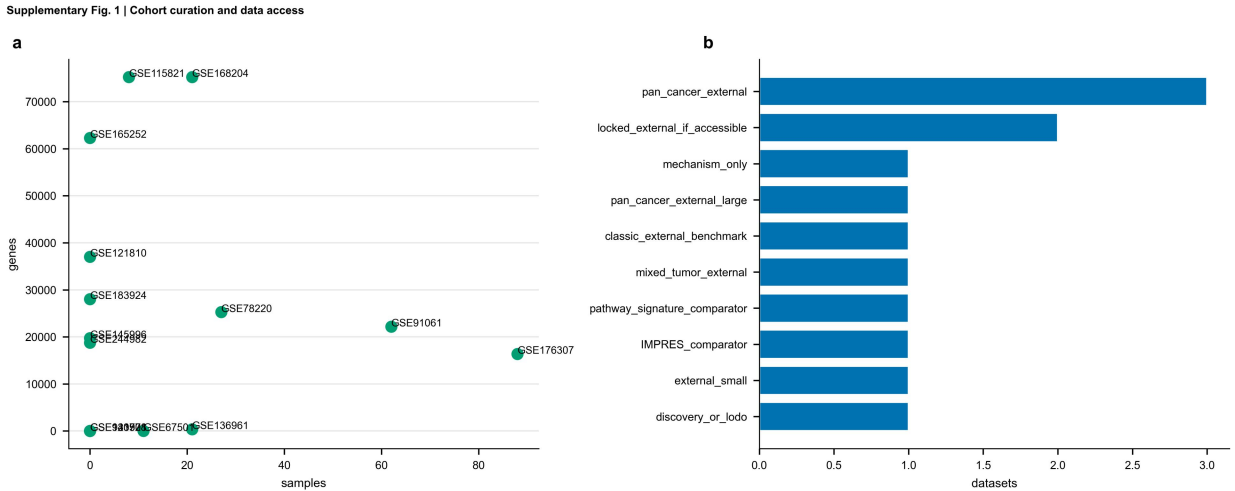
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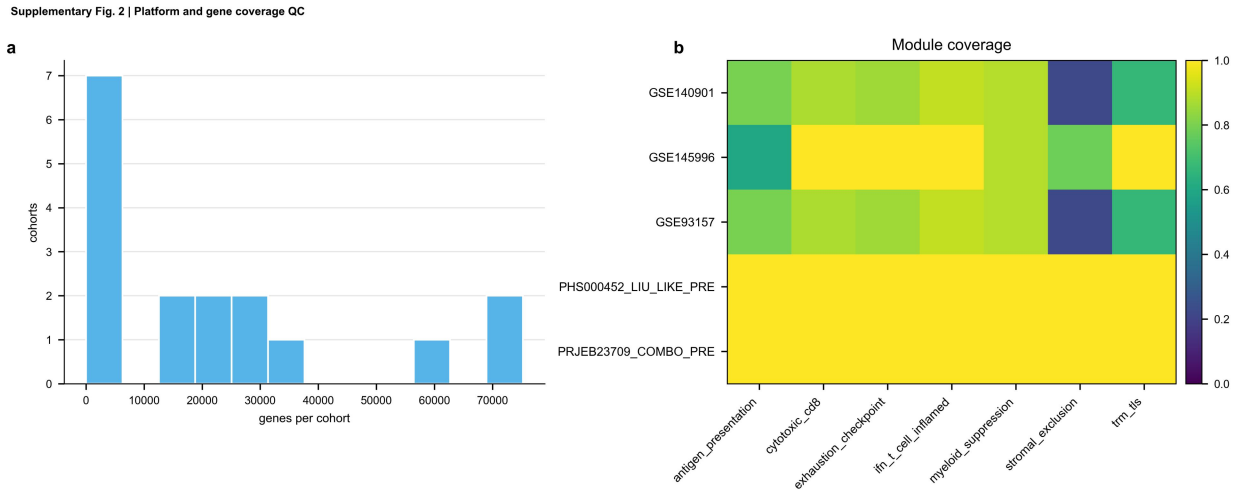
Supplementary Figures

Supplementary Figure 1 | Cohort curation and data access.



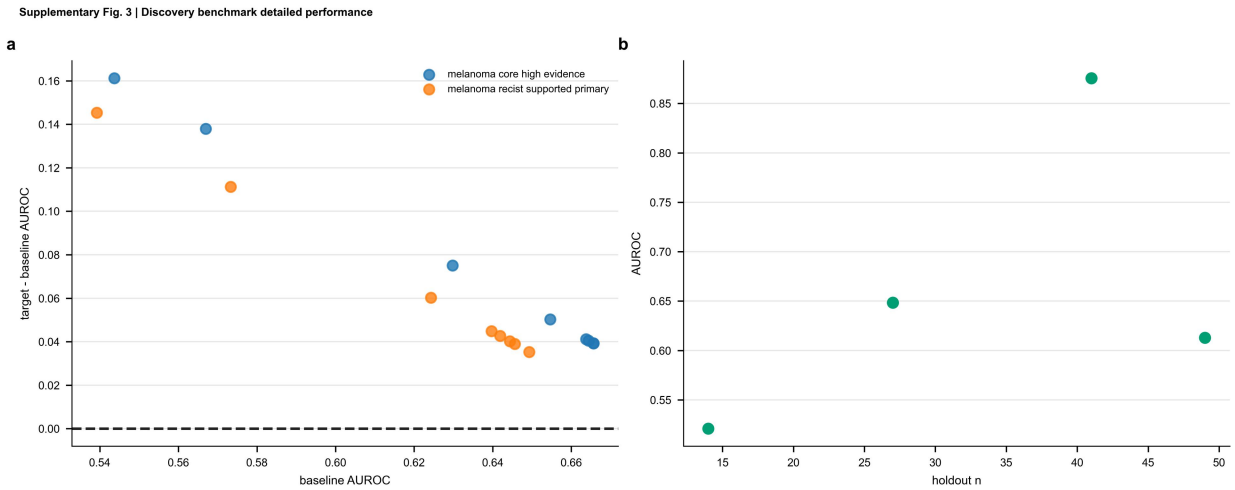
Extends the cohort-curation component of Figure 1 by showing sample size, data access status, cohort role, and inclusion boundary for each registered data source. This figure corresponds to Supplementary Tables 1-5 and documents the traceable evidence for cohort registration, access-status assessment, manual curation, and external cohort availability.

Supplementary Figure 2 | Platform and gene coverage QC.



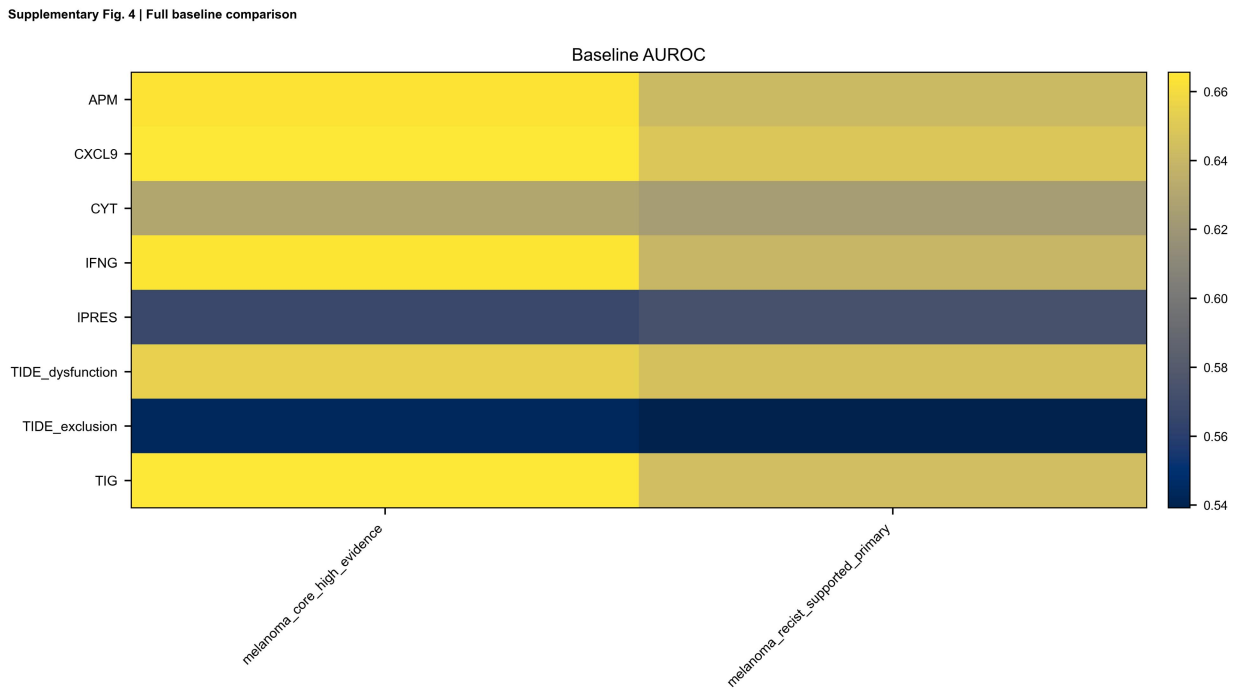
Extends Figure 1f by showing gene coverage across expression platforms, cohorts, and ecological modules. The figure identifies which cohorts can fully compute IFN/T inflamed, CD8 cytotoxic, checkpoint/exhaustion, antigen presentation, myeloid suppression, stromal exclusion, and TRM/TLS modules and which cohorts show panel gene loss. It corresponds to Supplementary Tables 3, 7, and 16.

Supplementary Figure 3 | Discovery benchmark detailed performance.



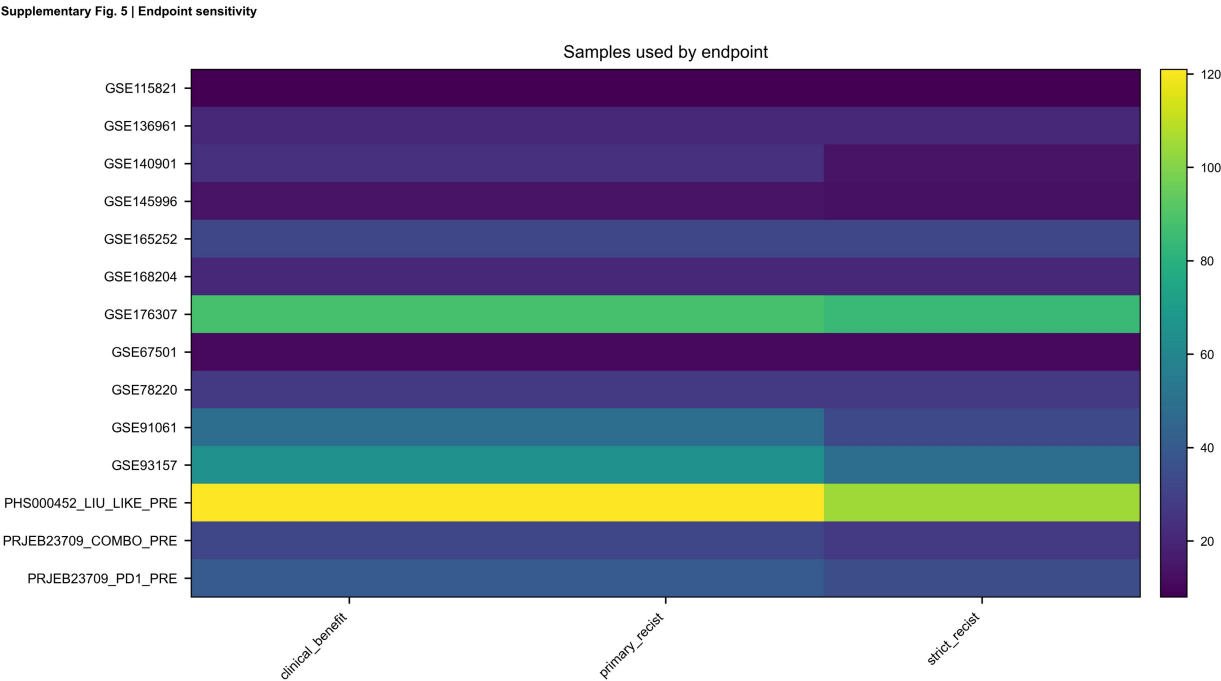
Extends the melanoma benchmark in Figure 3 by showing discovery/LODO AUROC, AUPRC, balanced accuracy, and holdout cohort heterogeneity. It supports the detailed numerical interpretation of primary melanoma performance and corresponds to Supplementary Tables 10 and 12.

Supplementary Figure 4 | Full baseline comparison.



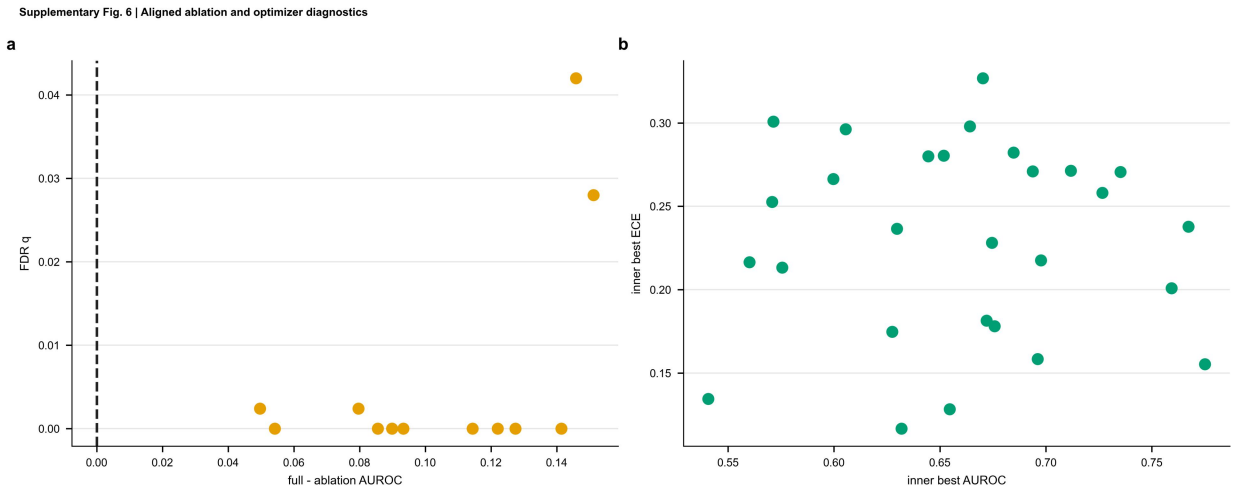
Shows EcoNiche-Opt against the broader baseline set, including expanded signatures or model variants beyond the strong-signature family. It supports the conclusion that EcoNiche-Opt's advantage is concentrated in family-level comparison and ecological-module structure and corresponds to Supplementary Tables 10-12.

Supplementary Figure 5 | Endpoint sensitivity.



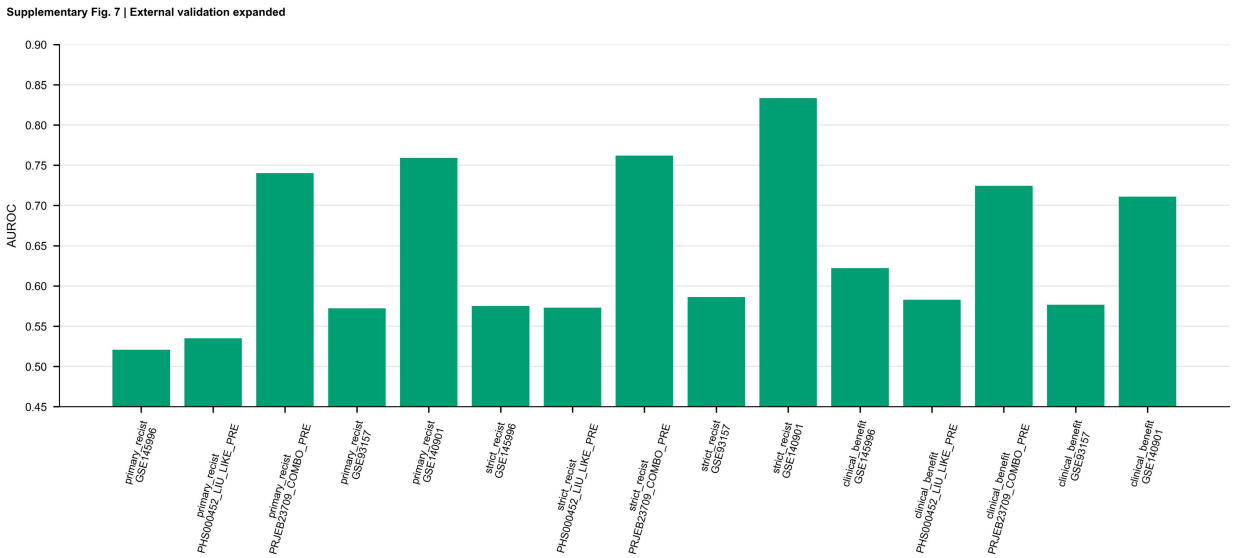
Extends Figure 4a by showing how strict RECIST, primary RECIST, and clinical-benefit endpoints affect sample inclusion, response balance, and model evaluation. It corresponds to Supplementary Table 6 and demonstrates why endpoint harmonization is required.

Supplementary Figure 6 | Aligned ablation and optimizer diagnostics.



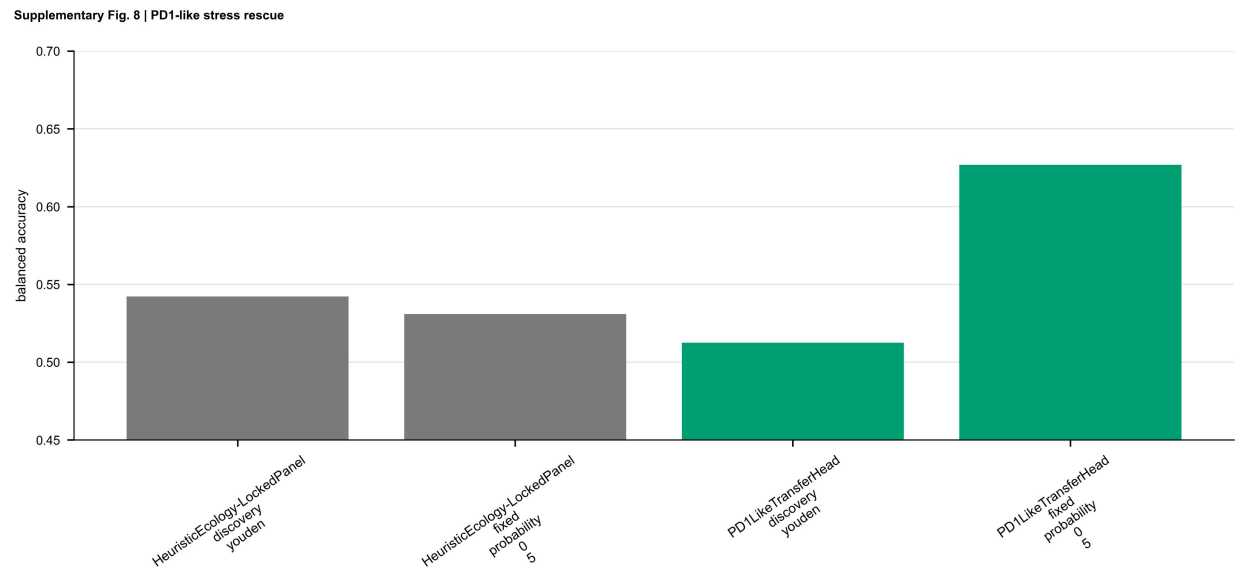
Extends Figures 2 and 4b by showing aligned locked-panel component ablation together with optimizer diagnostics for the ecological graph search space. It corresponds to Supplementary Tables 8, 9, and 13.

Supplementary Figure 7 | Expanded locked external validation.



Extends Figure 5b-c by showing performance of locked external and panel-transfer cohorts under primary RECIST, strict RECIST, and clinical-benefit endpoints. It corresponds to Supplementary Table 15.

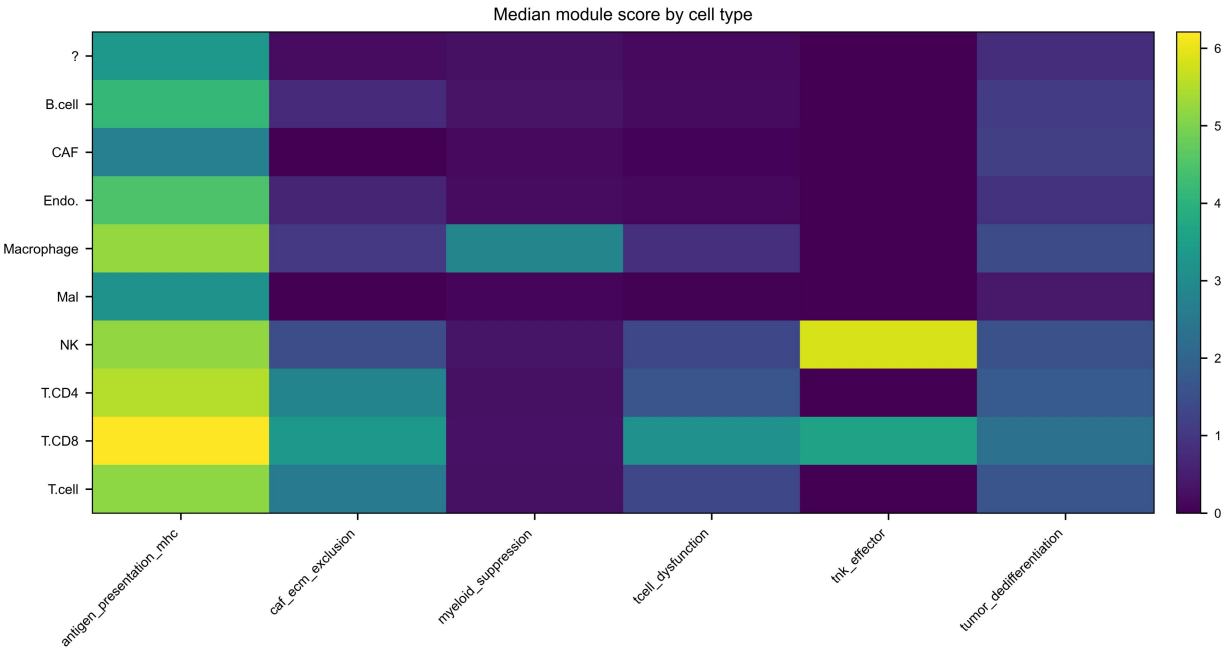
Supplementary Figure 8 | PD1-like and GPU biological-prior rescue.



Extends Figure 5e-f by showing strict RECIST PD1-like stress analysis in GSE145996 and PHS000452_LIU_LIKE_PRE. It includes the original locked-panel stress analysis, the frozen GPU lipid/PI3K rescue combo, the balanced-accuracy-guarded PLA2G2D/PIK3CD pair rescue, component ablation against the MAP4K1/TBX3/AXL base axis, and cBioPortal Liu/DFCI source cross-check evidence. It corresponds to Supplementary Tables 17 and 21-24.

Supplementary Figure 9 | Single-cell and ecological mechanism.

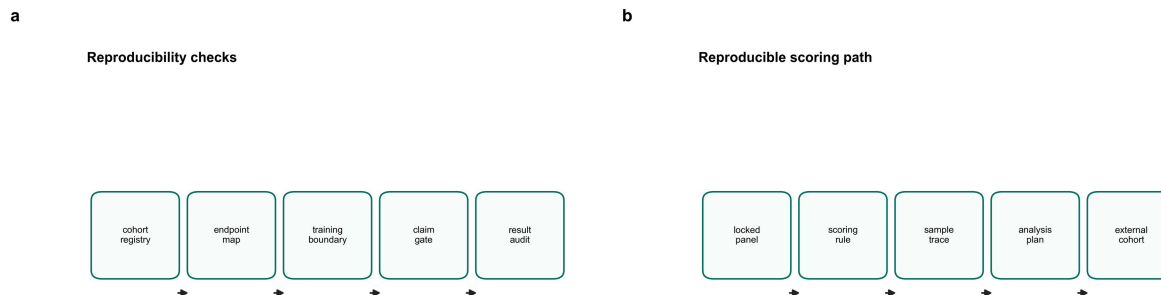
Supplementary Fig. 9 | Single-cell and ecological mechanism



Extends Figure 6a-b by showing module-score distributions across single-cell cell types and the interaction structure among ecological states. It corresponds to Supplementary Table 18 and supports the biological interpretability of the model.

Supplementary Figure 10 | Reproducibility path for EcoNiche-Opt.

Supplementary Fig. 10 | Reproducibility path for locked external scoring



Extends Figure 7 by showing how cohort curation, locked scoring rules, sample-level traceability, prespecified endpoint analysis, external scoring boundaries, and open reproducibility interfaces support independent rescoring. It corresponds to Supplementary Table 20.

Supplementary Methods

Cohort registration and access control

All data sources first underwent cohort registration and then proceeded to expression QC, label curation, and model evaluation. Controlled or unknown-access sources retain their access status and are not replaced with simulated data. Analyses depending on inaccessible data are reported as access-restricted rather than filled with substitutes.

Label traceability

Response-label curation requires each sample_id to be traceable to subject_id, treatment, timepoint, response_raw, and evidence source. CR/PR/SD/PD, R/NR, and DCB/NDB labels are mapped to three endpoints while preserving the original response_raw annotation.

Leakage-safe evaluation

Training, feature selection, gene-direction estimation, thresholding, calibration, and model selection use training data only. Holdout, locked external, and panel-transfer cohorts do not participate in model selection.

Claim gate

Superiority claims are used only when paired bootstrap or DeLong-compatible comparisons pass FDR correction. Family-level and individual-signature claims are reported separately. Perturbation rankings are hypothesis-only outputs.

Reporting and reproducibility

The study follows prediction-model reporting logic for data sources, sample inclusion, endpoint definitions, model formulas, training/validation splits, performance metrics, calibration, decision-curve analysis, external validation, code availability, and data availability. The open reproducibility layer documents how the same frozen algorithm preserves module scores, interaction features, thresholds, and probability outputs across Python/R environments, allowing readers to distinguish discovery training, external scoring, and performance evaluation. Figures 1-7, Supplementary Figures 1-10, and Supplementary Tables 1-24 form the reporting evidence map.